

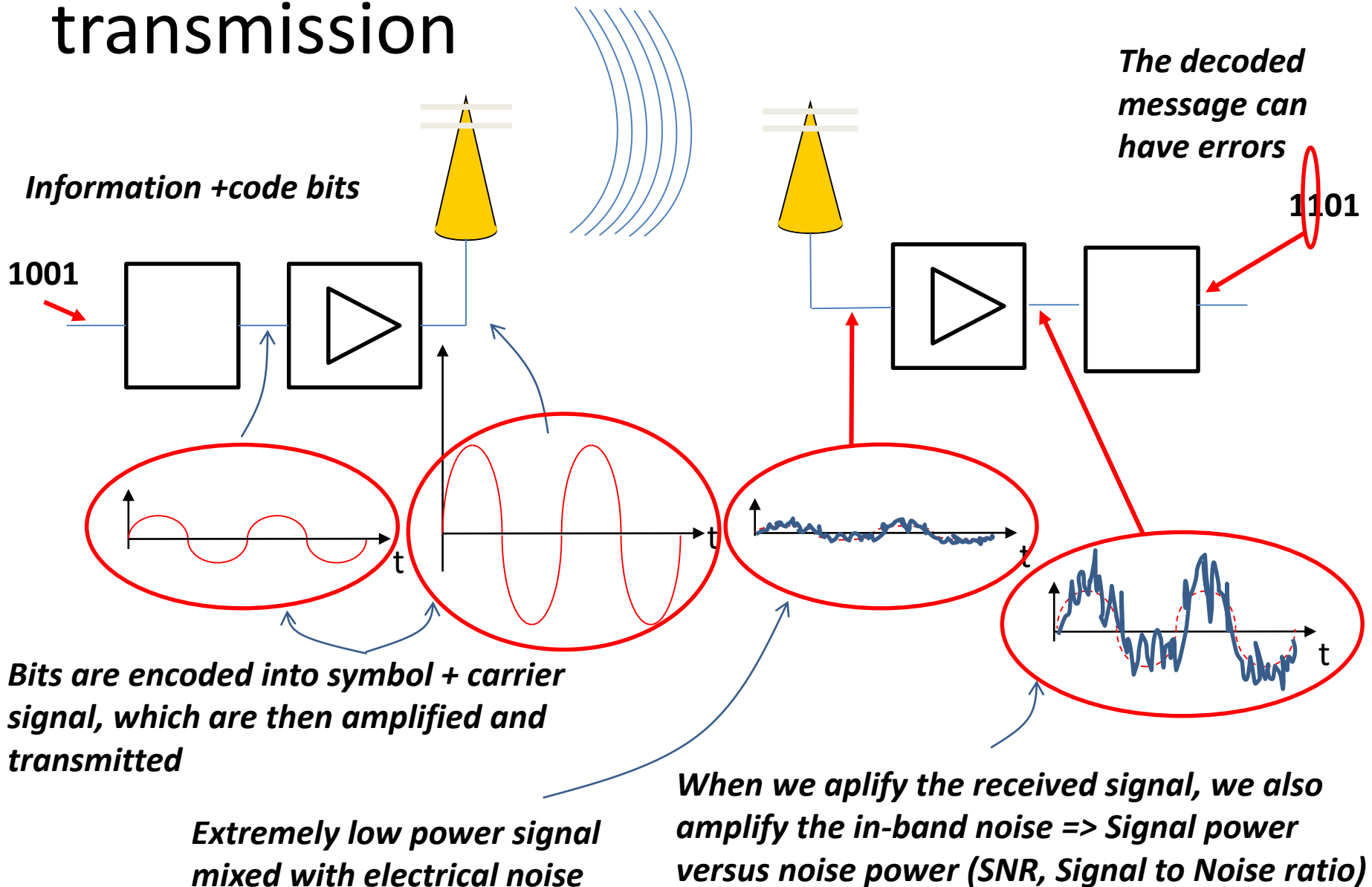
Mobile Internet

2.- Wireless Communications

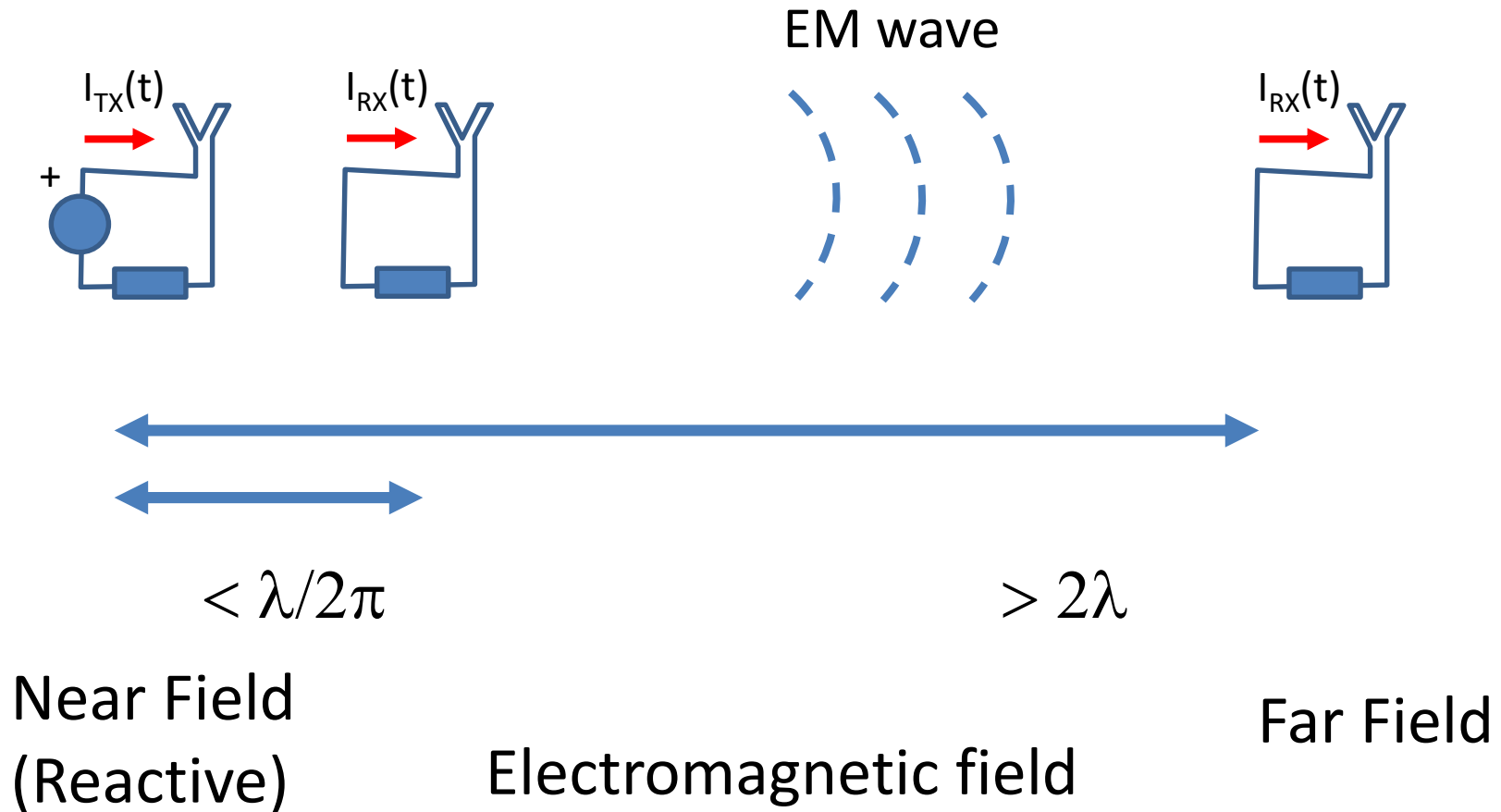
Radio transmission

- Mobility is linked to radio (wireless) transmission
 - Cellular Nets: 2G, 3G, 4G, 5G, etc
 - WLANs: 802.11xx (WiFi)
 - LowPower: BLE, 802.15.4 (ZigBee), RFID&NFC, LoraWAN, NBloT, etc

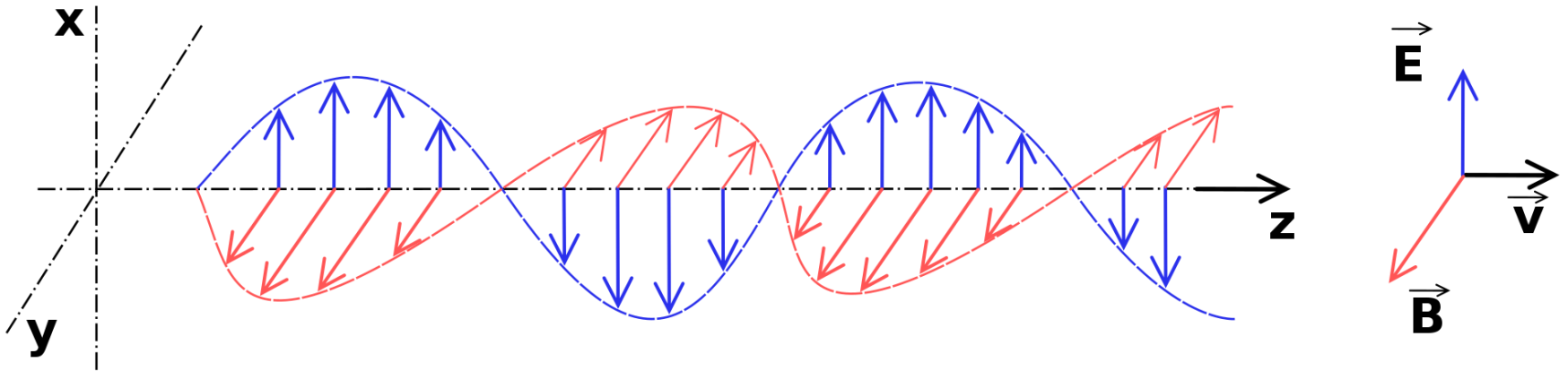
Wireless transmission



Physical principles: Near/Far EM Field



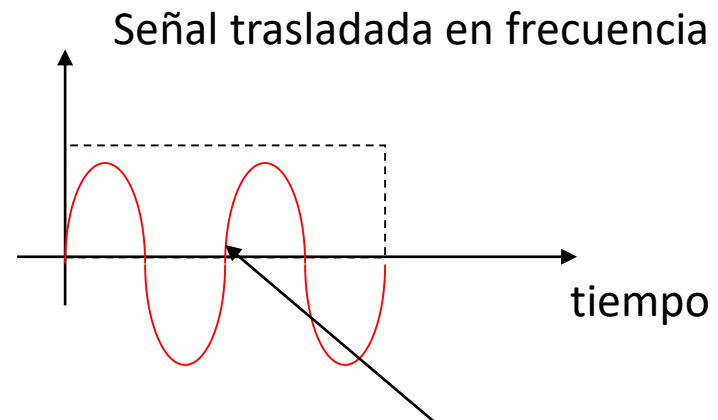
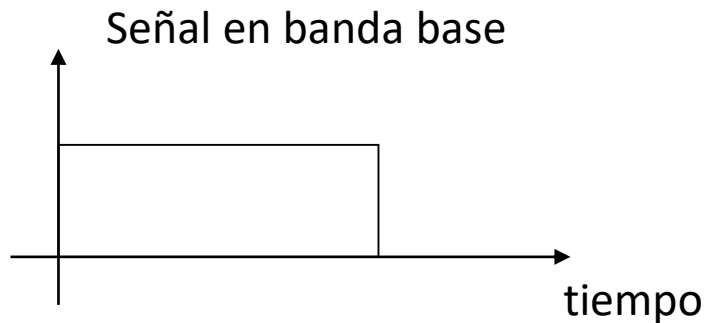
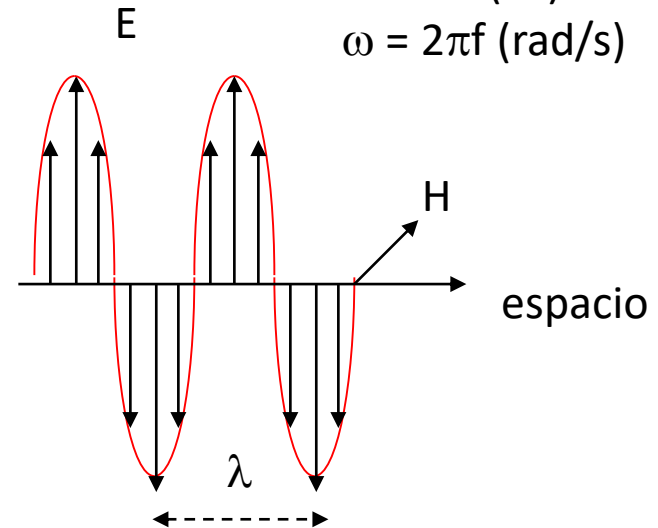
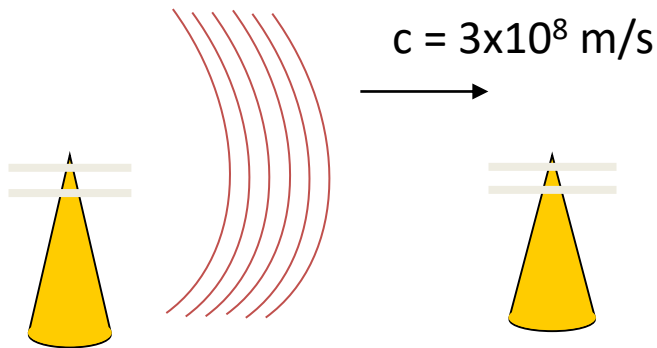
Electromagnetic (EM) wave



Source: https://en.wikipedia.org/wiki/Electromagnetic_radiation#/media/File:Onde_electromagnetique.svg

Wireless transmission

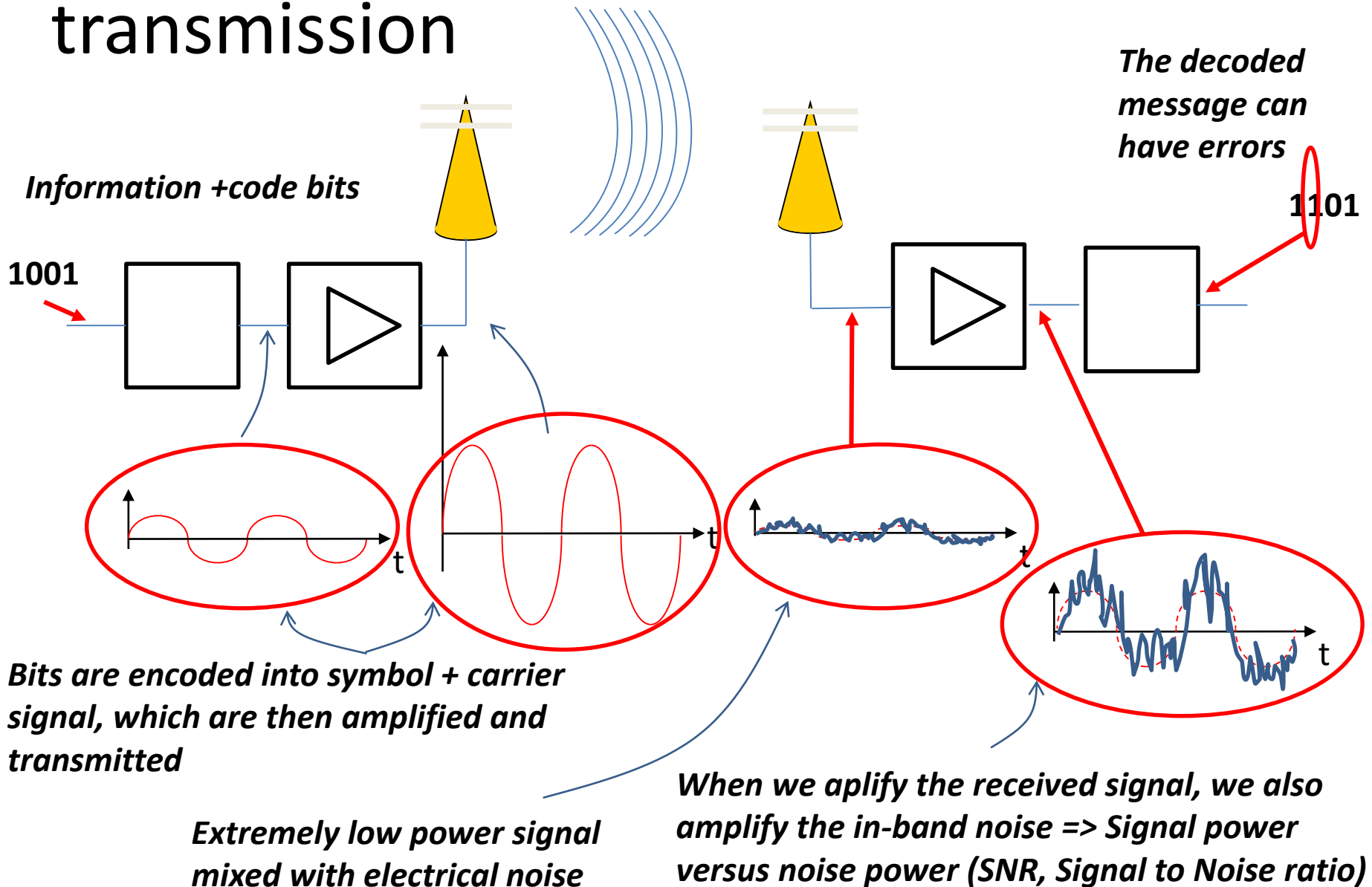
$$\lambda = c/f$$
$$f \text{ en Hz (s}^{-1}\text{)}$$
$$\omega = 2\pi f \text{ (rad/s)}$$



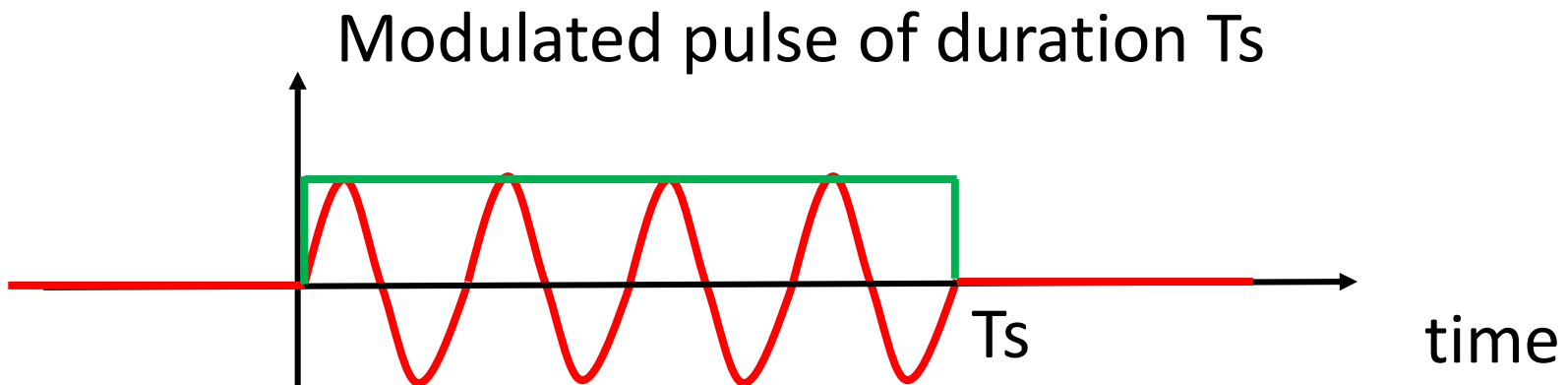
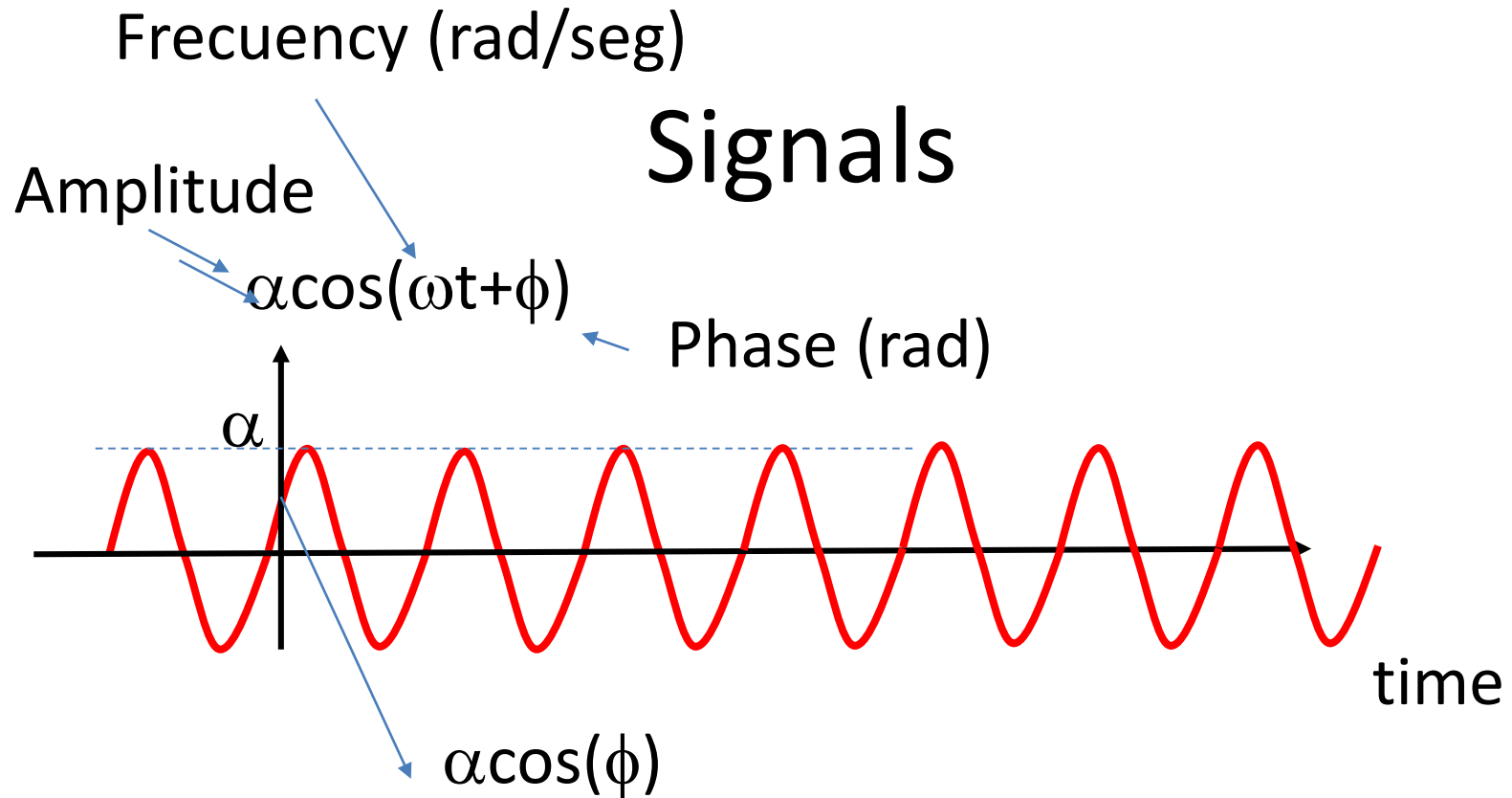
$$\text{Potencia} = k |E|^2$$

Frecuencia portadora

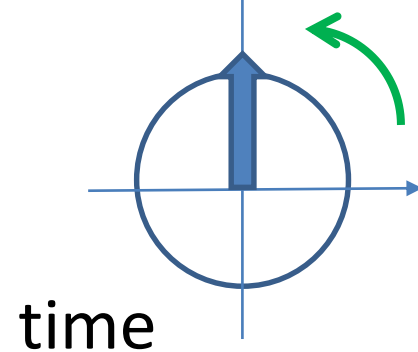
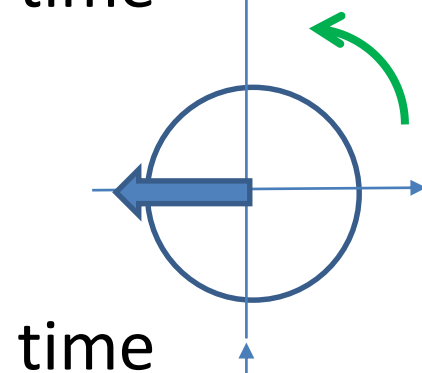
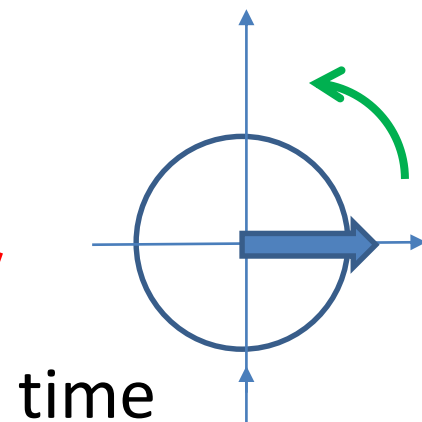
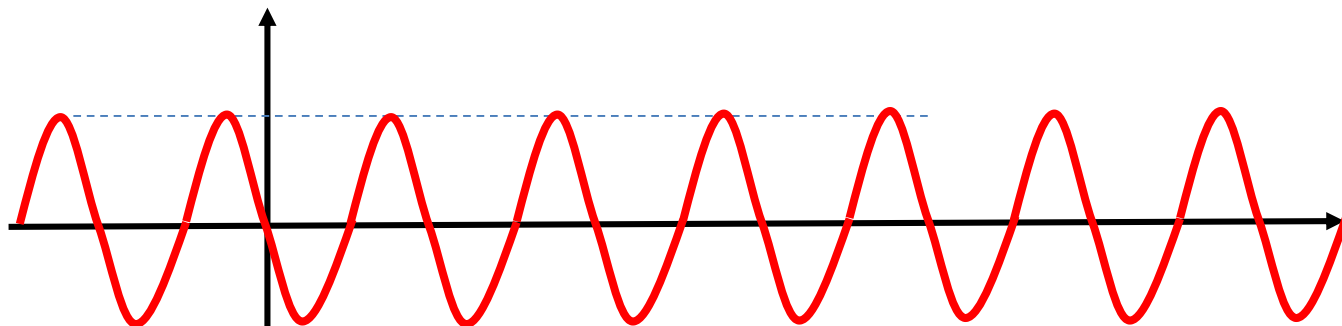
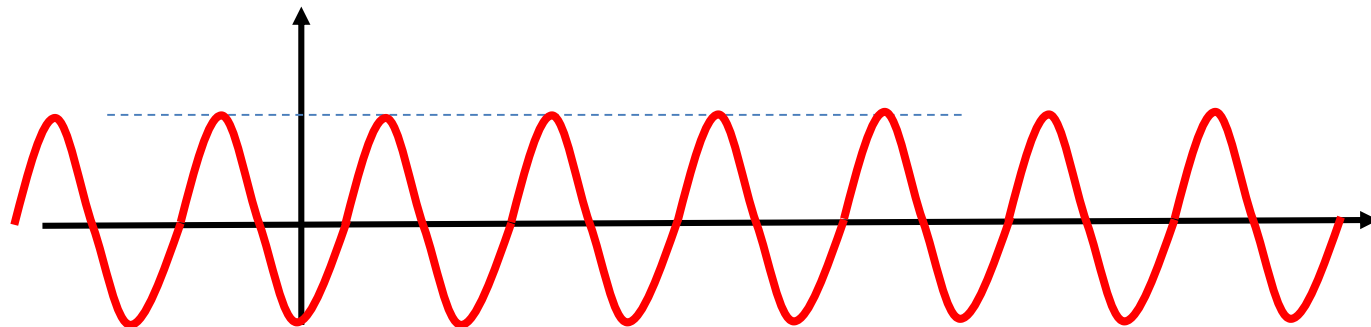
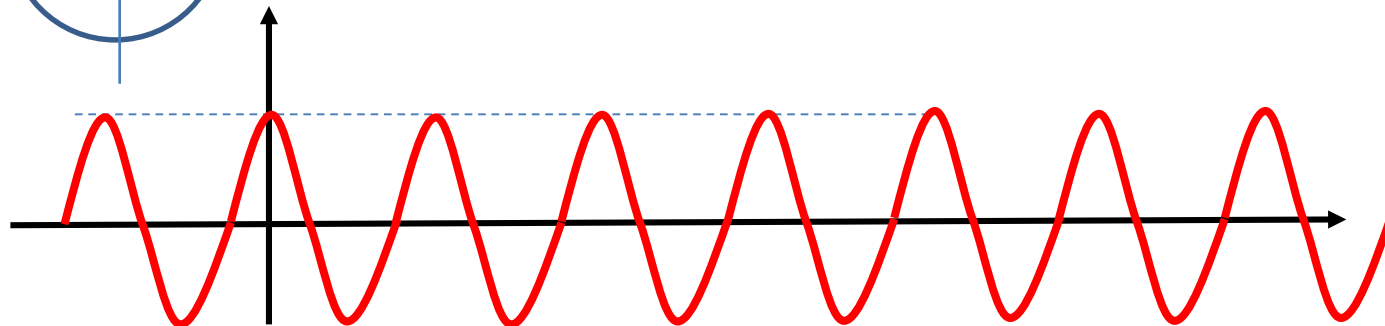
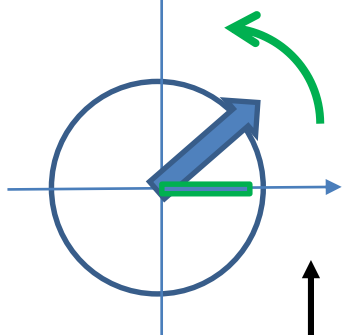
Wireless transmission



Signals

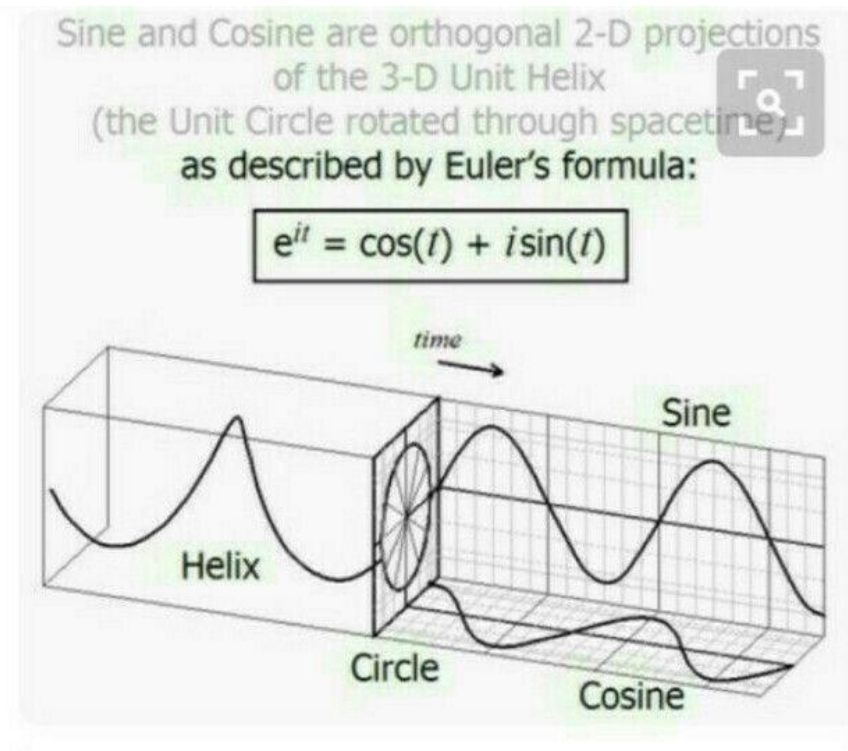


Signals

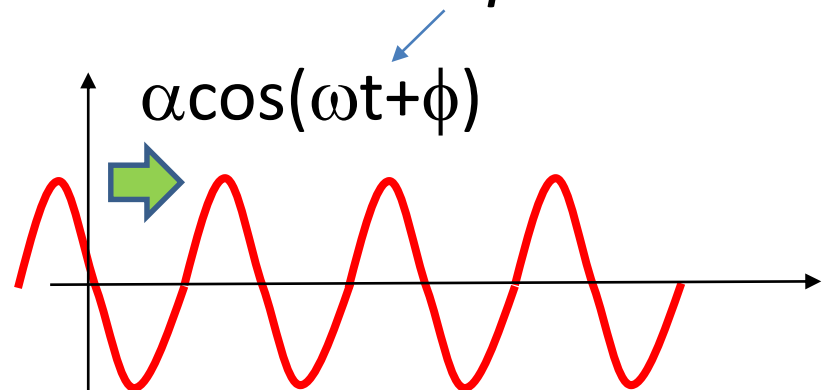
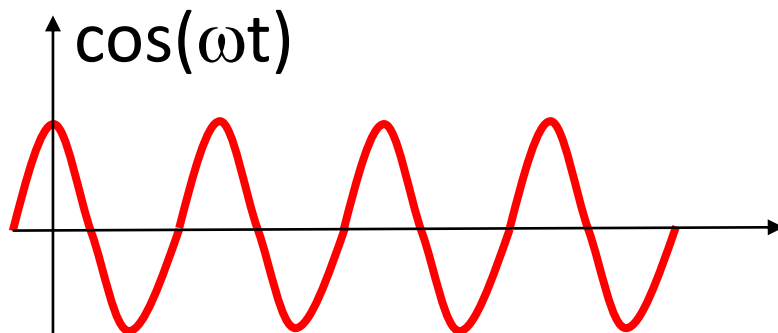
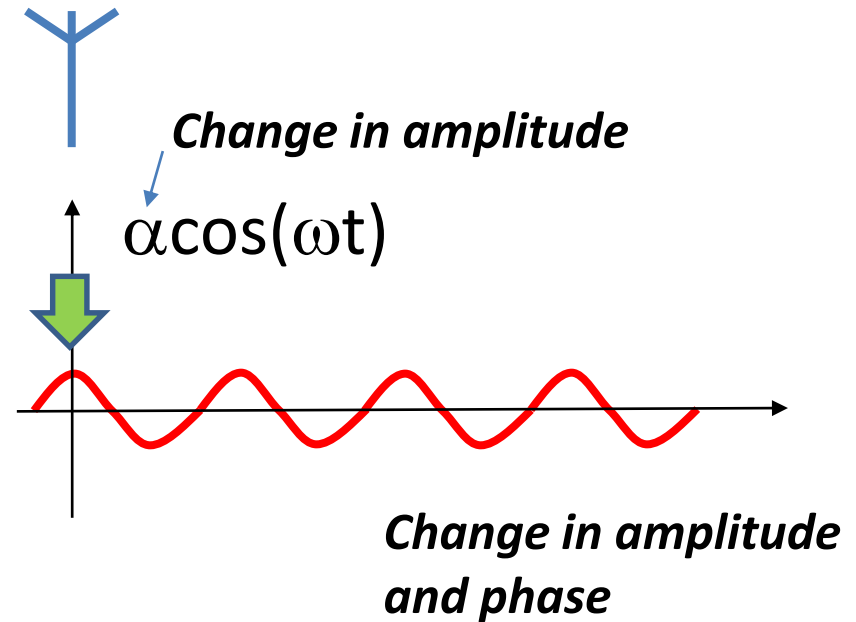
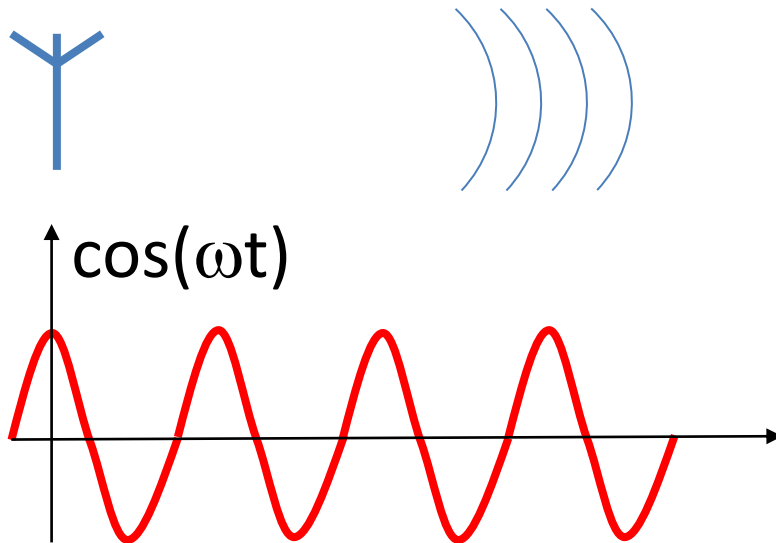


Representing signals by means of complex numbers

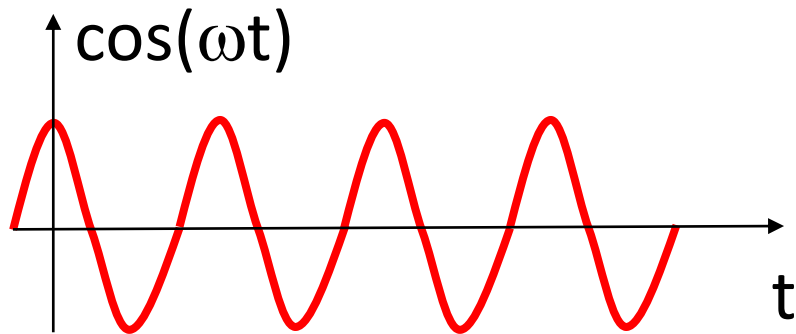
Signals and transmissions channels are modeled very often using **complex numbers**. This is done as we can model attenuation and phase changes by a **single multiplicative complex coefficient**.



Signals and transmissions channels are modeled using **complex numbers** to model attenuation and phase changes by a single multiplicative complex coefficient

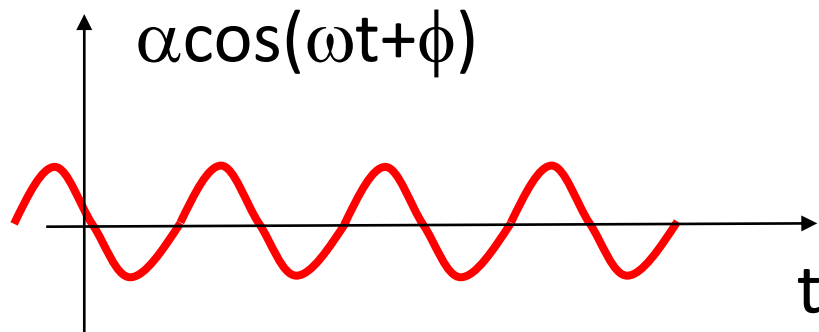


Signals and transmissions channels are modeled using **complex numbers** to model attenuation and phase changes by a single multiplicative complex coefficient



$$s(t) = \cos(\omega t) + i \sin(\omega t) = e^{i\omega t}$$

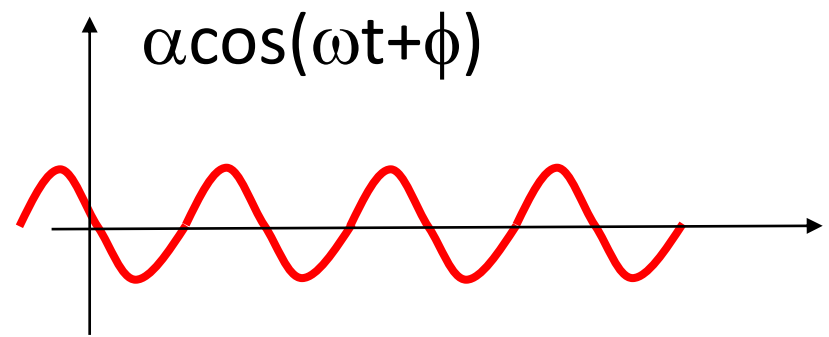
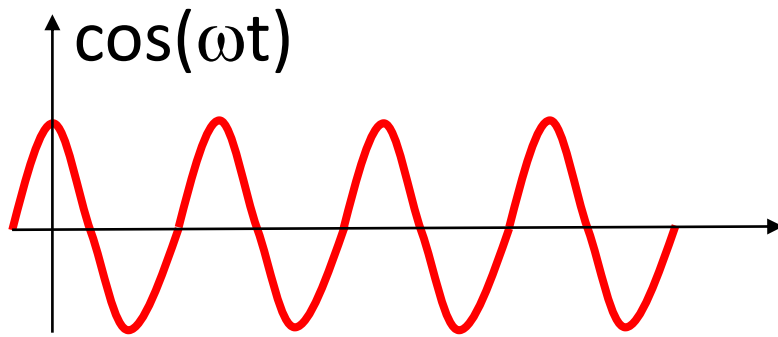
↑
 $\text{Re}[s(t)]$



$$r(t) = \alpha \cos(\omega t + \phi) + i \alpha \sin(\omega t + \phi) = e^{i(\omega t + \phi)}$$

↑
 $\text{Re}[r(t)]$

Signals and transmissions channels are modeled using **complex numbers** to model attenuation and phase changes by a single multiplicative complex coefficient

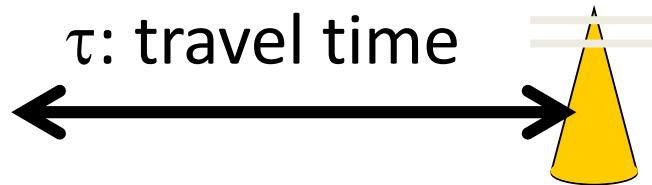
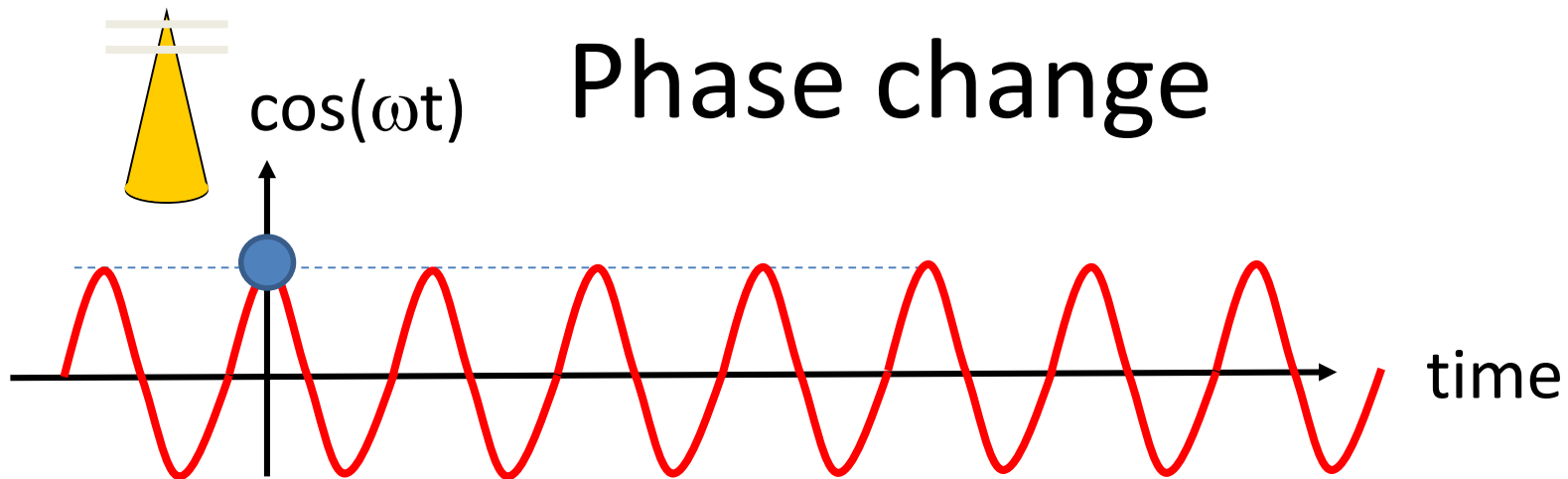


$$h = \alpha e^{i\phi}$$

$$r(t) = s(t)h = e^{i\omega t} \alpha e^{i\phi} = \alpha e^{i(\omega t + \phi)}$$

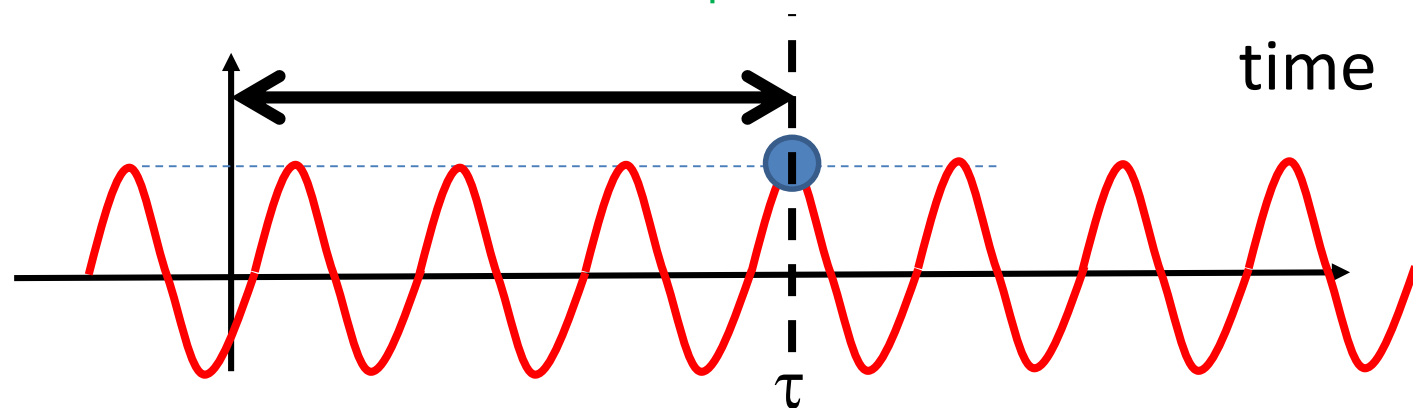
Channel effect: modify amplitude and phase

Phase change

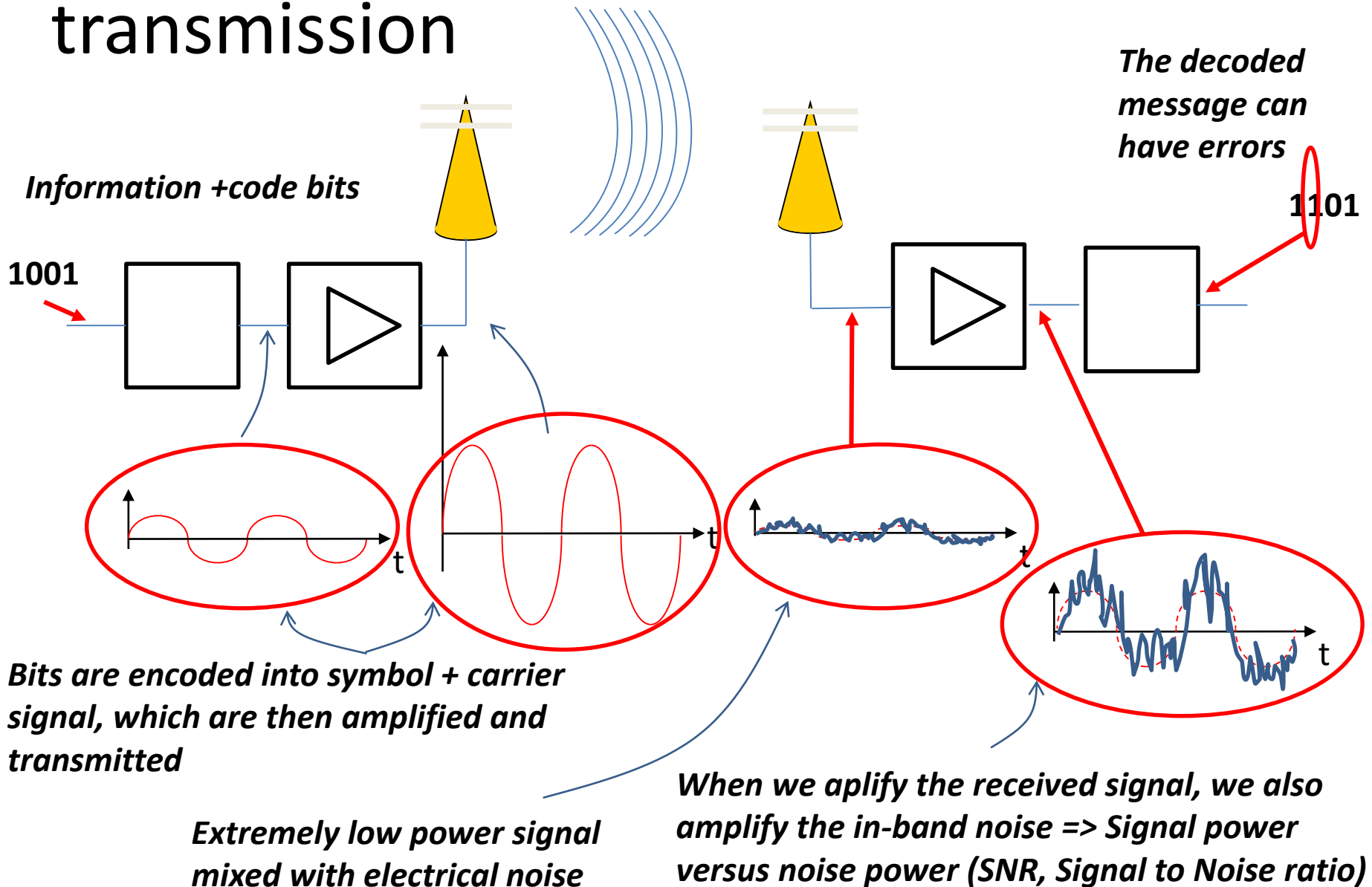


$Ej: d=3 \text{ metros} \Rightarrow$
 $\tau = 10 \text{ ns}$
 $\omega = 2 \pi 2.4 \text{ G rad/s}$
 $\phi = -2 \pi 2.4 10 = -150 \text{ rad}$
 $= -0.69 \text{ rad}$

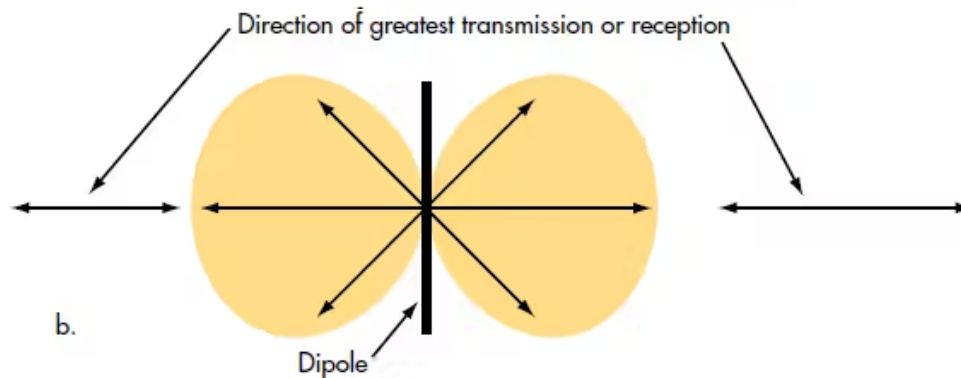
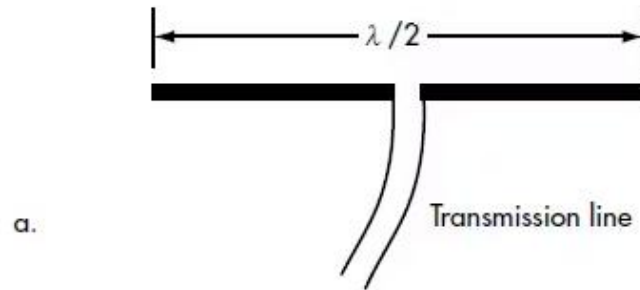
$$\cos(\omega(t-\tau)) = \cos(\omega t - \omega\tau) = \cos(\omega t + \phi) =$$



Wireless transmission

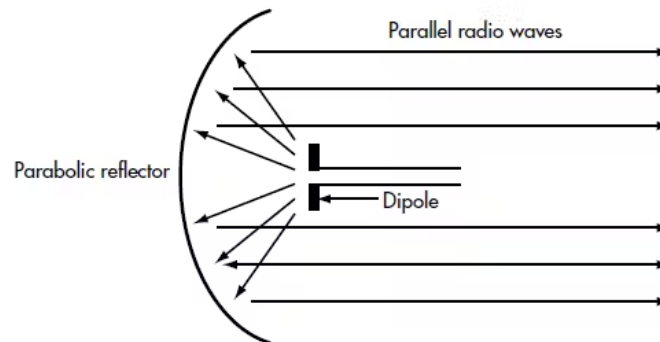
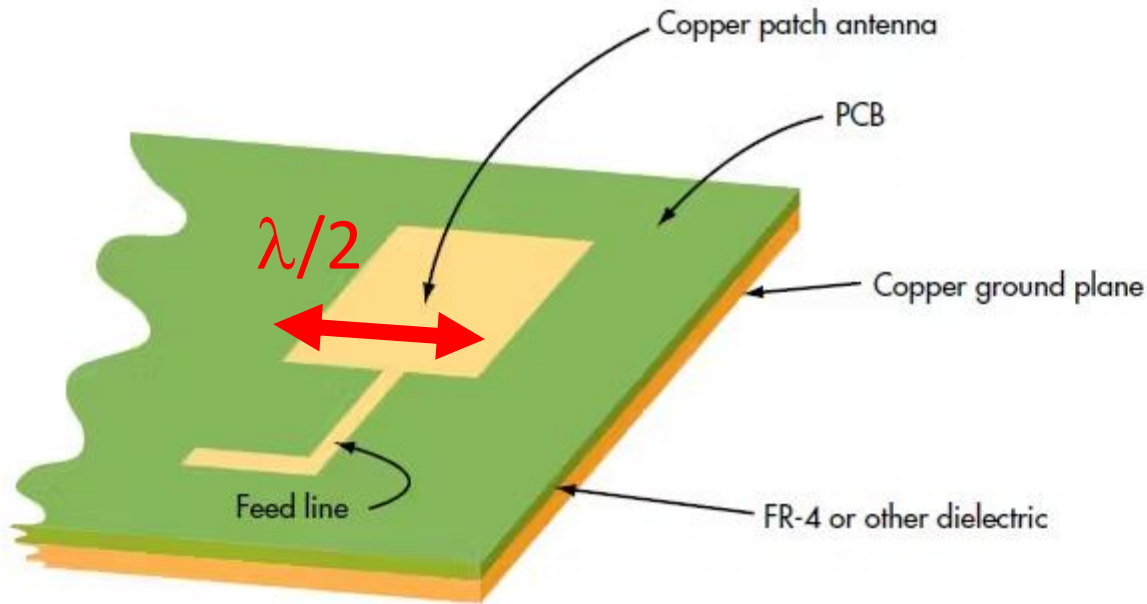


e.g. Dipole antenna



2. Dipole construction consists of two $\lambda/4$ elements end-to-end with a feed in the center by a transmission line (a). At resonance, the antenna appears to be a $73\text{-}\Omega$ resistor. A dipole's horizontal radiation pattern looks like a figure 8 from above (b). In 3D, the pattern is shaped like a doughnut with maximum radiation perpendicular to the length of the antenna.

Microstrip antenna ($> 1\text{GHz}$) & parabolic reflector



(B)
Transmission

Reminder: $\log_{10}$ counts 0's

$$\log_{10}(1000)=3$$

$$\log_{10}(10)=1$$

$$\log_{10}(1)=0$$

$$\log_{10}(0.001)=-3$$

$$\log_{10}(2)=0.3010\dots$$

$$\log_{10}(4)=0.6020\dots$$

$$\log_{10}(5)=1-0.3010\dots=0.6989\dots$$

$$\mathbf{x \text{ dB} = 10 \log_{10}(x)}$$

- Used to express very large/very small numbers (no units!)
- e.g: 100 000 = 50 dB
- e.g: 0.000 01 = -50 dB

Reminder: dB

- Loss:
 - P_{tx} / P_{rx} (No units)
 - $10 \log(P_{tx} / P_{rx})$ (decibels, dB)
- Eg:
 - P_{tx} : 100 mW,
 - P_{rx} : 10 nW
 - Loss: $10.000.000 = 70$ dB

Reminder: dBm

- *Power* in dBm:
 - $10 \log(P/1 \text{ mW}) \quad (\text{dBm})$
- Eg:
 - $P = 100 \text{ mW} = 20 \text{ dBm}$
 - $P = 10 \text{ nW} = -50 \text{ dBm}$
 - $\text{Loss (dB)} = P_{\text{tx}} (\text{dBm}) - P_{\text{Rx}} (\text{dBm}) = 70 \text{ dB}$

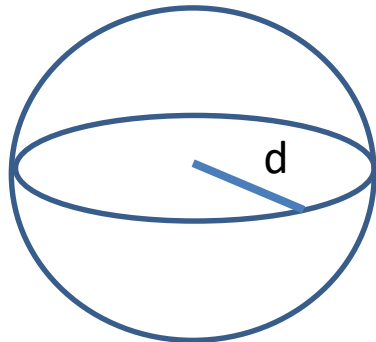
Wireless transmission (free space)

- Friis' transmission equation (free space path loss)

$$P_r = P_t \underbrace{G_t G_r}_{\text{Antenna isotropic gains in the direction to the other antenna(Tx and Rx)}} \left(\frac{\lambda}{4\pi d} \right)^2$$

Carrier wavelength

Distance



$$A = 4\pi d^2$$

Antenna isotropic gains in the direction to the other antenna(Tx and Rx)

Wireless Transmission (Land)

$$\frac{P_r}{P_t} = k \frac{1}{d^2}$$

In land transmissión, we have different attenuation exponents with distance

$$\frac{P_r}{P_t} = k \frac{1}{d^\alpha} \quad 1.5 \leq \alpha \leq 6$$

In many examples we will use $\alpha \approx 4$

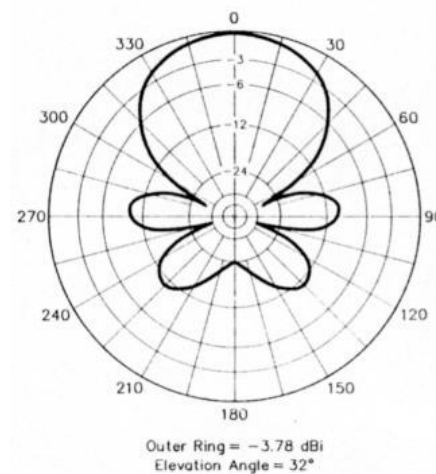
Wireless Transmission (Land)

- Strong attenuation with distance

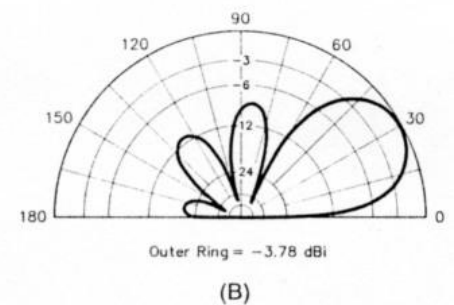
$$\frac{P_r}{P_t} = k \frac{1}{d^\alpha} \quad 1.5 \leq \alpha \leq 6$$

In many examples we will use $\alpha \approx 4$

- Antenna directivity



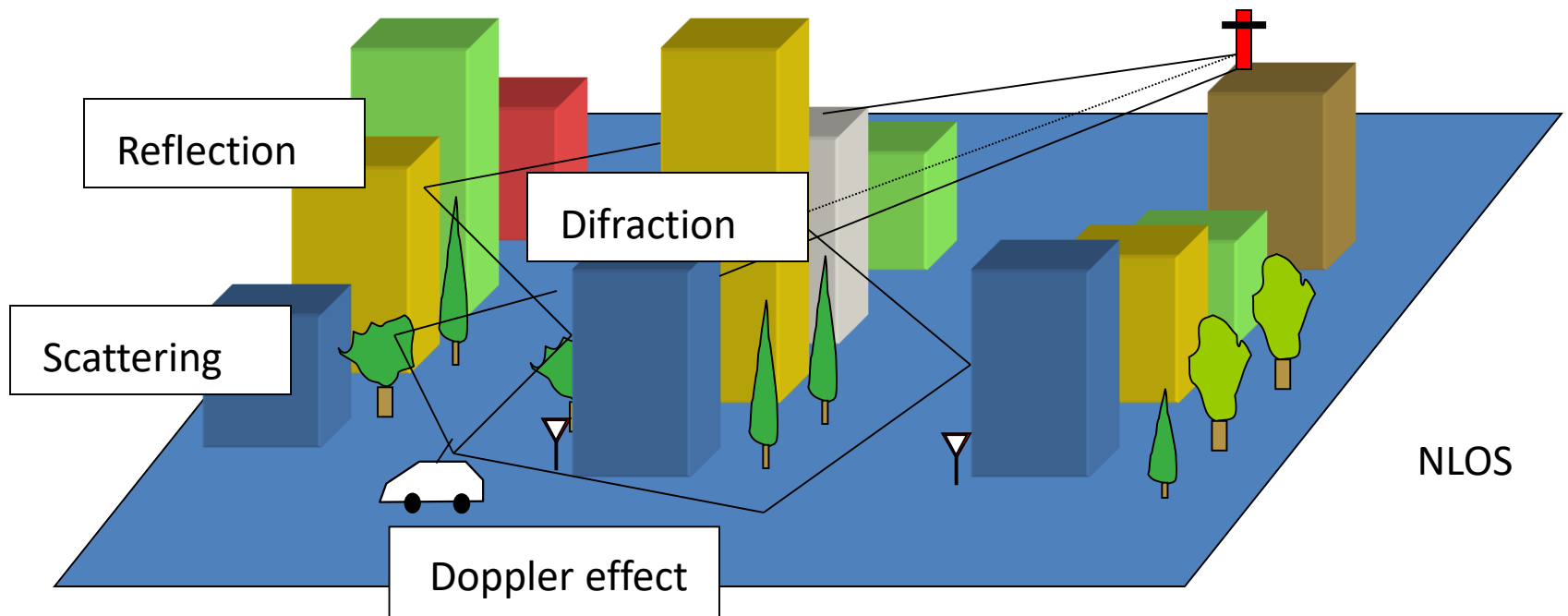
(A)



(B)

Wireless transmission

- Fading, shadowing, *multipath fading*, Doppler effect, changing channel conditions, etc



Wireless transmission

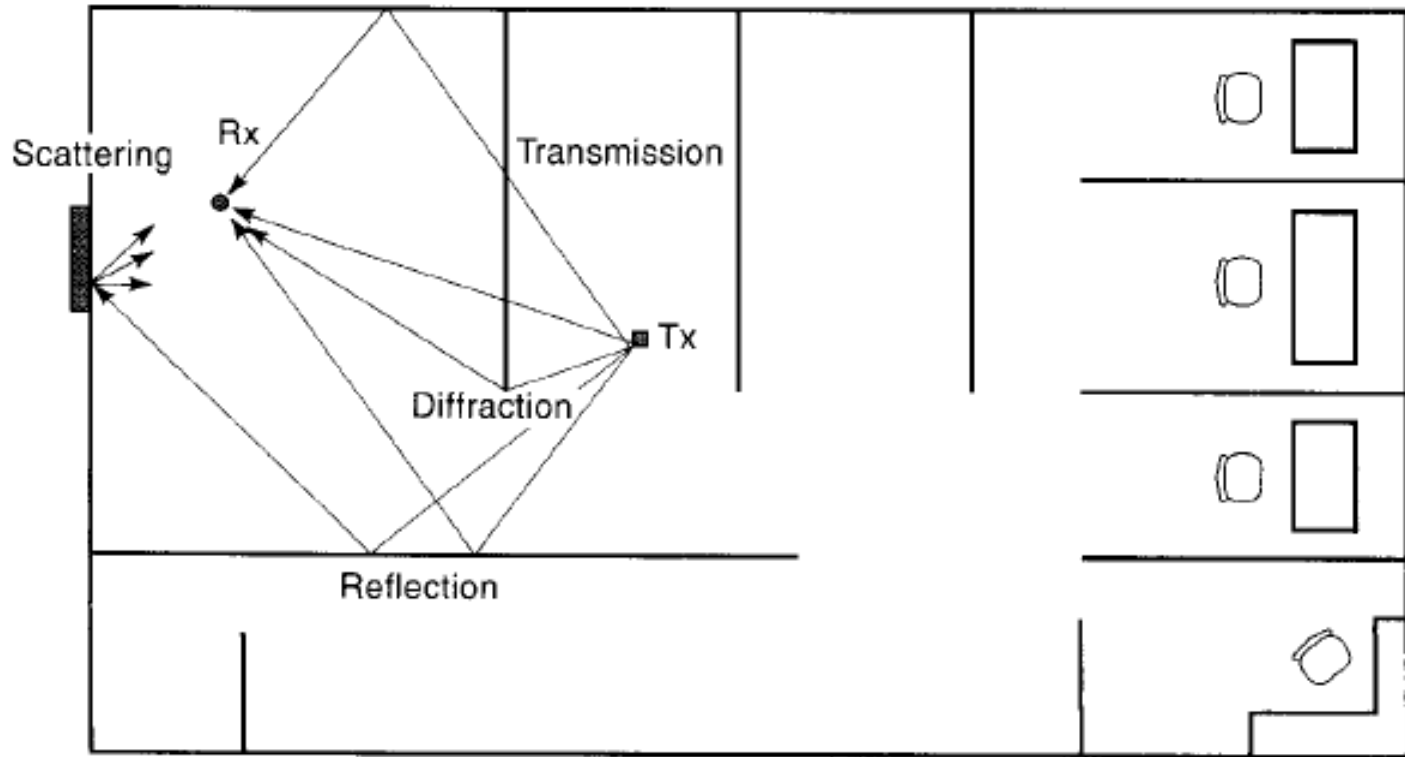
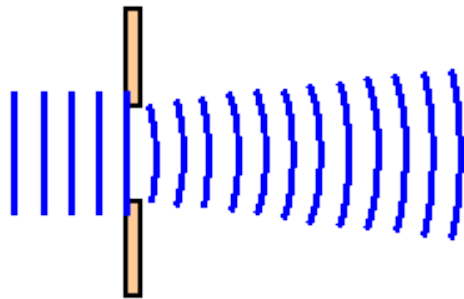


Figure 2.1 Radio propagation mechanisms in an indoor area.

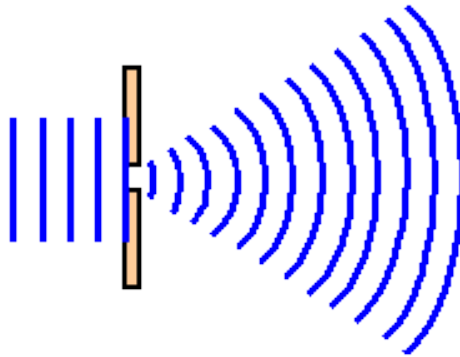
Diffraction

Better coverage for lower frequencies
(i.e. larger wavelengths)

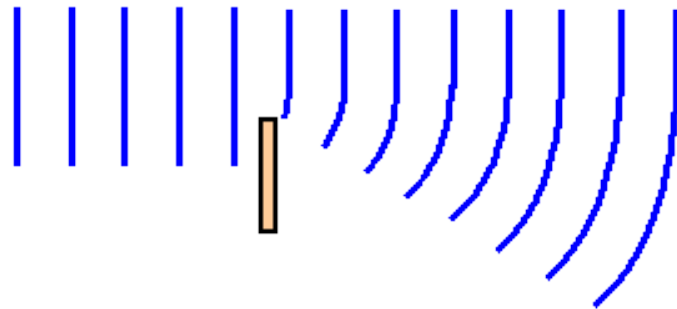
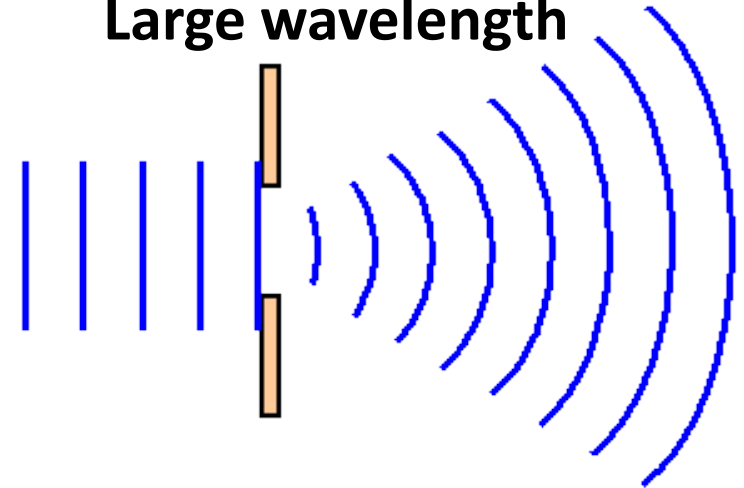
(relative)
small wavelength



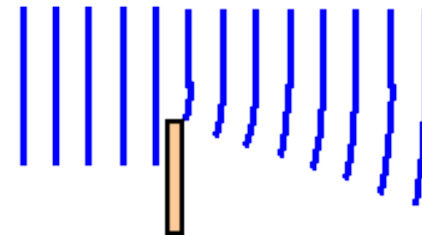
(relative)
Large wavelength



(relative)
Large wavelength



Large wavelength



Small wavelength



WINDOW MOB
2-layered energy-efficient window with metal frame.



WALL MOB
45 cm multi-layer wall made of reinforced-concrete and brick.



DOOR MGB
3-layered energy-efficient door with metal frame.



WINDOW MTF
2-layered energy-efficient window with wooden frame.



WINDOW OOB
1-layered thin glass window with wooden frame.

45 cm concrete and brick wall

2-layer energy efficient window

1-layer thin glass window

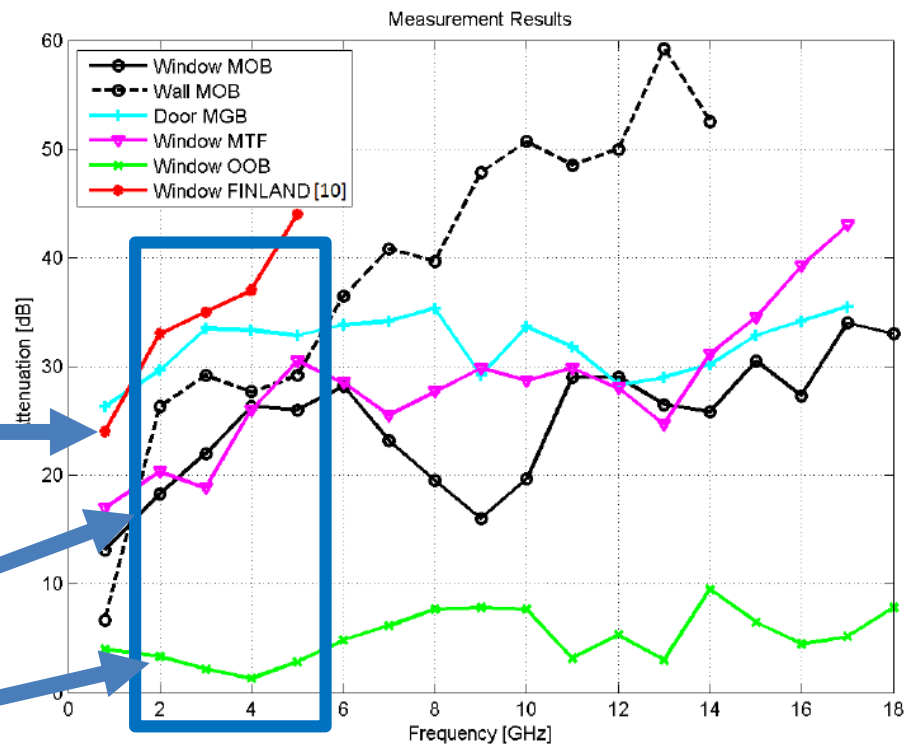
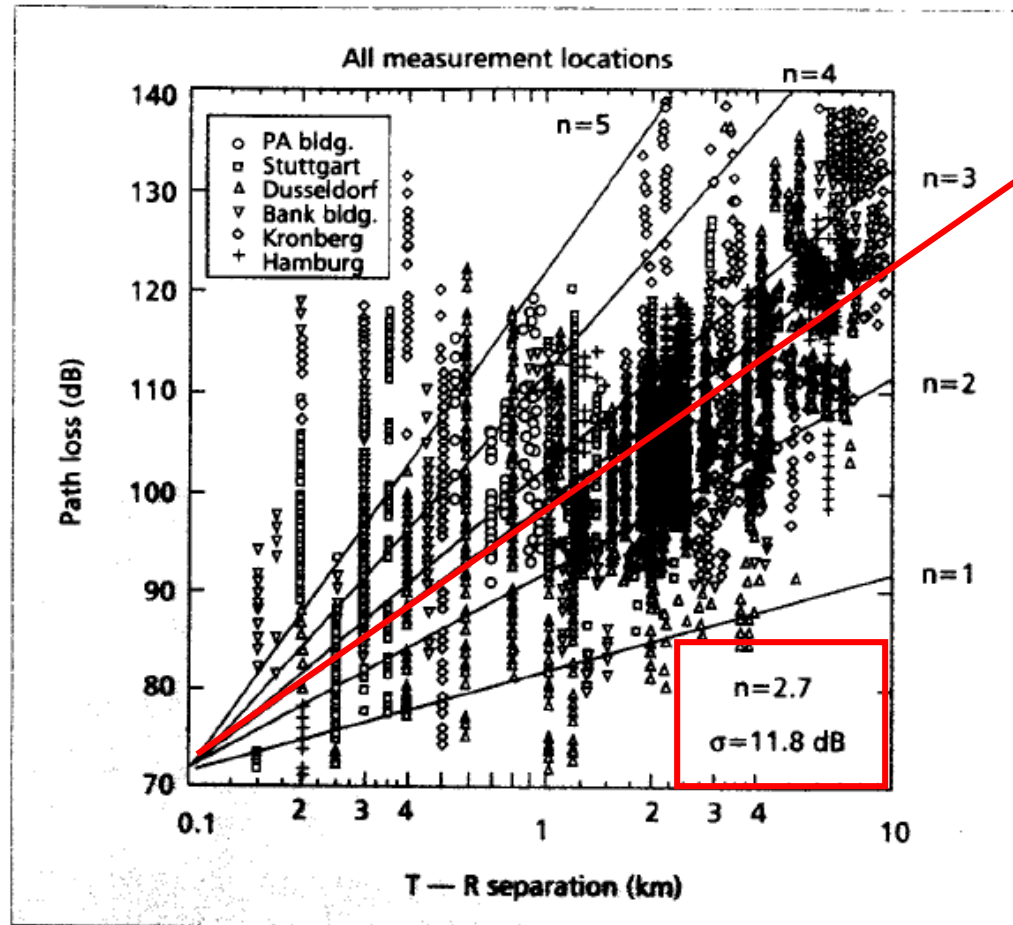


Fig. 6. Measured attenuation for the different scenarios in the range from 800 MHz to 18 GHz.

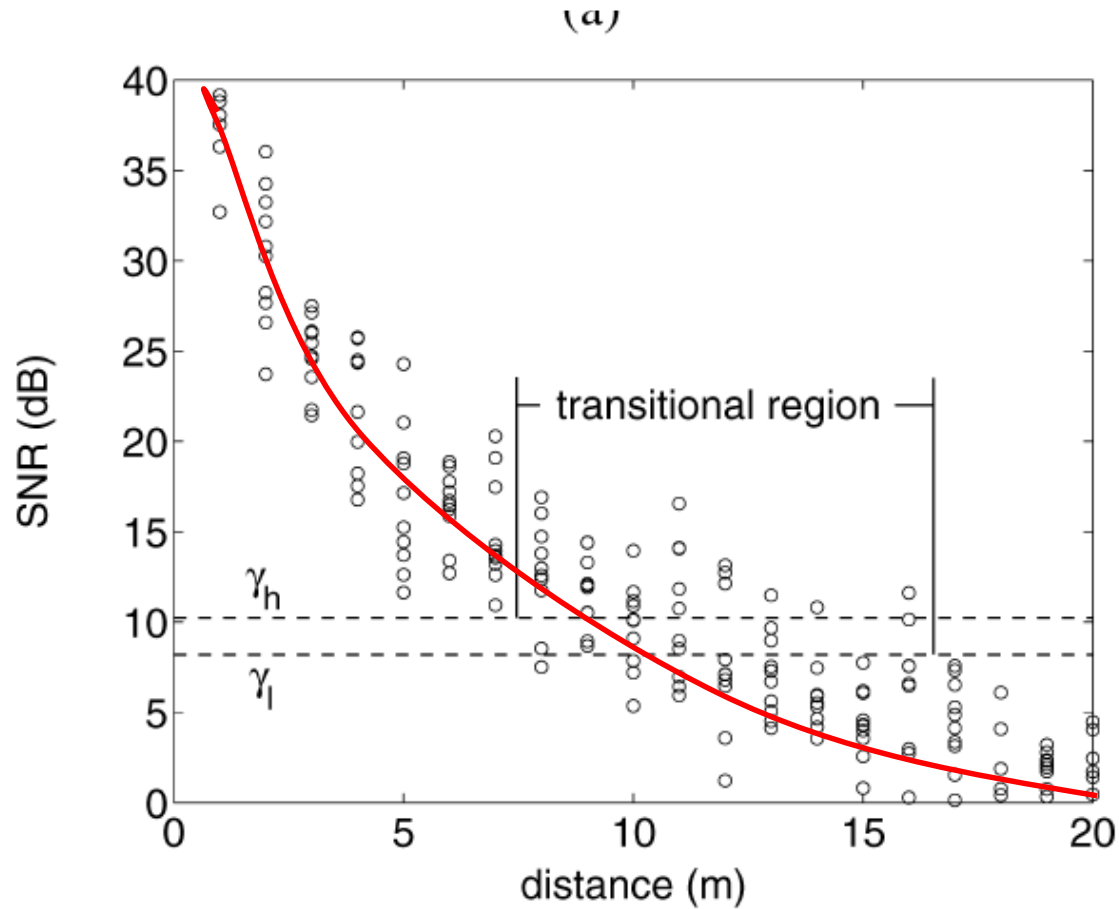
Atenuation



■ **Figure 3.** Scatter plot of path loss vs. distance for measurements in five German cities. The path loss is referenced to a free space reference measurement at $d_0 = 100 \text{ m}$. Note that σ has a value of 11.8 dB due to the large spread of PL values relative to the straight-line fit.

$1/d^n +$
Log-Normal...

Atenuation



Ref: M Zúñiga, B. Krishnamachari, "An Analysis of Unreliability and Assymetry in Low-Power Wireless Links", 2007

$$10 \log P_r/1\text{mW} = 10 \log P_t/1\text{mW} + 10 \log k - \alpha 10 \log d$$

cte

-4 = cte - alpha 0
 -31 = cte - alpha 10
 -27 = -alpha 10 =>
 cte = -4
 alpha = 27/10 = 2.7

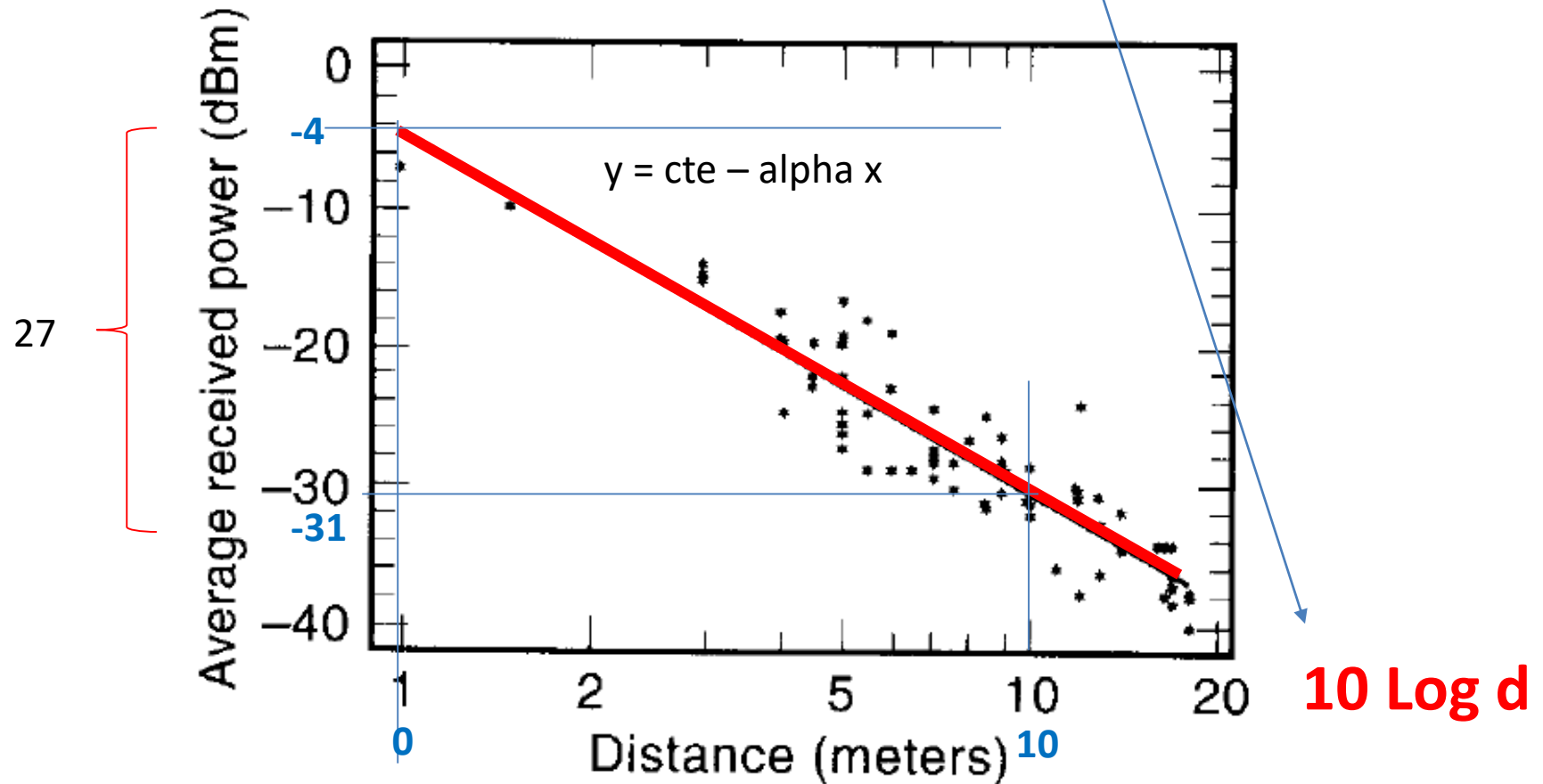
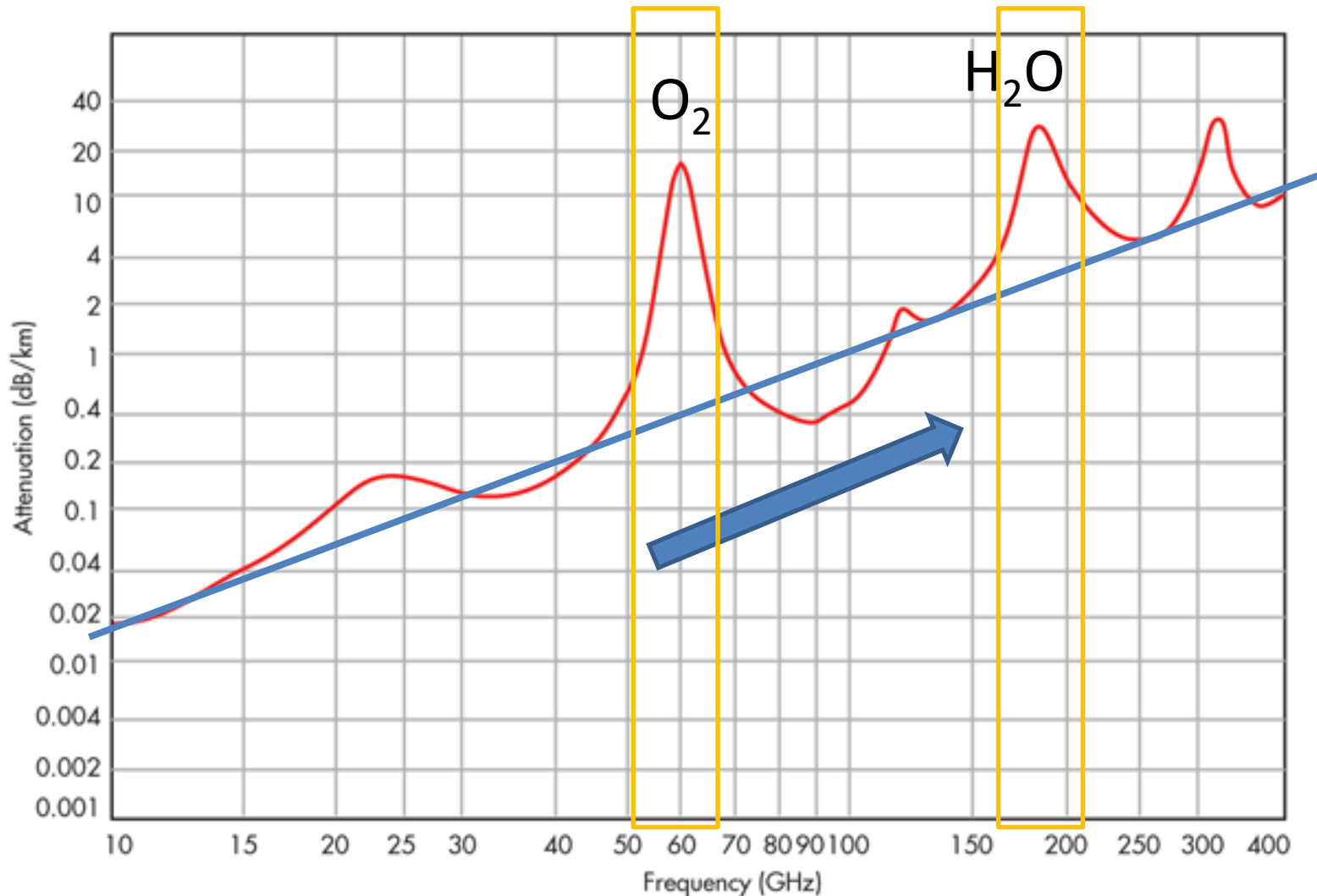
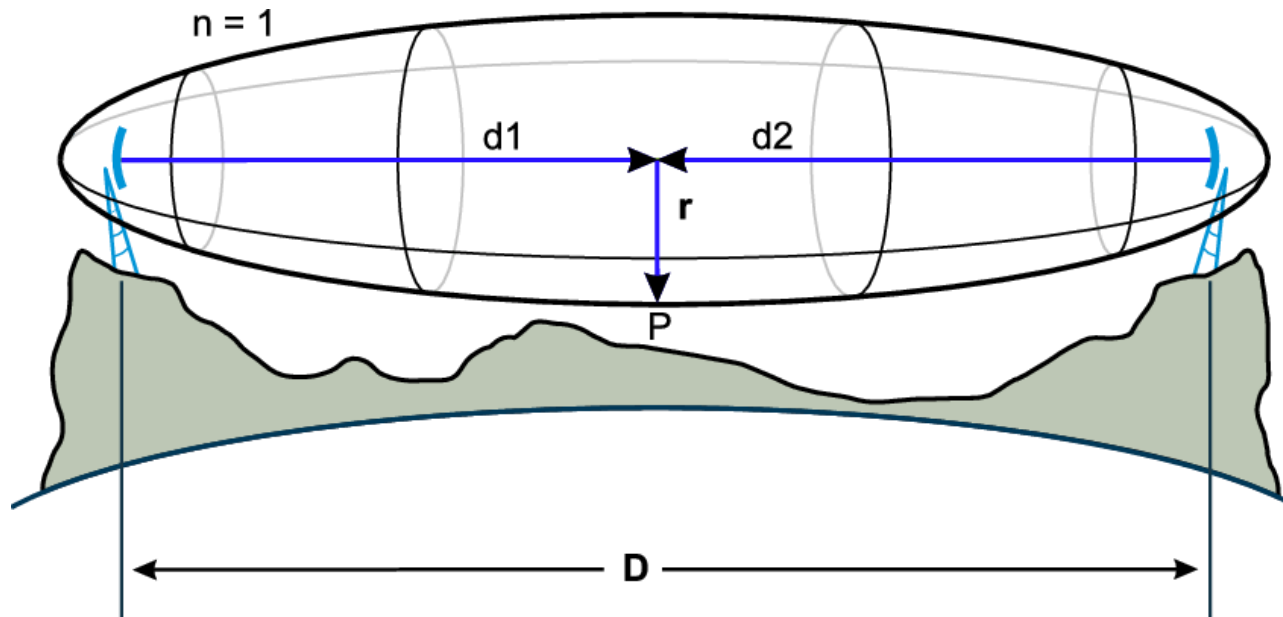


Figure 2.4 Measured received power and a linear regression fit to the data.

Attenuation and frequency



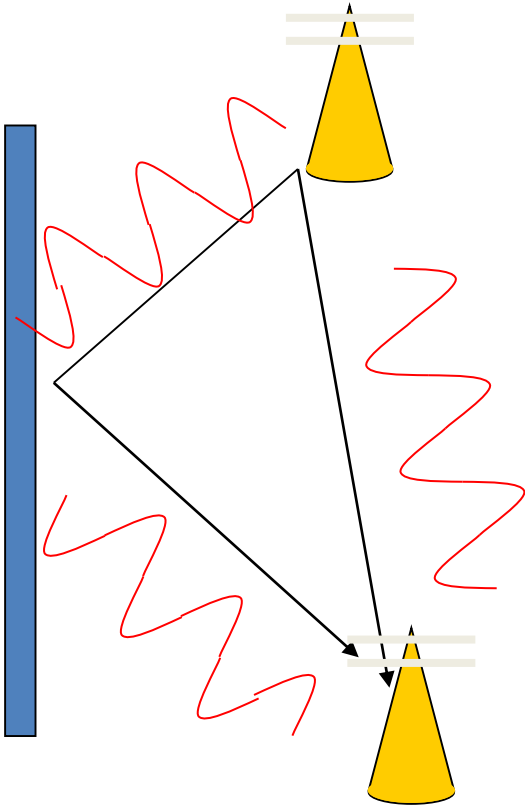
Fresnel zone



1st Fresnel zone at $d_1 = d_2 = D/2$

$$F_1 = 8,656 \sqrt{\frac{D(Km)}{f (GHz)}}$$

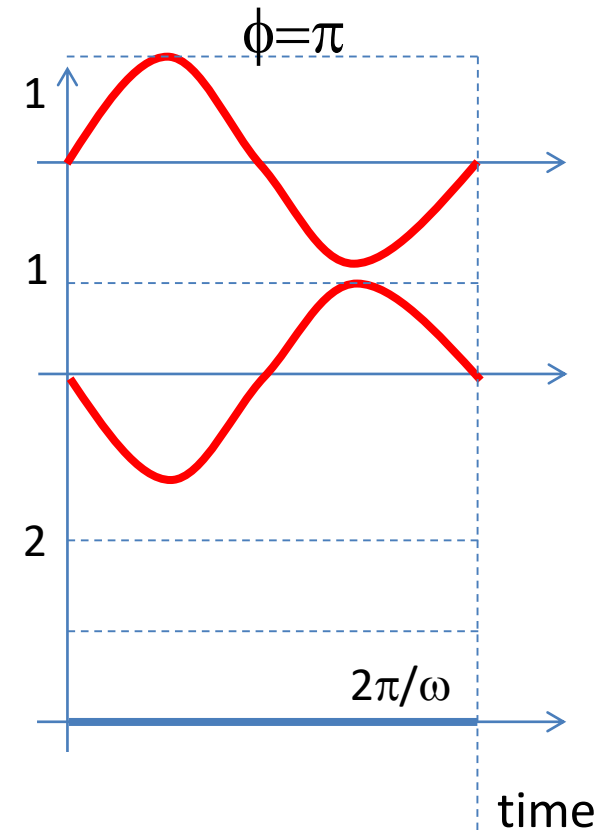
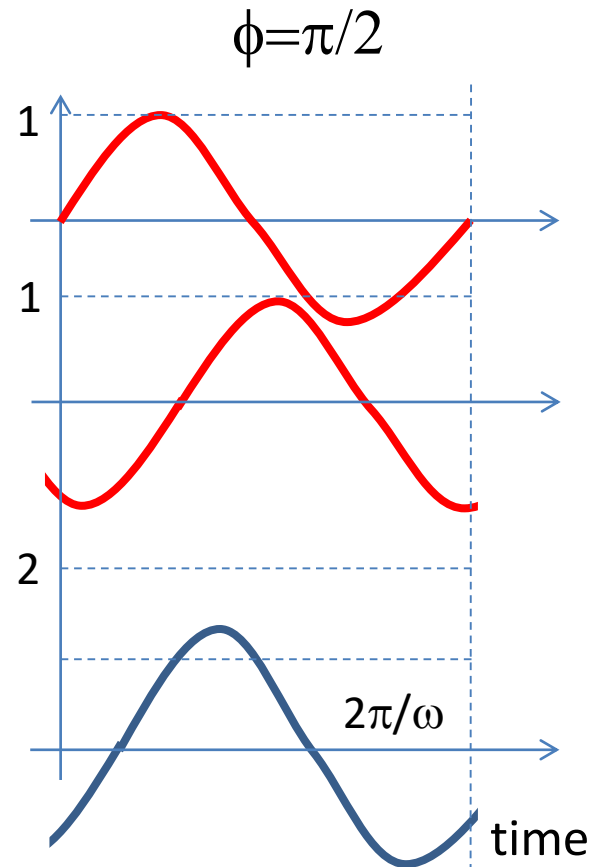
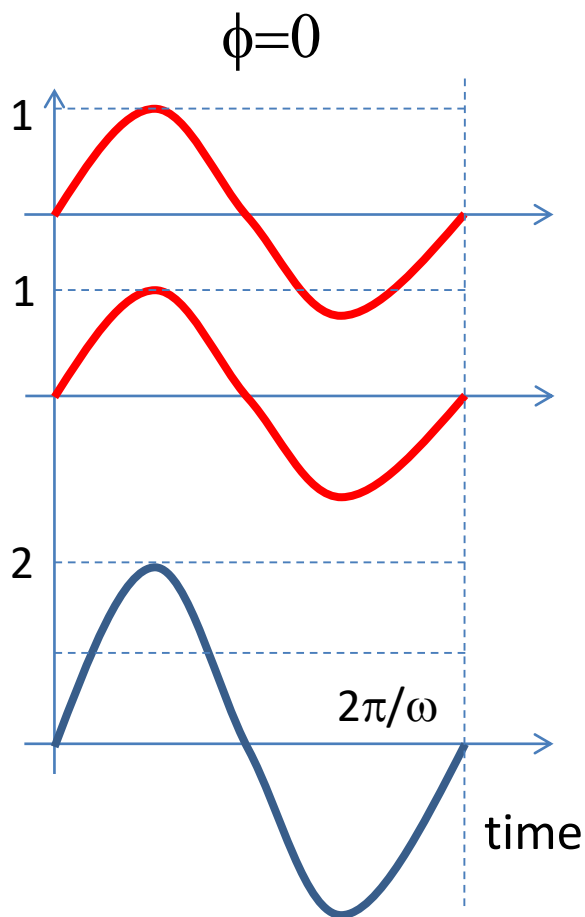
Multipath fading



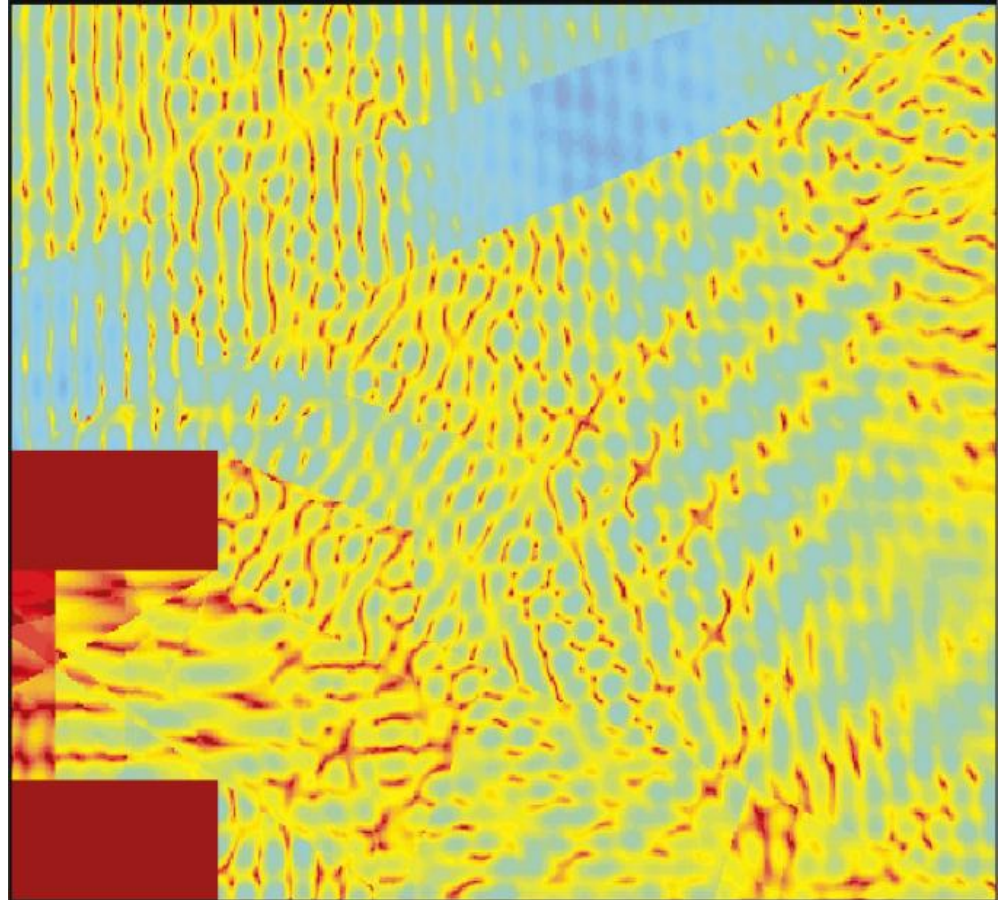
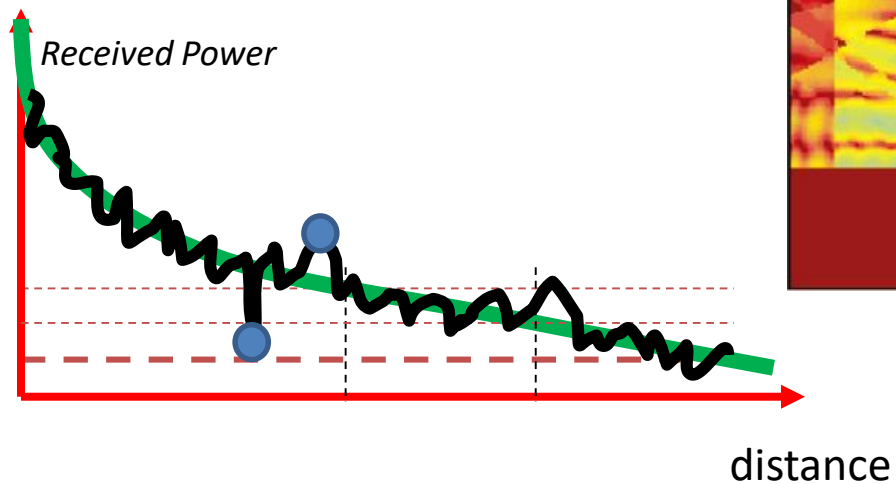
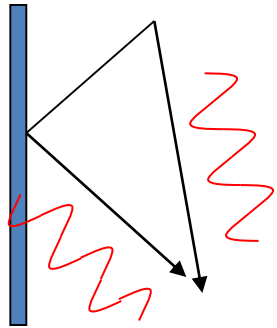
Different paths =>
different travel times =>
different phases

Adding sinusoids with different phases

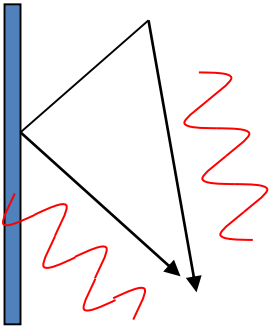
$$\sin(\omega t) + \sin(\omega t + \phi) = 2 \cos(\phi/2) \sin(\omega t + \phi/2)$$



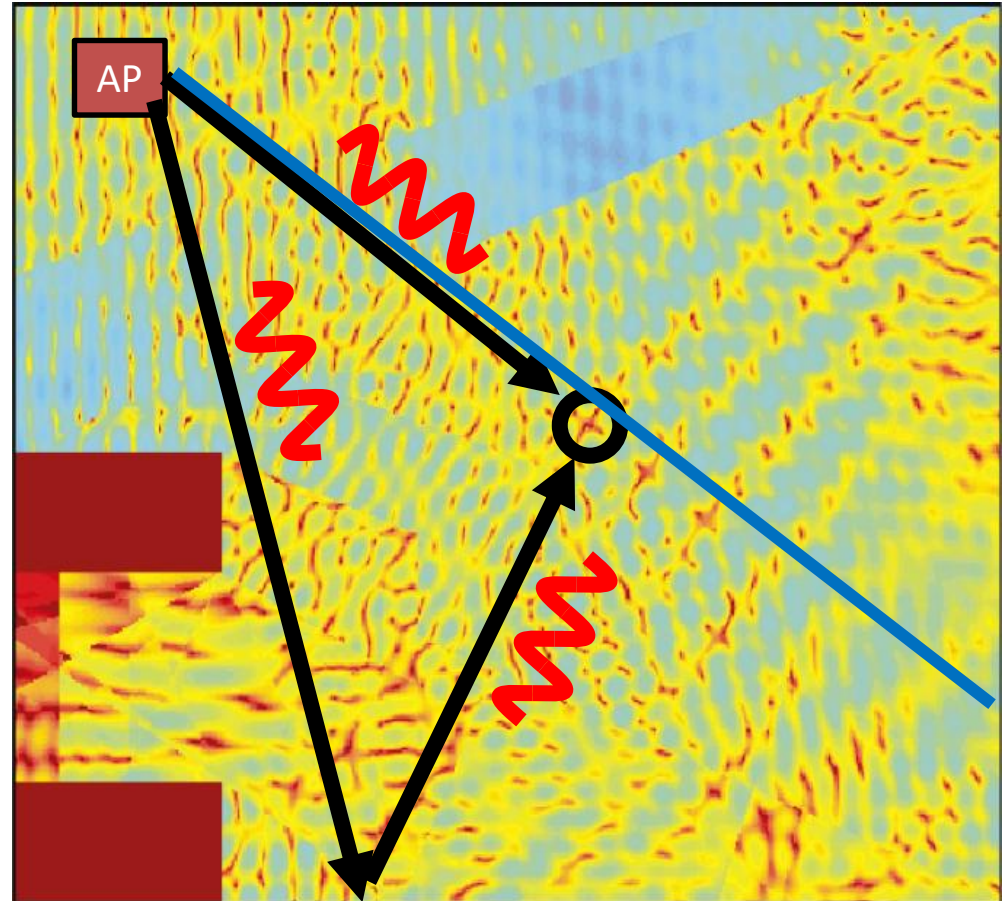
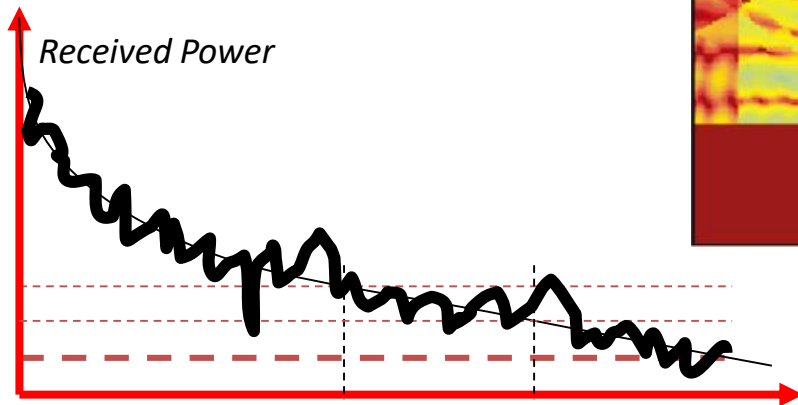
Multipath Fading



Multipath Fading

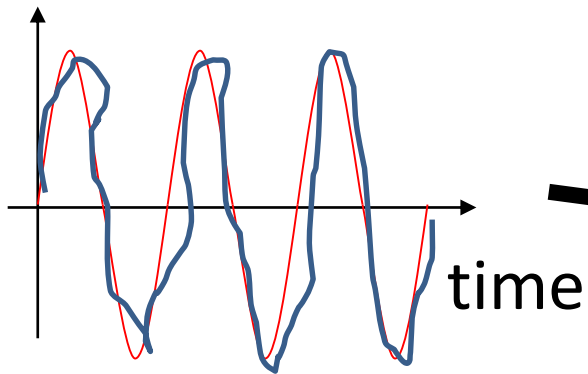


Different paths,
Different travel paths
Different phases

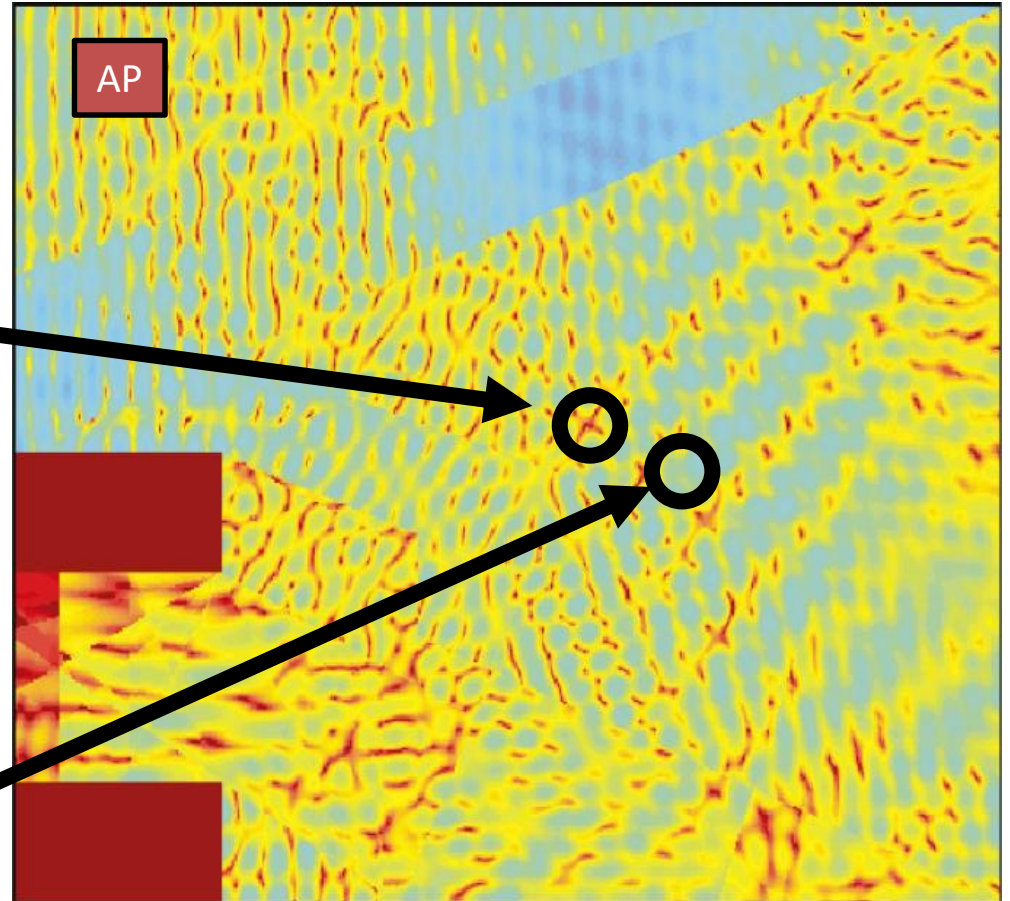
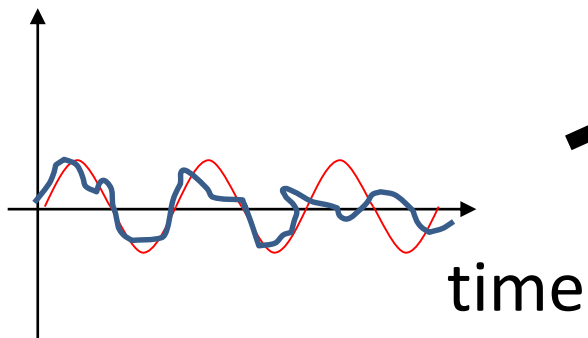


Signal to Noise ratio (SNR)

Received signal



Received signal



Wireless transmission

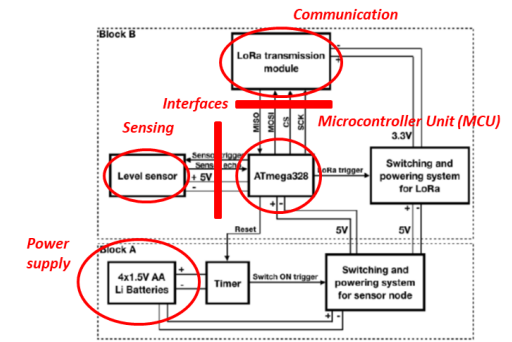
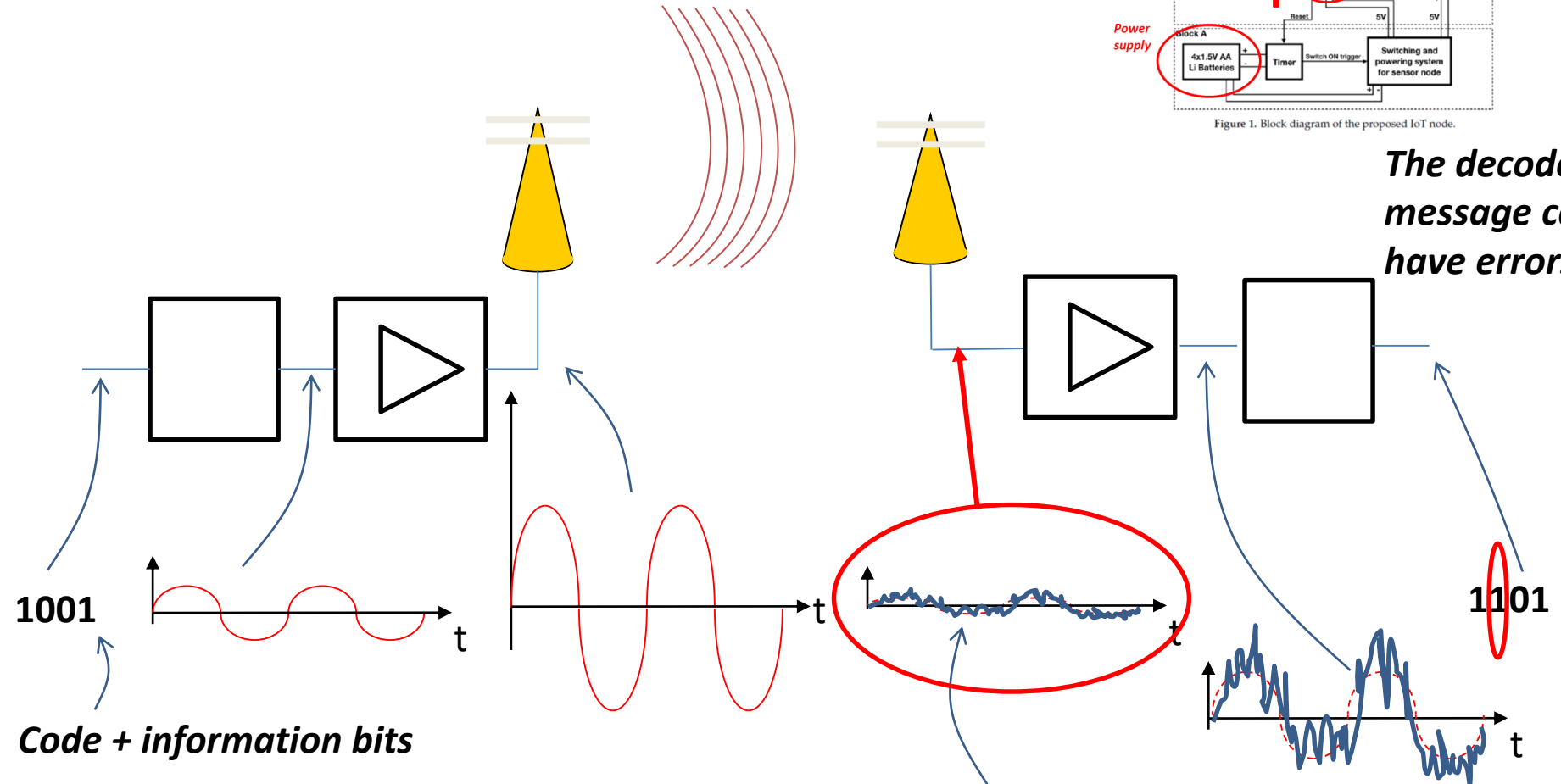


Figure 1. Block diagram of the proposed IoT node.

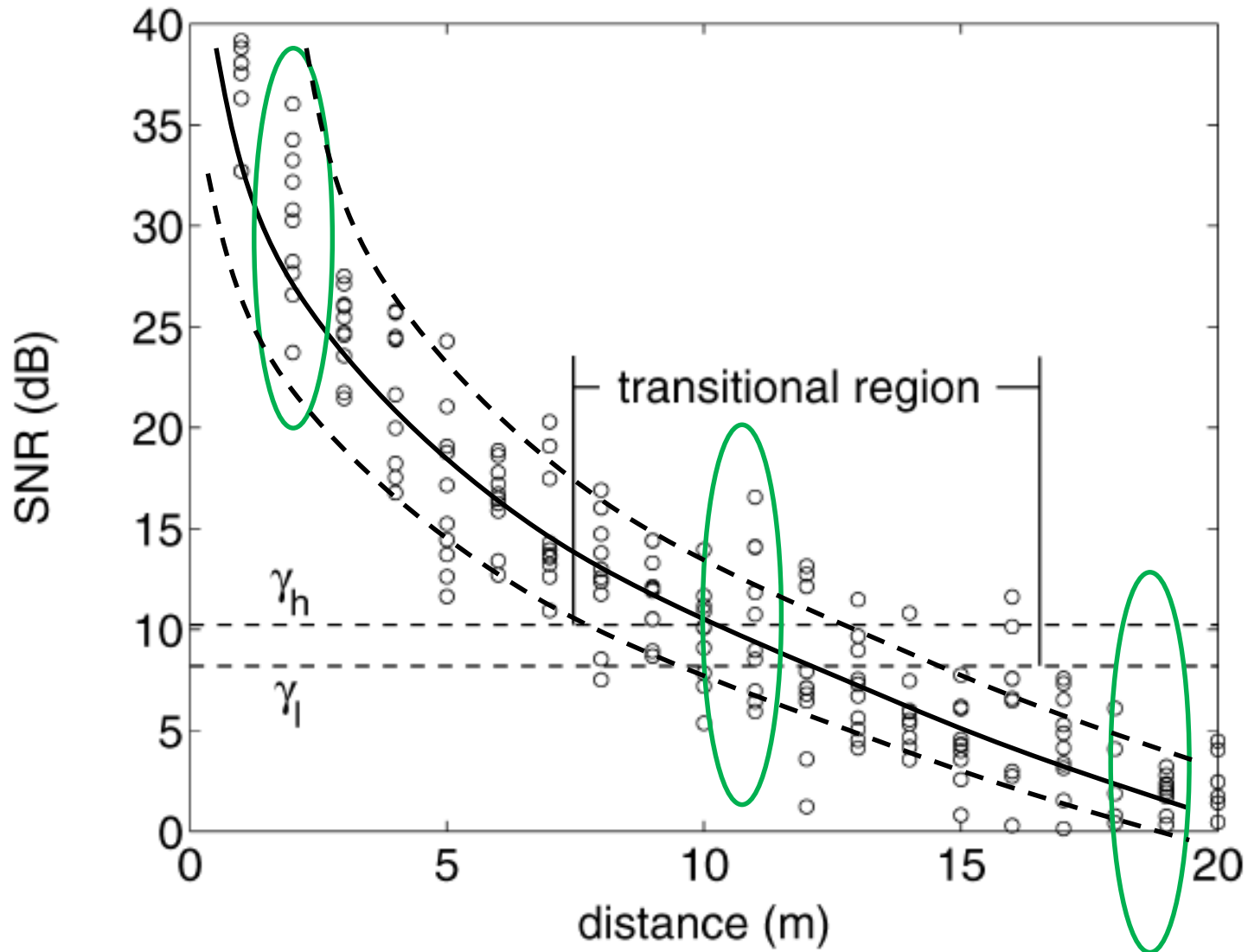
The decoded message can have errors



Extremely low power signal mixed with electrical noise => Signal power versus noise power (SNR, Signal to Noise ratio)

Atenuation

(a)



Multipath Fading: variation with space

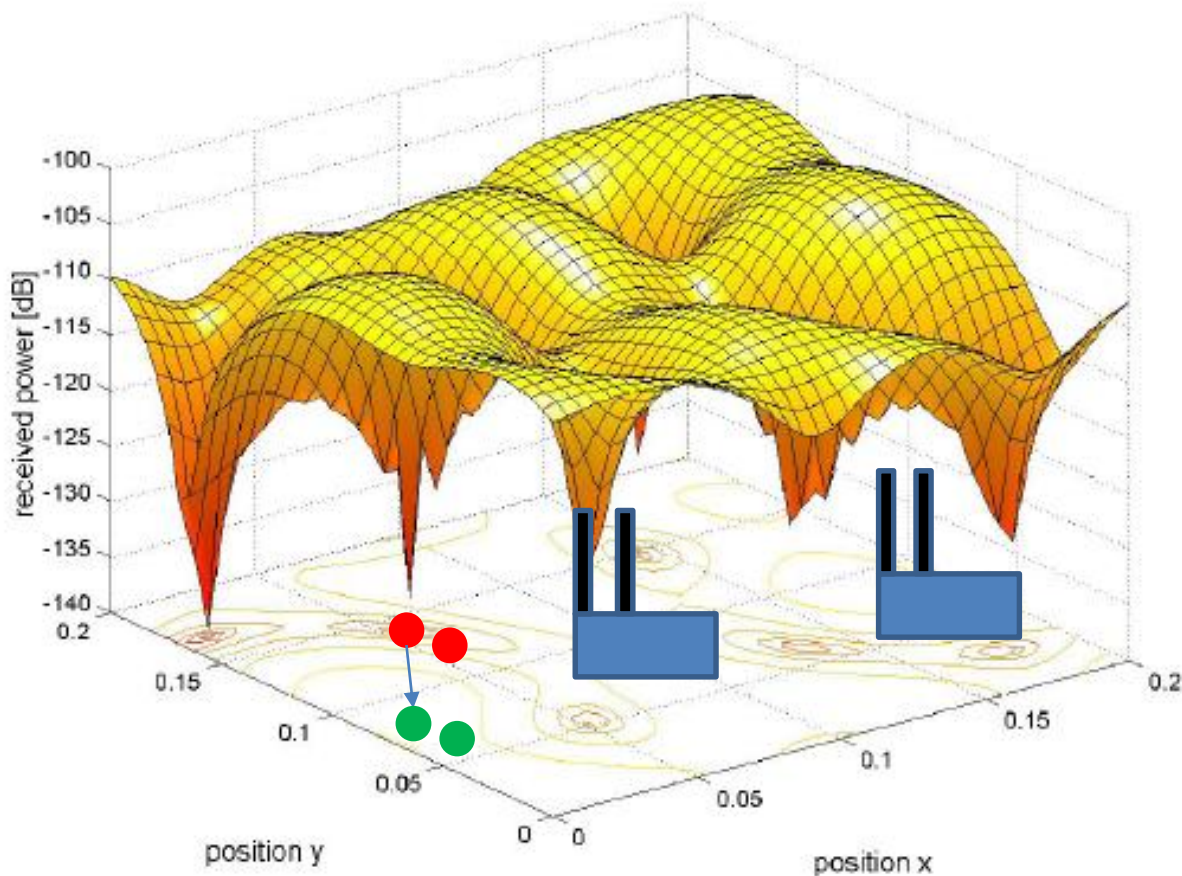


Figure 21: Typical signal quality fluctuations on wireless channel as a function of location [2].

Multipath Fading: variation with space

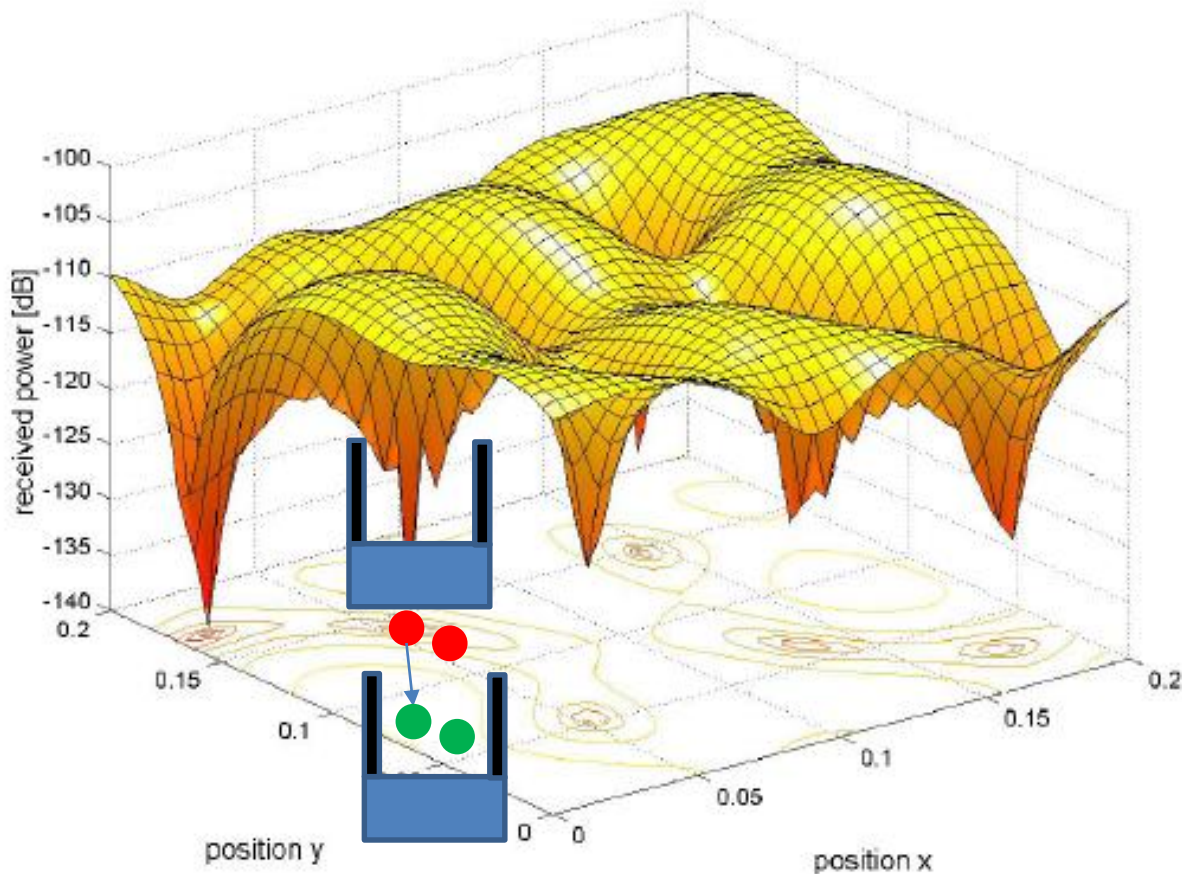


Figure 21: Typical signal quality fluctuations on wireless channel as a function of location [2].

Packet loss ratios

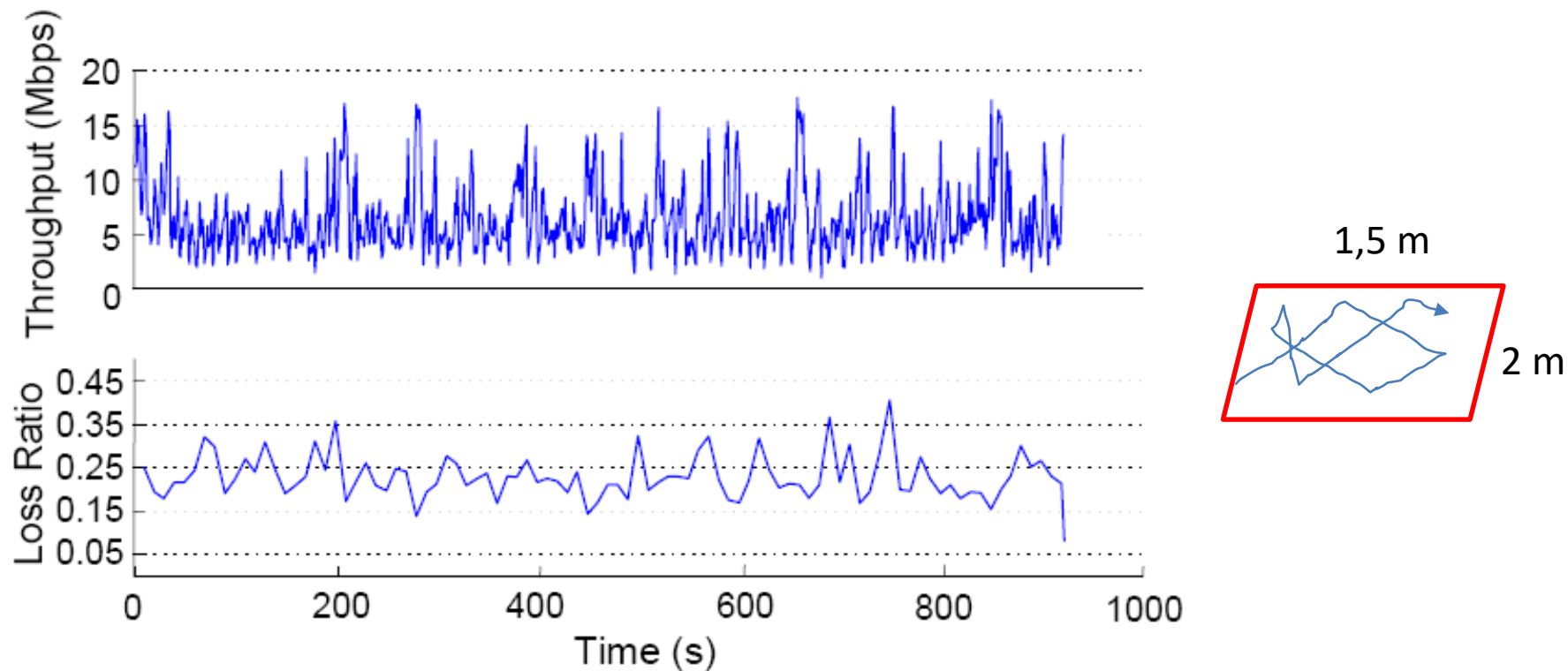
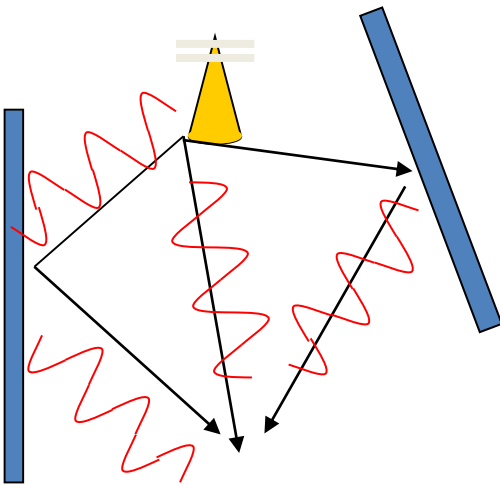


Figure 1-1: An example of poor performance in an 802.11a experiment that involved a moving transmitter and a stationary receiver. The top graph shows how throughput can fluctuate by almost an order of magnitude as the transmitter moves within a small 1.5 m $\times$ 2 m area (Figure 3-8). The bottom graph shows the corresponding frame loss rates averaged over one-second windows.

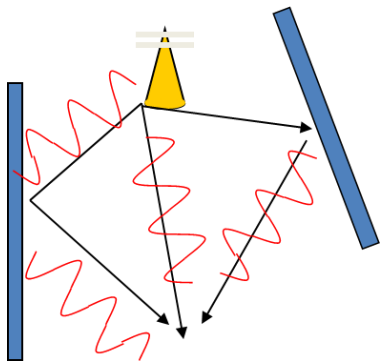
Delay spread in indoor/outdoor

The **delay spread** is the maximum difference between the arrival times of different multipath signals.



Δspace	Delay spread (Δtime)	Typical Scenario
3 m	10 ns	Indoor (PAN, WLAN)
30 m	100 ns	Indoor (WLAN)
300 m	1 μs	Outdoor (Cellular)
3 Km	10 μs	Outdoor (Cellular)

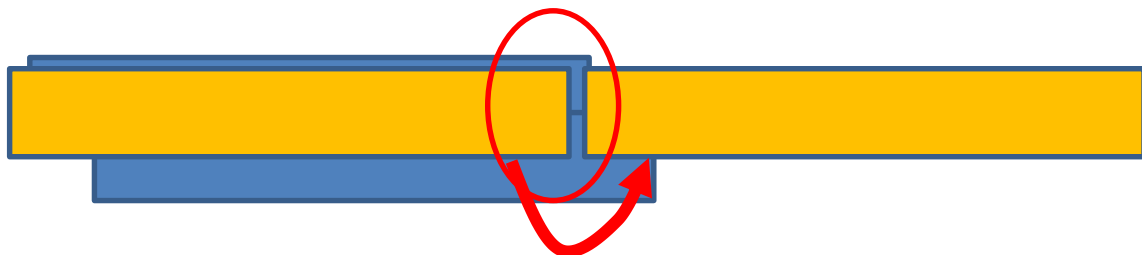
Symbol time = 500 nsec



Δspace	Delay spread (Δtime)	Typical Scenario
3 m	10 ns	Indoor (PAN, WLAN)
30 m	100 ns	Indoor (WLAN)
300 m	1 μs	Outdoor (Cellular)
3 Km	10 μs	Outdoor (Cellular)

Symbol time = 500 nsec =>
R = 2 Mbauds

indoor

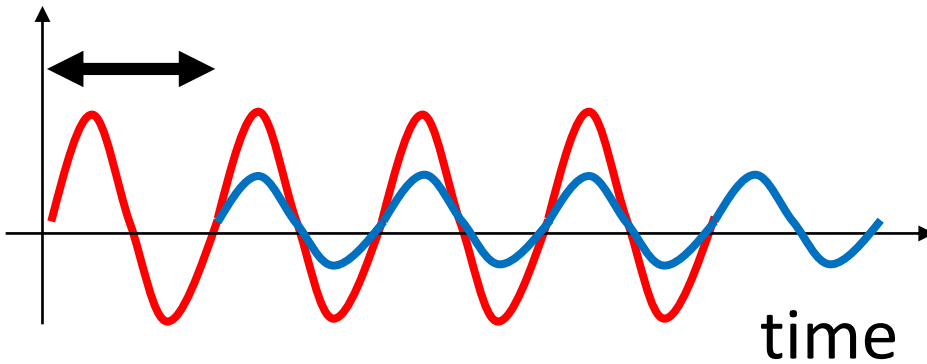


outdoor

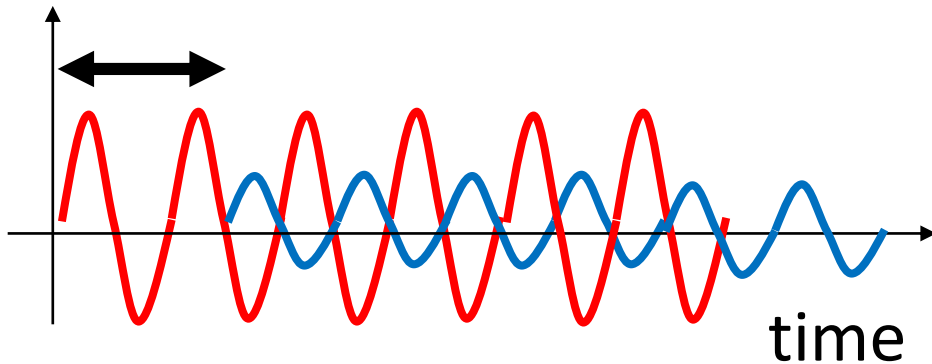


Same delay spread, different fc

Delay spread

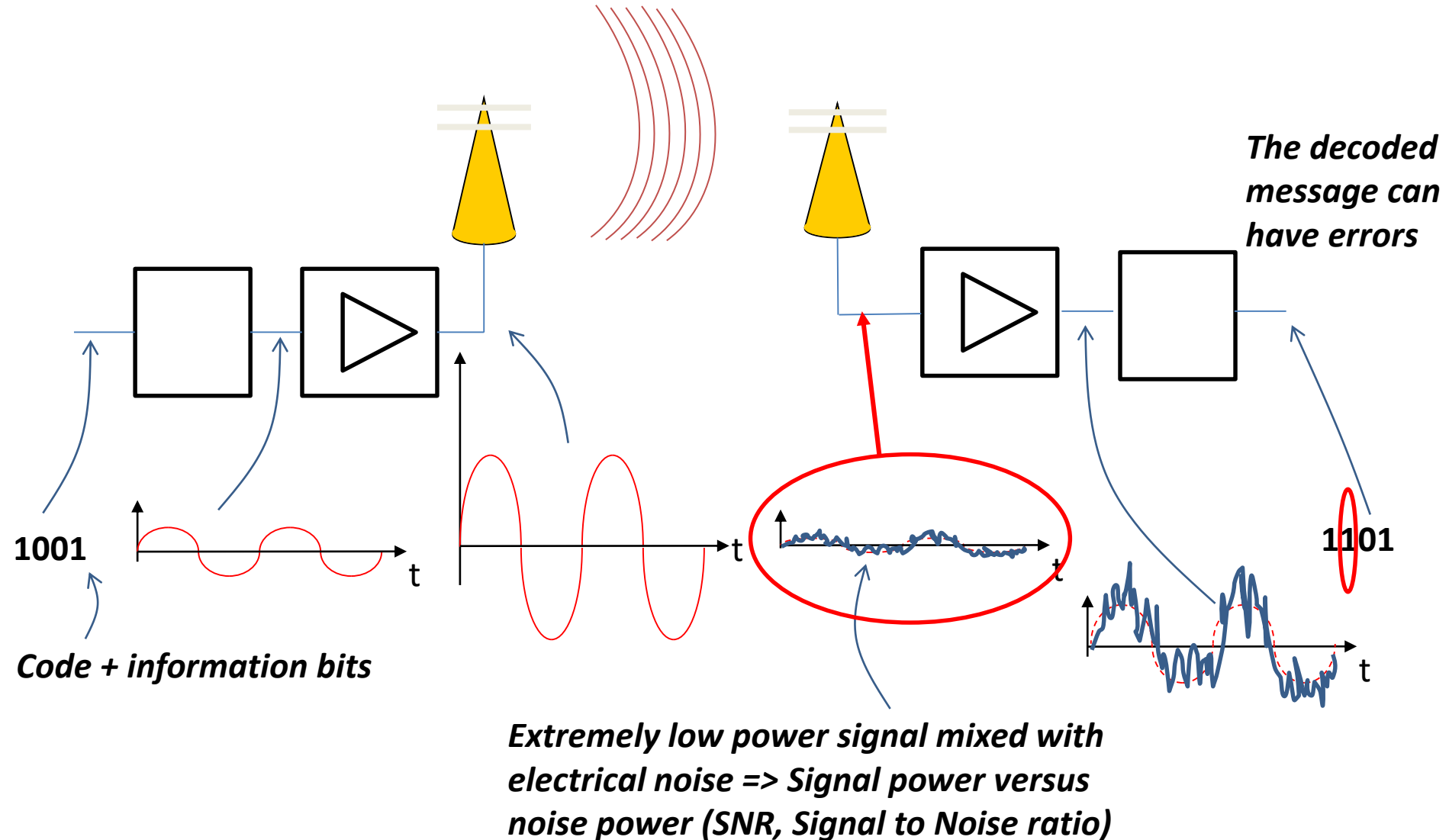


Constructive interference



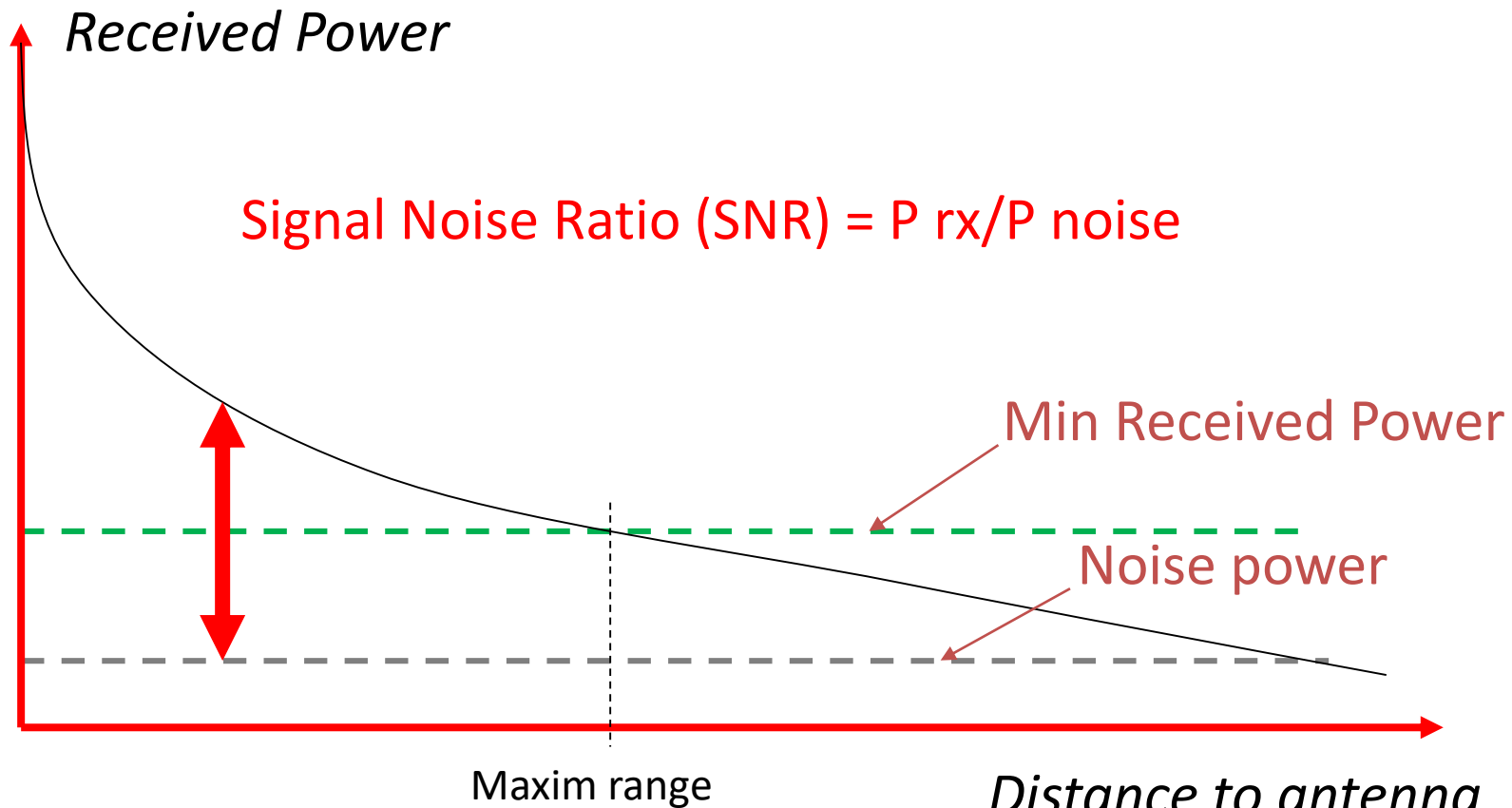
Destructive interference

Wireless transmission



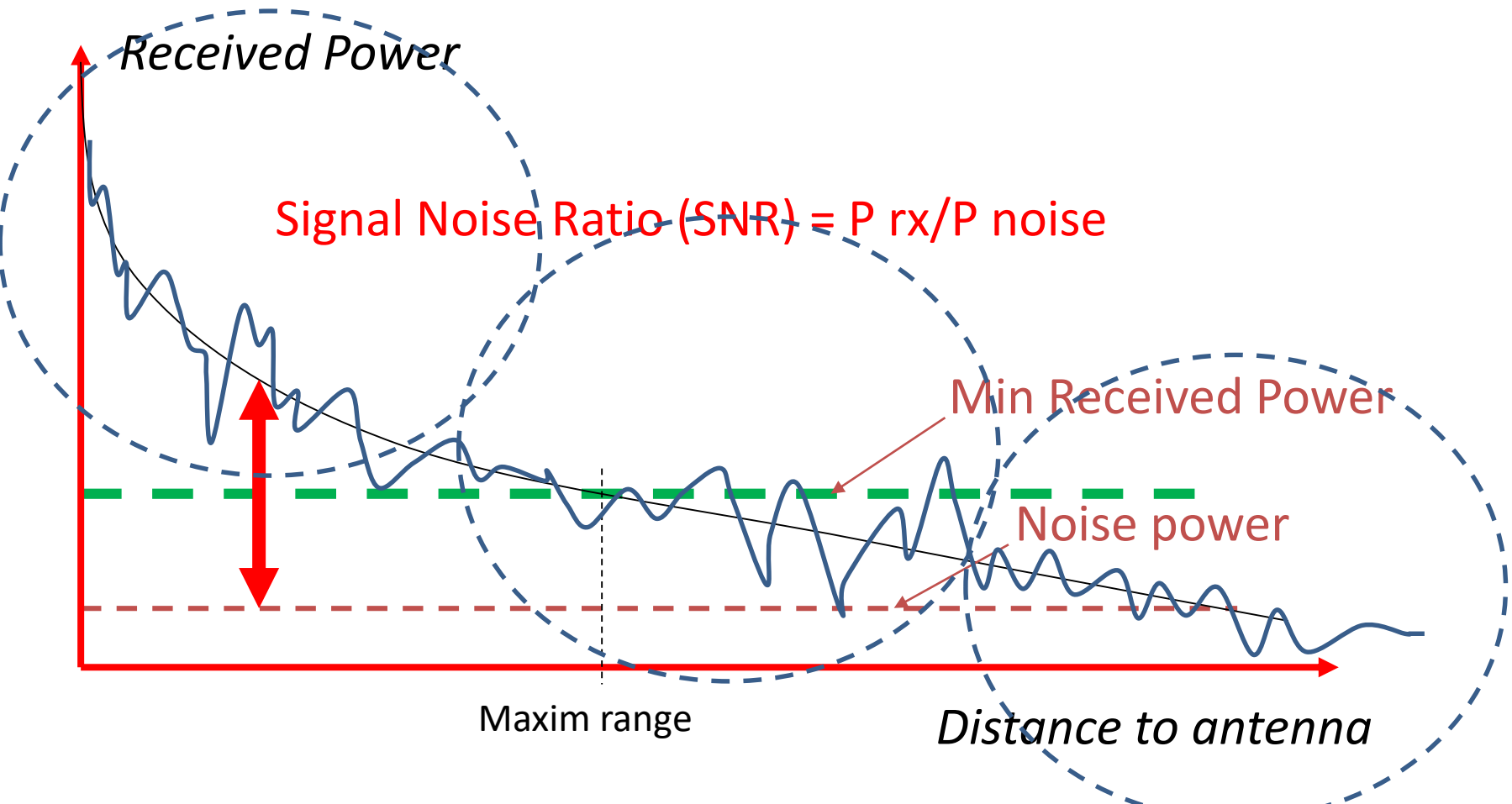
Radio transmission

Received power > minimum threshold



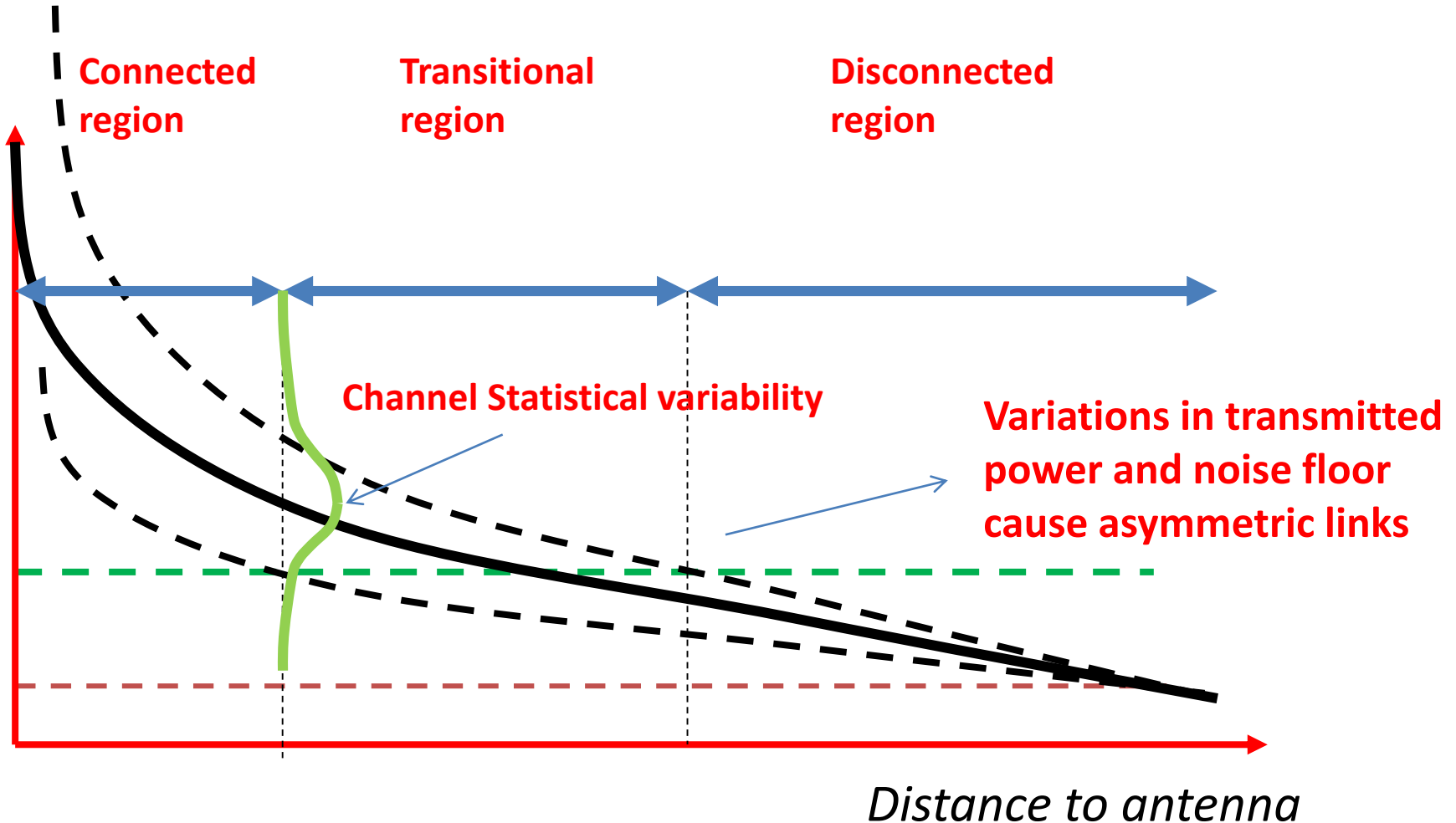
Radio transmission

Received power > minimum threshold



Radio transmission

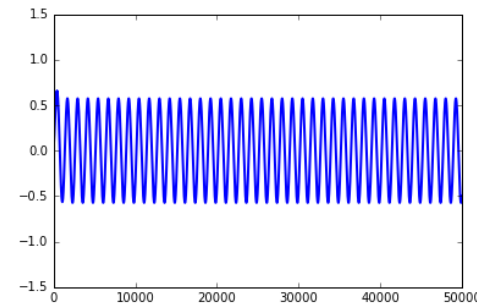
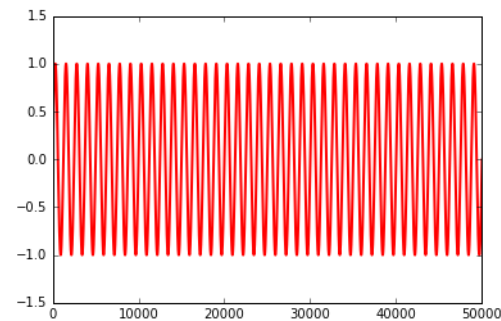
*Received
Power*



Frequency response

Frequency response and bandwidth

- Transmission systems introduce different attenuation levels depending on frequency

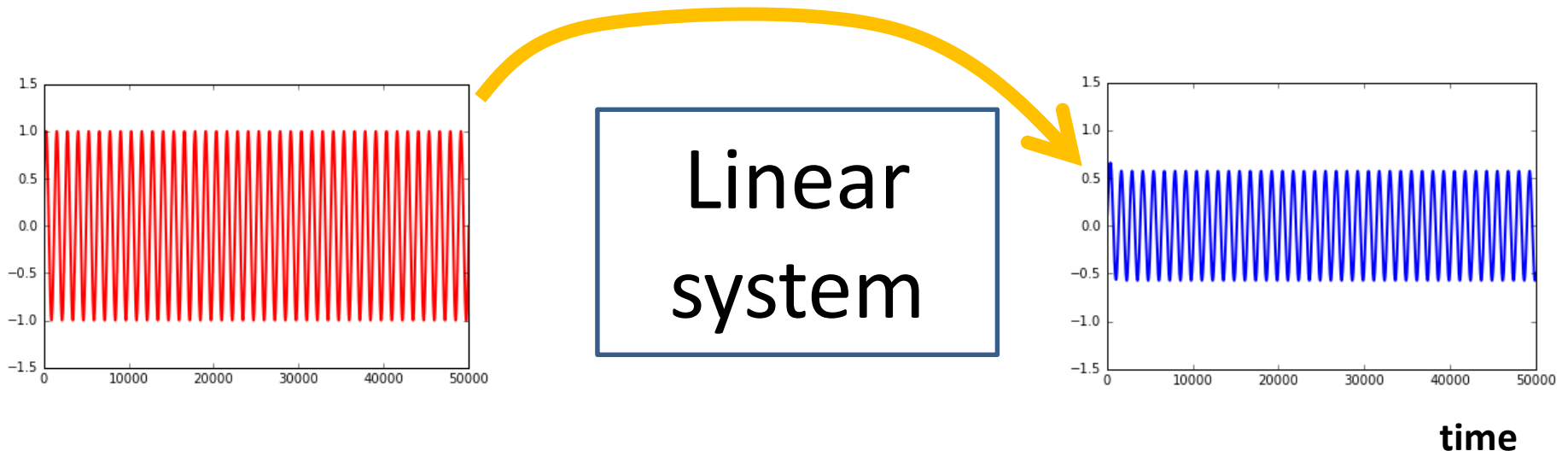


time

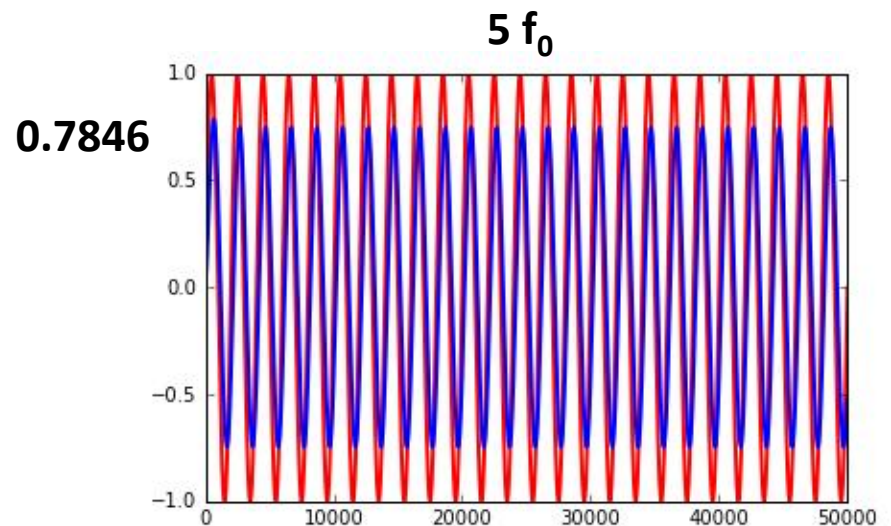
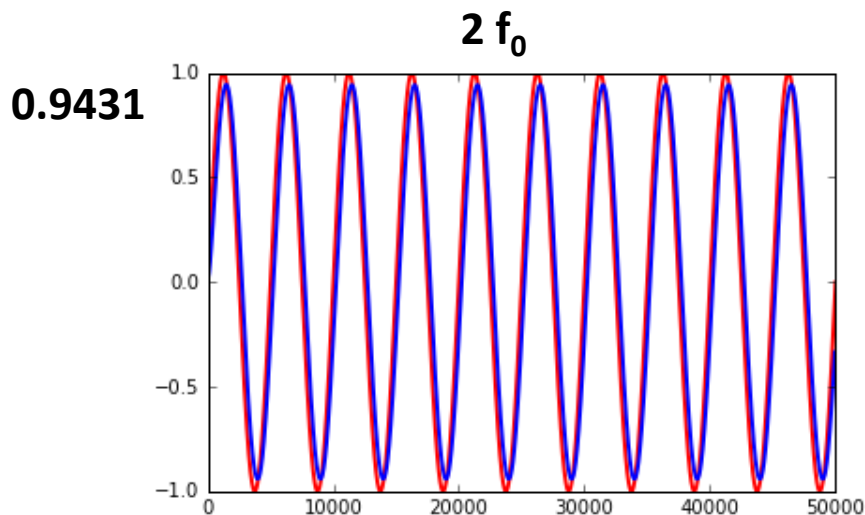
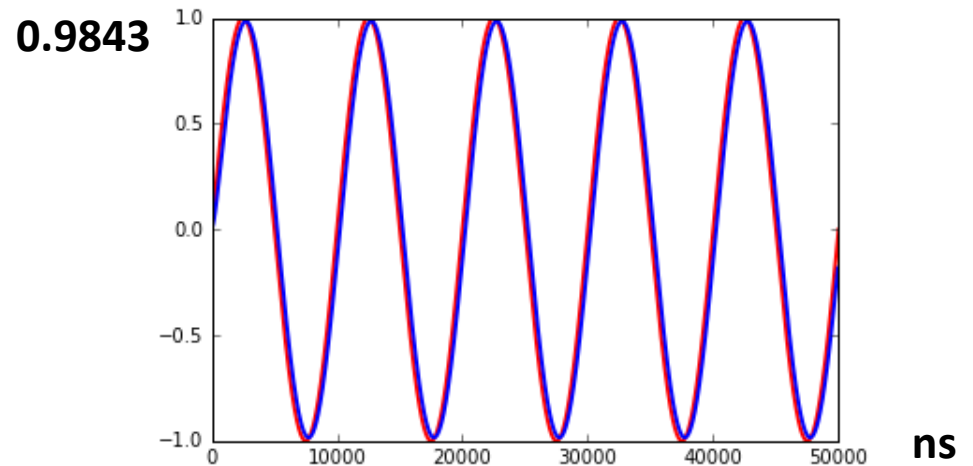
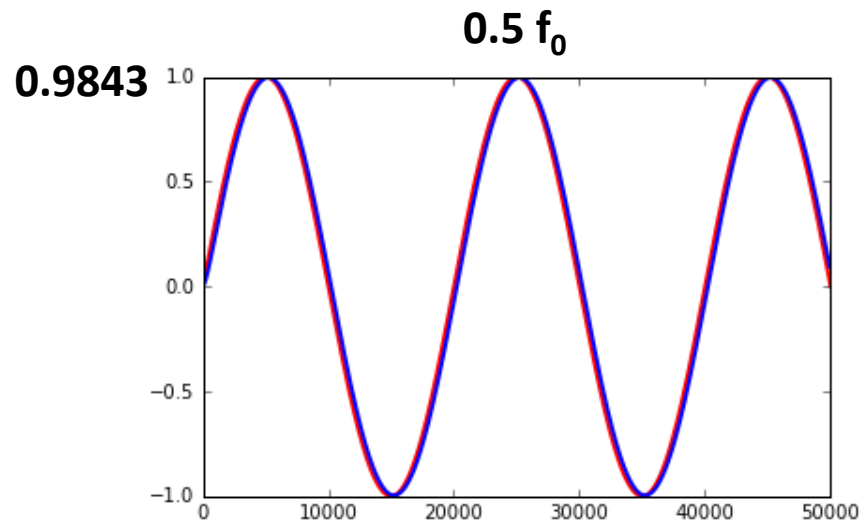
Response to sinusoidal signals

- When the input of a time-invariant linear system is a sinusoidal signal of a given frequency, the response is **another sinusoidal signal of the same frequency**, but different amplitude and phase (**depending on the frequency**)

$$\text{output amplitude} = \text{input amplitude} \times H(f)$$



Amplitude frequency response



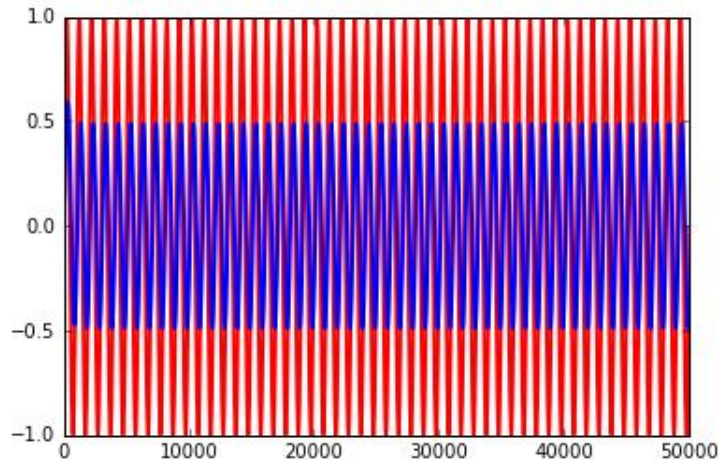
f0=100 KHz

time

Amplitude frequency response

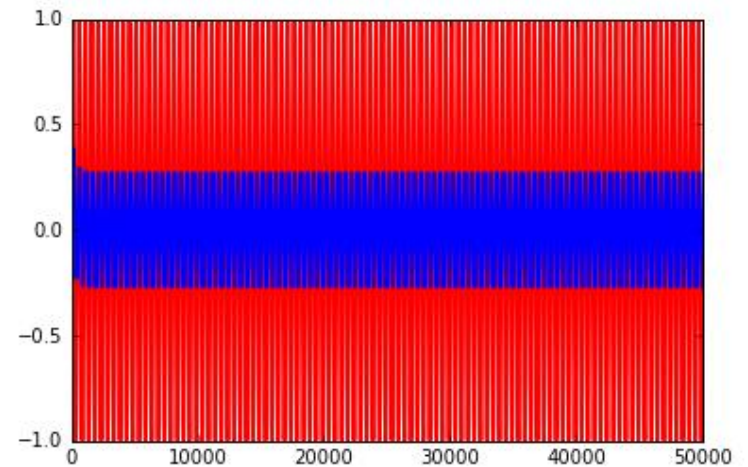
$10 f_0$

0.4886



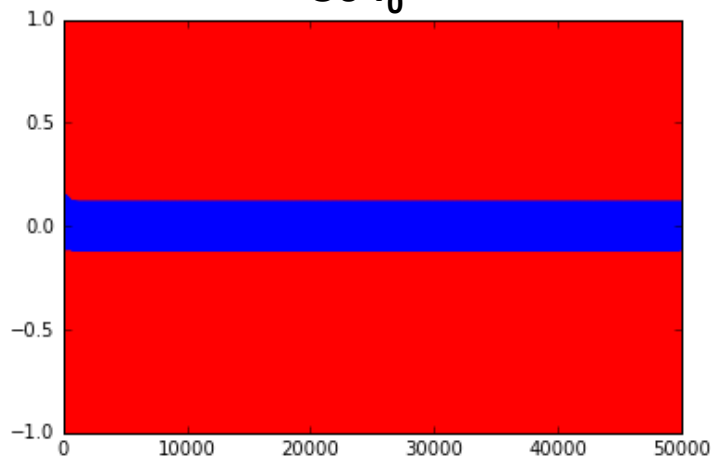
$20 f_0$

0.2707



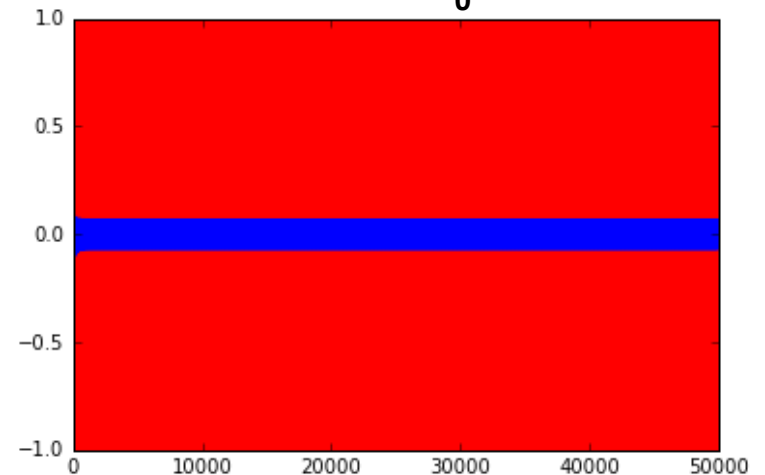
$50 f_0$

0.1153



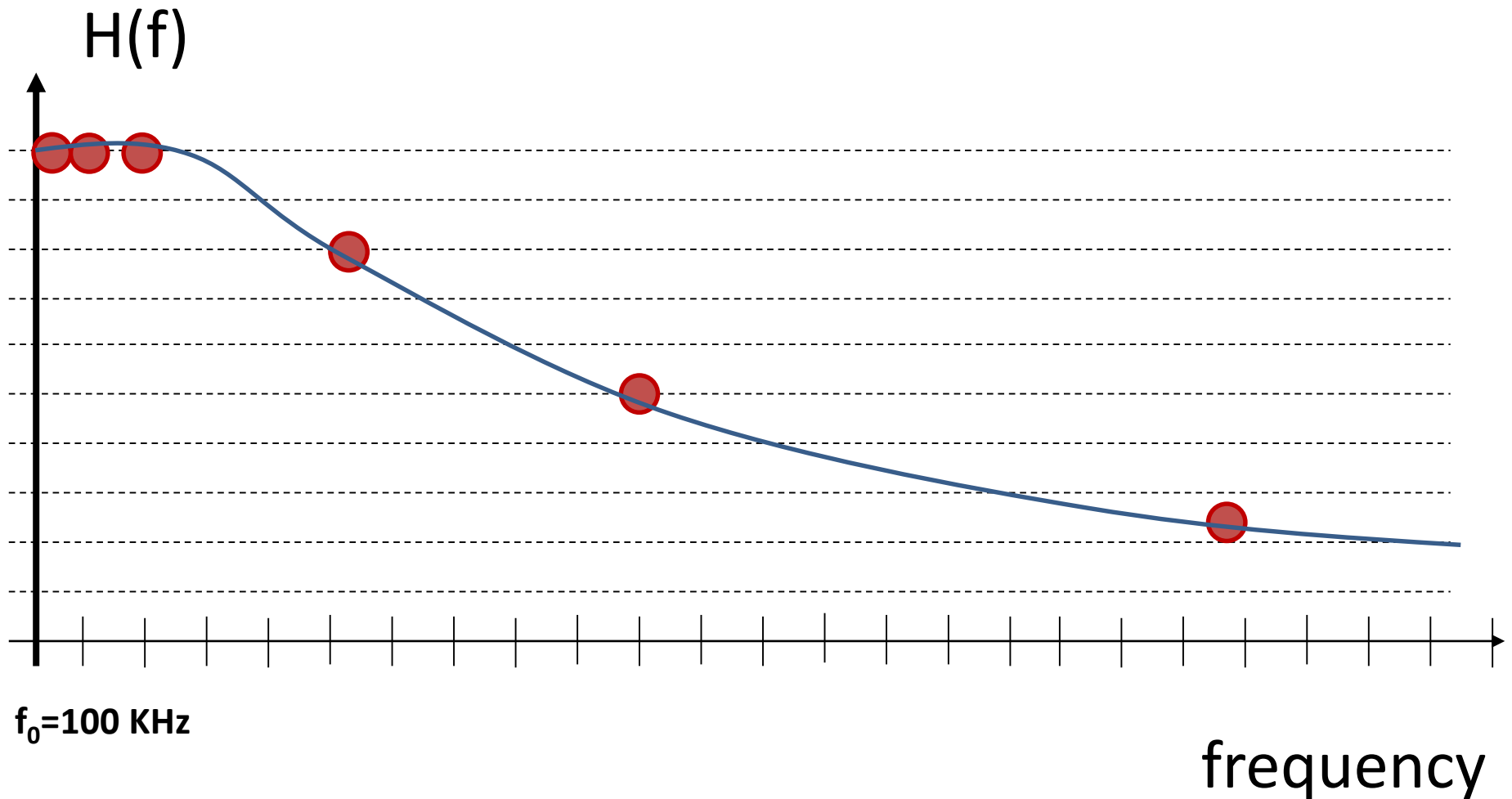
$100 f_0$

0.0649

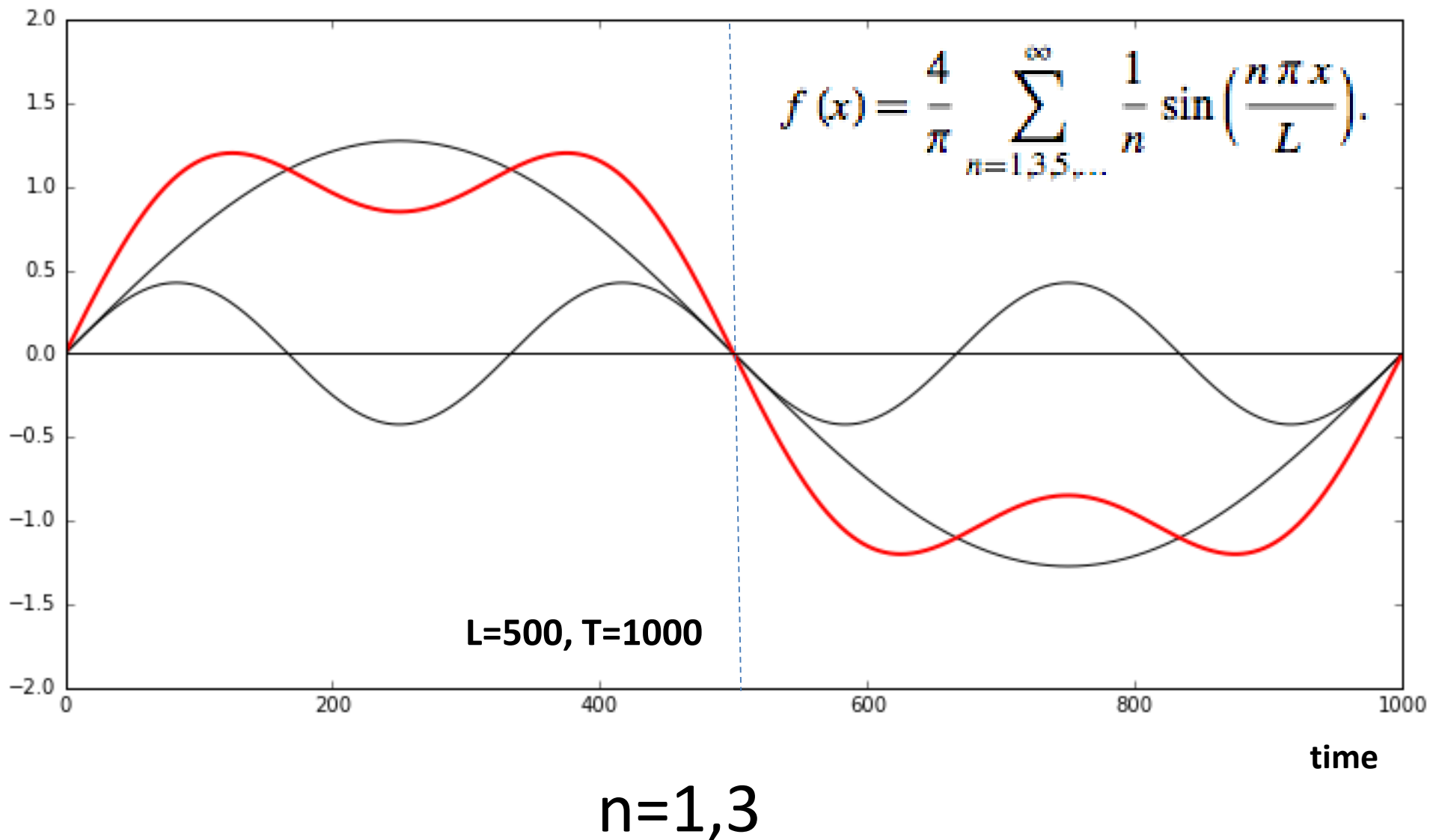


time

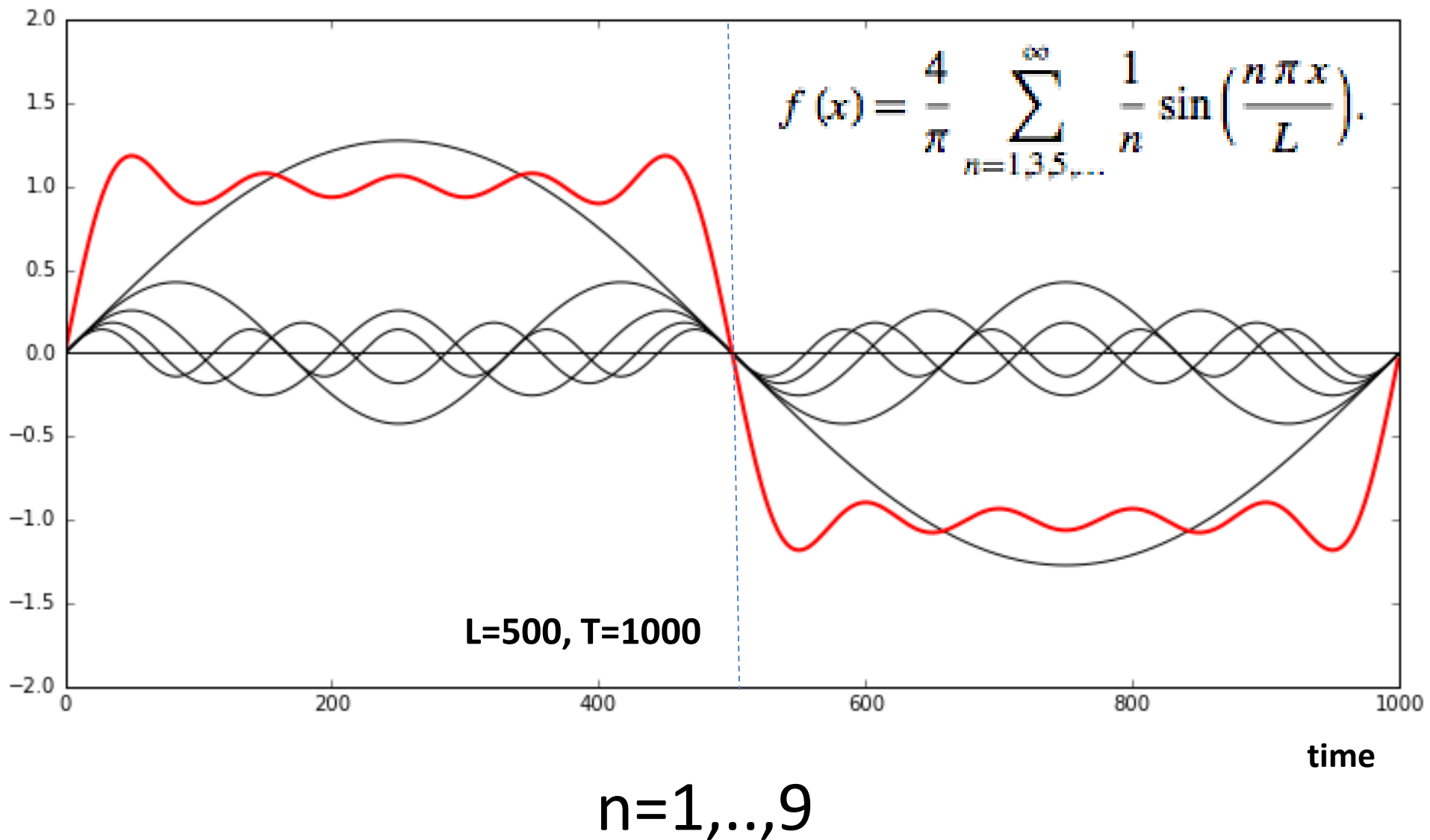
Amplitude frequency response



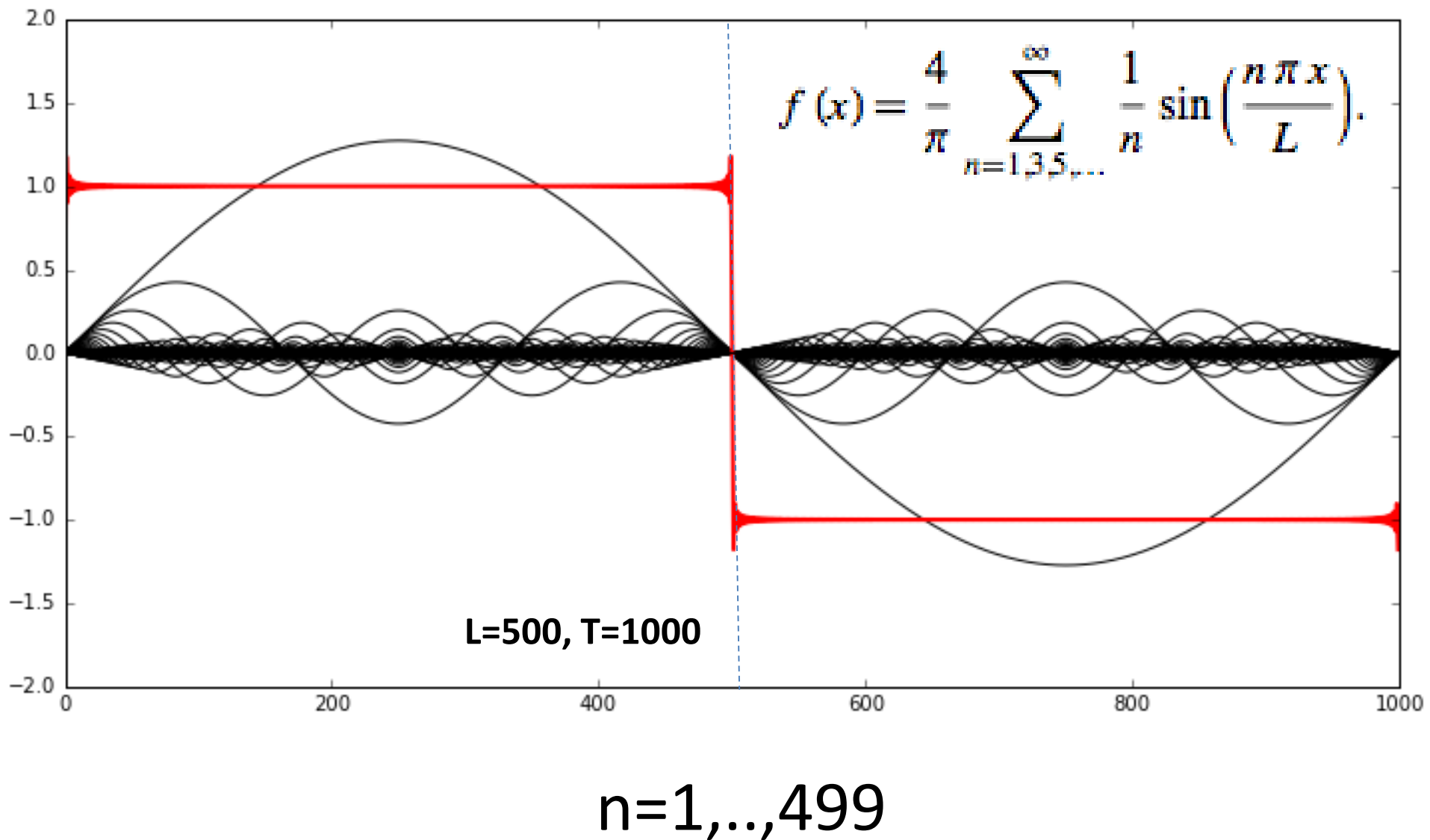
Fourier decomposition of a signal



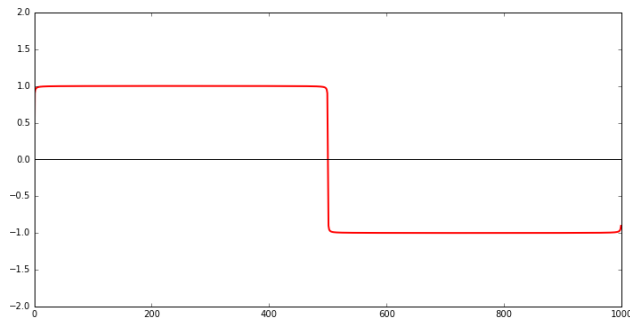
Fourier decomposition of a signal



Fourier decomposition of a signal

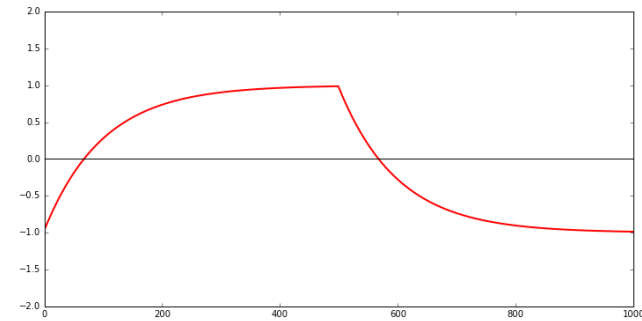


Output?



time

Linear
system



$$\alpha x(t) + \beta y(t)$$



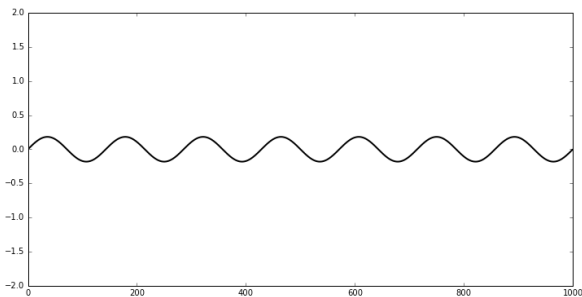
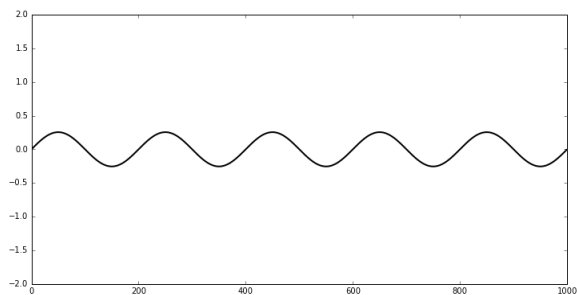
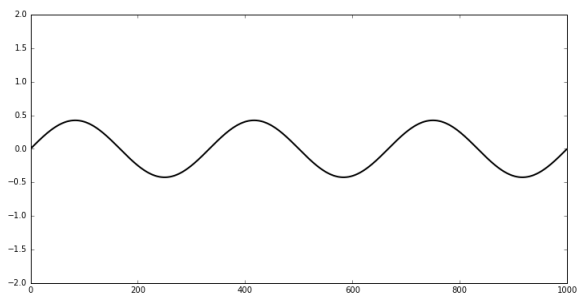
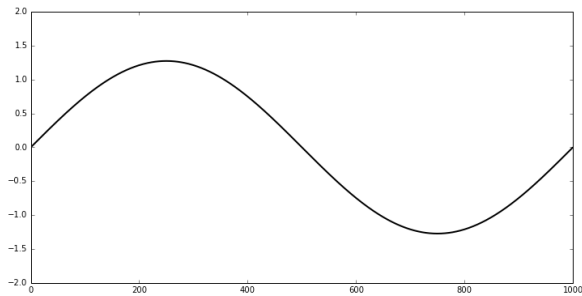
LS



$$\alpha L(x(t)) + \beta L(y(t))$$

Main idea behind signal filtering

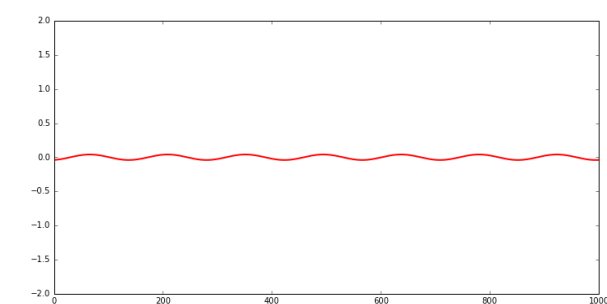
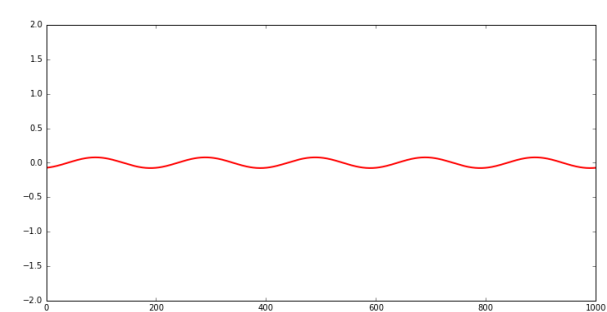
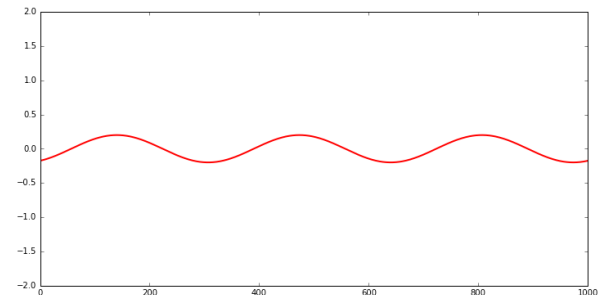
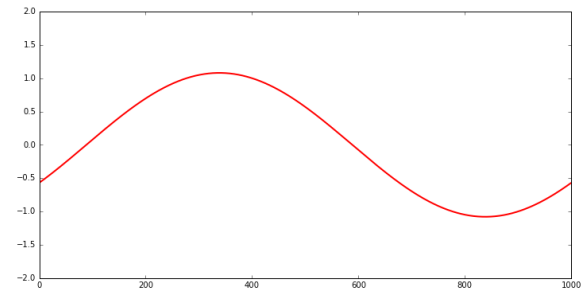
Exploit Fourier decomposition and superposition property of linear systems to predict the output signal



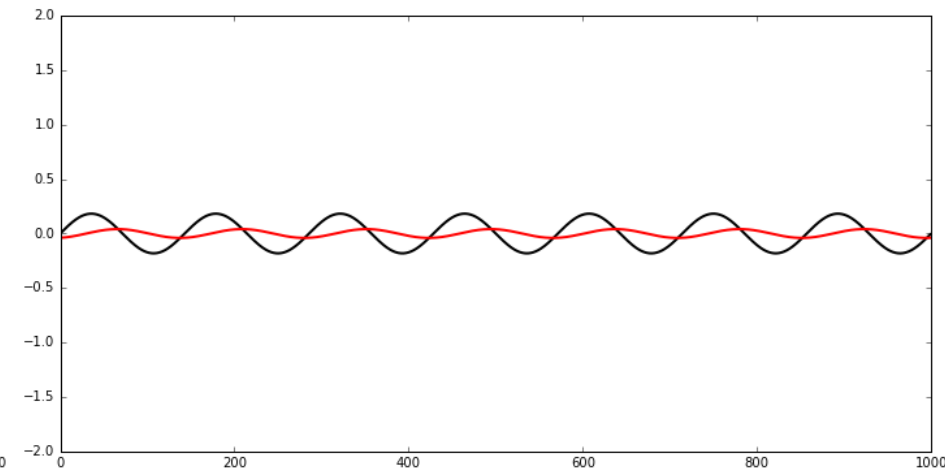
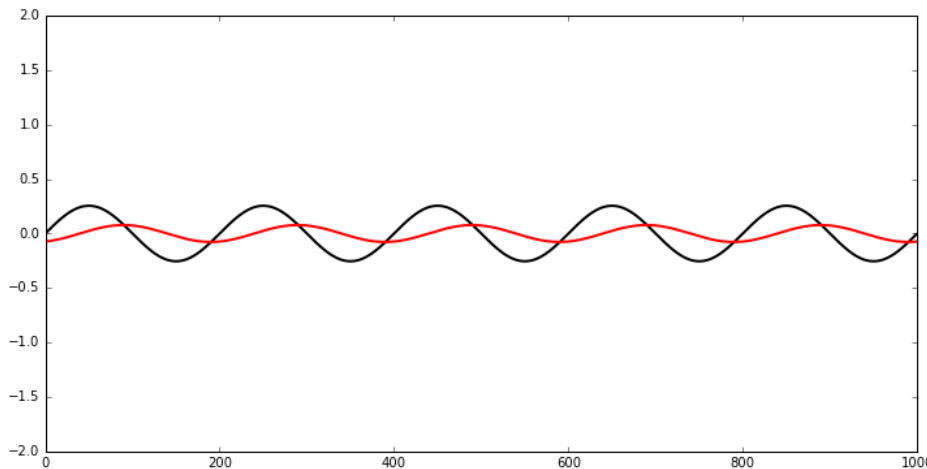
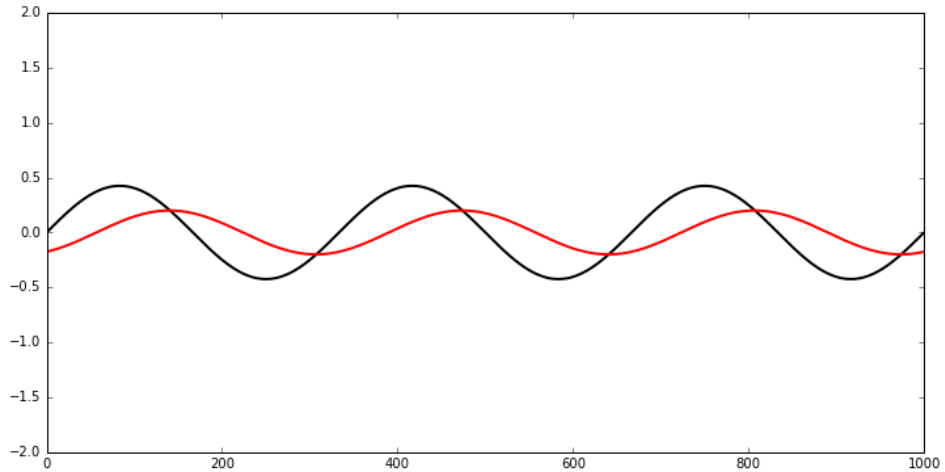
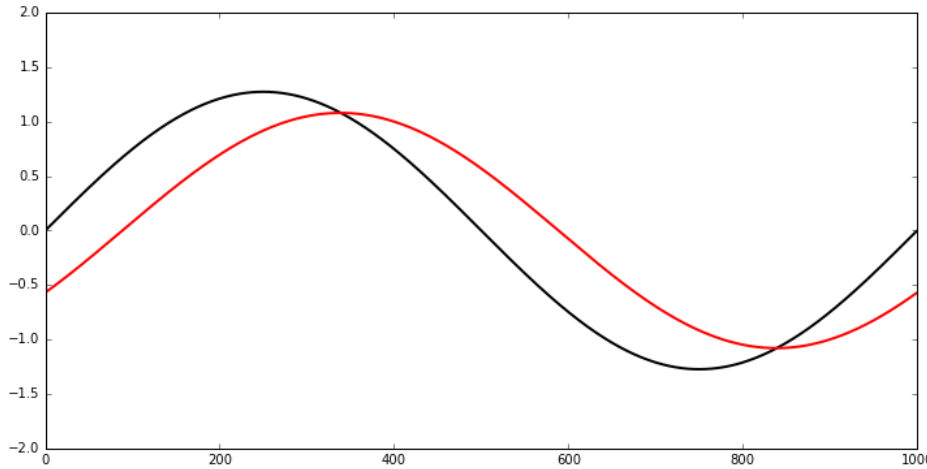
Linear
system

*Amplitude/phase
response depends on
frequency of Fourier
component*

time

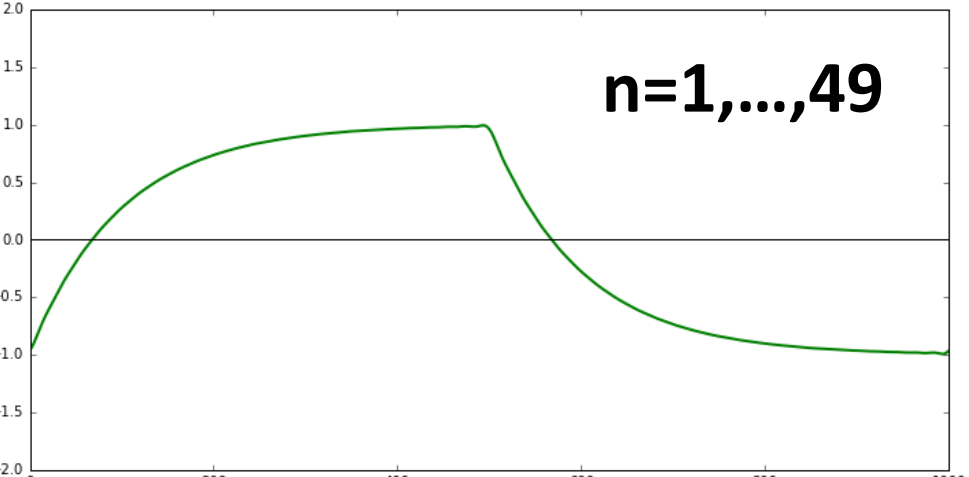
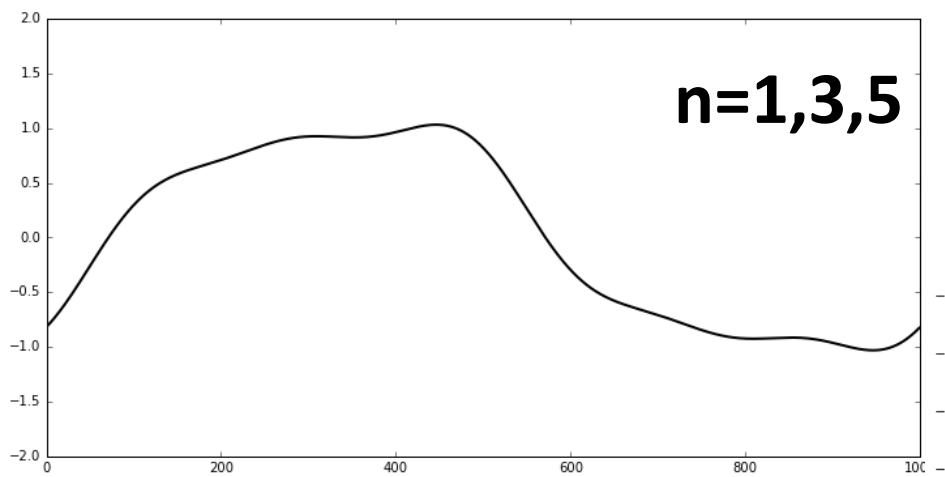
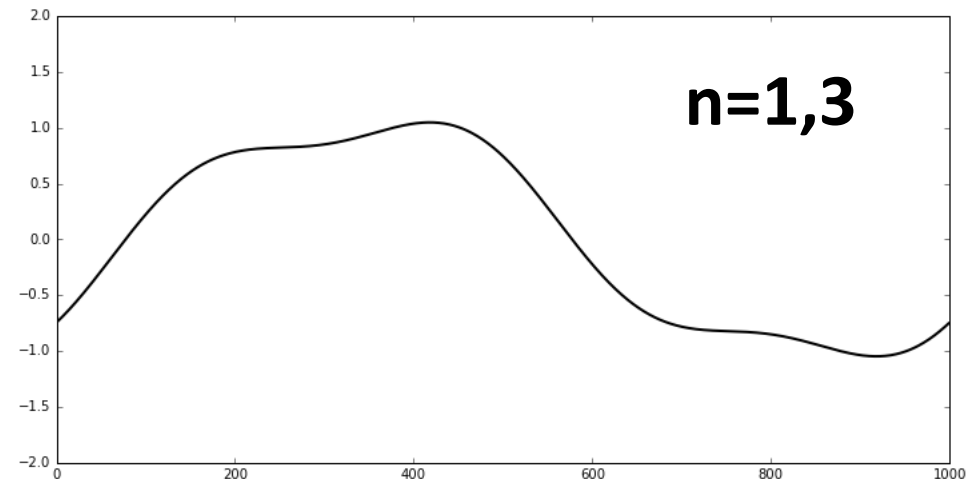
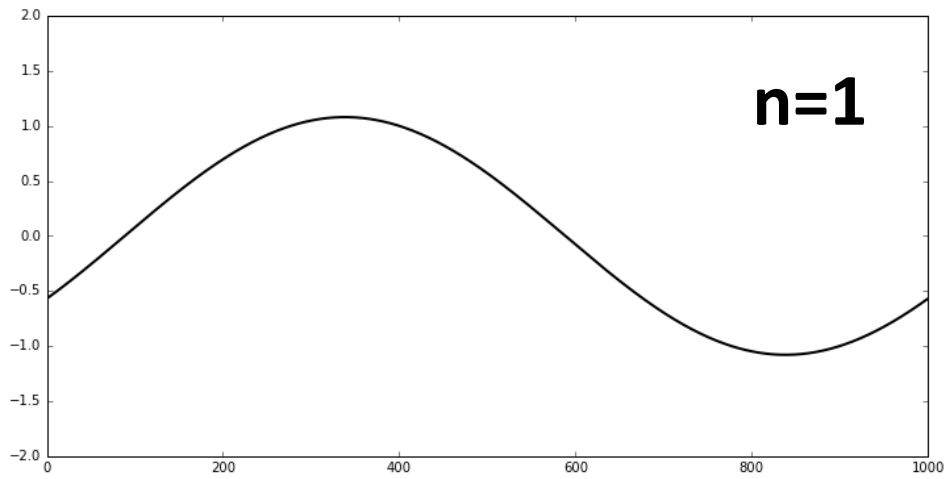


Amplitude/phase response depends on frequency of Fourier component

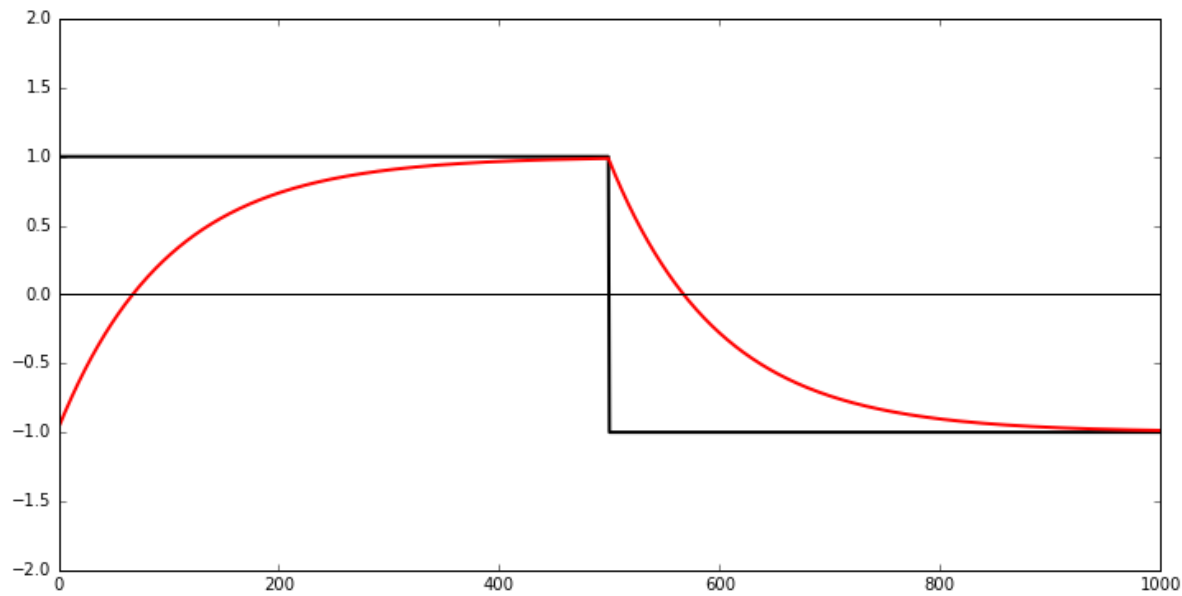


time

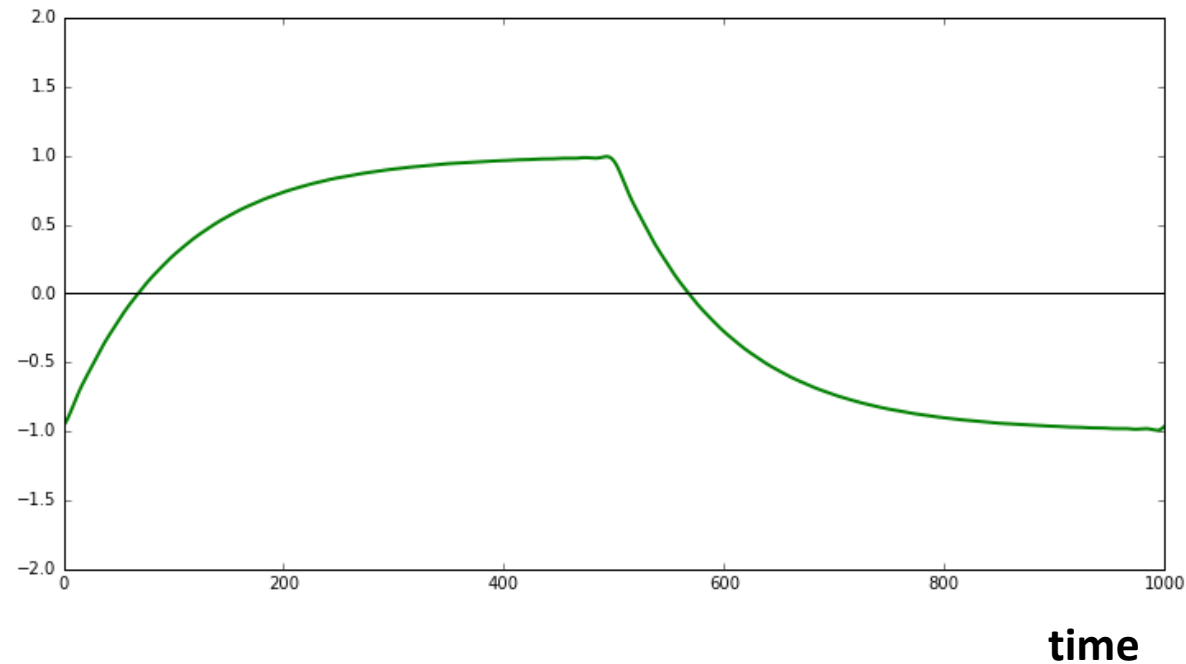
Superposition of output



time



**Input/output
signals**

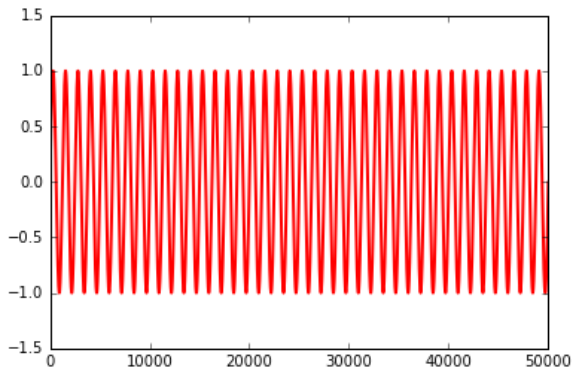


**Output using
superposition
to Fourier
Components
(n:1..,49)**

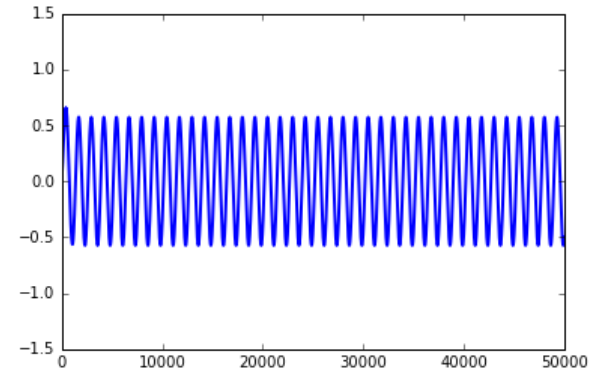
Noise & signal filtering

Filtering

- Low pass filters: attenuates/rejects high frequency components of a signal



Low pass
filter



time

Low-pass filters

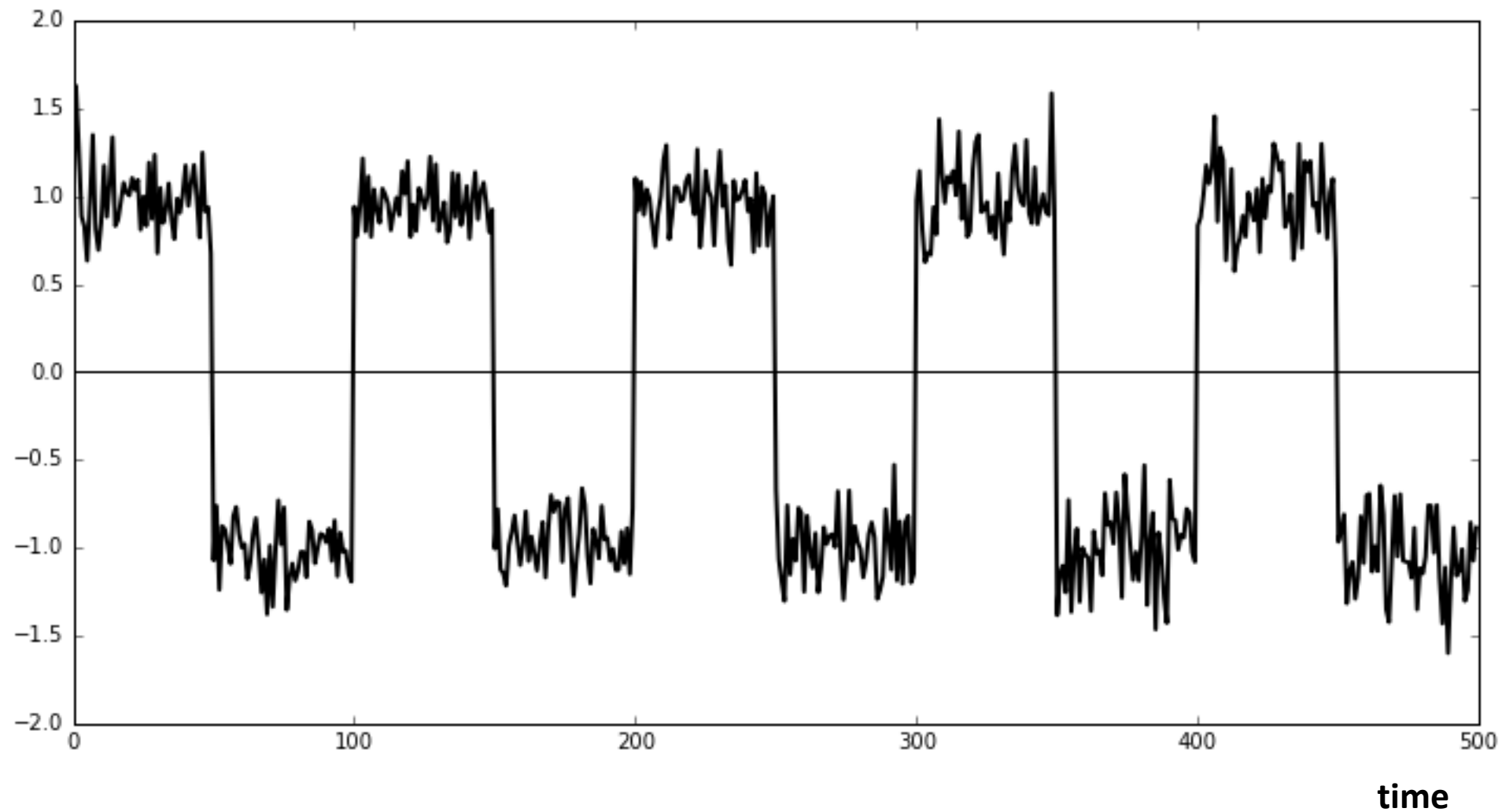
- Simple example of **low-pass FIR** (Finite Impulse Response) filter

$$y[t] = (1/N) * (x[t-(N-1)] + \dots + x[t-2] + x[t-1] + x[t])$$

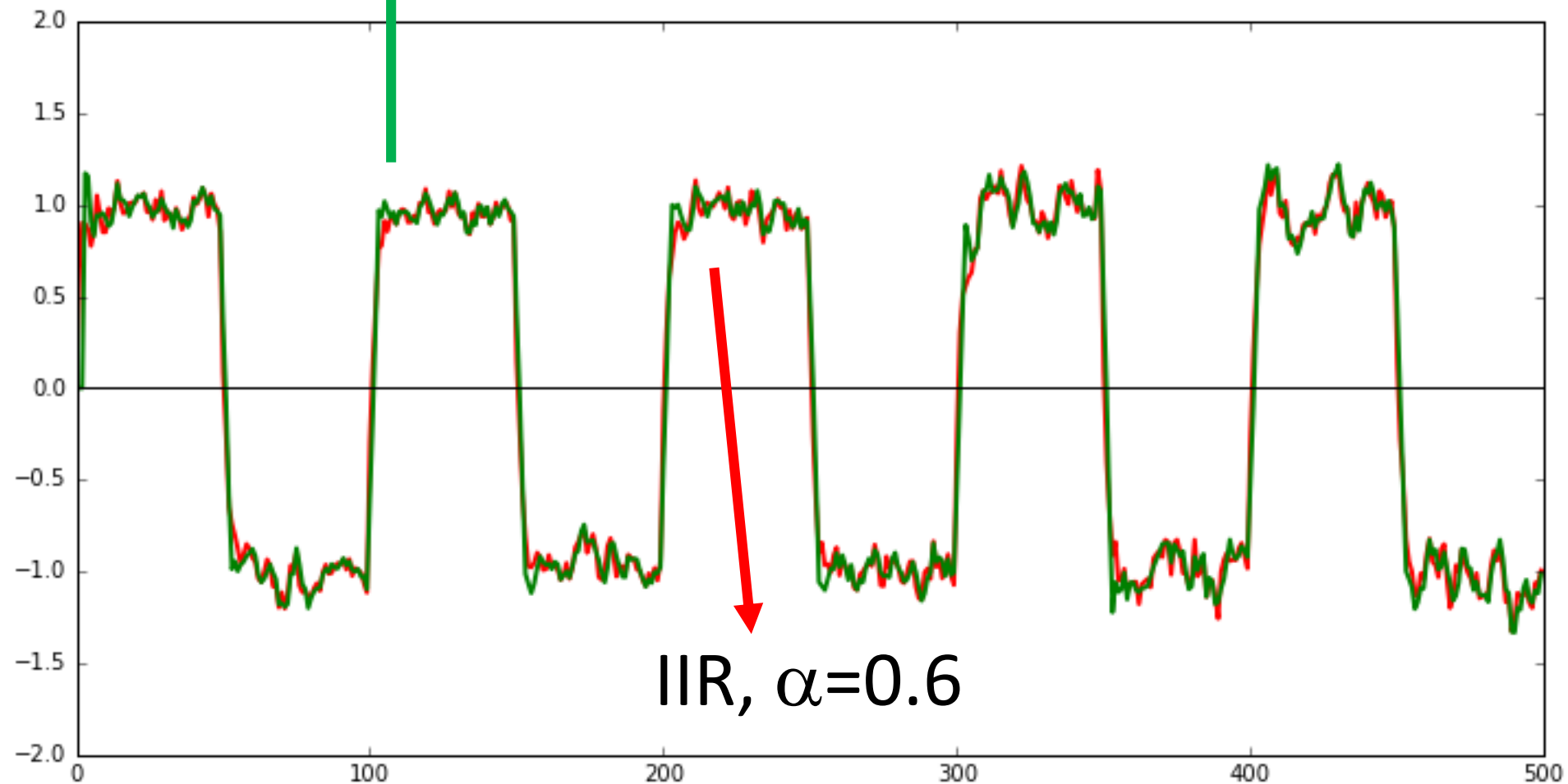
- Simple example of **low-pass IIR** (Infinite Impulse response) filter

$$y[t] = \alpha * y[t-1] + (1-\alpha) * x[t]$$

Filtering out noise



FIR, N=4

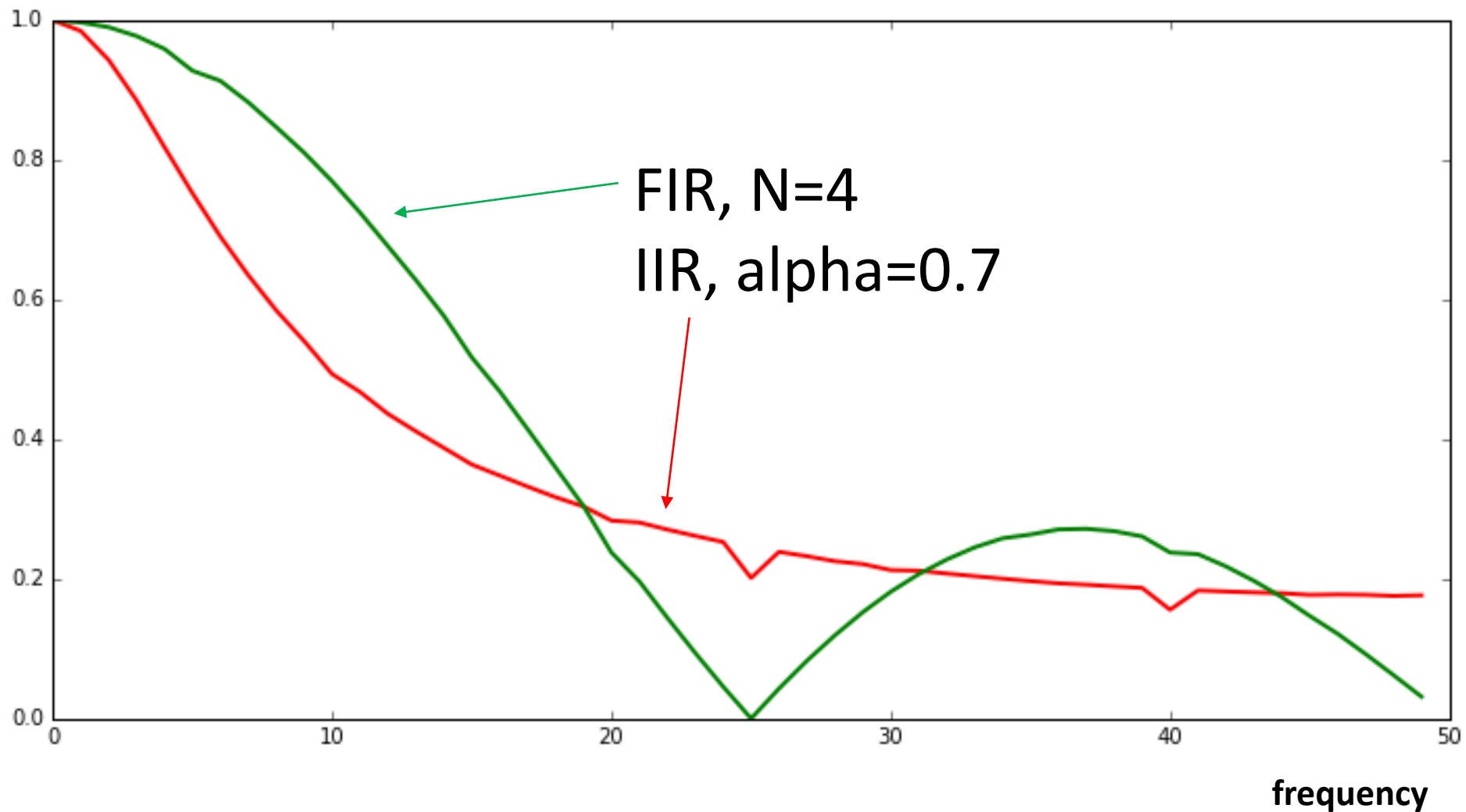


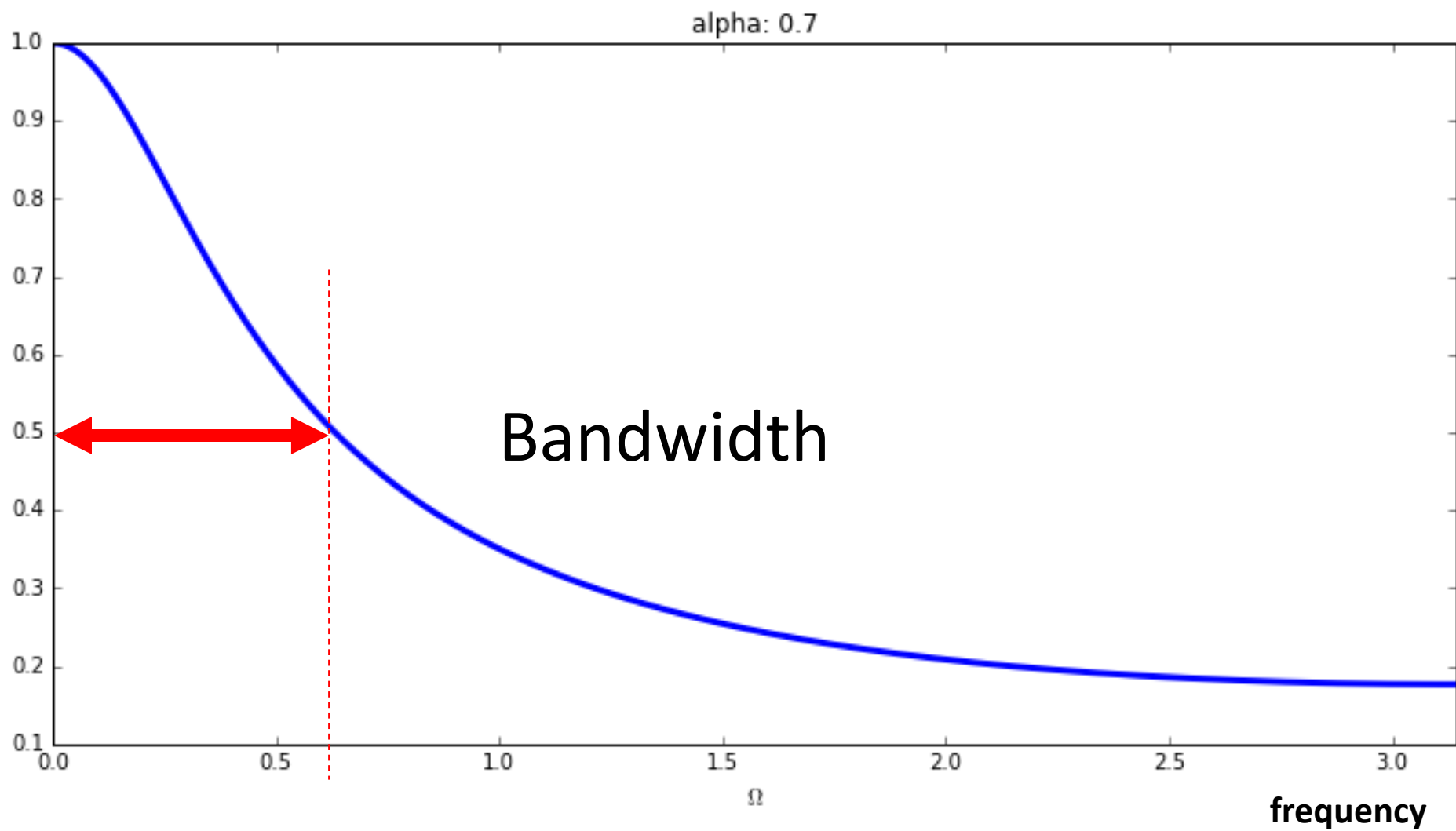
IIR, $\alpha=0.6$



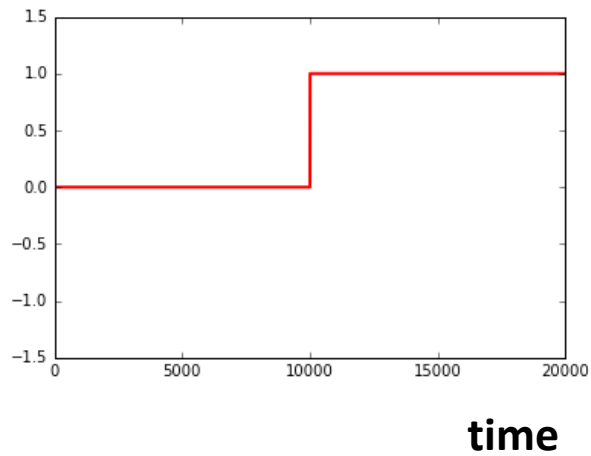
time

Simple evaluation of frequency response

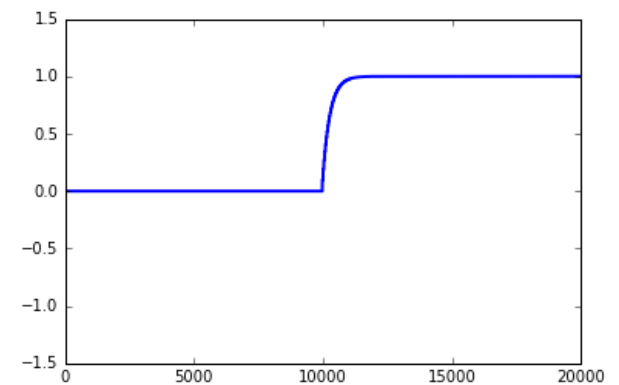




Time response

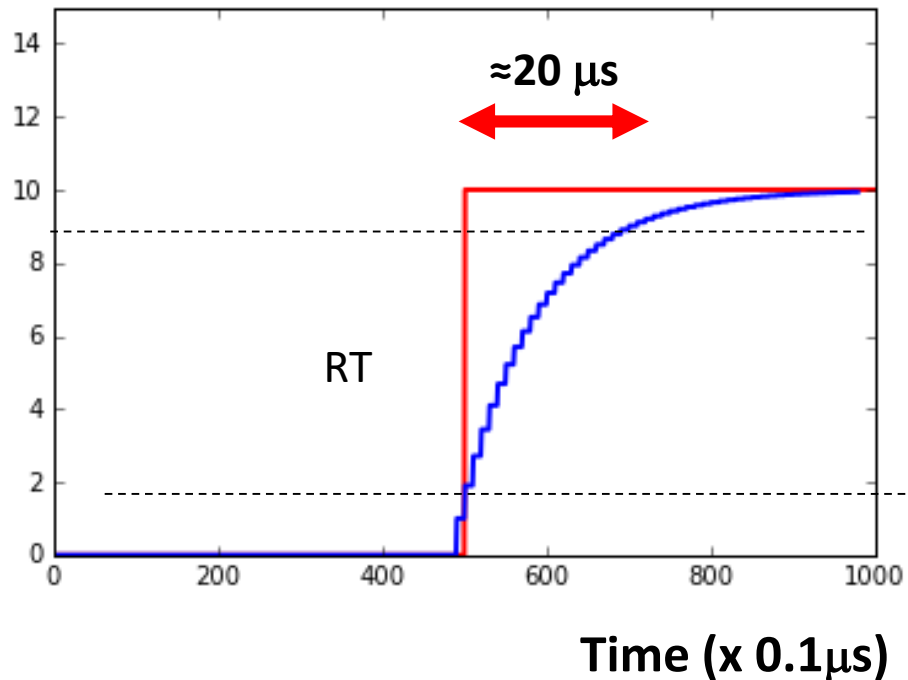


Low pass
filter

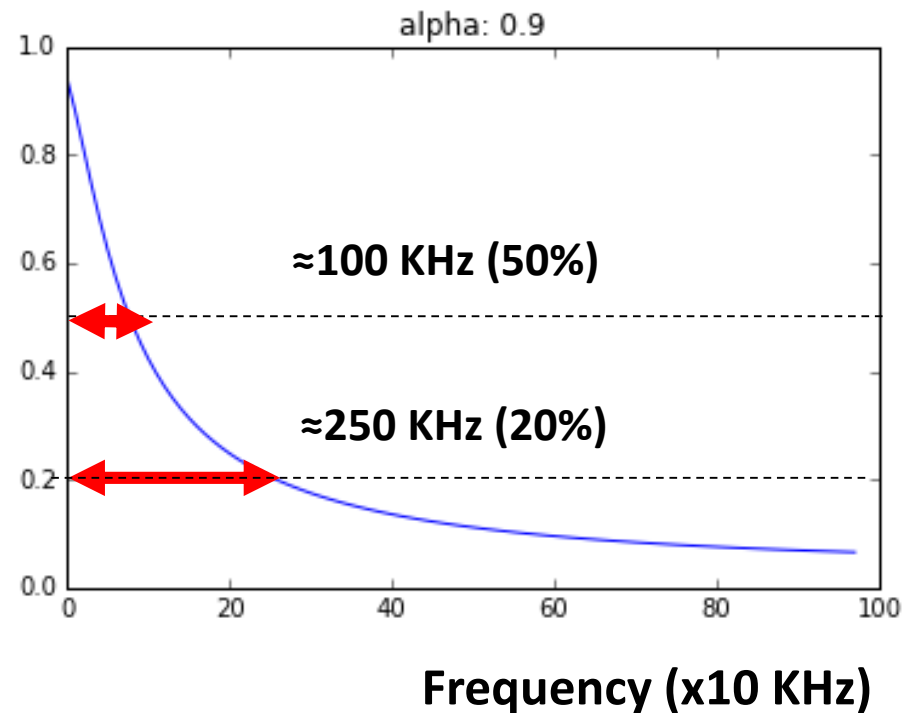


Time and frequency response

Rising Time (e.g. 10 to 90%)



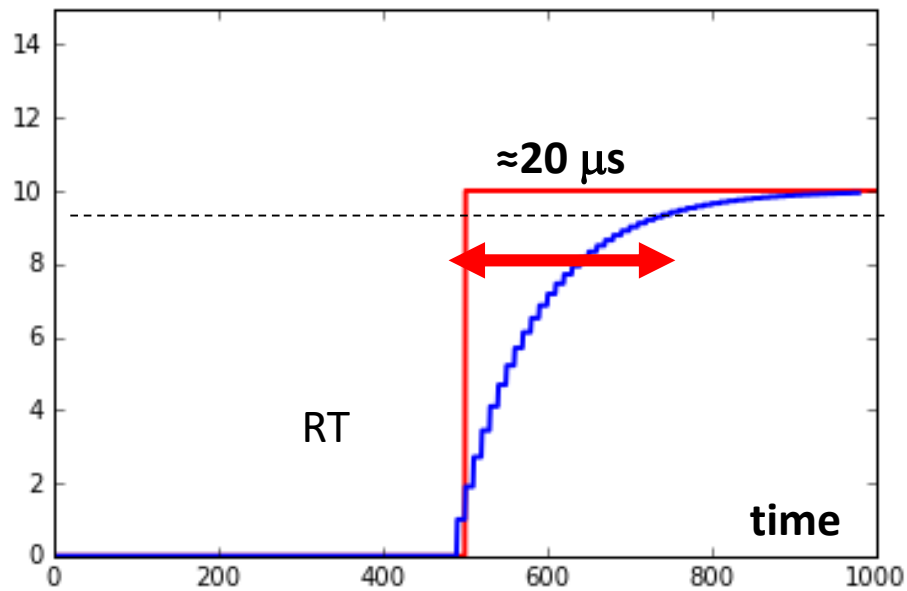
Bandwidth (e.g. 50%, 20%)



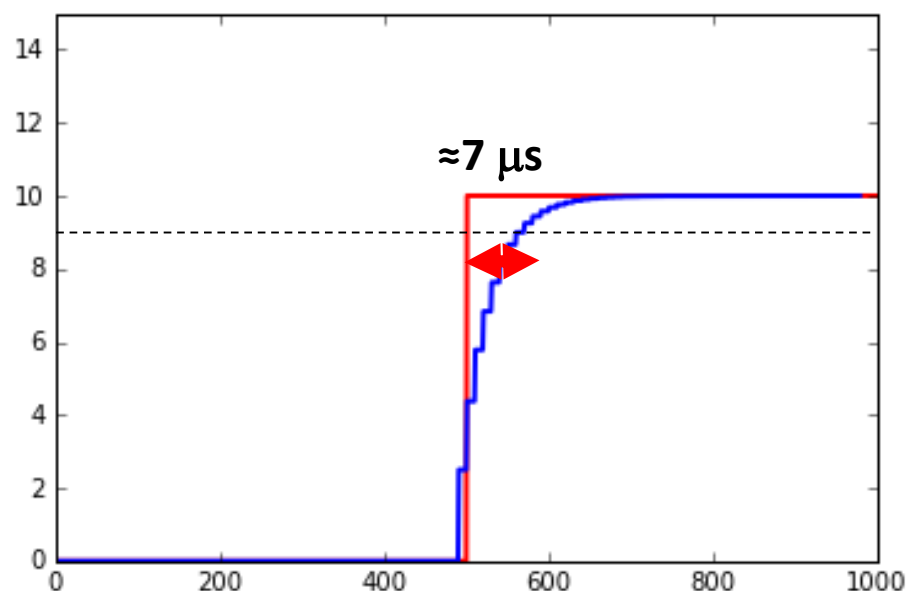
Time and frequency response

- There is a relationship between Bandwidth (Bw) and Rising Time (RT).
- The exact formula depends on the thresholds we use to define bandwidth and rising times
- In general:

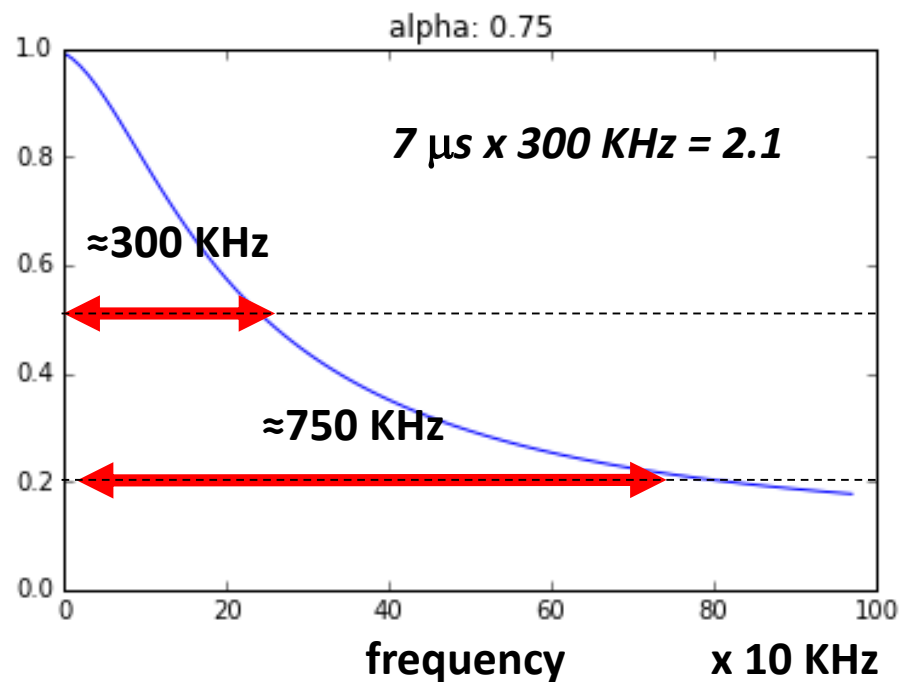
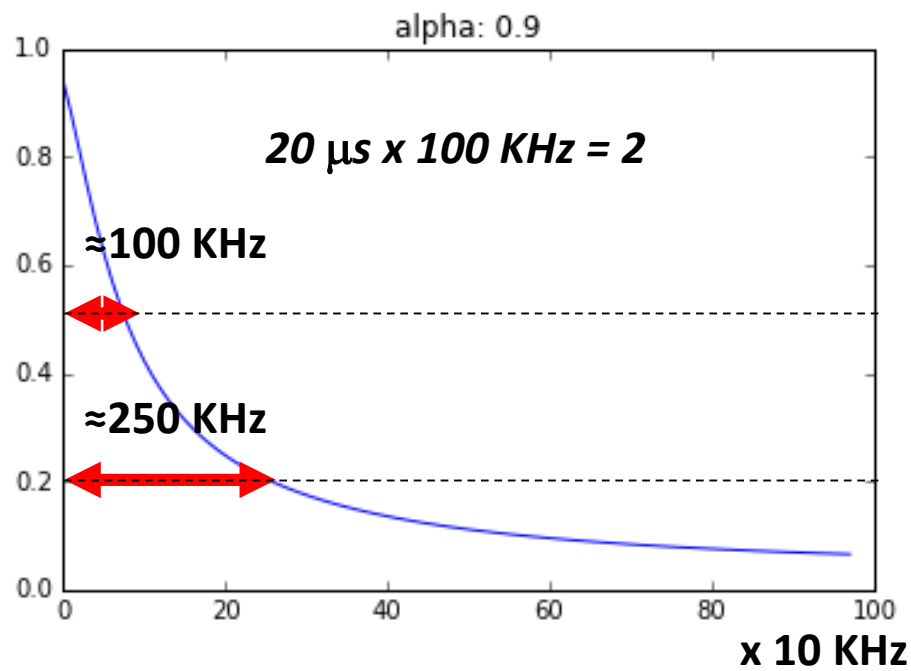
$$Bw = \frac{k}{RT}$$



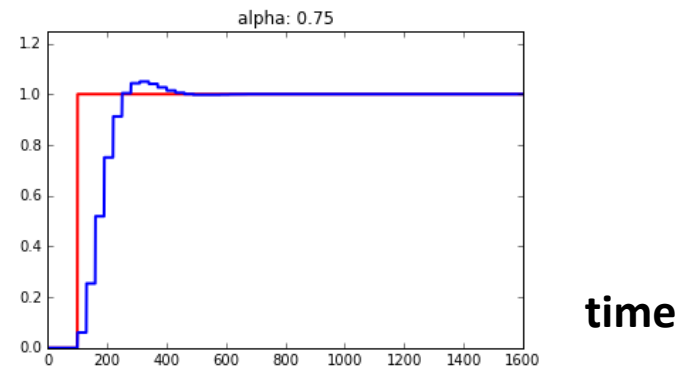
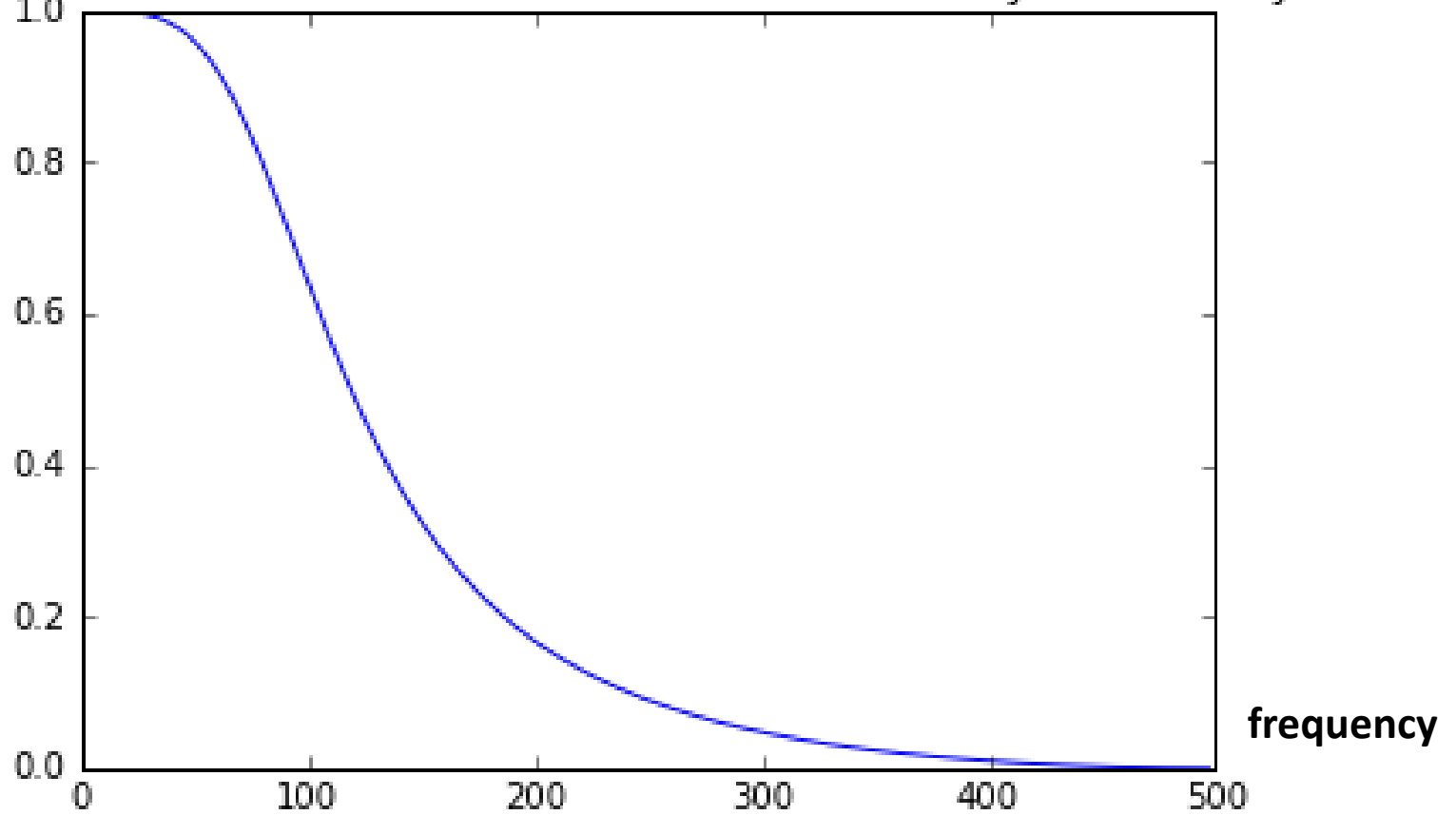
$\times 0.1 \mu s$



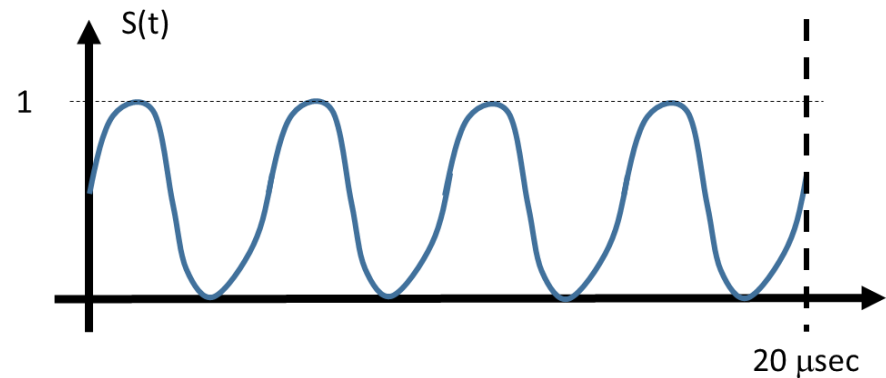
$\times 0.1 \mu s$



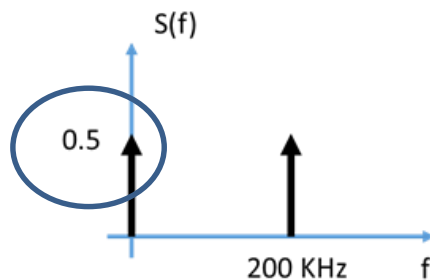
$$y[i]=0.0605*x[i]+0.121*x[i-D]+0.0605*x[i-2*D]+1.194*y[i-D]-0.436*y[i-2*D]$$



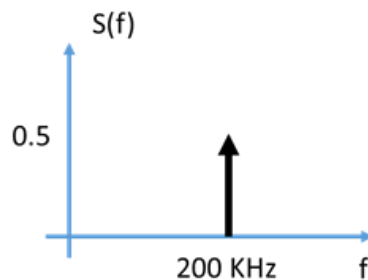
Un sensor nos da una señal como la mostrada en la figura:



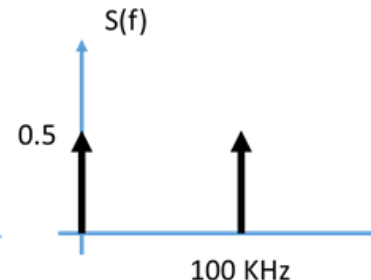
Explicar razonadamente cuál de las tres opciones corresponde al espectro de dicha señal:



Opción A

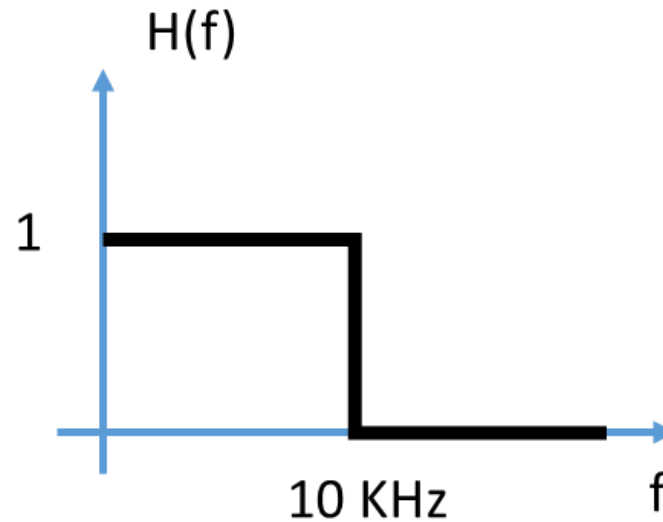


Opción B

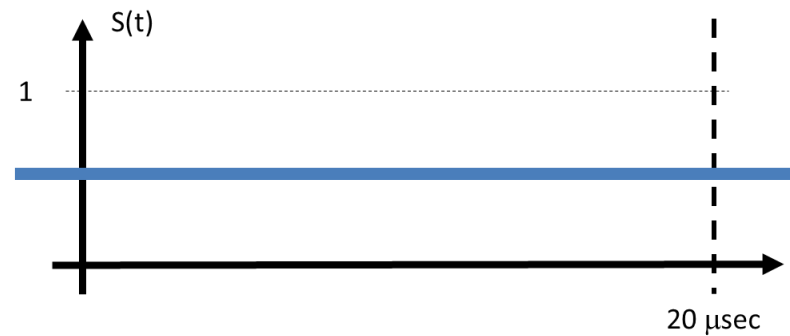


Opción C

Aplicamos el siguiente filtro a dicha señal:

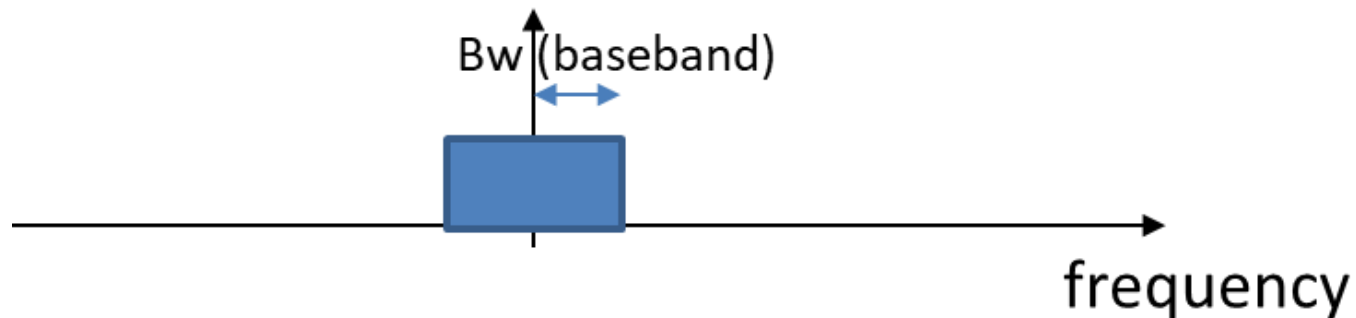


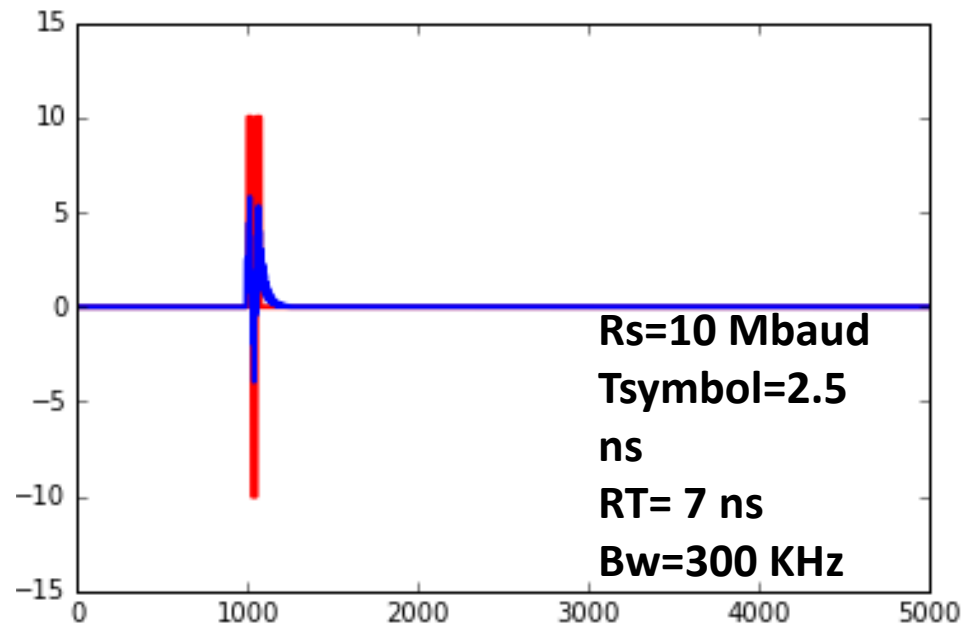
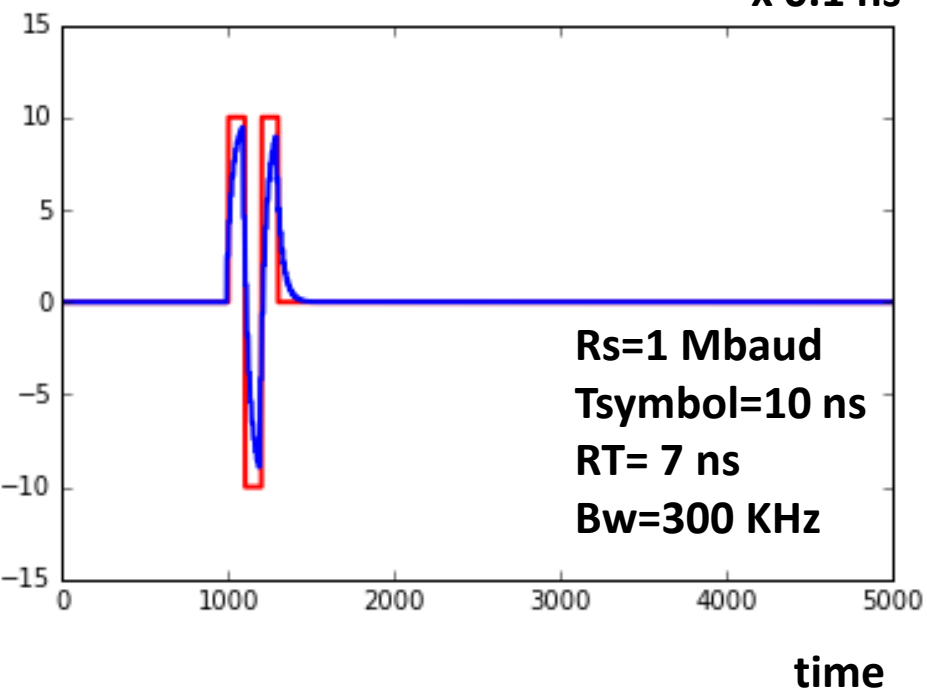
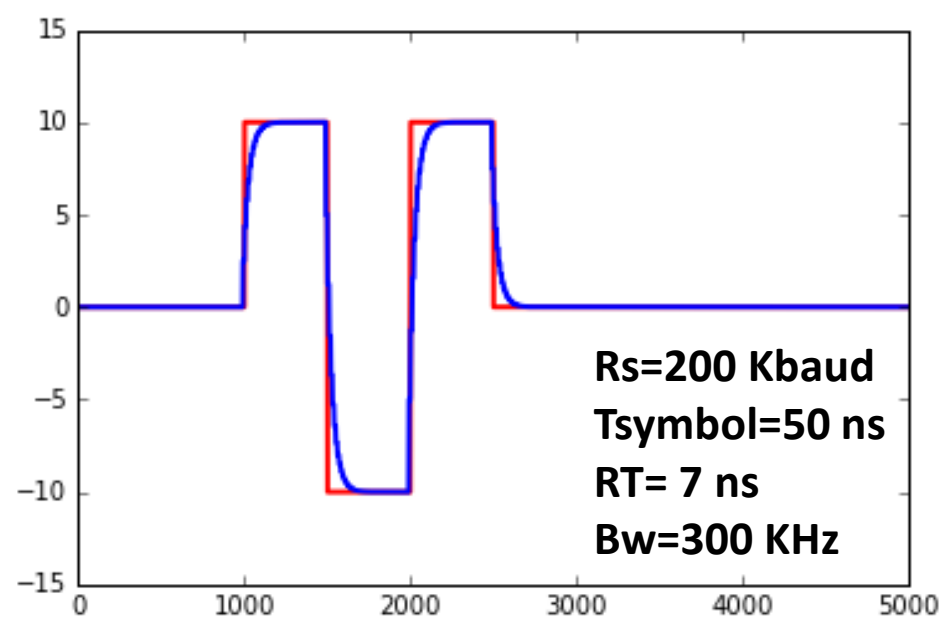
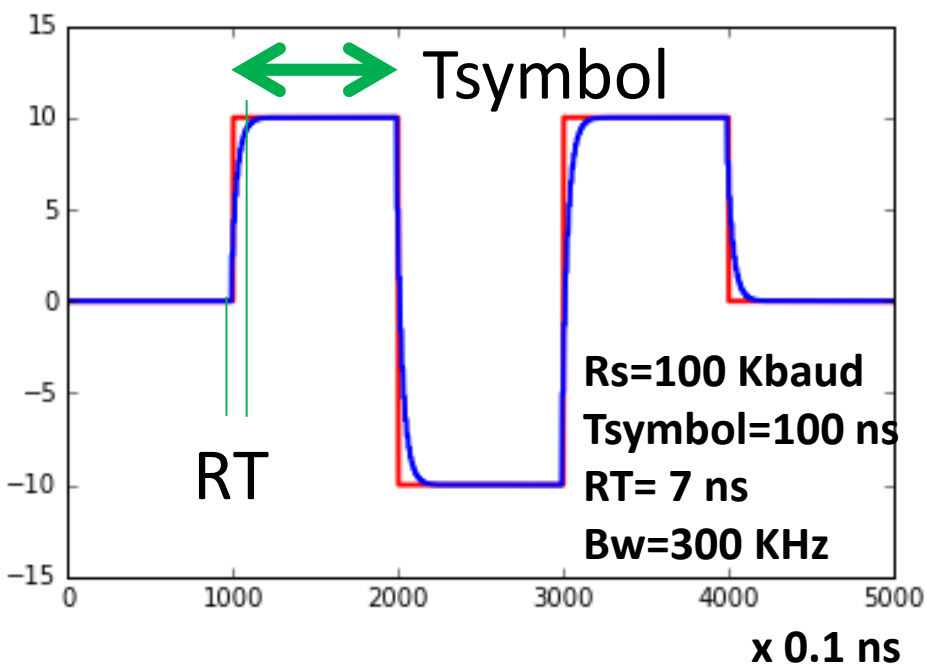
Dibuja la señal que obtendríamos como salida del filtro



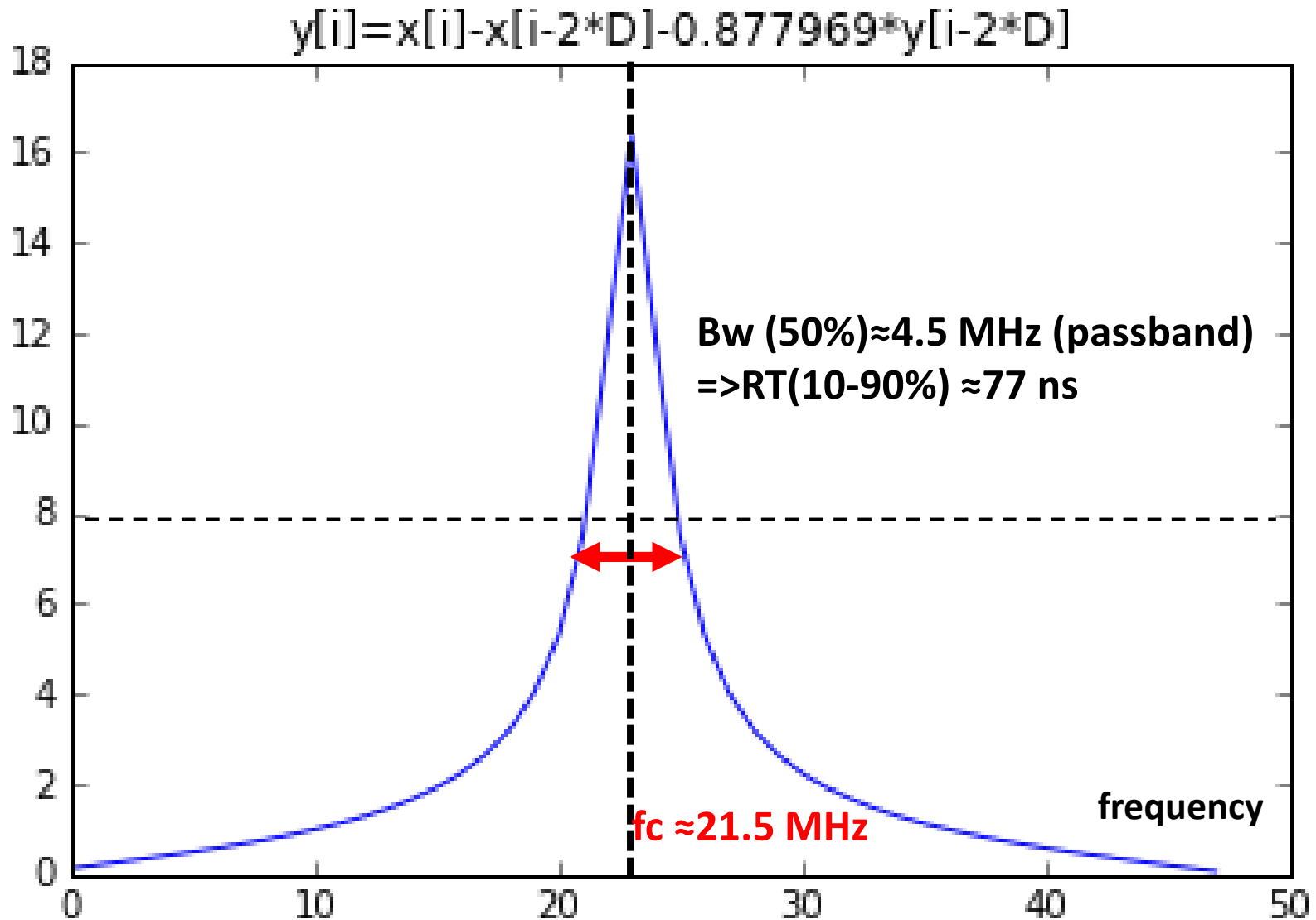
Bw and symbols per second

- Assuming ideal filtering in baseband, with a bandwidth of B_w it is possible to transmit $R=2*B_w$ symbols per second (baud).
- True rates in practice for a given band are usually lower (e.g. 10-50%).

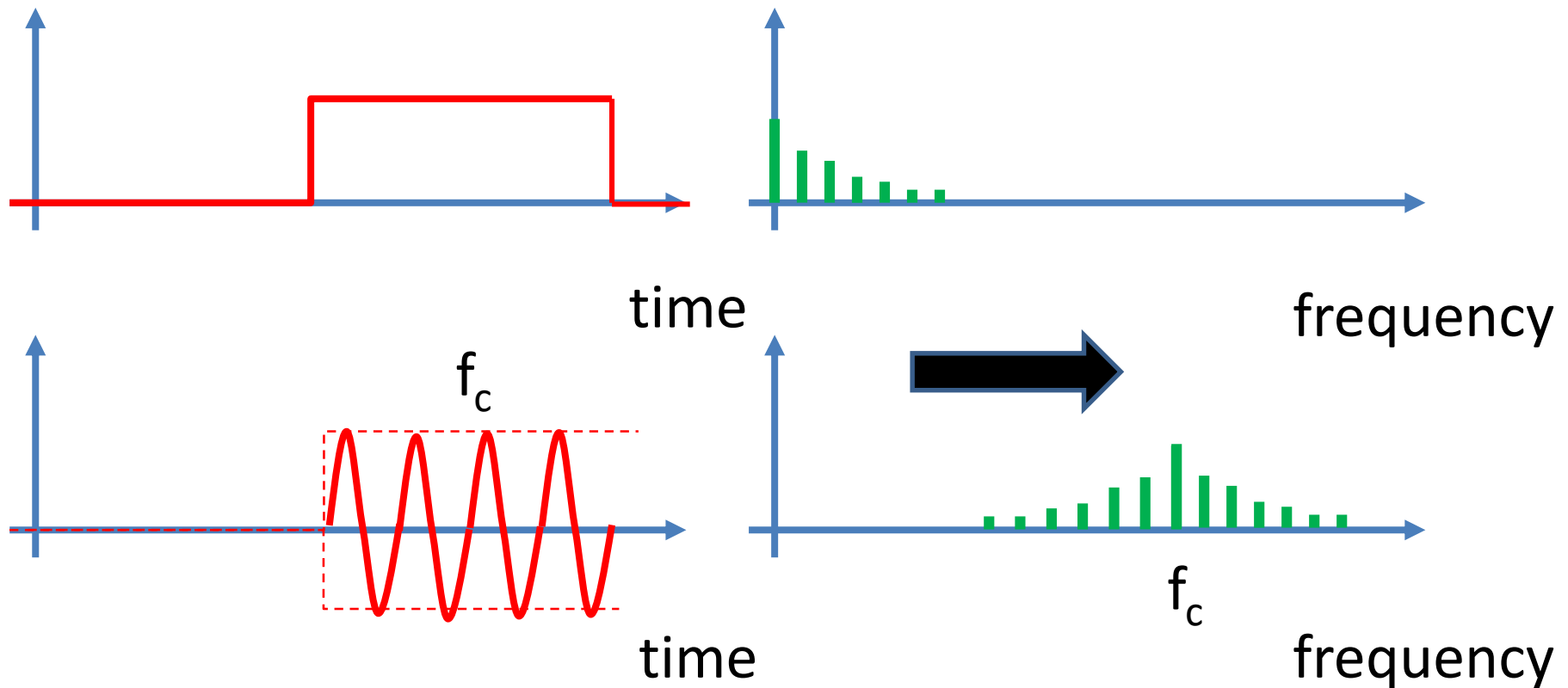


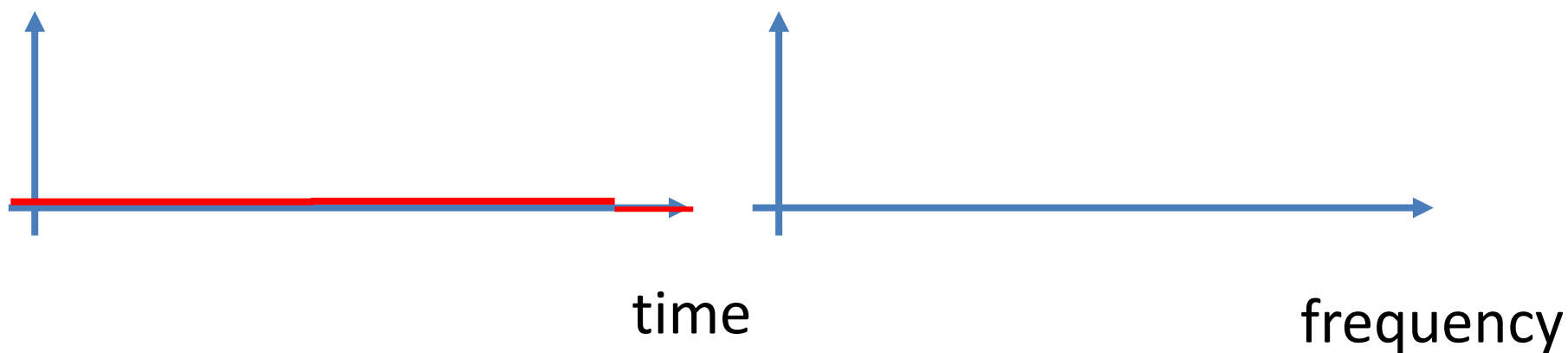
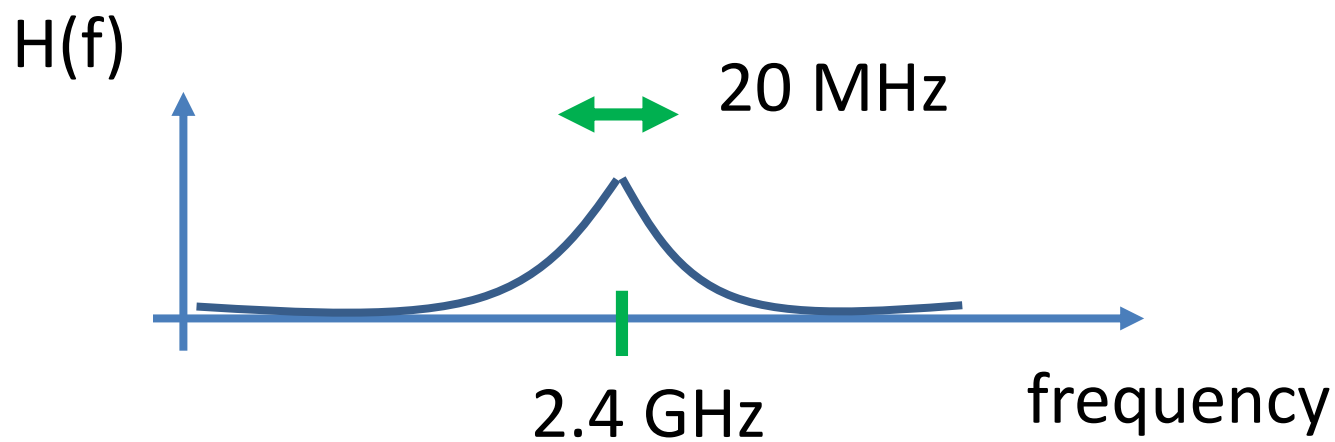
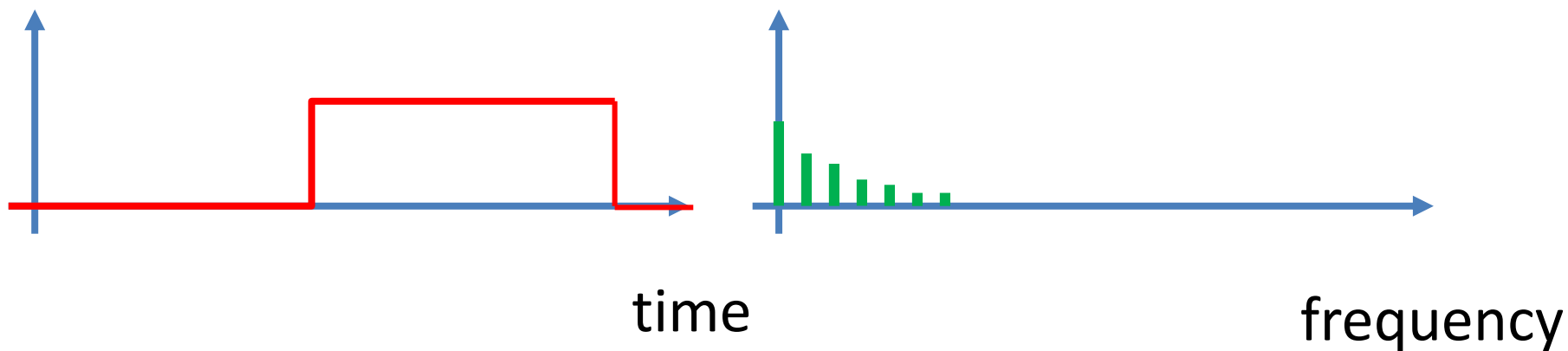


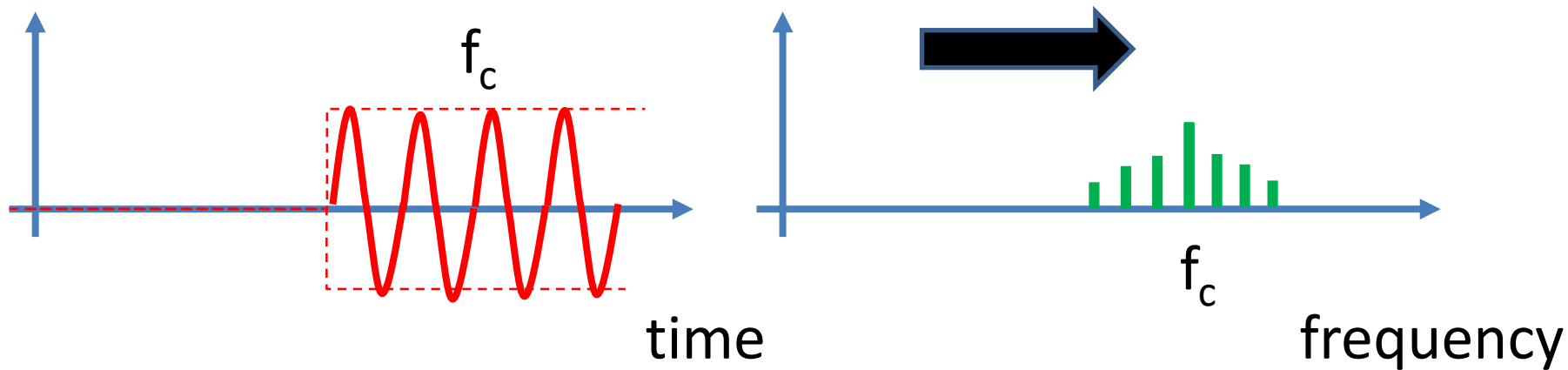
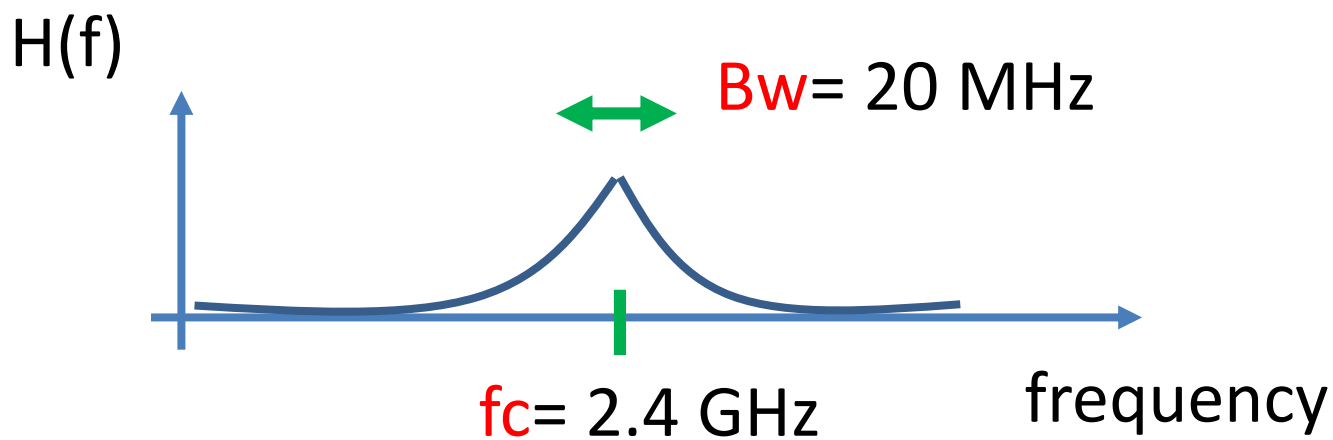
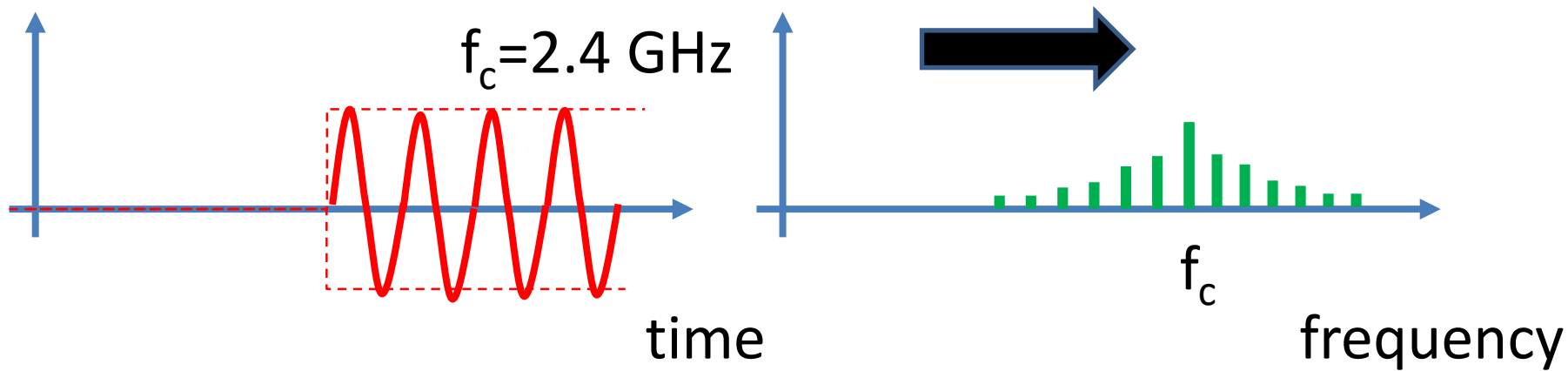
Bandpass filter

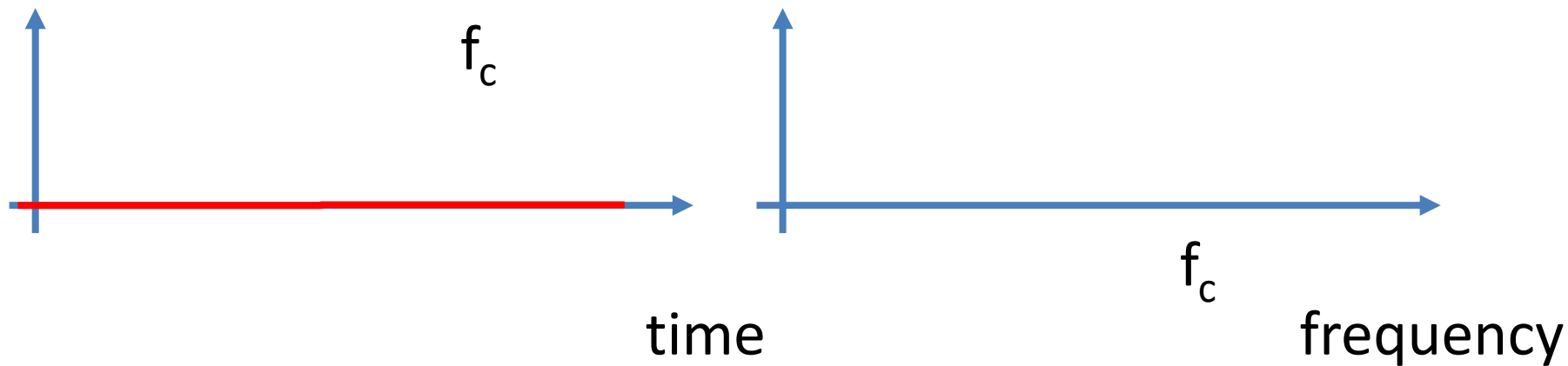
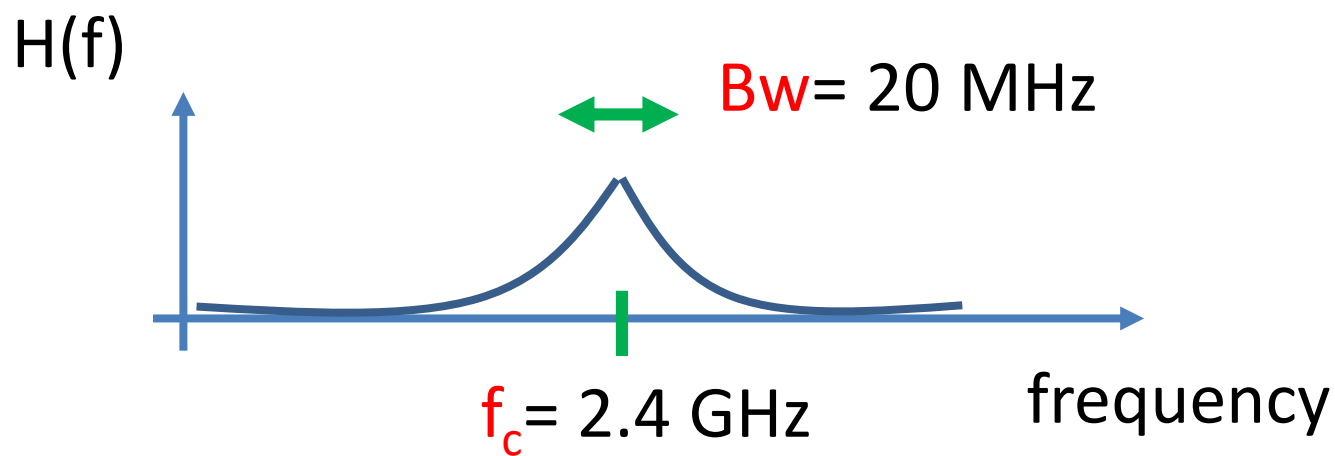
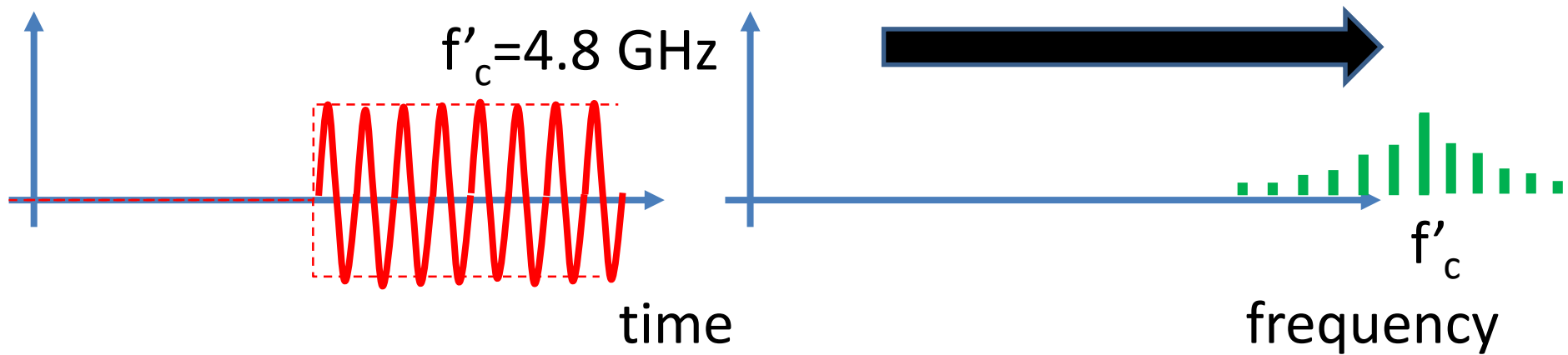


Modulation shifts the band of spectral components of a signal

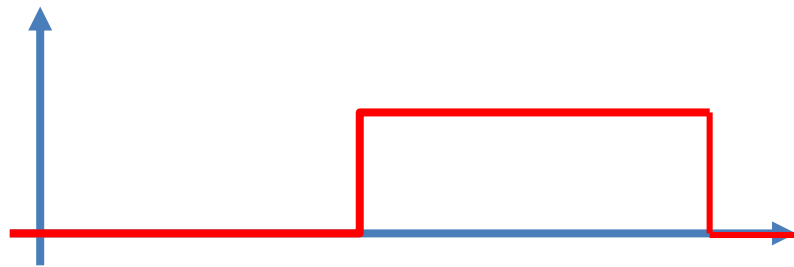








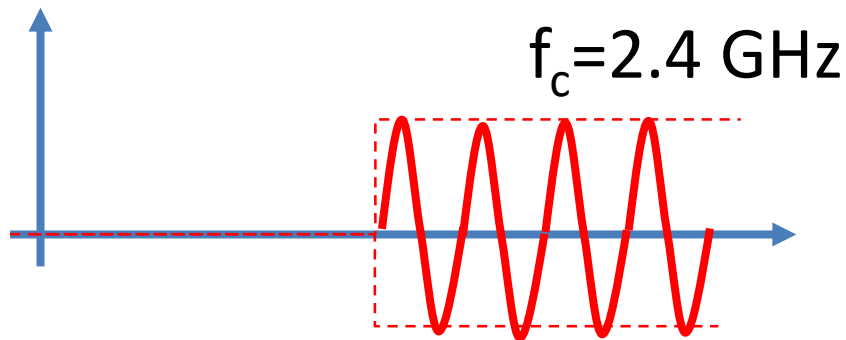
Sumamos las 3 señales...



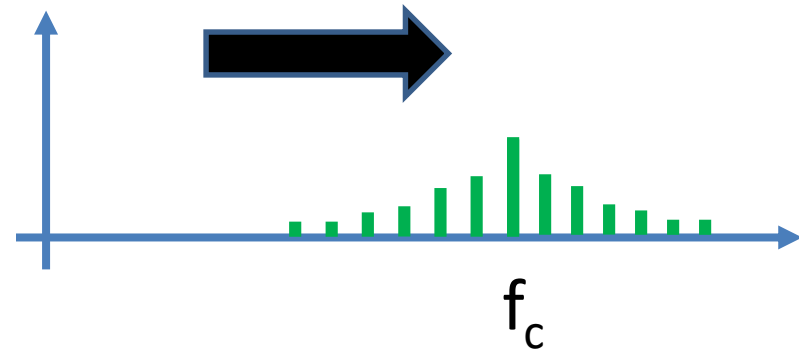
time



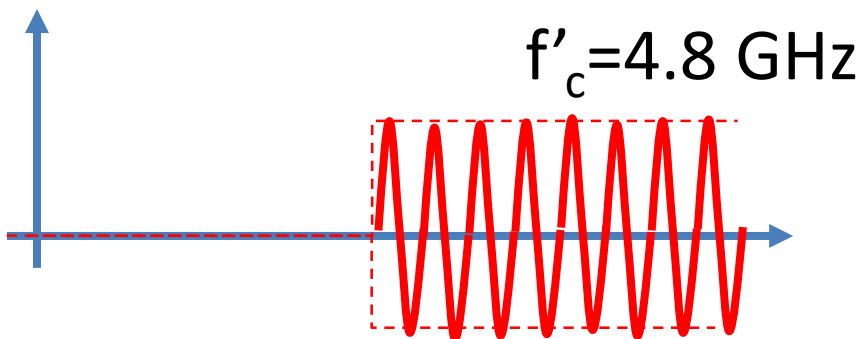
frequency



$f_c = 2.4 \text{ GHz}$



f_c

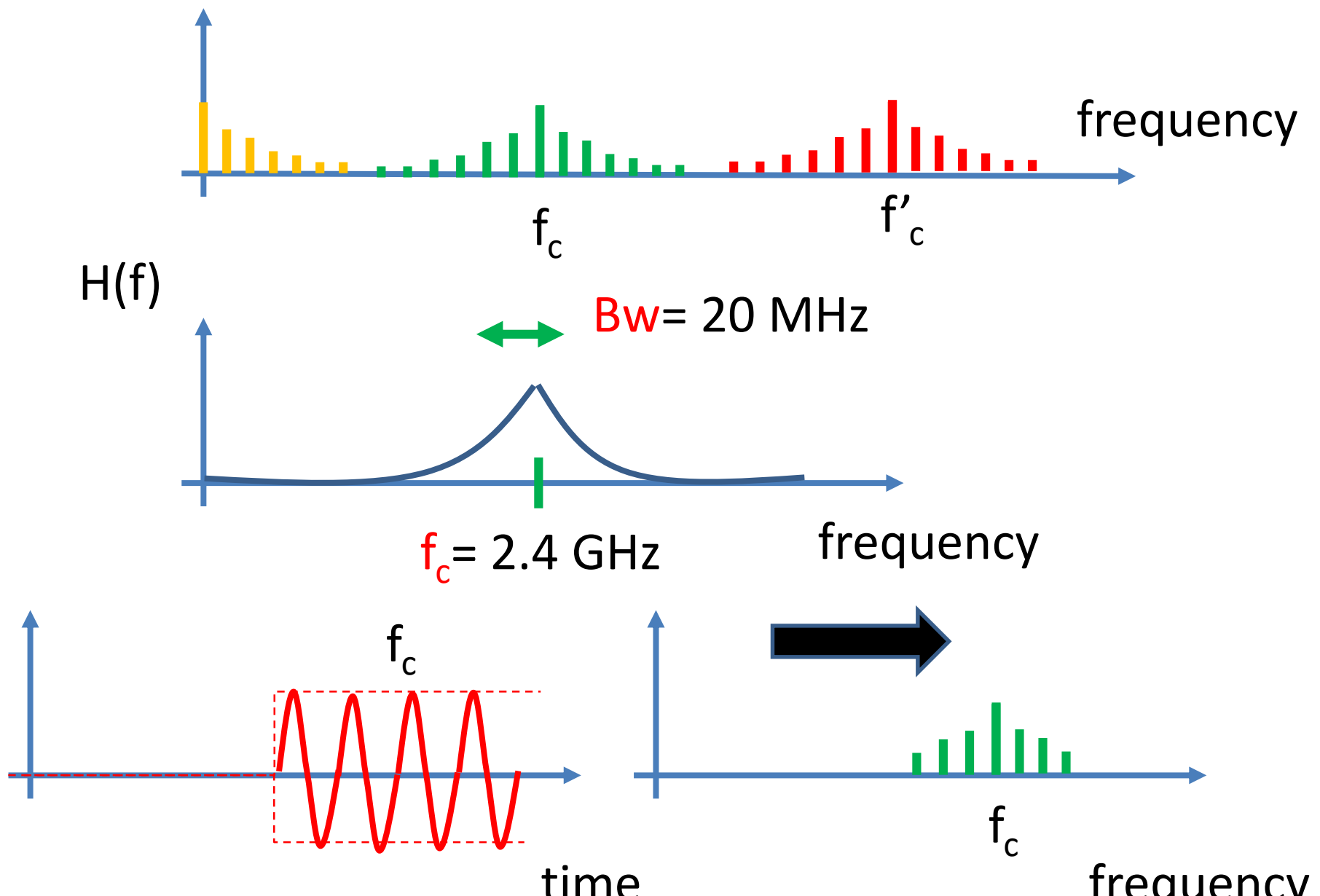


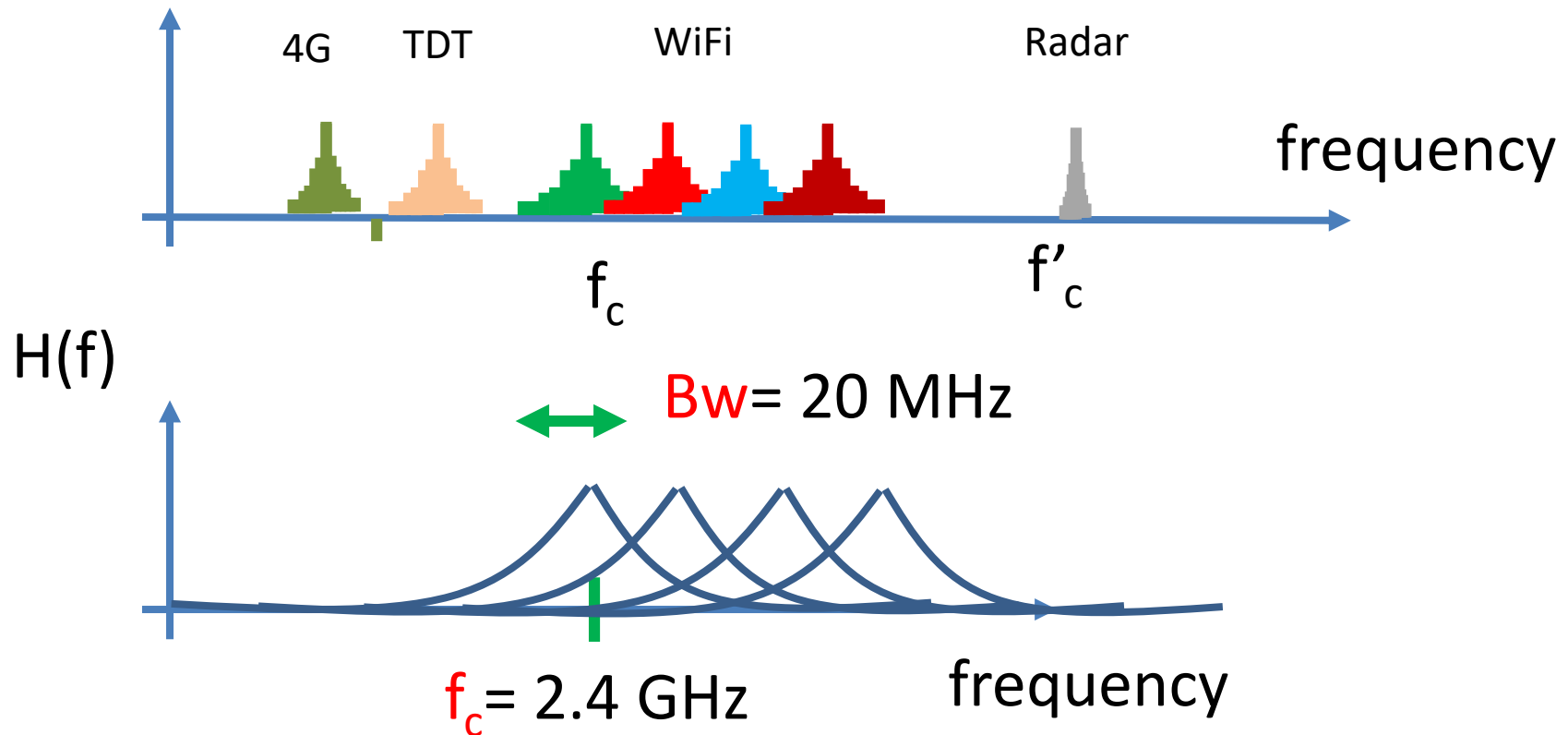
$f'_c = 4.8 \text{ GHz}$



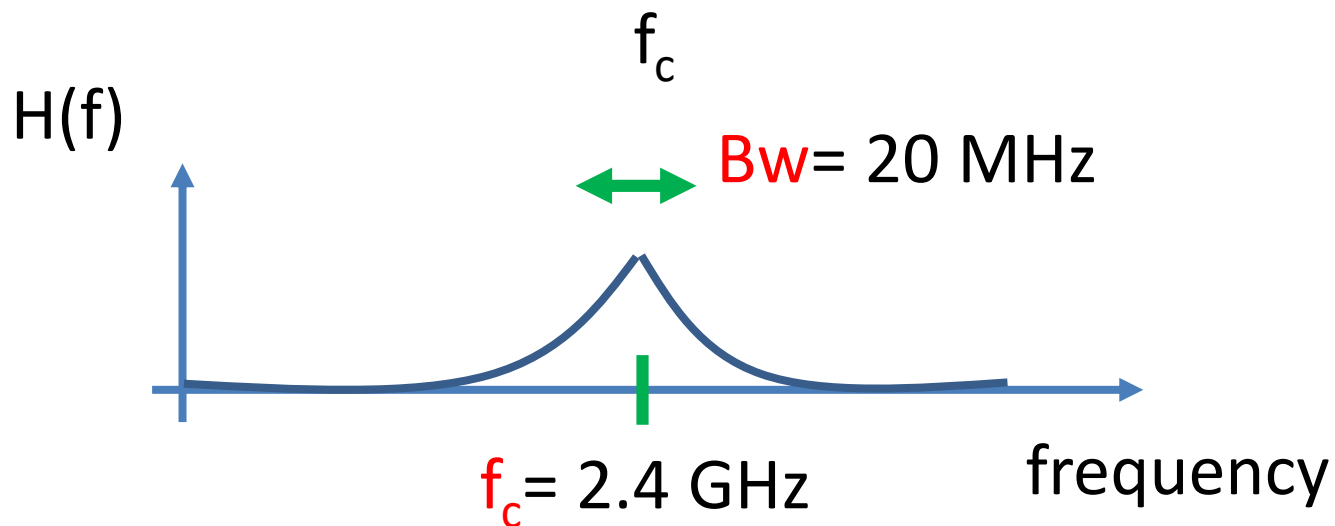
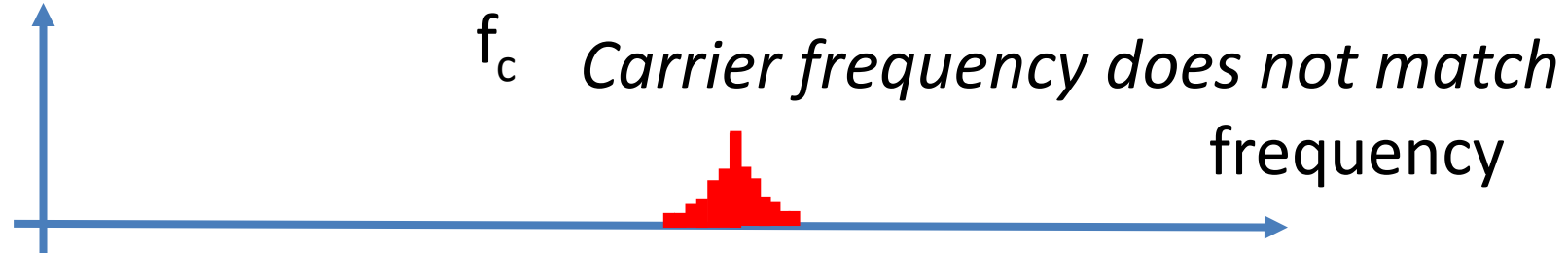
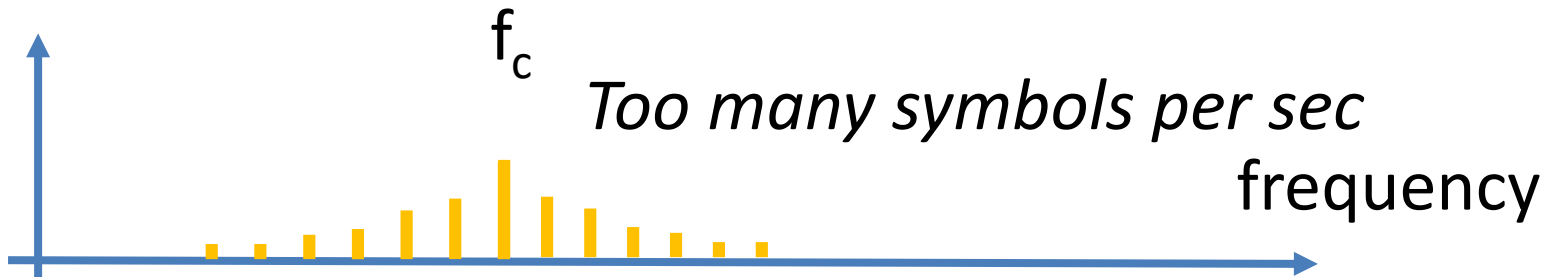
f'_c

Sumamos las 3 señales...



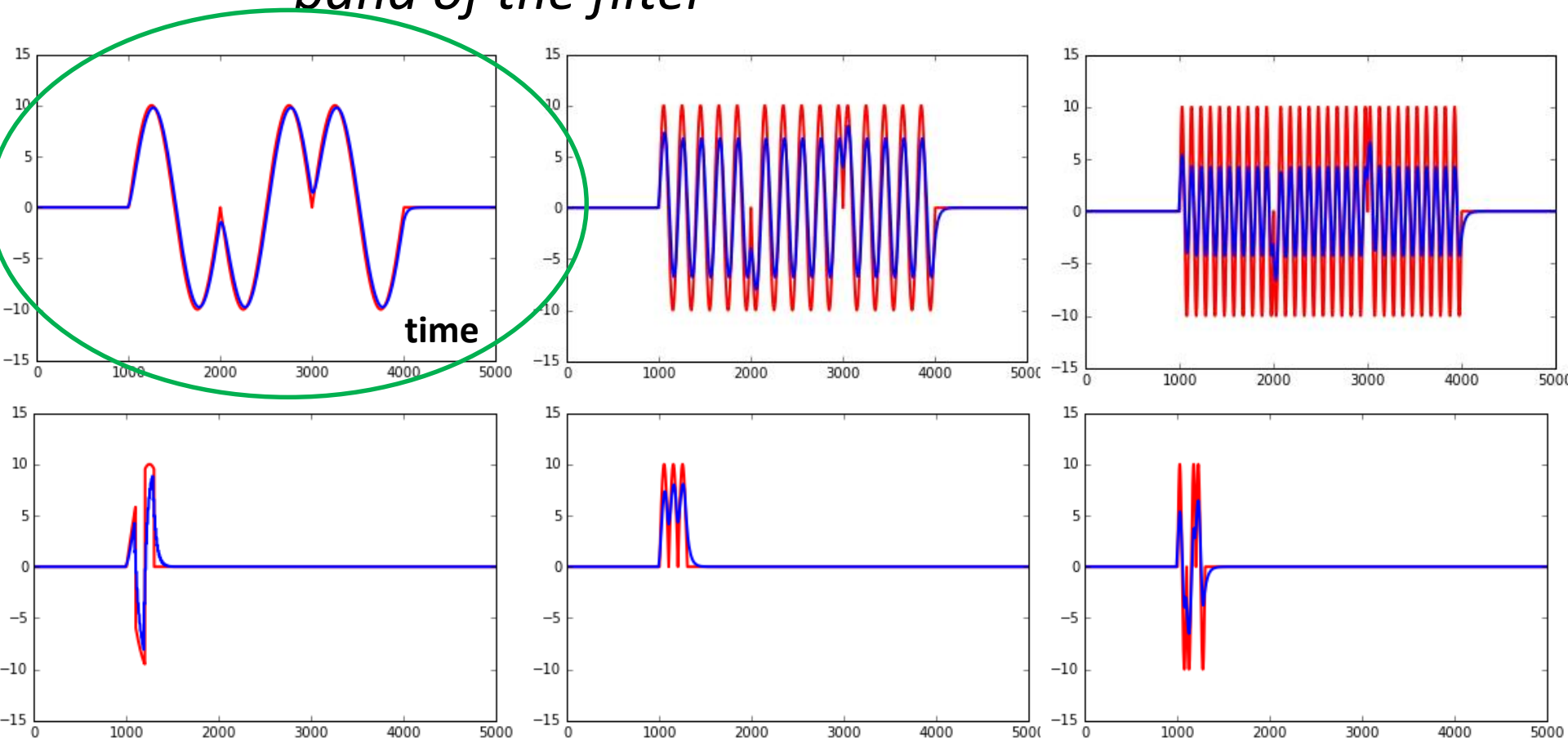


- (1) T_{symbol} cannot be smaller than k/Bw
- (2) The carrier frequency must be centered on the band of the filter



Putting all together, Passband

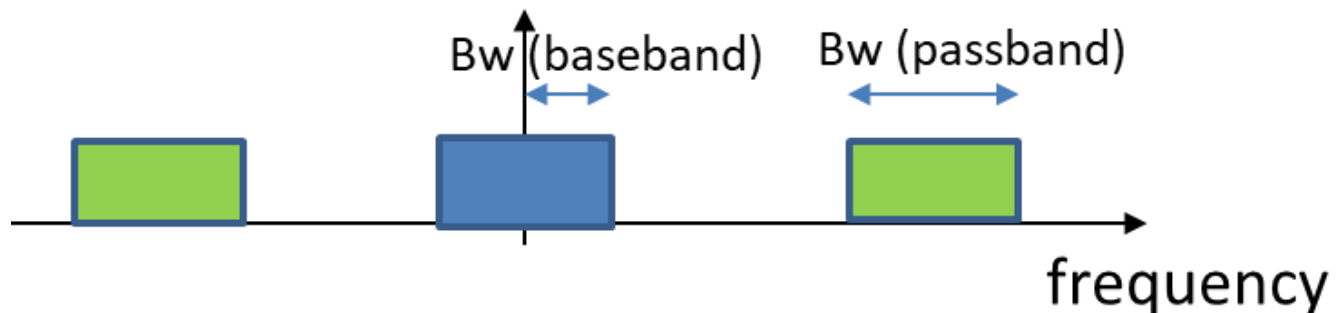
The carrier frequency must be centered on the band of the filter



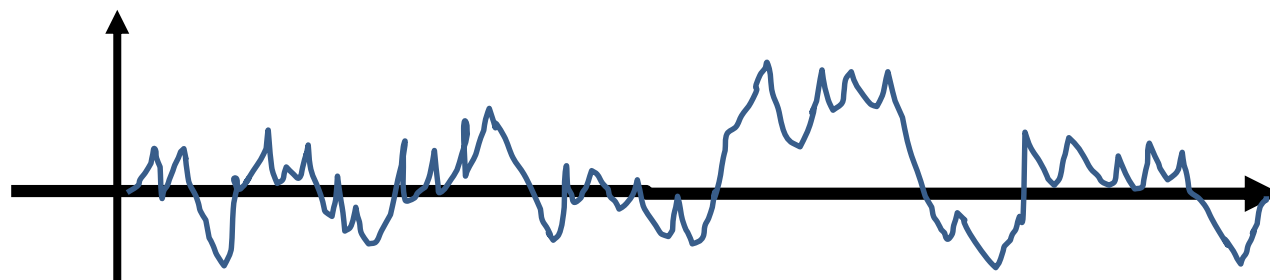
T_{symbol} cannot be smaller than k/Bw

Bw and symbols per second

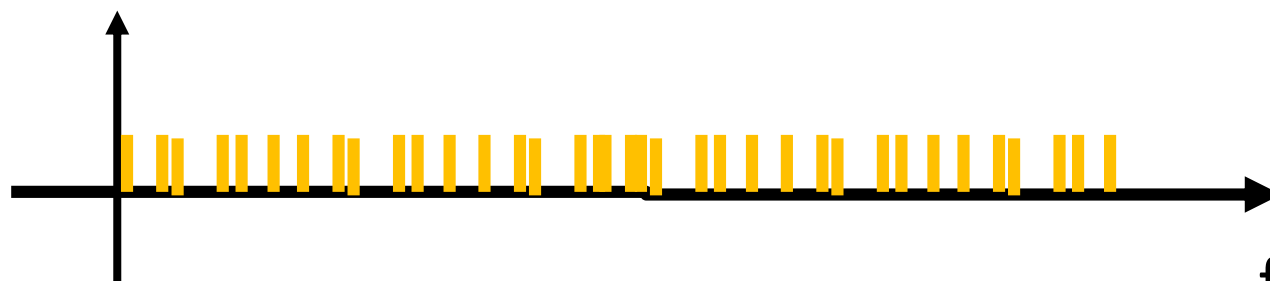
- **For radiofrequencies (passband)** it doubles, so **$R=Bw$** (e.g. 1kbaud of rate takes 1 kHz of band, in 1 MHz of band it is possible to transmit 1Mbaud).
- True rates in practice for a given band are usually lower (e.g. 10-50%).



White noise



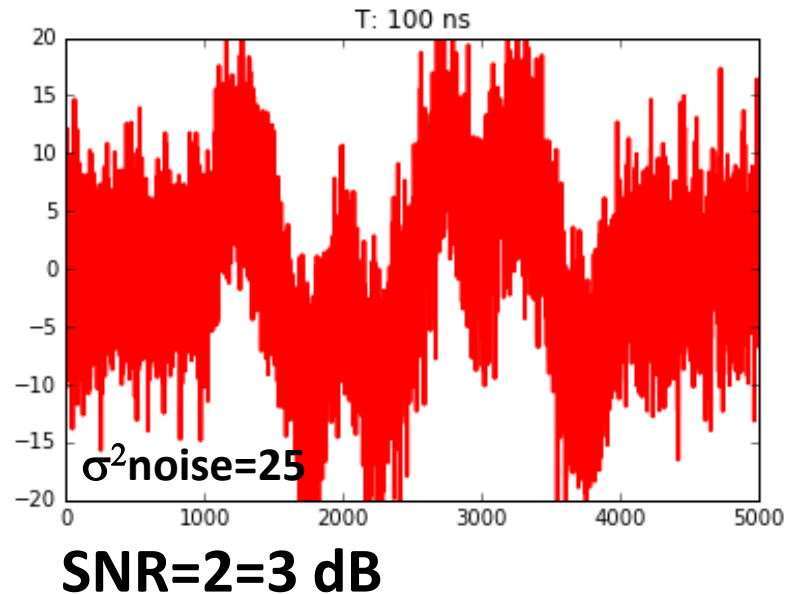
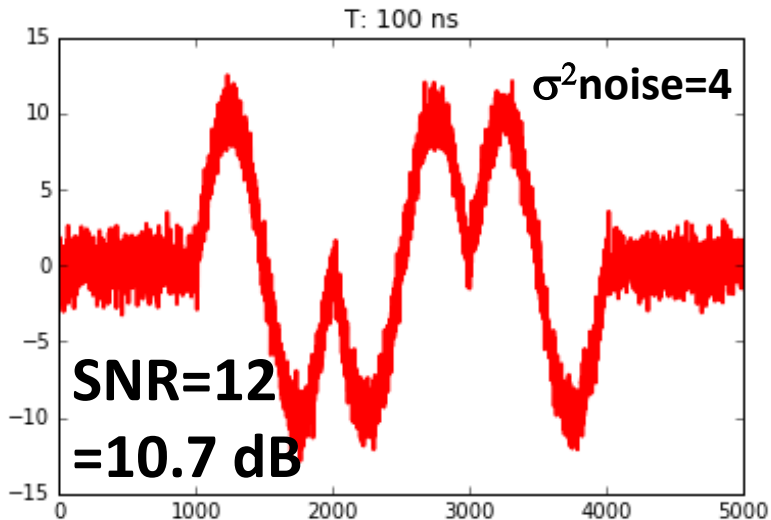
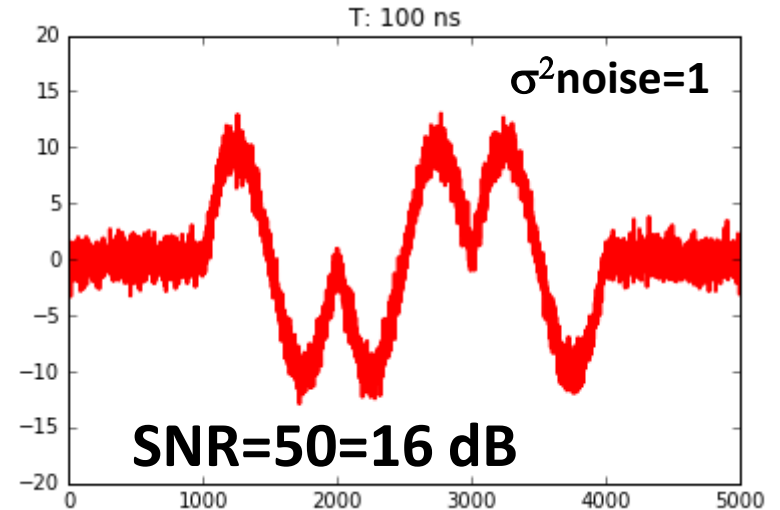
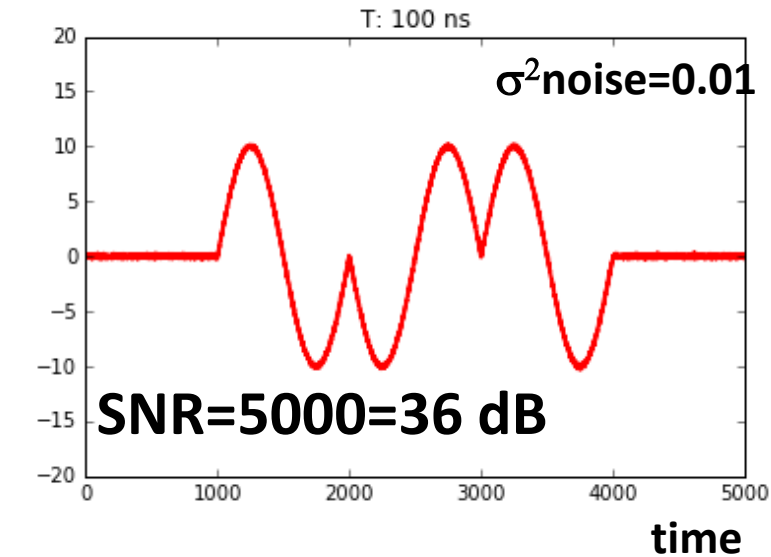
time



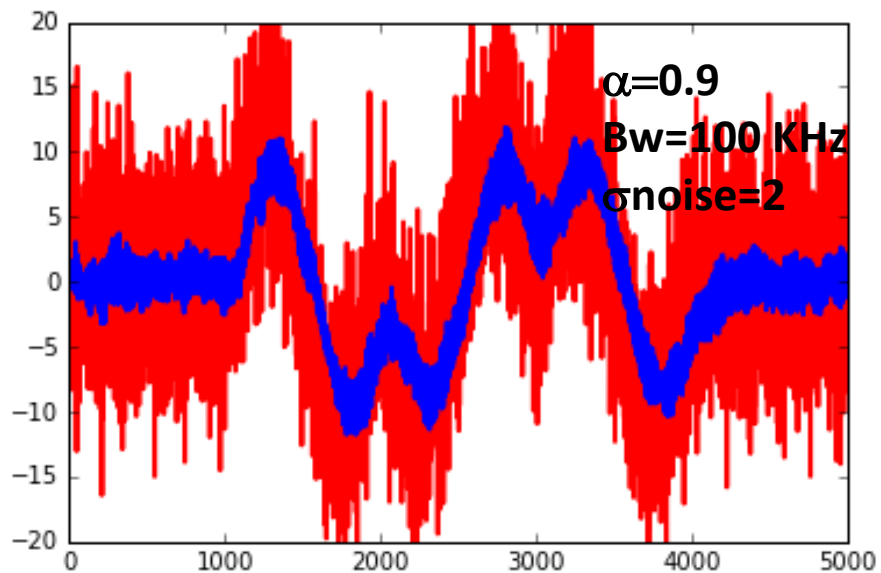
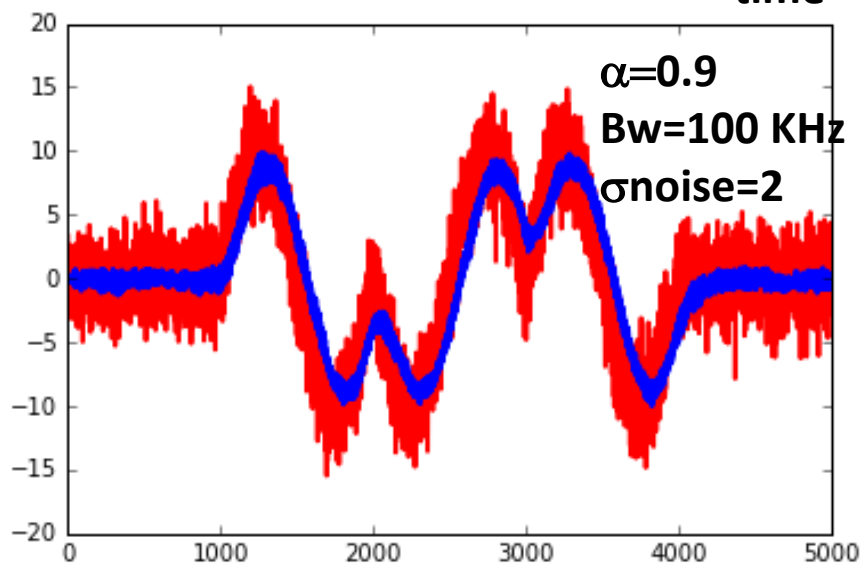
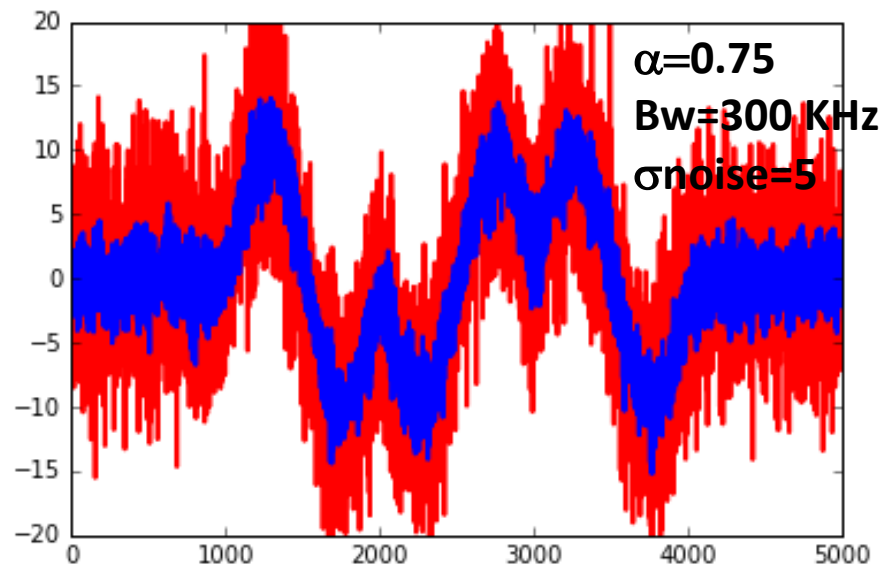
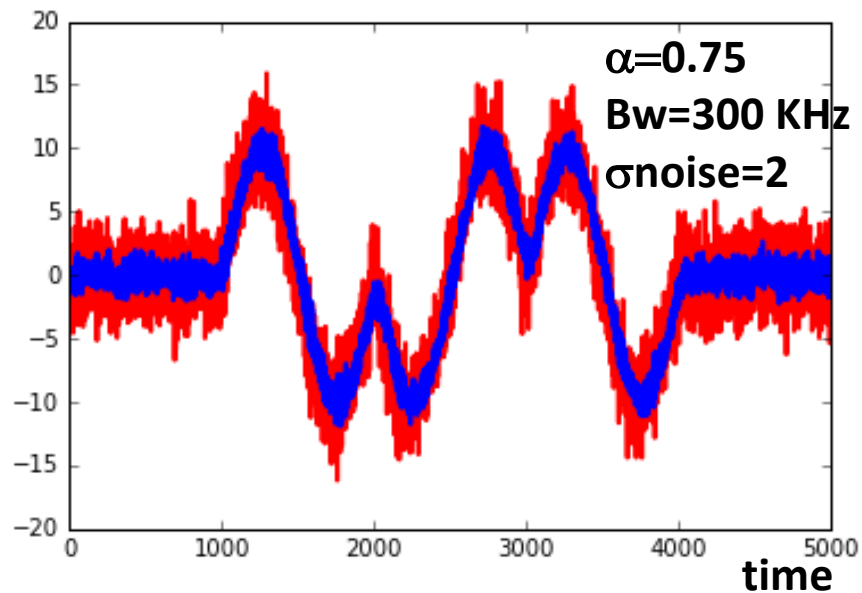
frequency

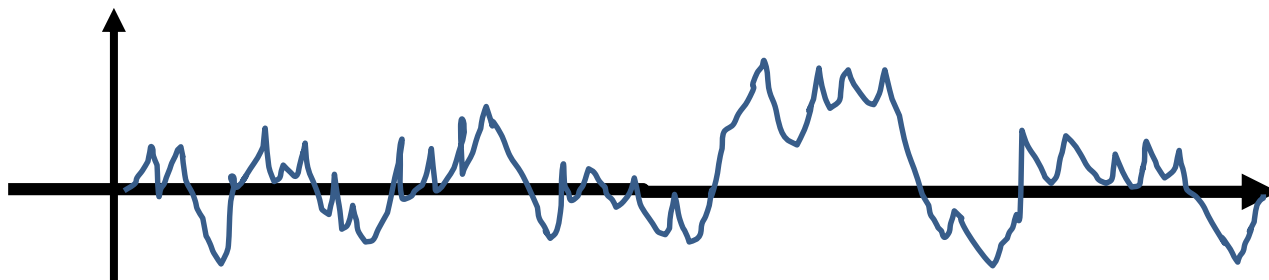
Signal power
 $=A^2/2=50$ ($A=10$)

“White” Noise

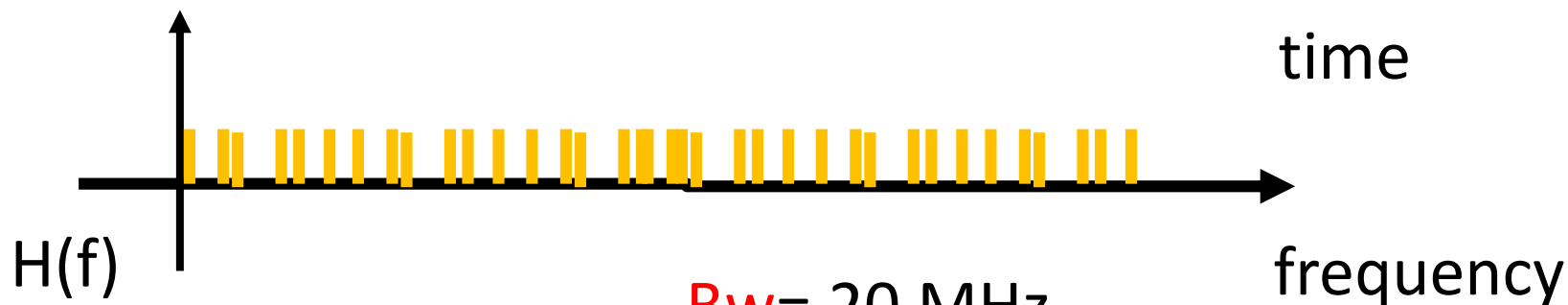


Noise filtering and Bw

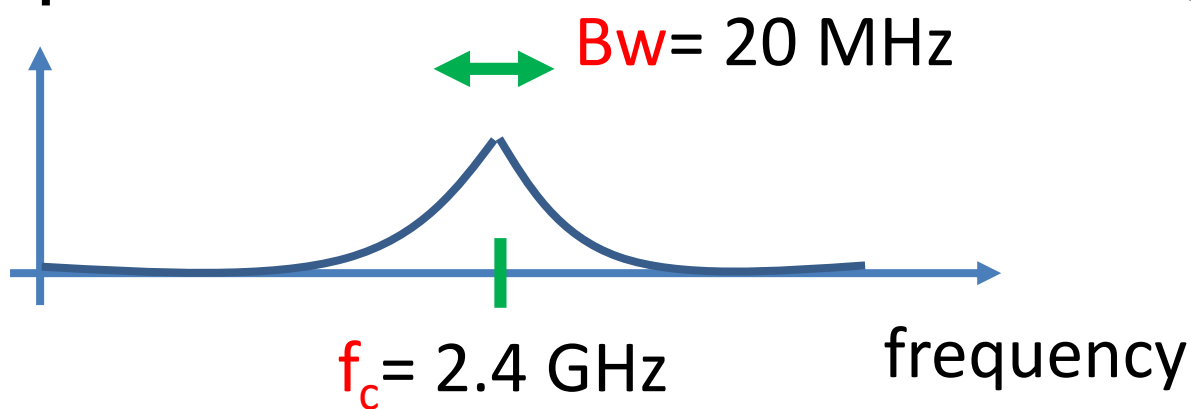




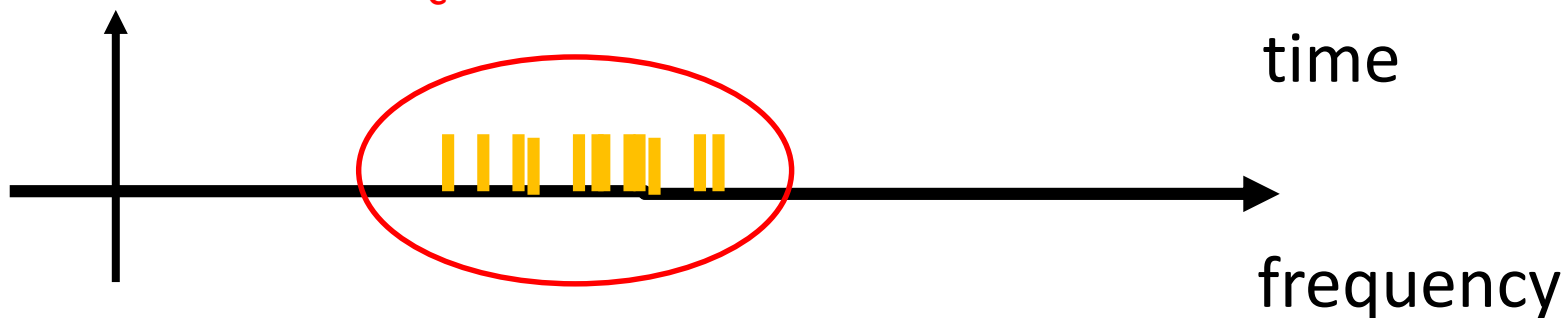
time



frequency



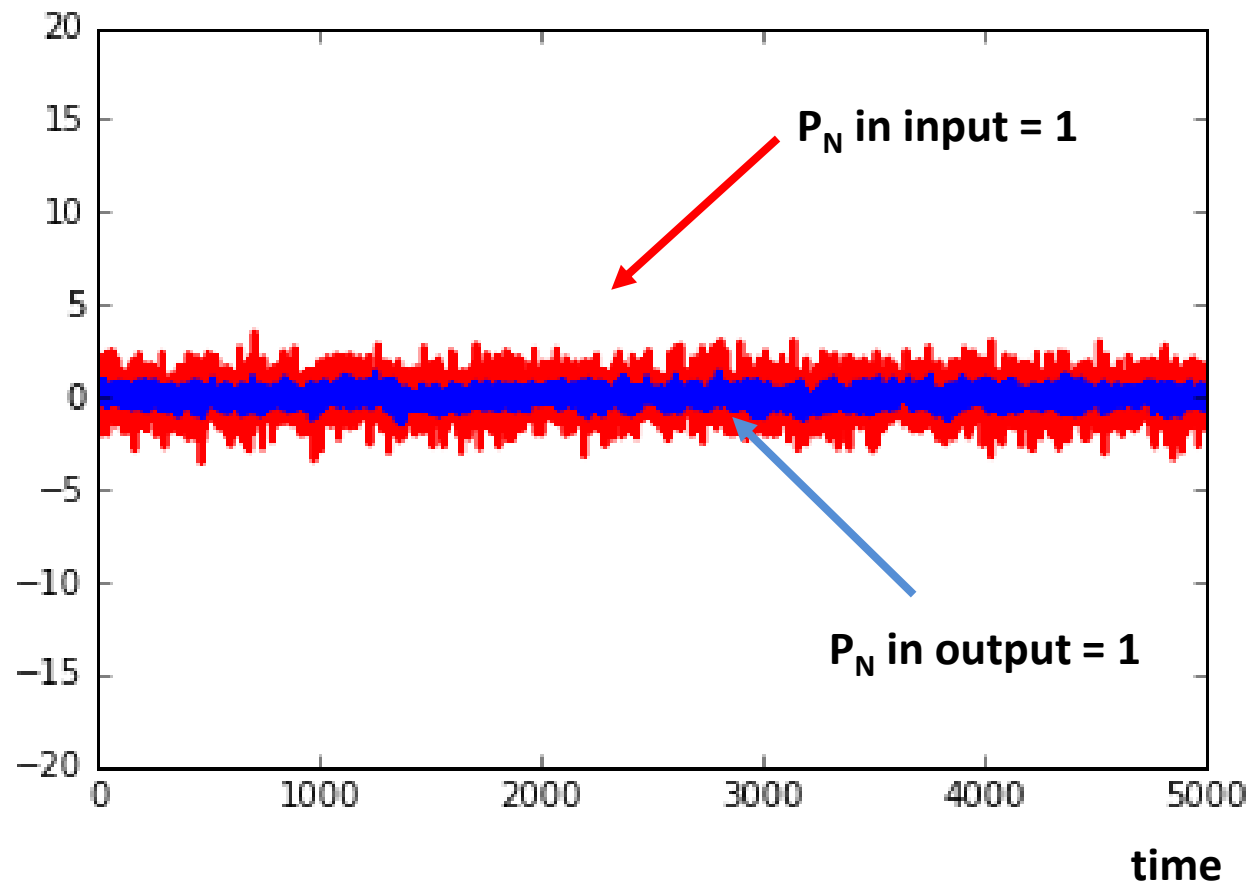
frequency



time

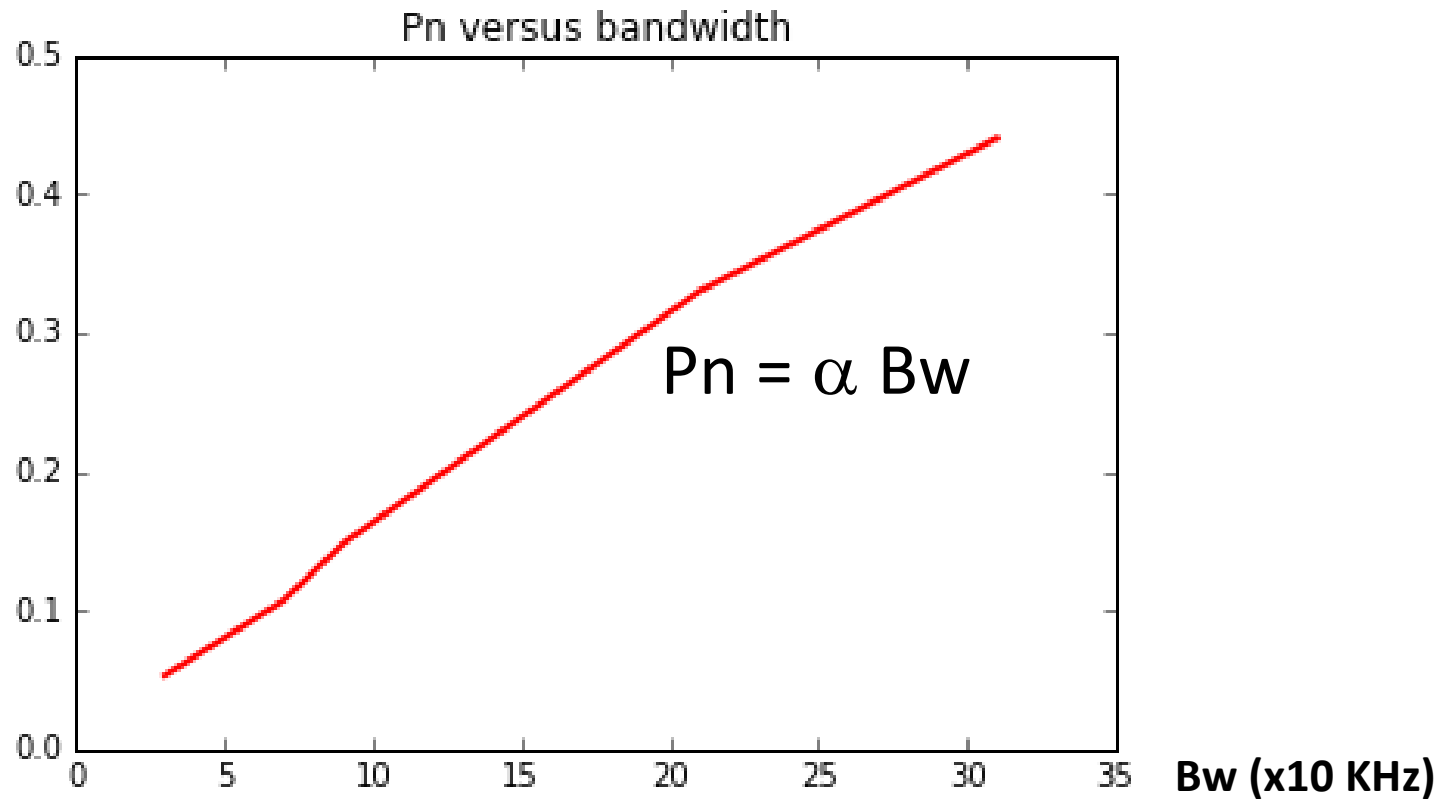
frequency

Noise power versus bandwidth



Noise power versus bandwidth

P_N in output



Electronic Noise

*Densidad espectral
de potencia de ruido*

- In **electronic circuits**

$$P_N (\text{output}) = N_0 \times Bw$$

$$N_0 = -174 \text{ dBm/Hz} \approx 4 \times 10^{-18} \text{ mW/Hz} \\ \approx 4 \times 10^{-21} \text{ W/Hz}$$

E.g. Wifi: $Bw=20 \text{ MHz} \Rightarrow P_N \approx -101 \text{ dBm}$

For a BPSK or QPSK receiver $SNR_{\min}=7\text{dB}$, in theory the minimum receive power is -94 dBm , although in practice it must be higher (e.g. -84 dBm)

E_b/N_0 and R/Bw

$$SNR = P_{RX} / P_N$$

We can express P_{RX} and P_N as:

$$\begin{cases} P_{RX} = E_b R \\ P_N = N_0 Bw \end{cases}$$

Does not include code bits!!

Obtaining

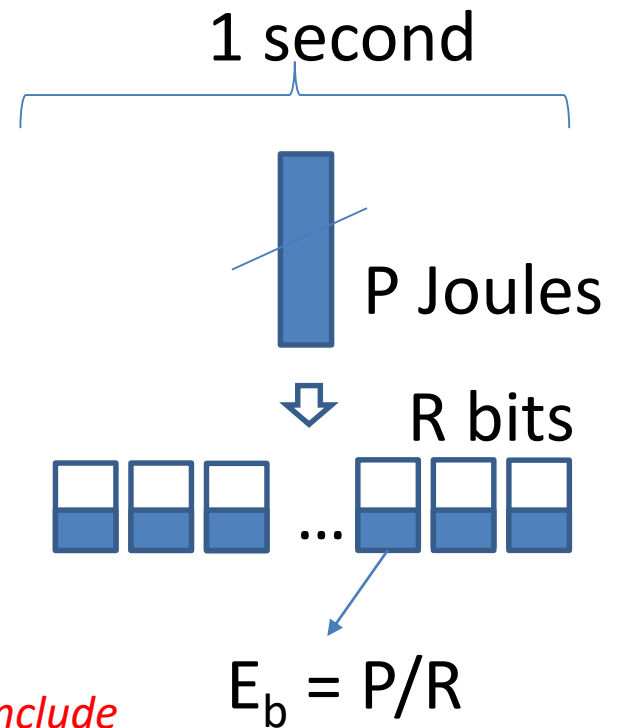
$$SNR = (E_b/N_0) (R/Bw)$$

Spectral efficiency

No units: (b/s)/(1/s)

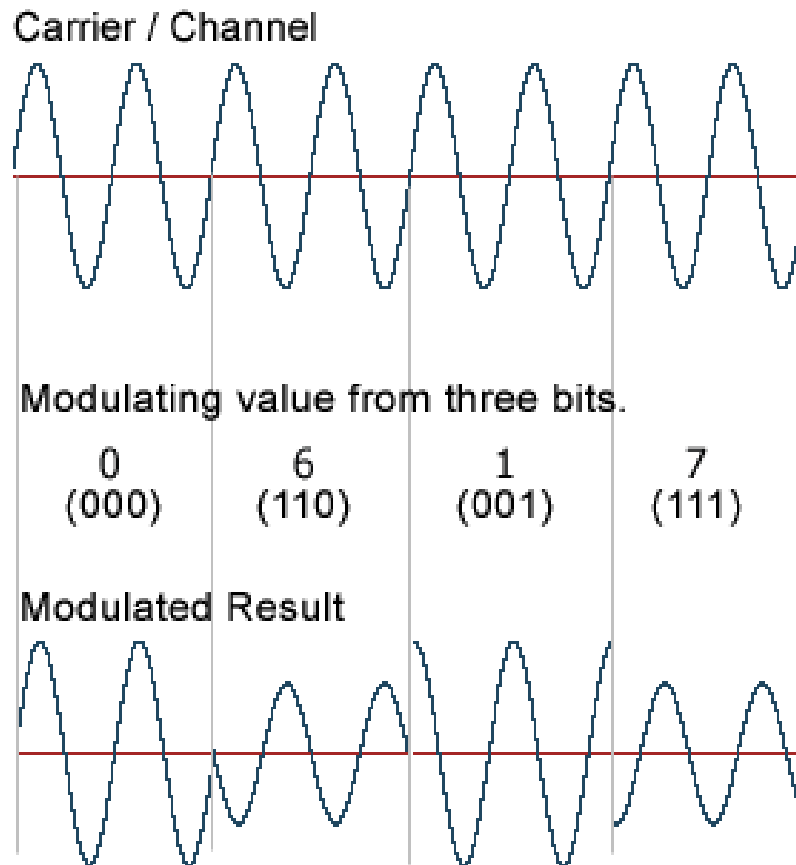
Energy per bit to noise power spectral density ratio

No units: J/(W/s)



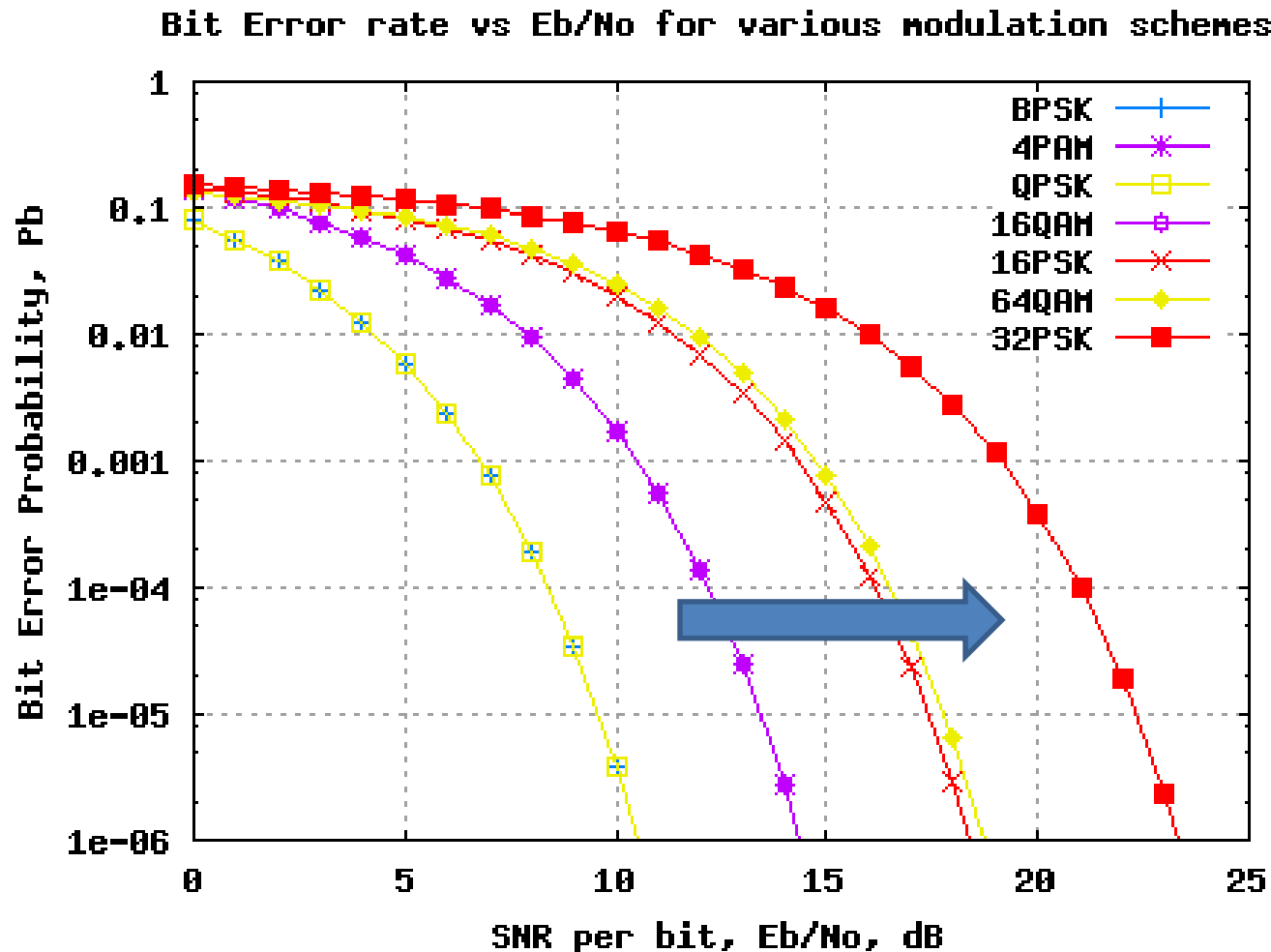
Multilevel modulation

DIGITAL QAM (8QAM)



Note: Only four (0, 6, 1 and 7) out of the eight possible modulation states (0-7) are shown in this illustration.

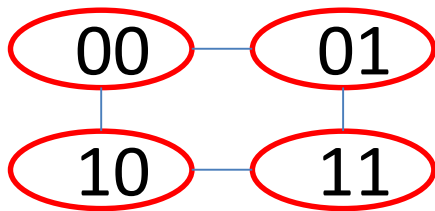
Multilevel modulation



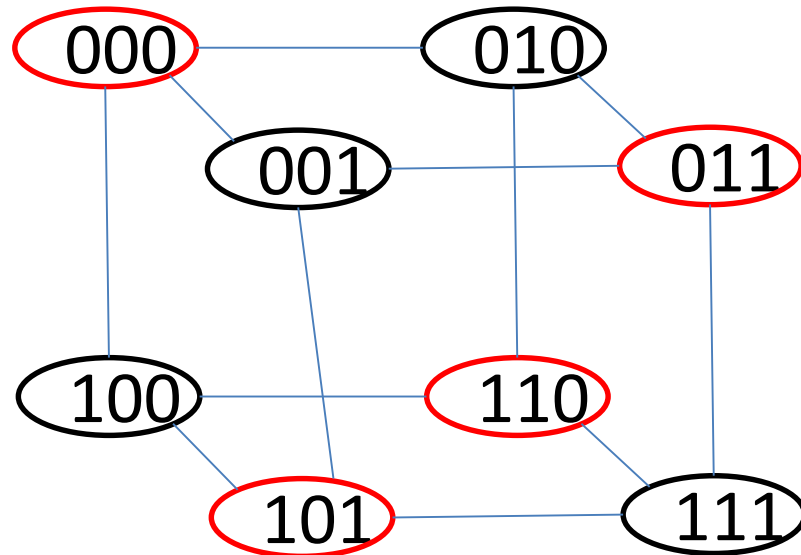
When we use multilevel modulation, we need higher SNR per bit for achieving the same BER

Error correcting codes

Distance of Hamming

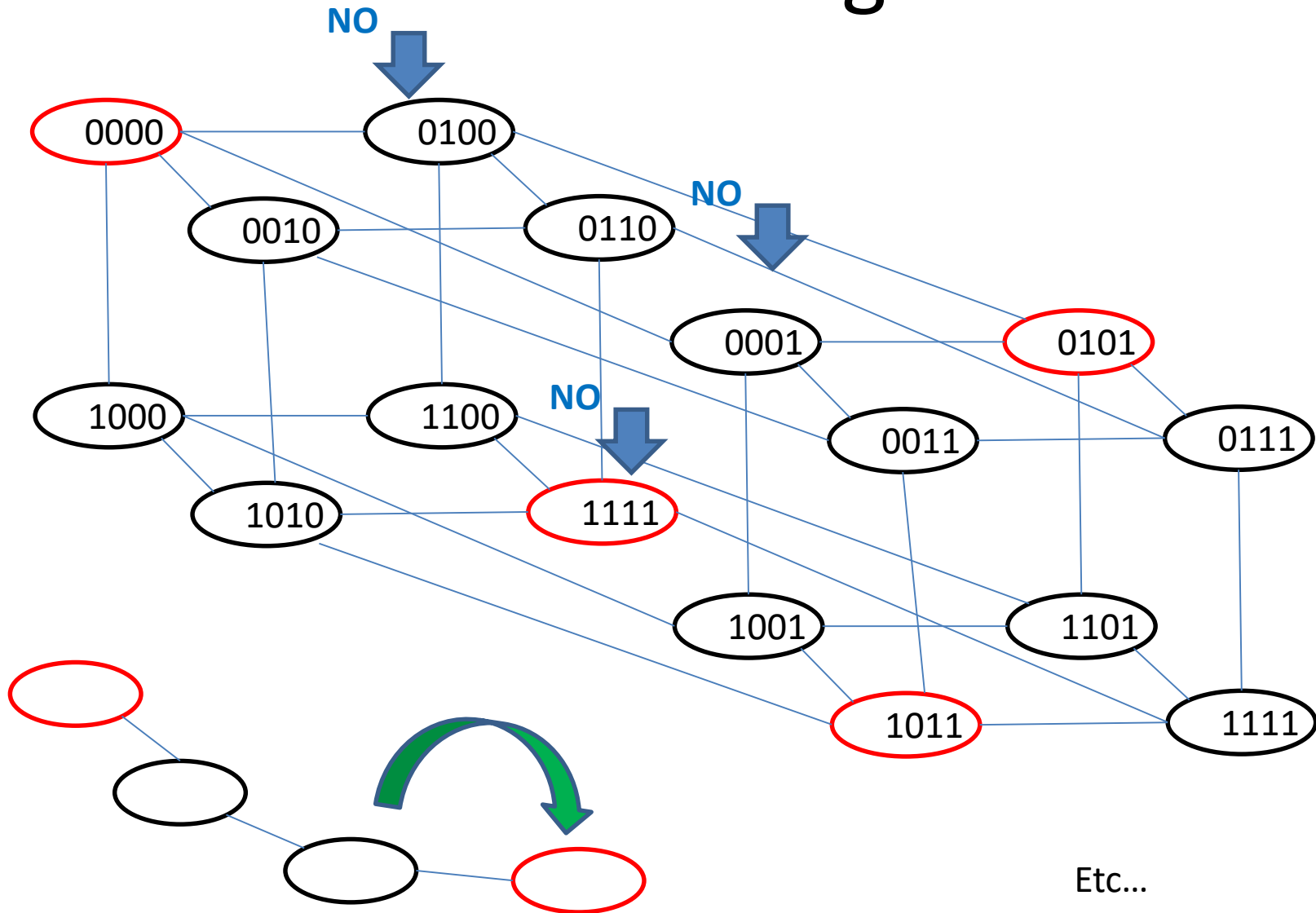


Error detection: No
Error correction : No



Error detection: 1 bit
Error correction : No

Error correcting codes



We need distance ≥ 3 for correcting 1 bit

HAMMING BINARY BLOCK CODE WITH k=4 AND n=7

- Message digits: $C_1 C_2 C_3 C_4$
- Code word $C_1 C_2 C_3 C_4 C_5 C_6 C_7$

Parity Check Equations:

$$C_1 \oplus C_2 \oplus C_3 \oplus C_5 = 0$$

$$C_1 \oplus C_3 \oplus C_4 \oplus C_6 = 0$$

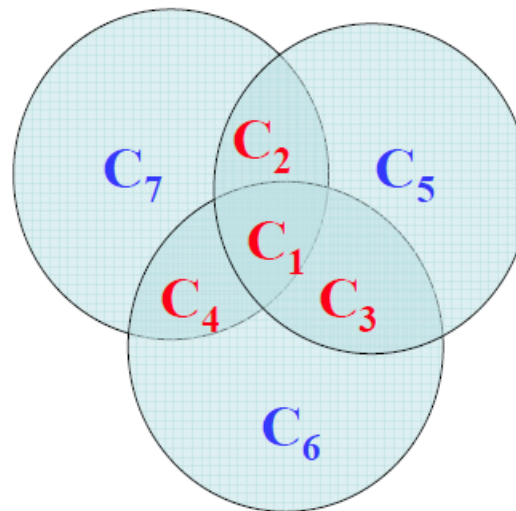
$$C_1 \oplus C_2 \oplus C_4 \oplus C_7 = 0$$

Parity Check Matrix:

1 1 1 0 1 0 0

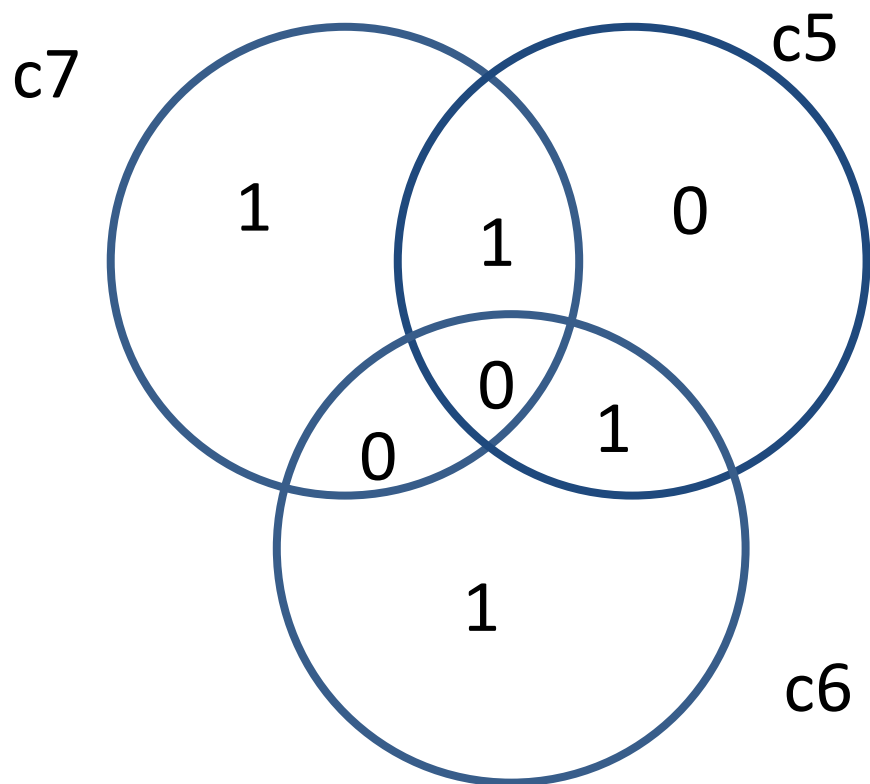
1 0 1 1 0 1 0

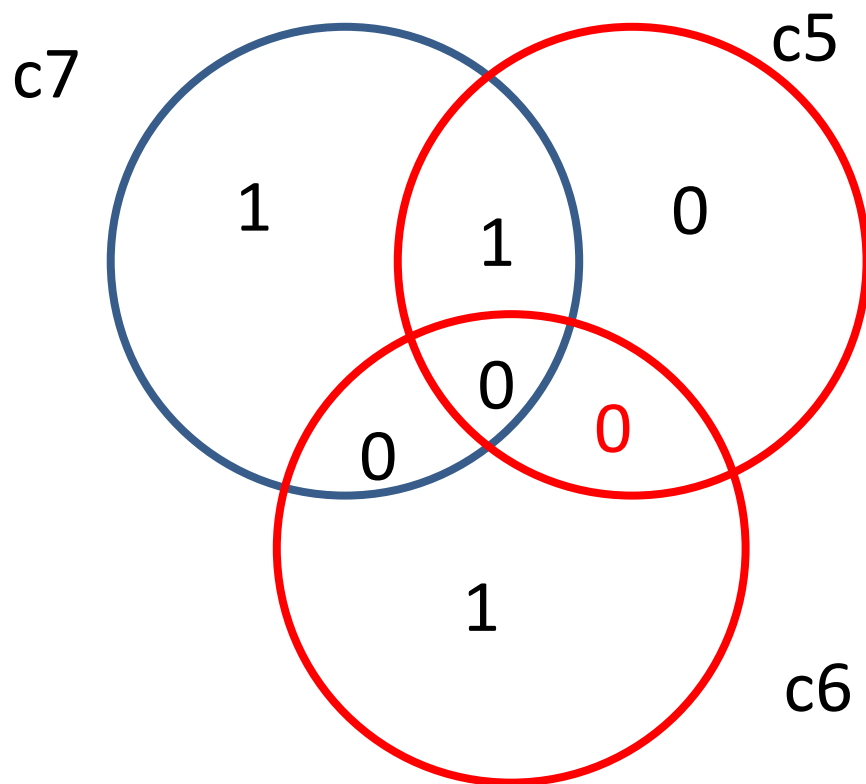
1 1 0 1 0 0 1



The circles represent the equations.

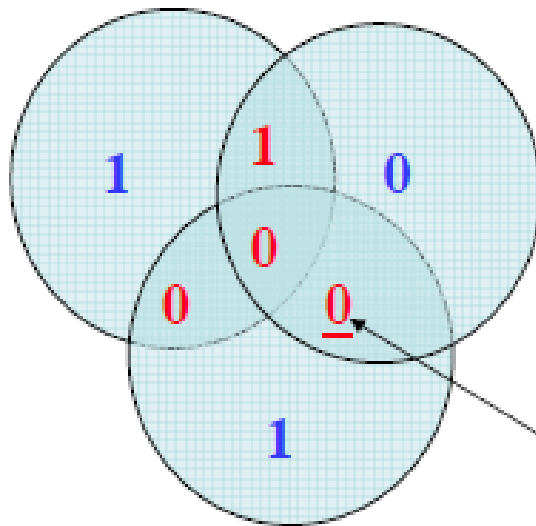
There is an **even** number of 1's in each circle.





HAMMING (7,4) CODE: DECODING

- Transmitted code word: 0 1 1 0 0 1 1
- Example 1:** Received block with one error in a message bit.
0 1 0 0 0 1 1



By counting 1's in each circle we find:

There is an error in right circle.

There is an error in bottom circle

There is no error in left circle.

Therefore the error is in the third digit!

Channel capacity

- Using *multilevel modulation* we increase R/Bw but we need more SNR for the same BER
- Using *error correcting codes* we decrease R/Bw but we need lower SNR for the same BER
- ***Fundamental limits for R/Bw versus E_b/N_0 for achieving arbitrary low levels of BER***
=> C. Shannon

Channel Capacity (AWGN)

$$\frac{C}{Bw} = \log_2 \left(1 + \frac{E_b R}{N_0 Bw} \right)$$

$$\frac{2^{R/Bw} - 1}{R/Bw} \leq E_b/N_0$$

N_0 : Spectral noise power density at receiver (W/Hz=J)

E_b : Energy per bit at receiver (J/b)

Bw : Bandwidth (Hz)

C, R : Capacity, Rate (information bits) (b/s);

$R < C$

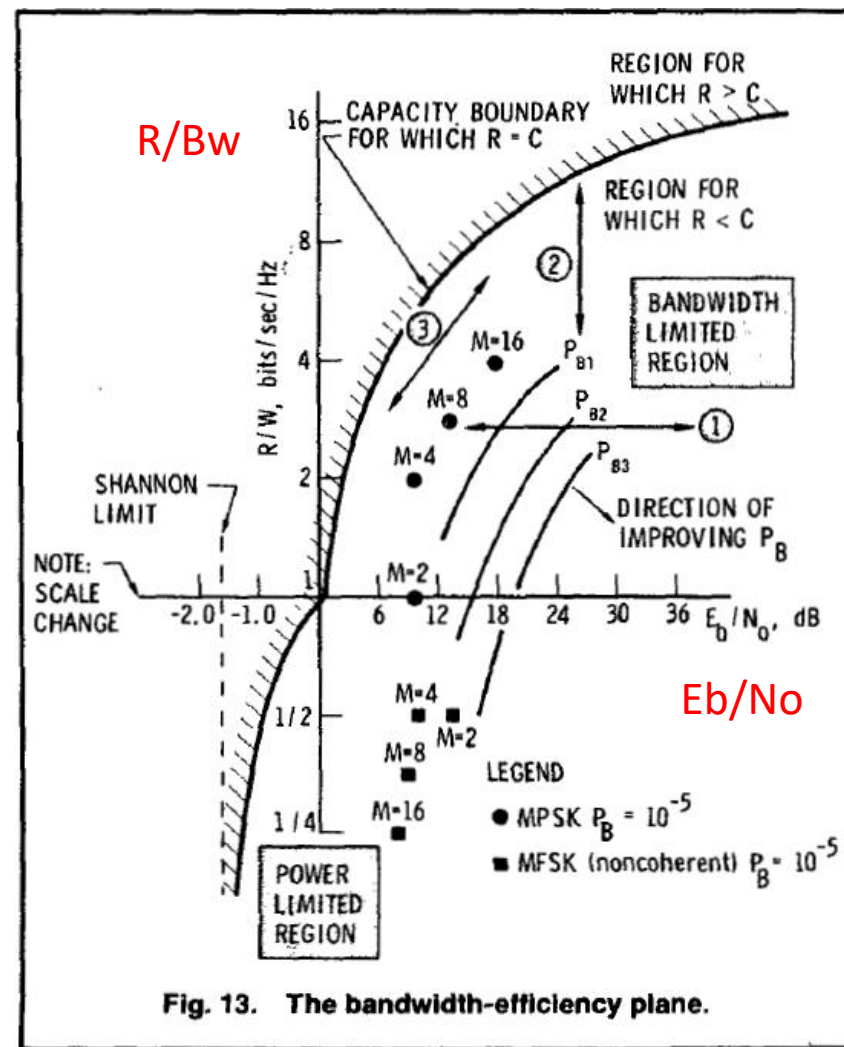
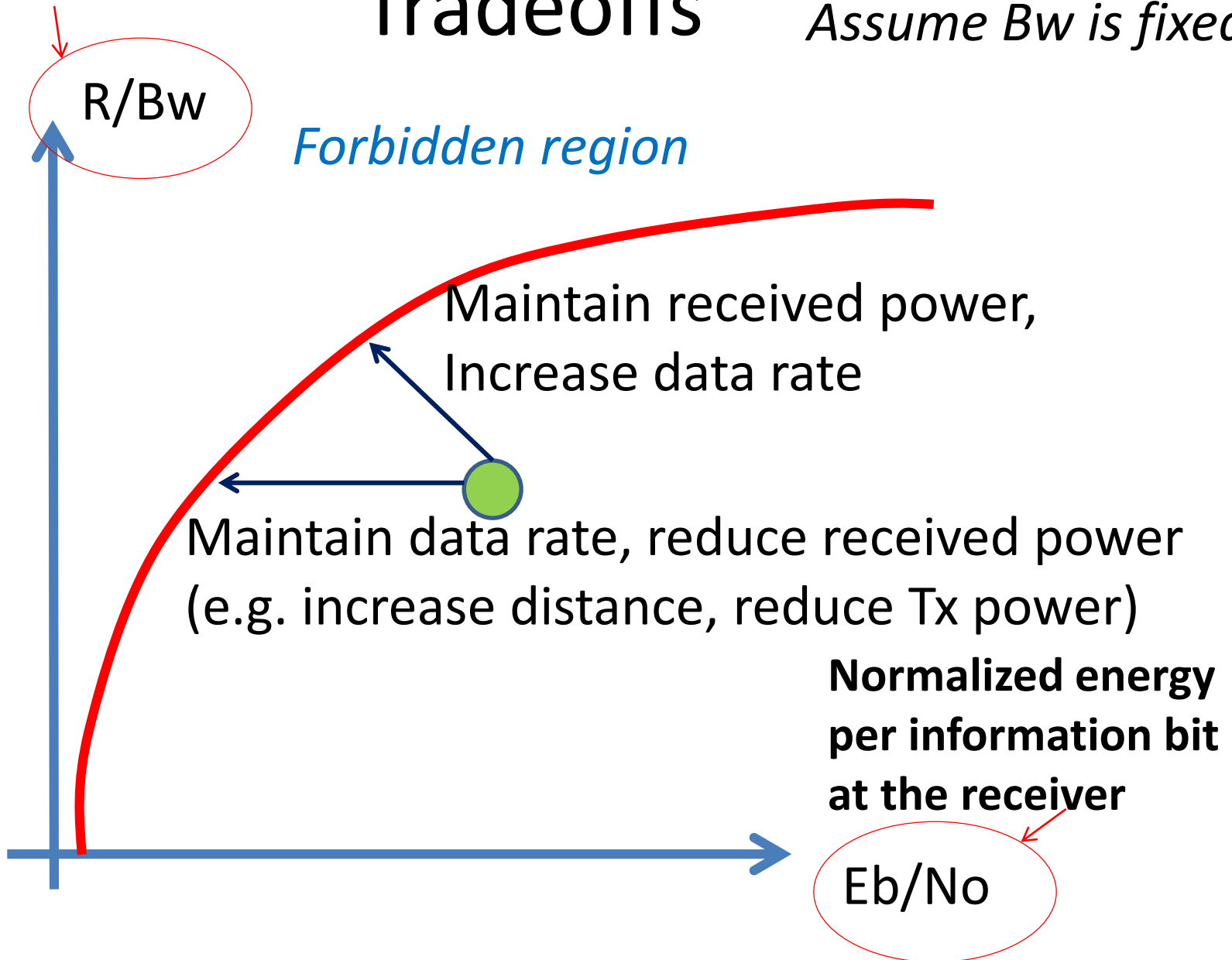


Fig. 13. The bandwidth-efficiency plane.

Spectral efficiency

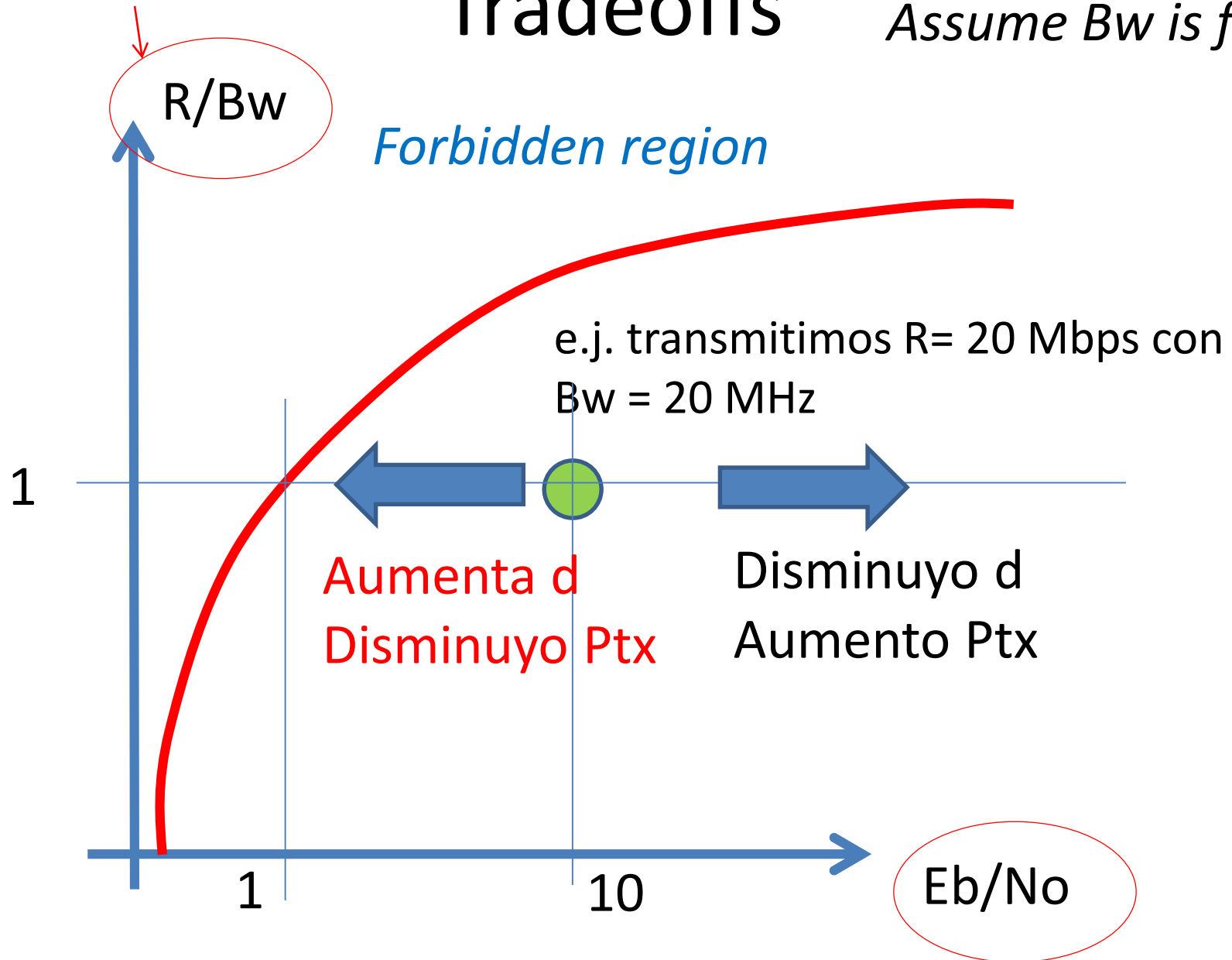
Tradeoffs

Assume Bw is fixed..



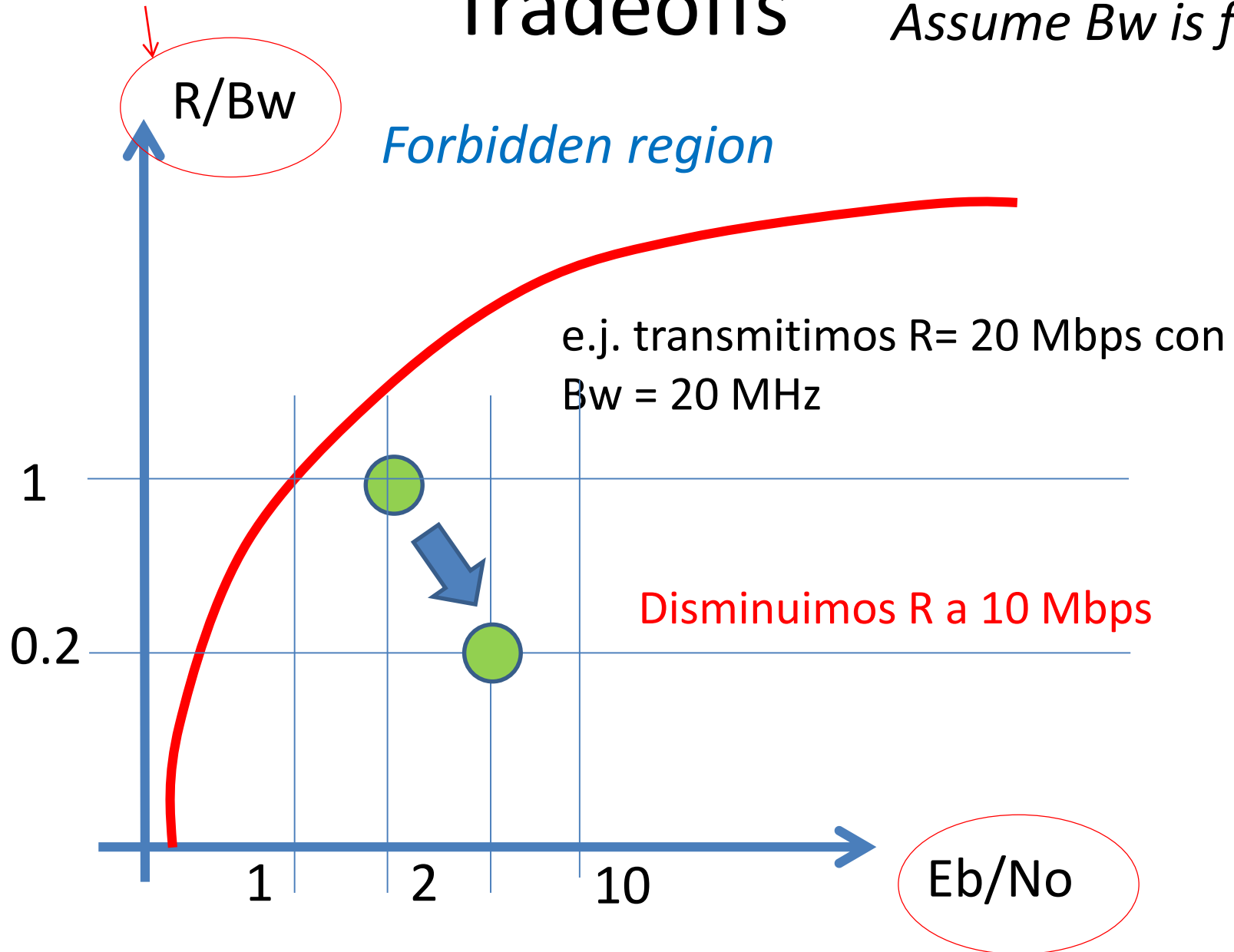
Tradeoffs

Assume Bw is fixed..

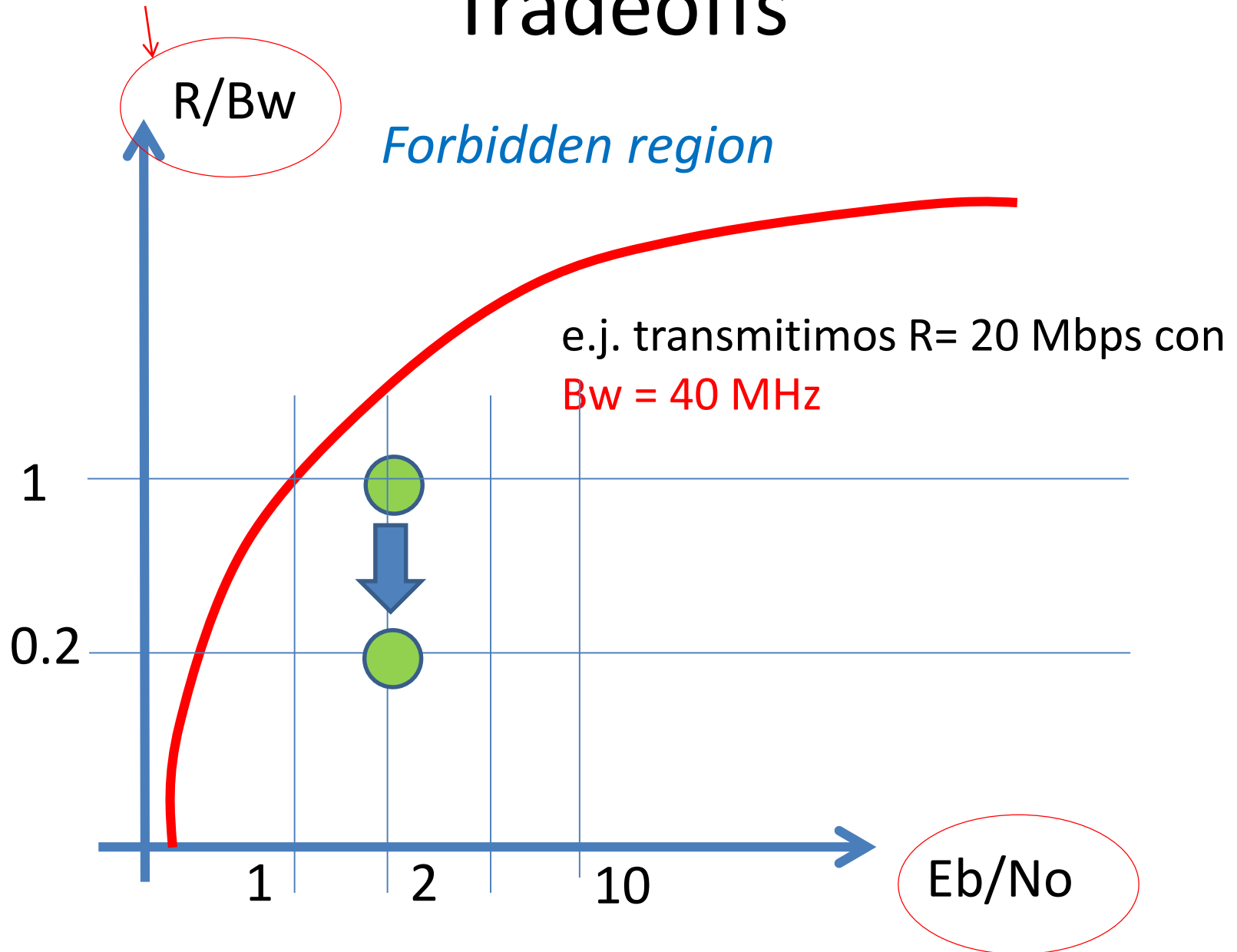


Tradeoffs

Assume Bw is fixed..



Tradeoffs



E_b/N_0 and R/Bw

$$SNR = P_{RX} / P_N$$

We can express P_{RX} and P_N as:

$$\begin{cases} P_{RX} = E_b R \\ P_N = N_0 Bw \end{cases}$$

Does not include code bits!!

Obtaining

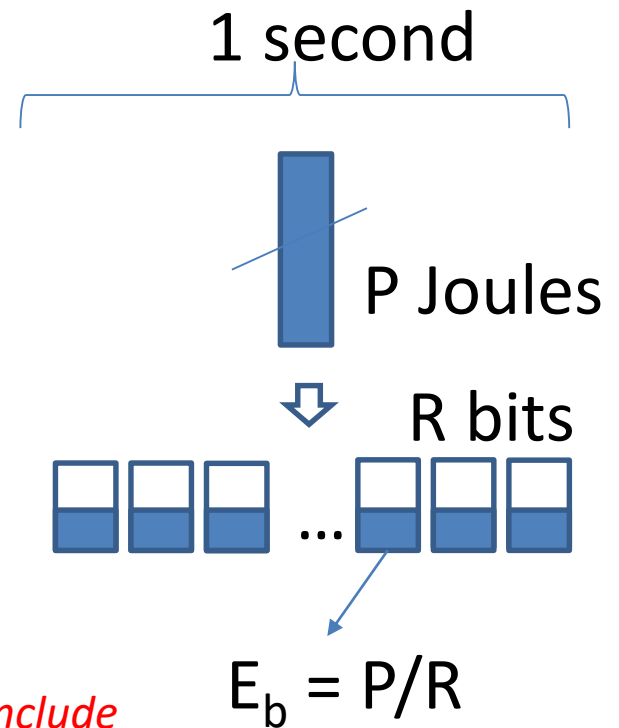
$$SNR = (E_b/N_0) (R/Bw)$$

Spectral efficiency

No units: (b/s)/(1/s)

Energy per bit to noise power spectral density ratio

No units: J/(W/s)



Tradeoffs

NO podemos tener simultáneamente:

- Una R muy grande
- Una d muy grande
- Una P_{tx} pequeña
- Un B_w pequeño

Baja potencia: e.g. NFC, BLE

Alta velocidad: e.g. Wifi

Gran alcance: e.g. Celulares, LowPowerWAN

e.g. 802.11a rate adaptation

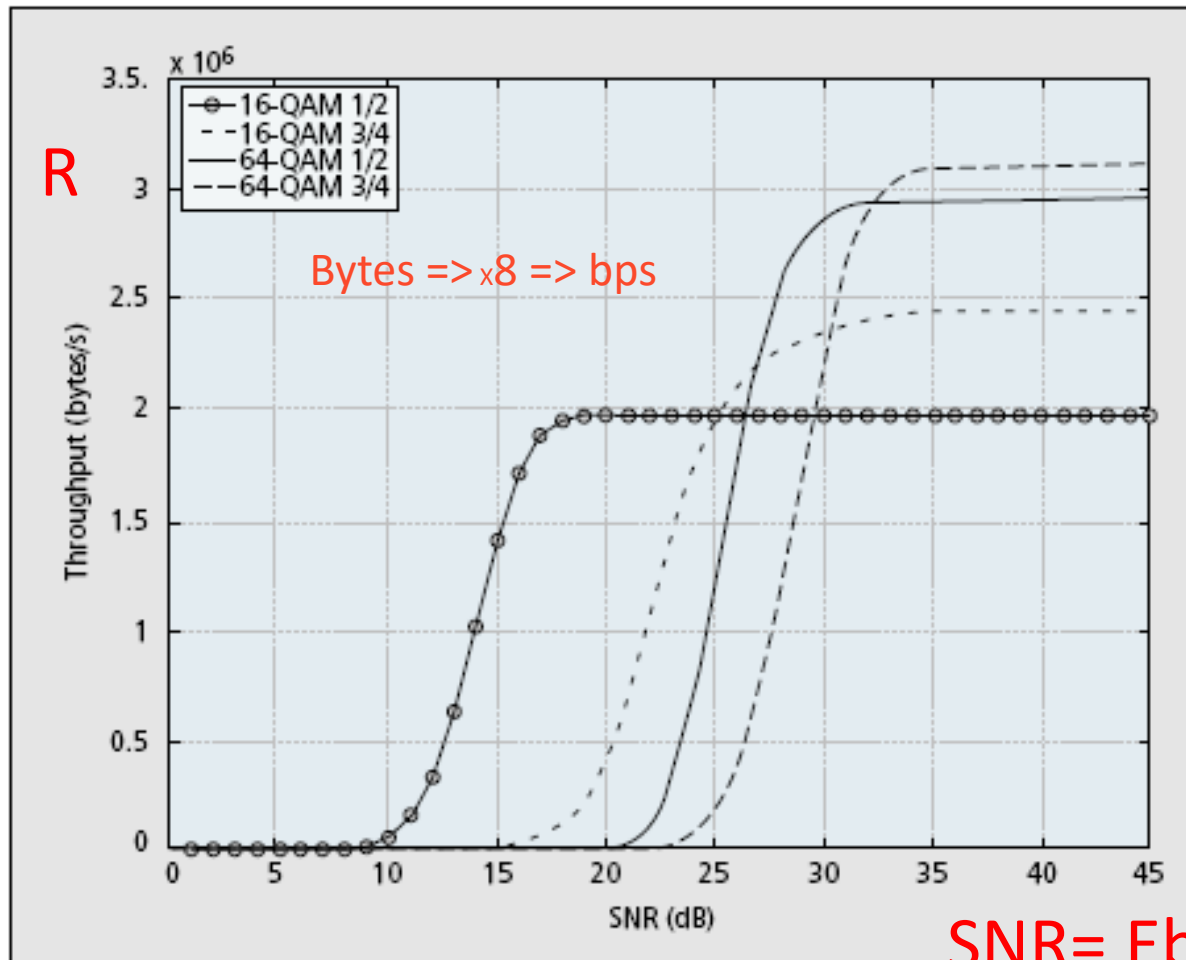


Figure 2. Throughput vs. SNR for some 802.11a modulation schemes.

$$\text{SNR} = E_b/N_o \quad R/B_w$$

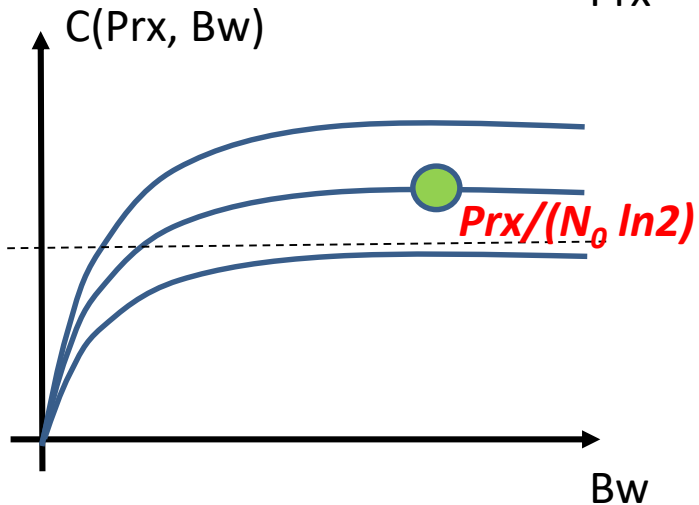
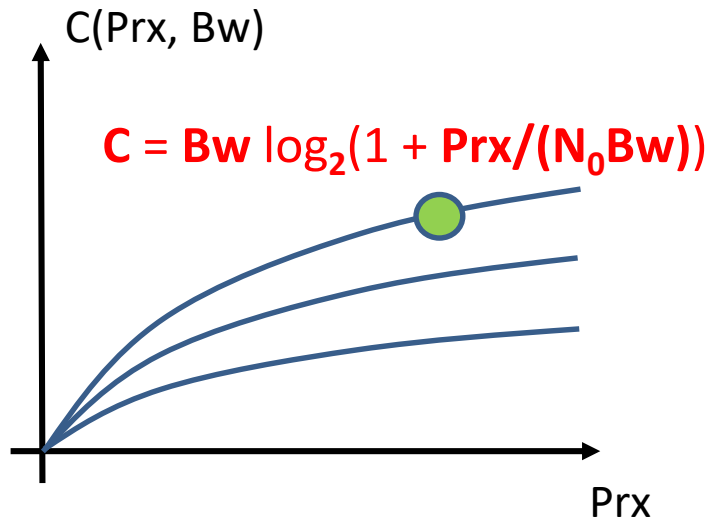
Advanced transmission techniques

- Modern wireless transmission systems use more sophisticated transmission techniques in order to increase data rate or range:
 - **OFDM**: Solves some problems that appear when we use higher bandwidth in a wireless channel (e.g. 802.11agn, 4G)
 - **Multiple Antenna In, Multiple Antenna Out (MIMO)**: increased capacity, meaning faster transmission or larger range (e.g. 4G, 802.11n)

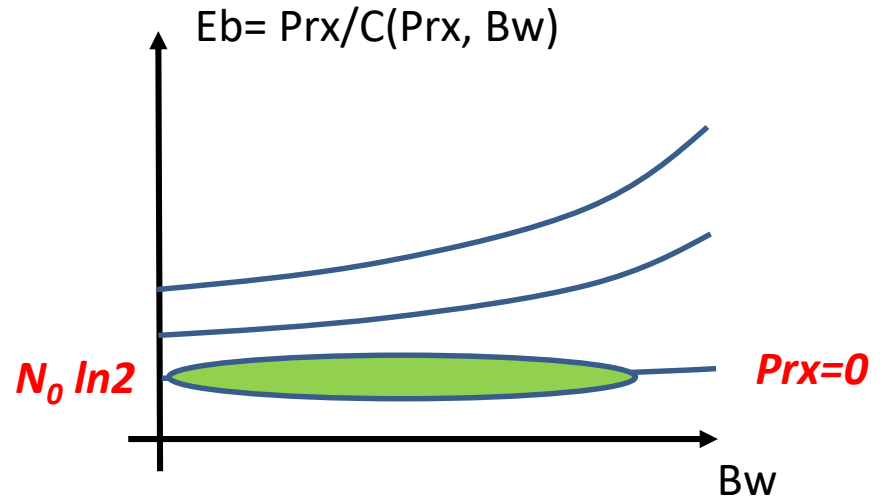
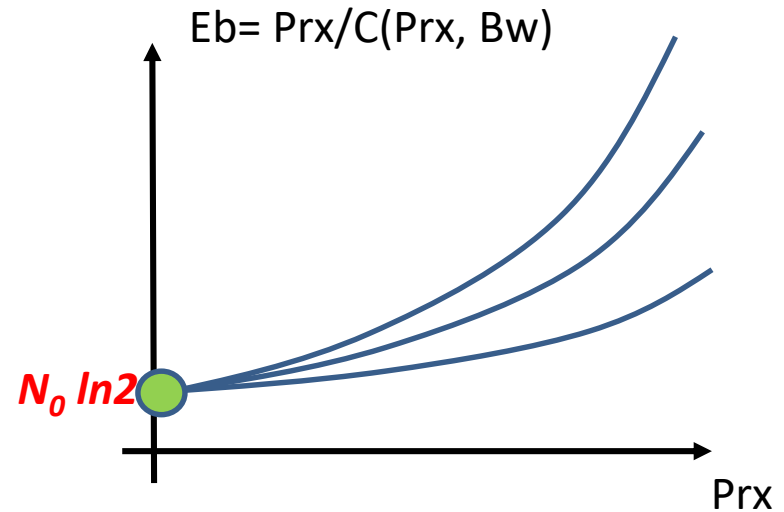
Optimizing R or Eb

- In many communication systems (e.g. WiFi, celular networks) our goal is to **maximize R**. We can play with Bw and Prx (i.e. Ptx and d)
- In IoT we usually want to **minimize Eb**, given that we want simple receivers (i.e. no complex error correction nodes, No MIMO, etc)

Max R & min Eb



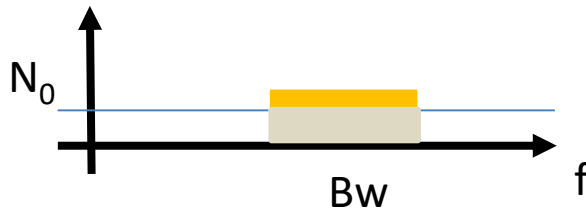
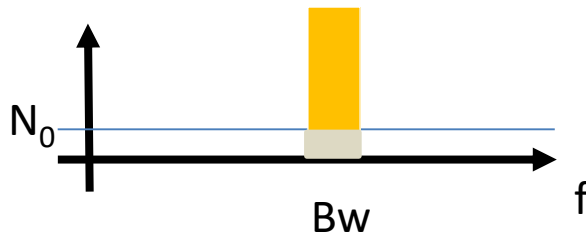
Max C
s.t. $Bw < B_{\max}$, $\text{Prx} < P_{\max}$



Min E_b
s.t. $Bw < B_{\max}$, $\text{Prx} < P_{\max}$

Low Prx and Low $E_b \Rightarrow$ low R

Maximizing R (e.g. WiFi, 4G, 5G)



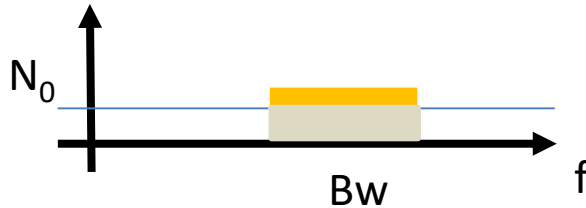
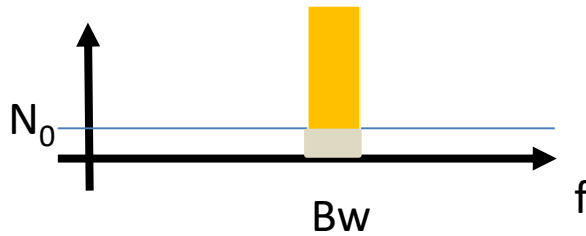
Wideband increases SNR, but we can use stronger codes

Using Shannon's capacity formula we can see that for maximizing R the optimal policy is to have high Bw and high Prx, but there is a saturation in the effect of increasing Bw. **In practice, use high enough Bw and high Prx (i.e. small d)**

Wideband (Broadband) Wireless

- In order to have high transmission rates (R), it is desirable to use channels with high bandwidths (B_w). We call these systems “**wideband wireless communication systems**”
- Examples:
 - 4G can use channels of $B_w = 5, 10, 15$ and 20 MHz
 - 802.11bga use $B_w = 20$ MHz, but 802.11n can use $B_w = 40$ MHz.
- However, **the multipath fading introduces complexities when we use wideband transmission**

Minimizing E_b (e.g. IoT)



Narrowband reduces SNR $\Rightarrow$
we have lower Bit Error Rate
for the same modulation

Using Shannon's capacity formula we can see that for minimizing E_b the theoretical optimal policy is to have low R independently of $Bw \Rightarrow$

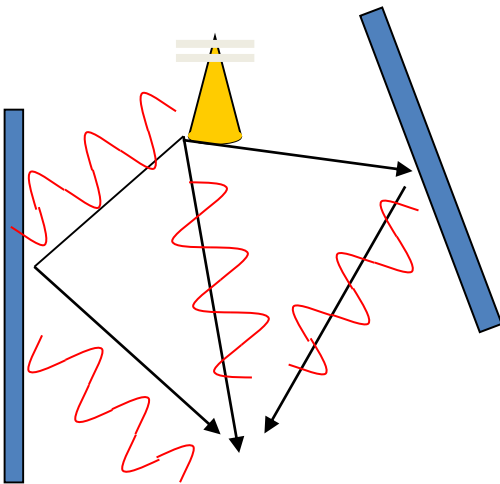
In practice, use low (conservative modulation) R and small Bw

Bw and R in different technologies

Tech	Bw	R	R/Bw	Ptx	d
4G (2x2)	5, 20 MHz	20 Mbps	1	100 mW	1Km
802.11b	20 MHz	54 Mbps	3	60 mW	100 m
802.11n (2x2)	20 MHz	150 Mbps	7	60 mW	100 m
BLE	2 MHz	125 Kbps	0.2	10 mW	10 m
802.15.4	2 MHz	115 Kbps	0.2	1 mW	10 m
LoRA	125 KHz	50 Kbps (xDC)	0.4	25 mW DC=1%	1 Km
NBIoT	180 KHz	60 Kbps (xDC)	0.3	100 mW DC nolimit	1 Km

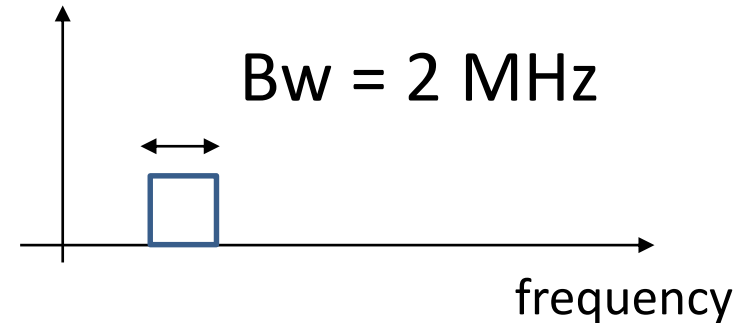
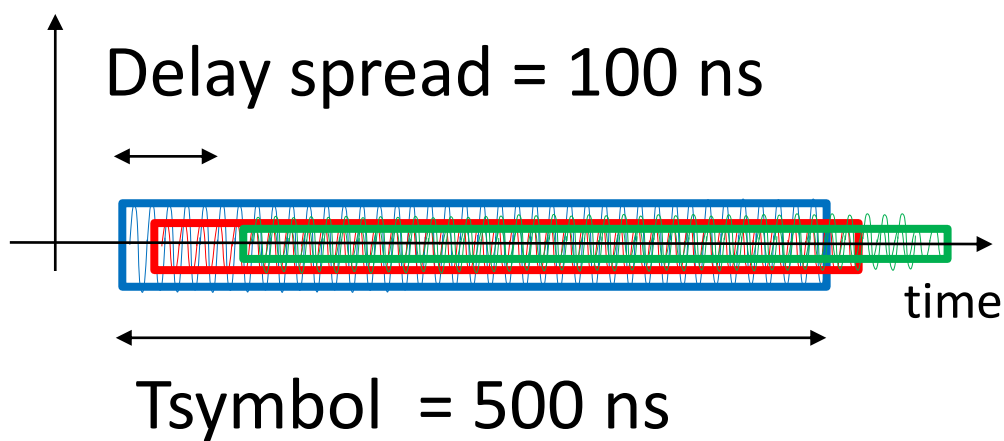
Delay spread in indoor/outdoor

The **delay spread** is the maximum difference between the arrival times of different multipath signals.

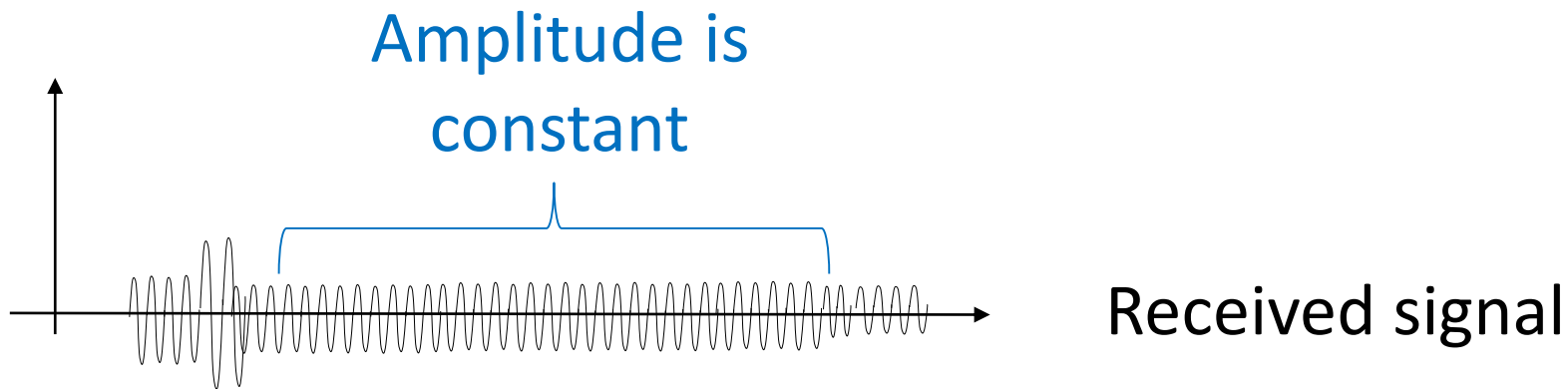
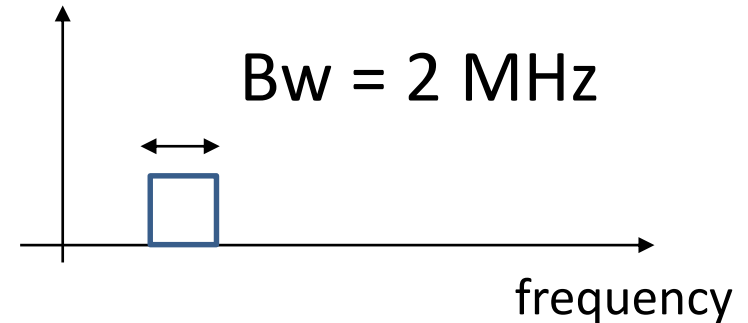
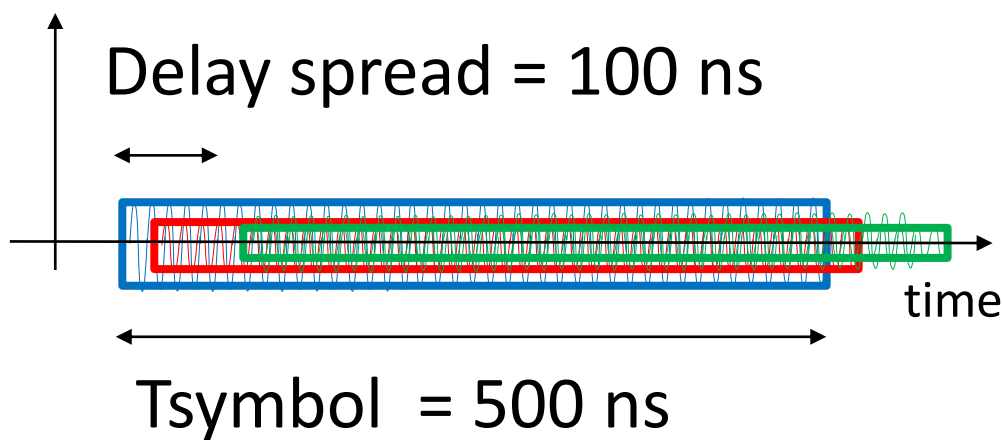


Δspace	Delay spread (Δtime)	Typical Scenario
3 m	10 ns	Indoor (PAN, WLAN)
30 m	100 ns	Indoor (WLAN)
300 m	1 μs	Outdoor (Cellular)
3 Km	10 μs	Outdoor (Cellular)

Delay spread $\ll$ symbol time
e.g.: $\Delta\text{space}=30\text{ m}$, $Bw=2\text{ MHz}$

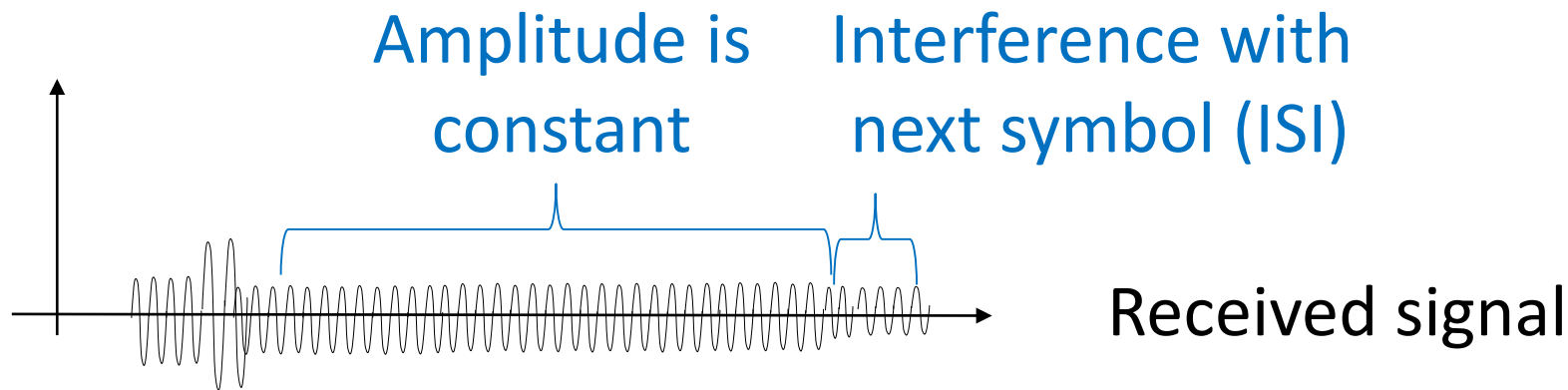
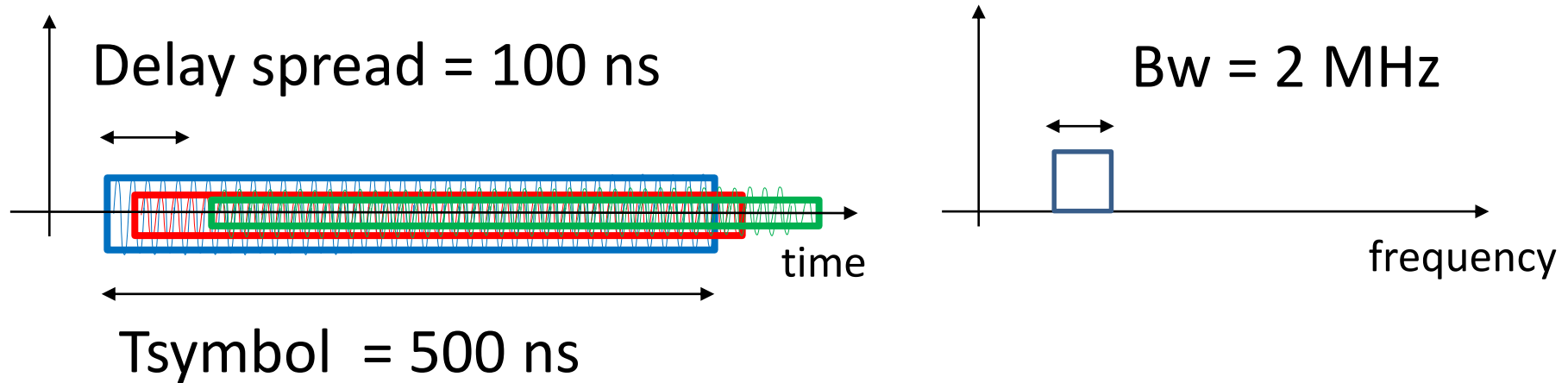


Delay spread $\ll$ symbol time
e.g.: $\Delta\text{space}=30\text{ m}$, $Bw=2\text{ MHz}$



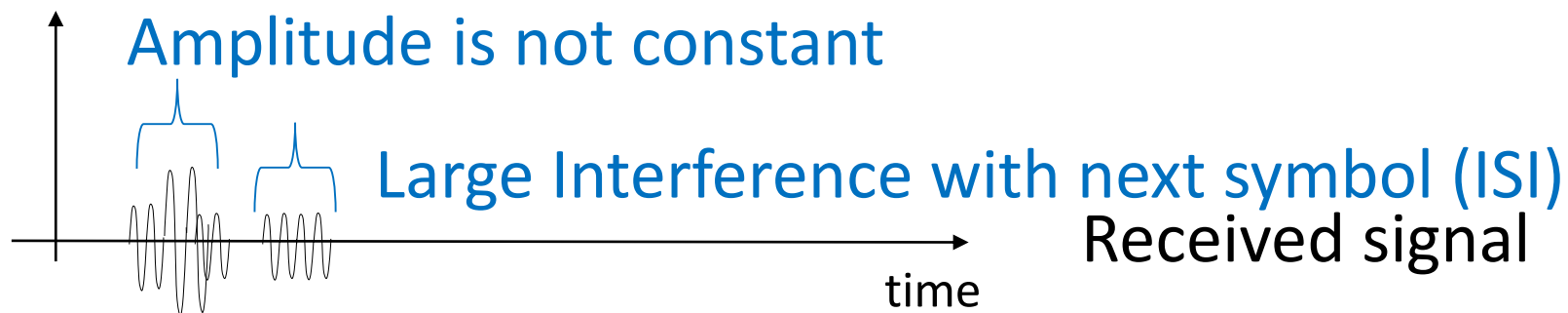
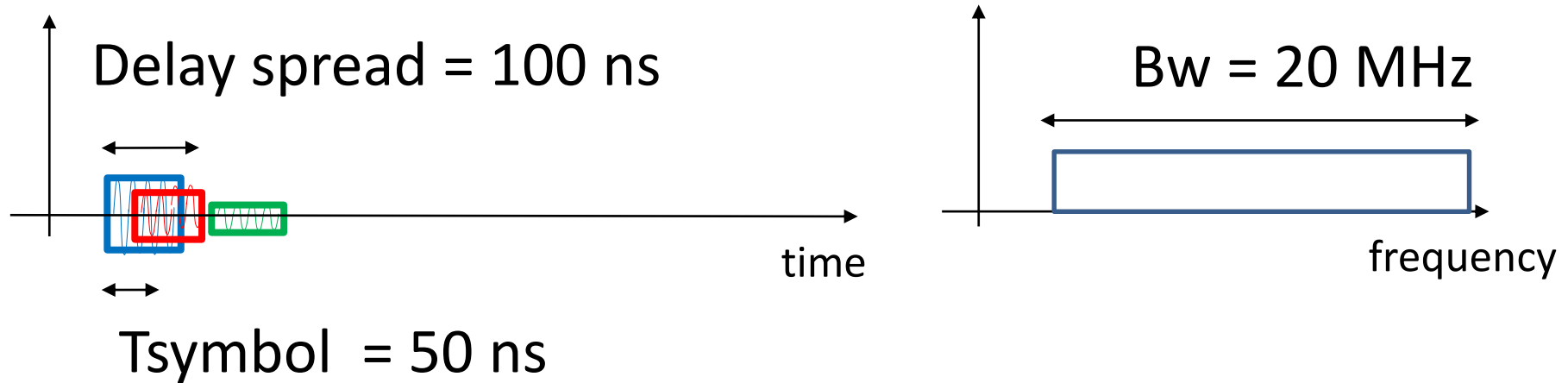
Delay spread $\ll T_{\text{symbol}} = 1/Bw$, Small Bw

Delay spread $\ll$ symbol time
e.g.: $\Delta\text{space}=30\text{ m}$, $Bw=2\text{ MHz}$



Delay spread $\ll T_{\text{symbol}} = 1/Bw$, Small Bw

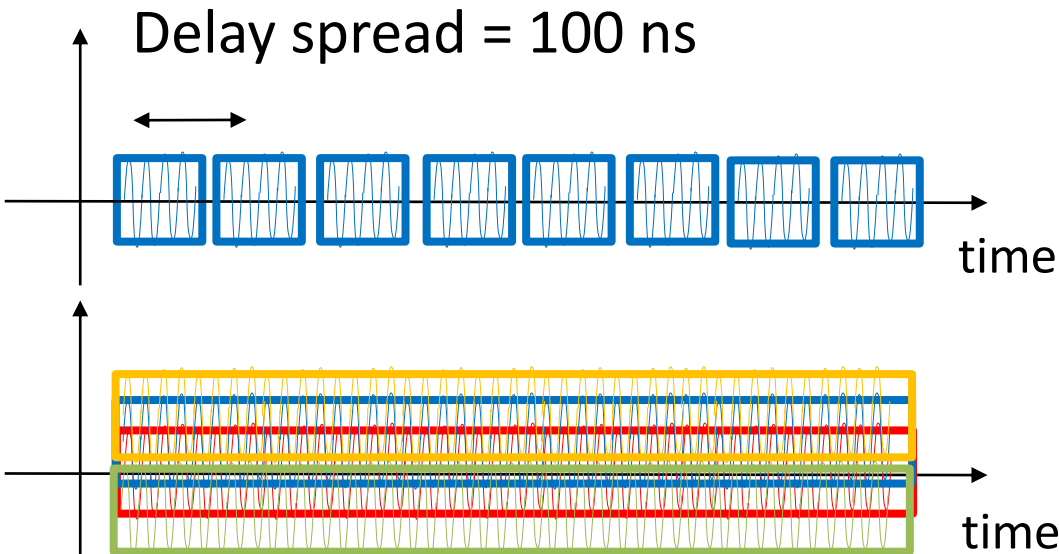
Delay spread $>$ symbol time
e.g.: $\Delta\text{space}=30\text{ m}$, $Bw=20\text{MHz}$



Delay spread $>$ $T_{\text{symbol}} = 1/Bw$, Large Bw

Multicarrier modulation

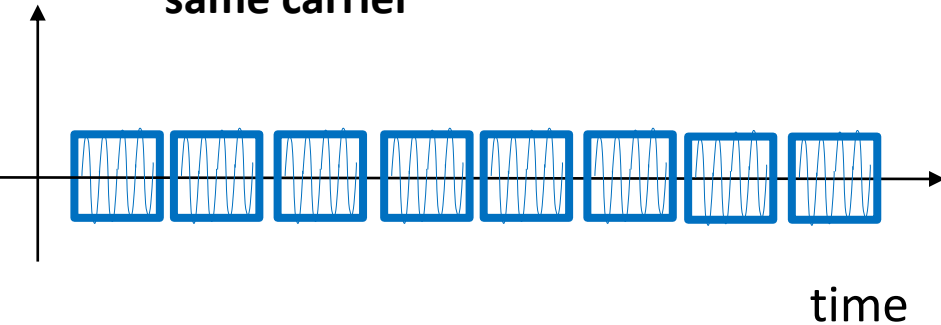
- Idea: instead of transmitting consecutive many short symbols using a common carrier, **transmit many simultaneous long symbols using multiple carriers.**



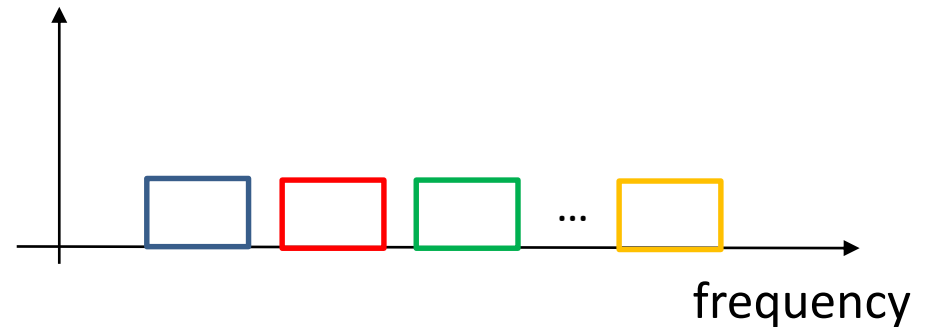
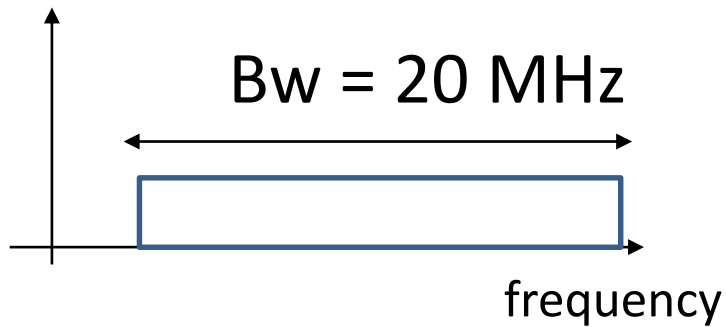
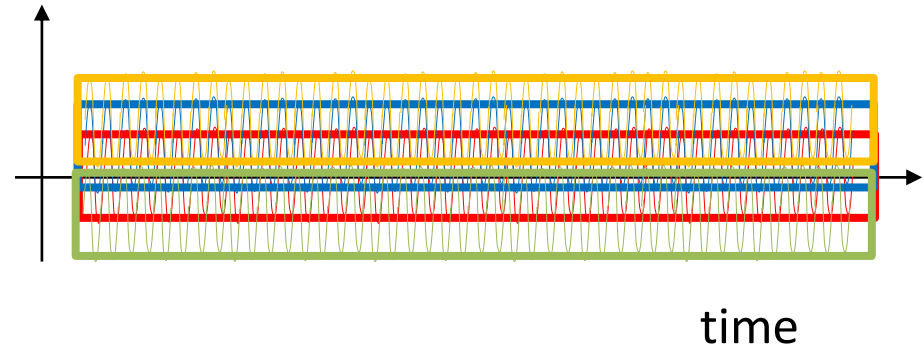
e.g. Transmit 10 symbols
Using 10 different carriers.
Each symbol is 10 times
longer, meaning that is
less vulnerable to the effects
of the delay spread

Multicarrier modulation

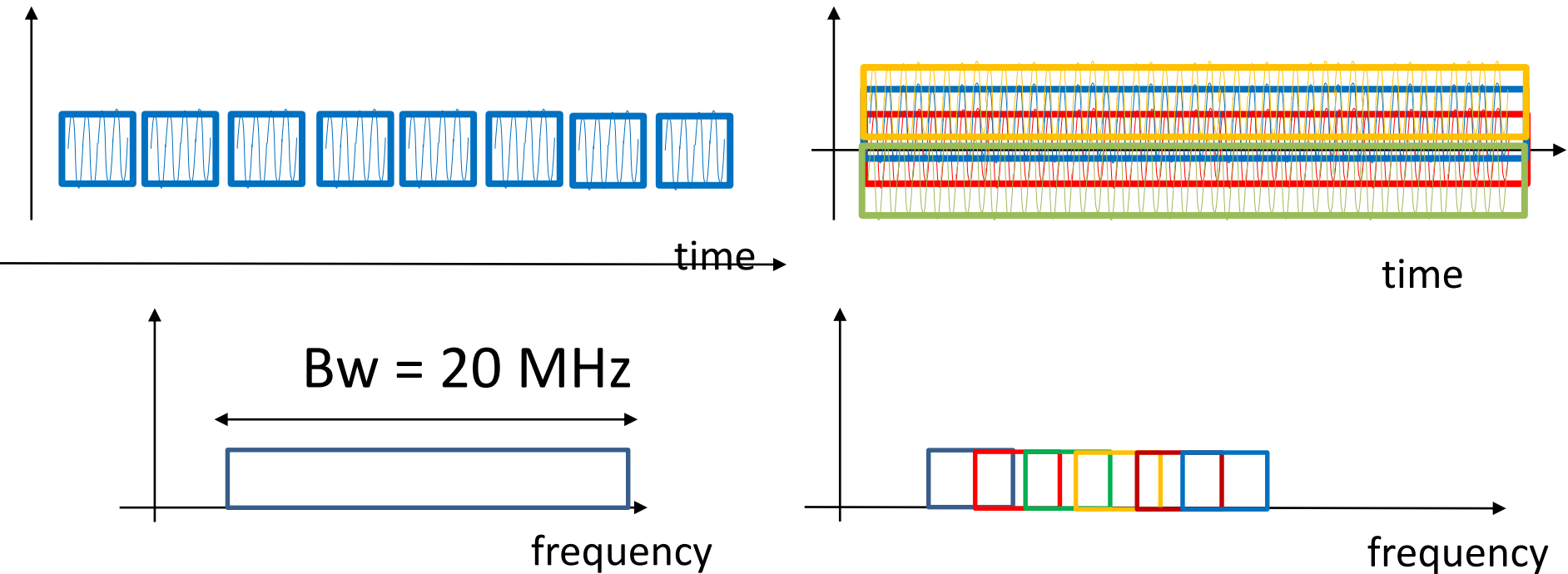
Consecutive short symbols,
same carrier



Simultaneous long symbols,
different carriers

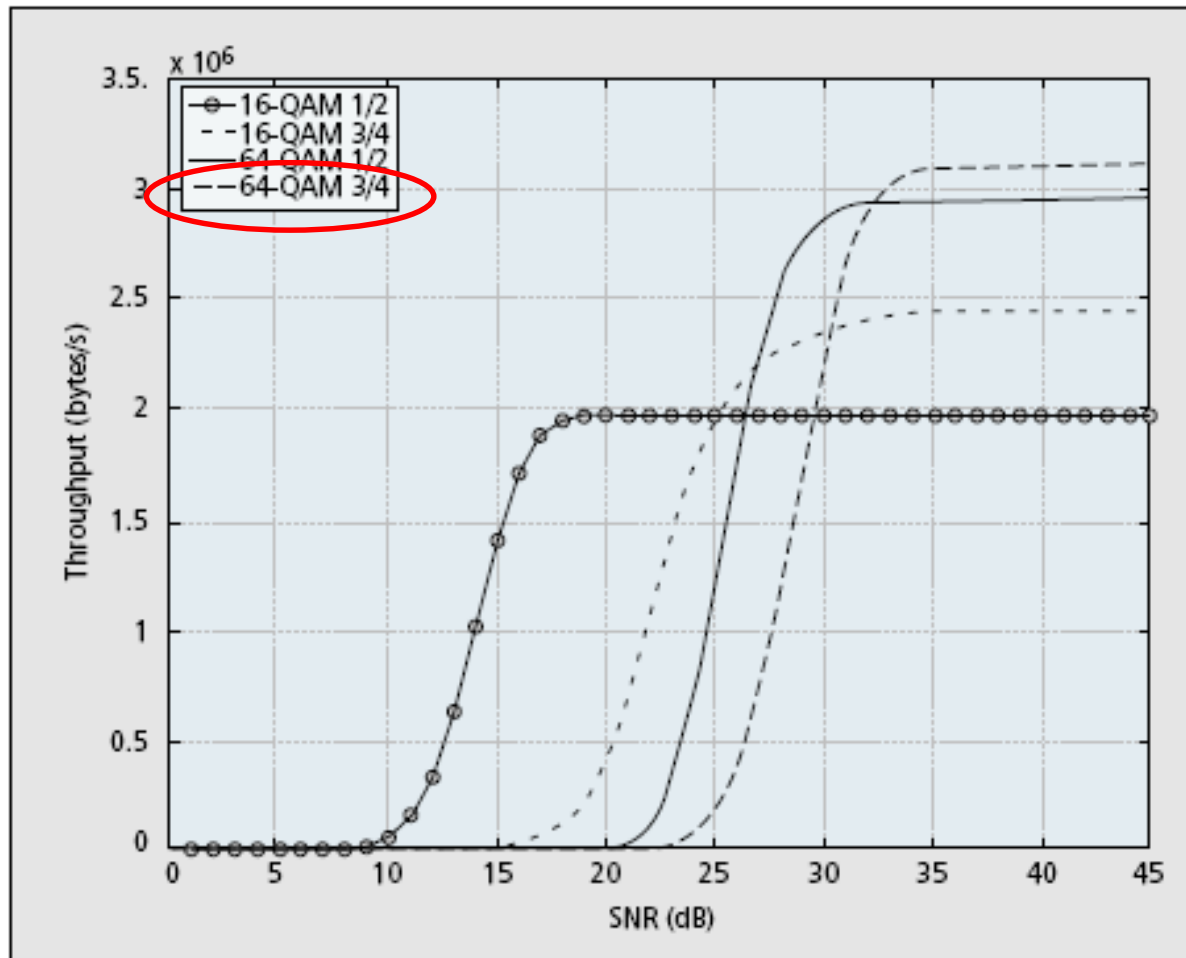


Overlapping channels (OFDM)



What is the **minimum separation between carriers**? It can be shown that it is $\Delta f_c = 1/T_{\text{symbol}}$. In this case the different subchannels overlap, but still we can recover correctly the individual symbols. This is called **OFDM (Orthogonal Frequency Division Modulation)**

802.11a OFDM



■ Figure 2. Throughput vs. SNR for some 802.11a modulation schemes.

e.g. 802.11a @54 Mbps

- OFDM with 64 subcarriers and 48 data subcarriers
- Coding rate: $\frac{3}{4}$ bit/codebit
- 6 coded bits per subcarrier symbol (64 QAM)
- Subcarrier symbol time: 4 μ sec

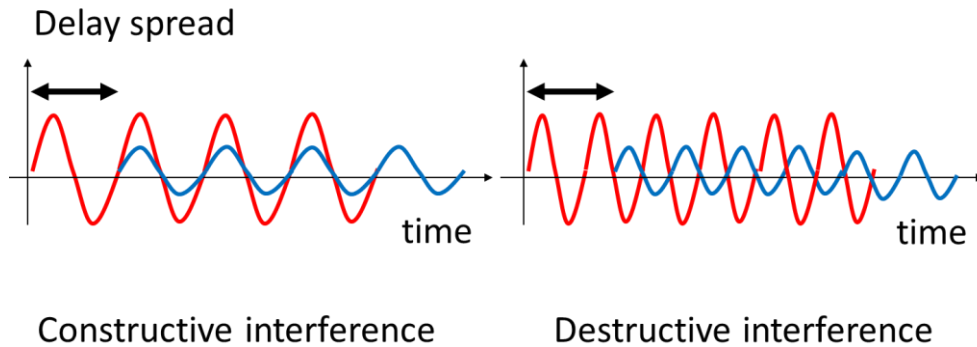
(Note: $Bw=20$ MHz, $Bw_subcarrier = 20/64$ MHz $\Rightarrow$ $T_{symbol} > 3.2$ μ sec)

R= 48 subcarrier
 x 1 subcarrier symbol/4 μ secs
 x 6 codebits/subcarrier symbol
 x $\frac{3}{4}$ bit/codebit x = 54 Mbps

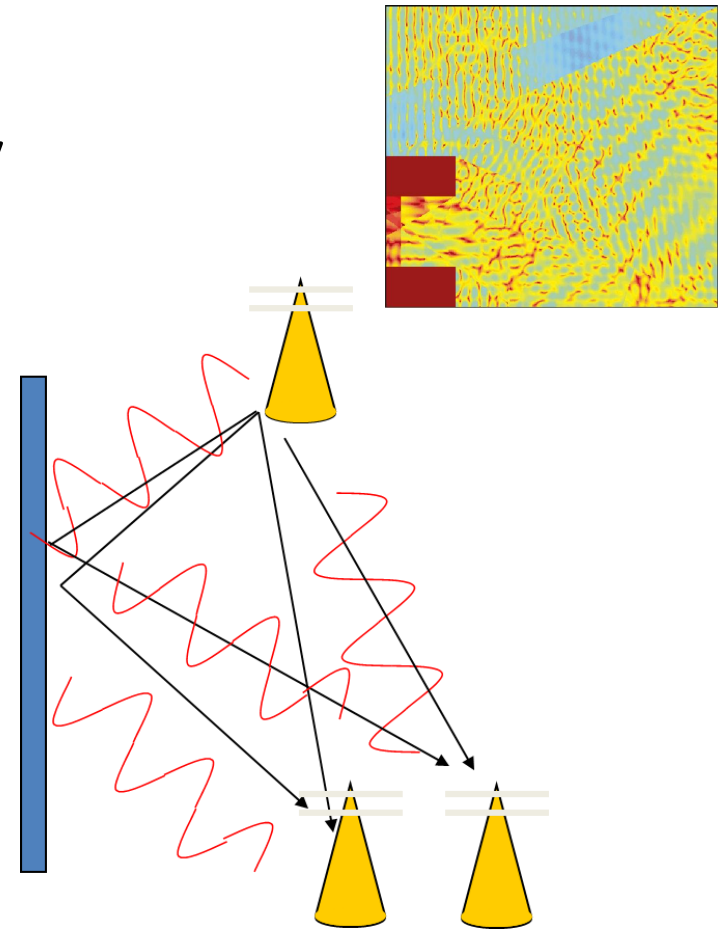
Diversity

- The main technique used in wireless systems to cope with variable wireless channels is diversity.
- Diversity is the spreading of information with some redundancy across multiple independently faded channels.
- When this is done, it is unlikely that a deep fade on a single channel will prevent successful communication.

Diversity



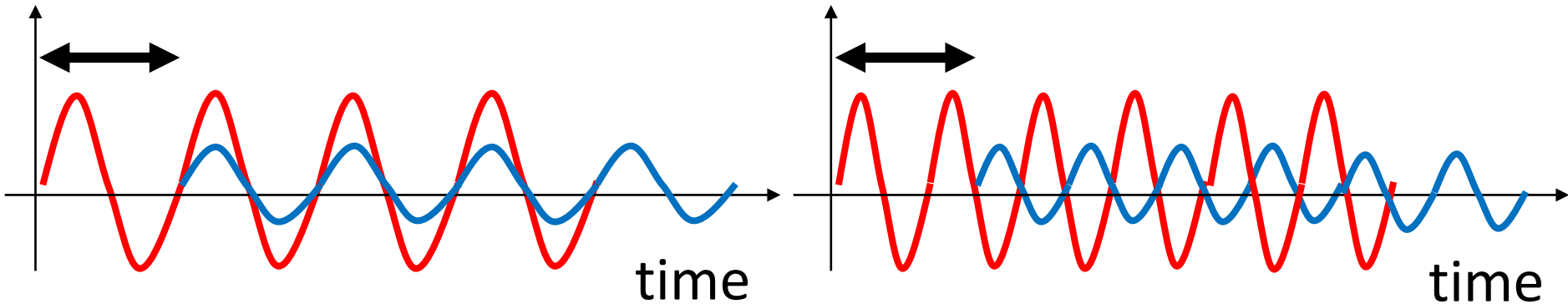
Use several channels with
different carriers



Use more than one
antenna for Tx and/or
Rx

Same delay spread, different fc

Delay spread



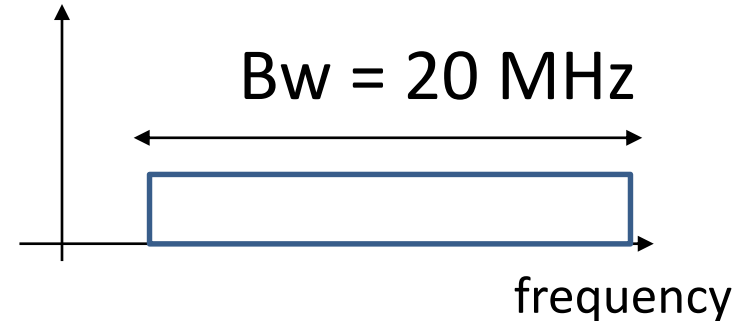
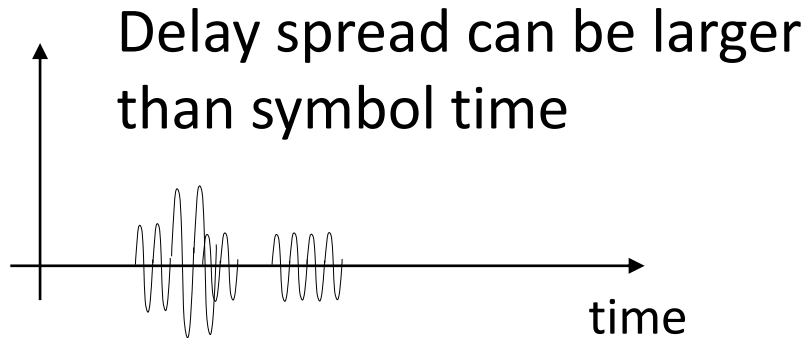
Constructive interference

Destructive interference

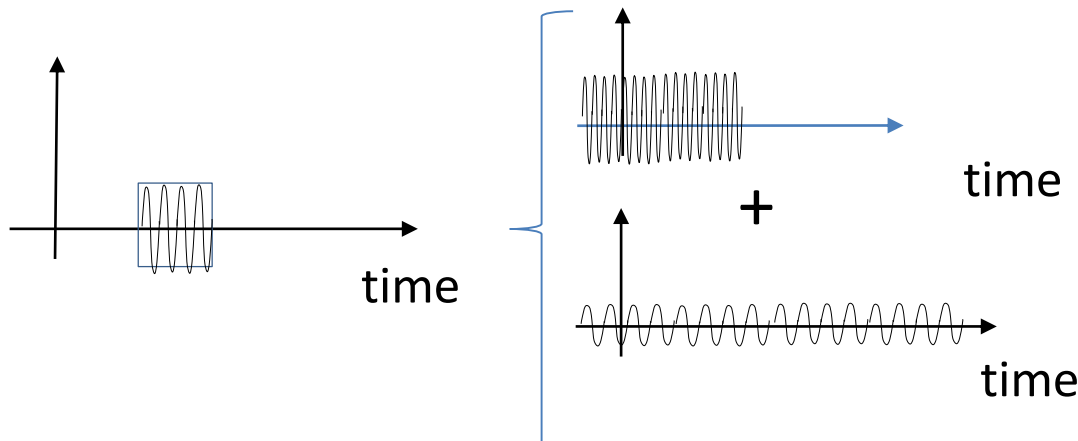
OFDM: diversity

- The different subchannels can suffer different multipath fading.
- The explanation to this comes from the fact that **the same delay spread can lead to a destructive or constructive interference depending on the frequency of the carrier.**
- We can use some subcarriers to estimate the multipath fading using pilot tones

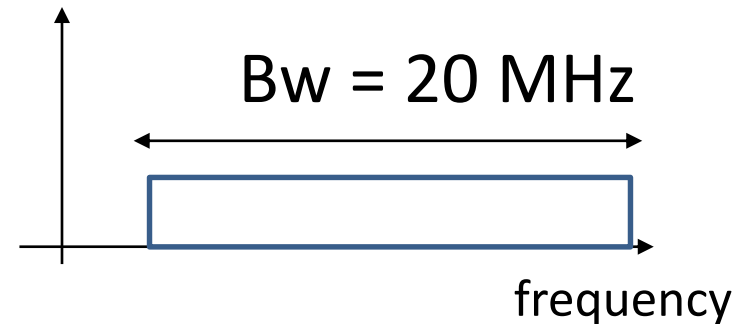
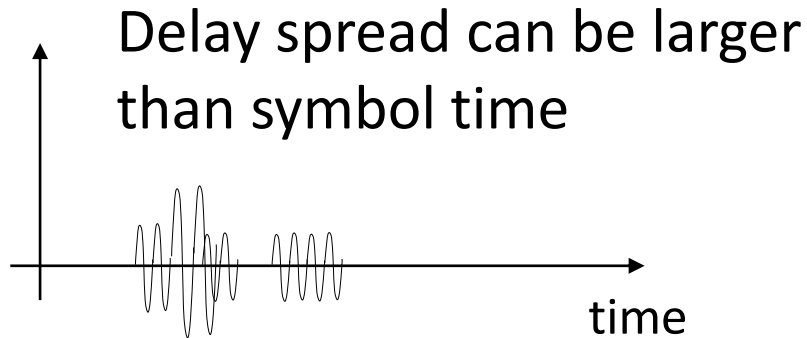
Wideband Wireless channels



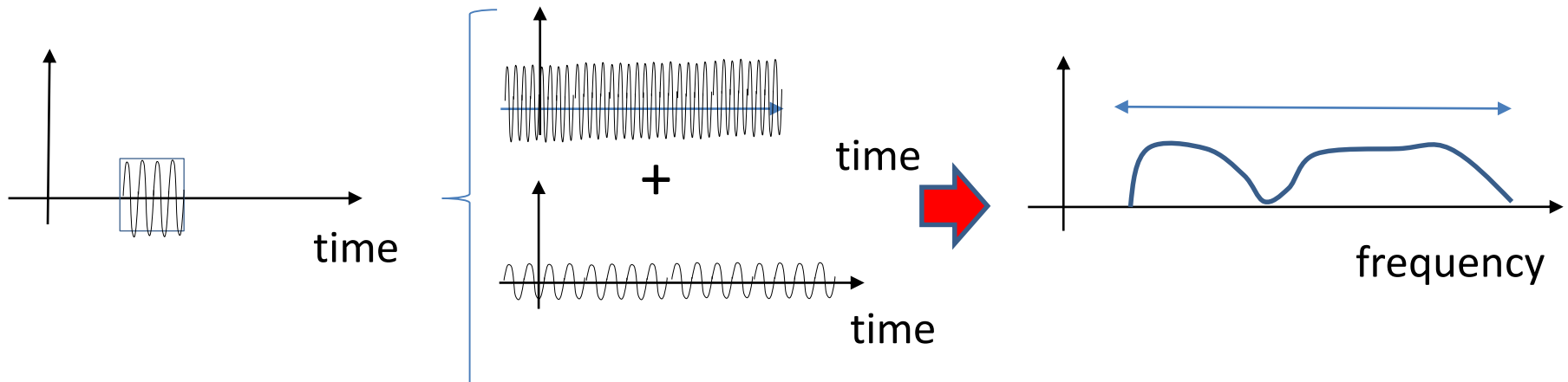
The Fourier components of short symbols span many different frequencies and each component can suffer different multipath attenuations



Wideband Wireless channels



This means that wideband wireless channels suffer a **frequency selective multipath fading**



802.11a PHY

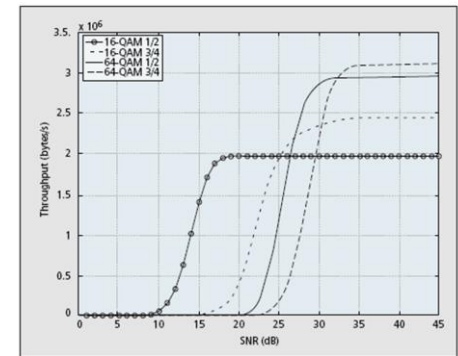


Figure 2. Throughput vs. SNR for some 802.11a modulation schemes.

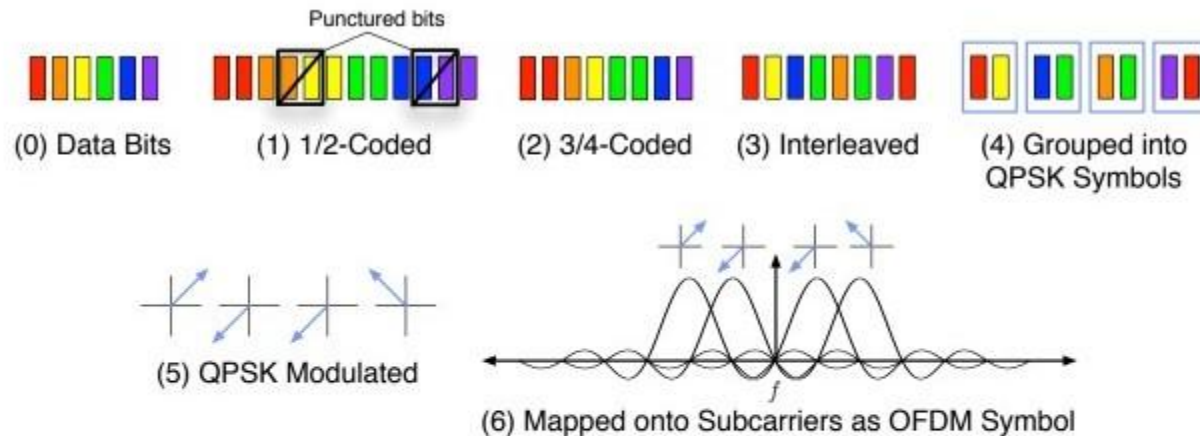
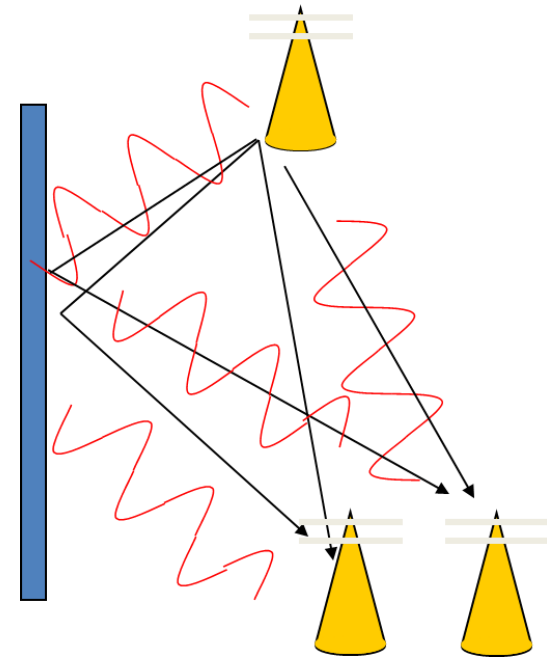
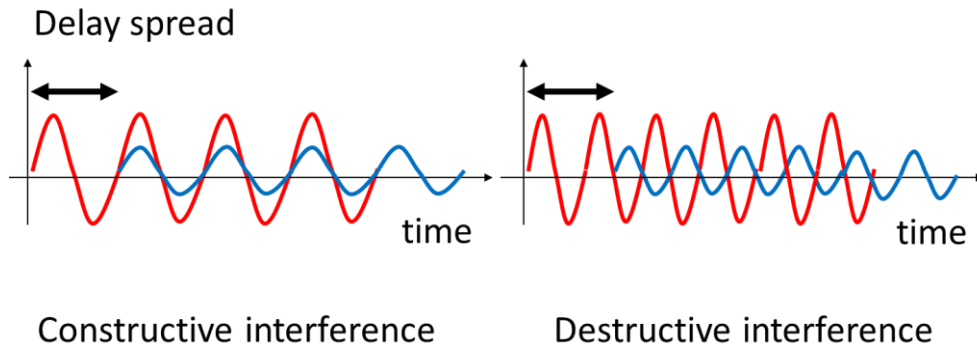


Figure 1: A graphical view of the OFDM encoding process for the 18 Mbps rate (QPSK, 3/4) of 802.11a. The data bits (0) are encoded by a rate-1/2 convolutional code (1) and then optionally punctured by dropping certain bits for higher coding rates (here, 3/4) that send fewer redundant bits (2). The remaining bits are interleaved (3) to spread the redundancy across subcarriers and protect against frequency-selective fades. These bits are grouped into symbols (4) based on the modulation (QPSK encodes 2 bits per symbol), modulated (5), and finally mapped onto the different subcarriers to form an OFDM symbol (6).

Diversity



Use several channels with
different carriers

Use more than one
antenna for Tx and/or
Rx

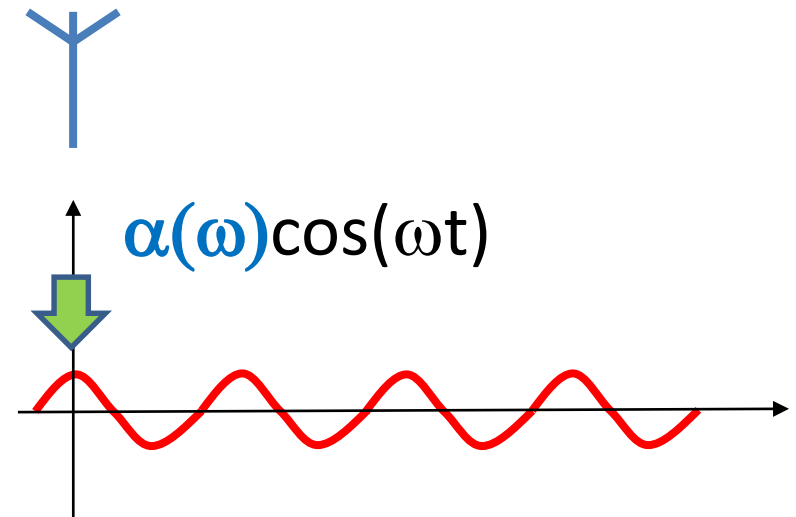
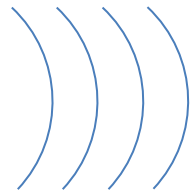
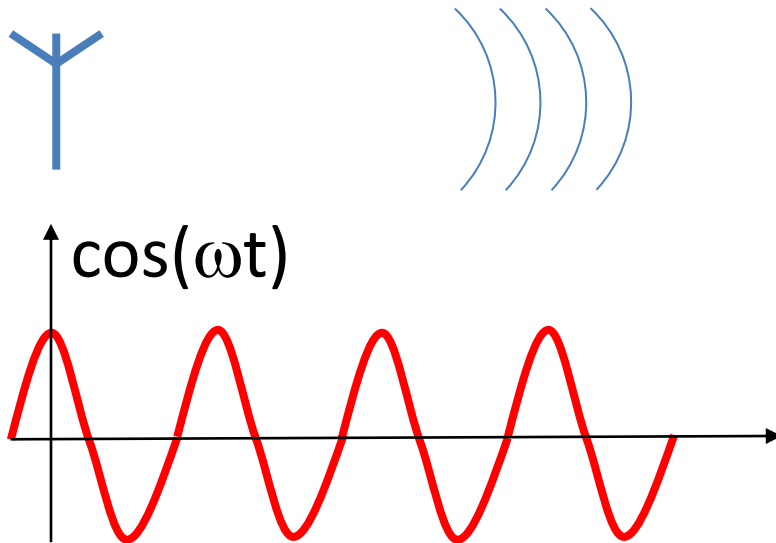
Interesting reading

802.11 with Multiple Antennas for Dummies

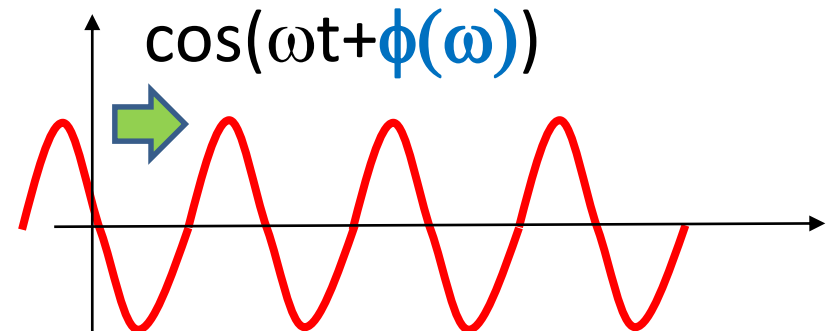
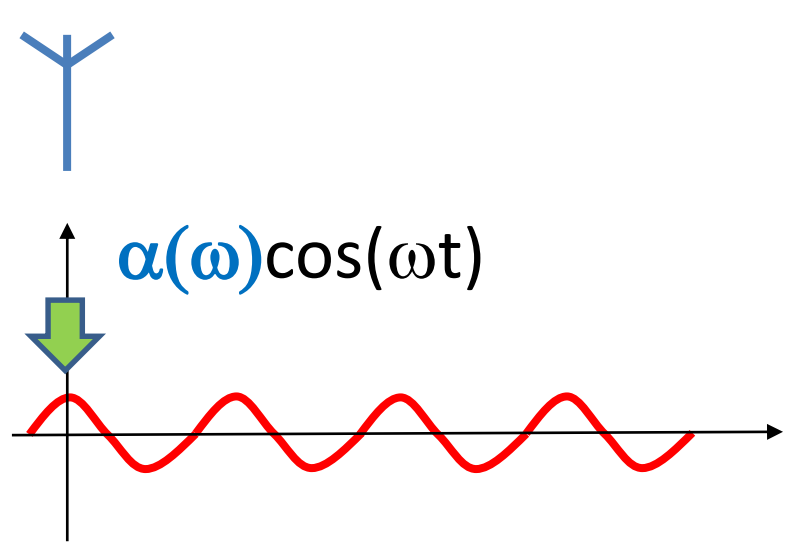
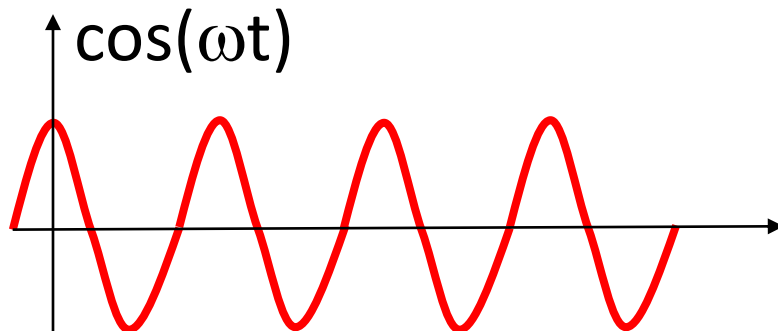
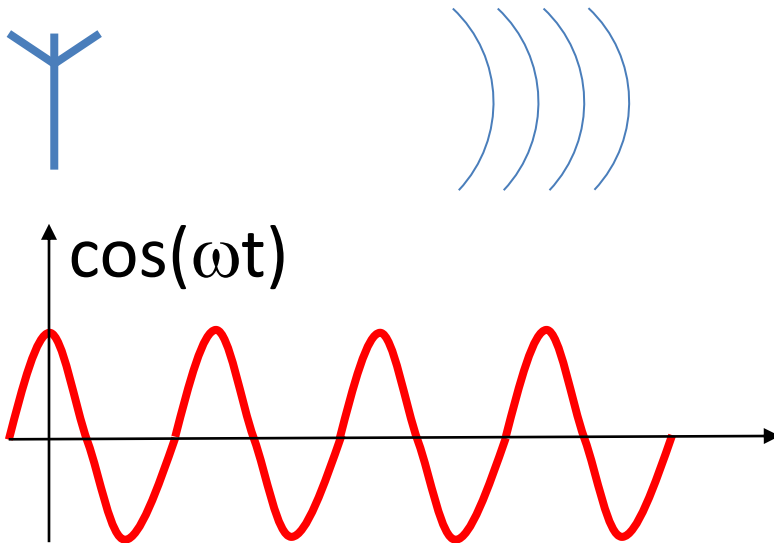
Daniel Halperin*, Wenjun Hu*, Anmol Sheth[†], and David Wetherall*[†]

University of Washington* and Intel Labs Seattle[†]

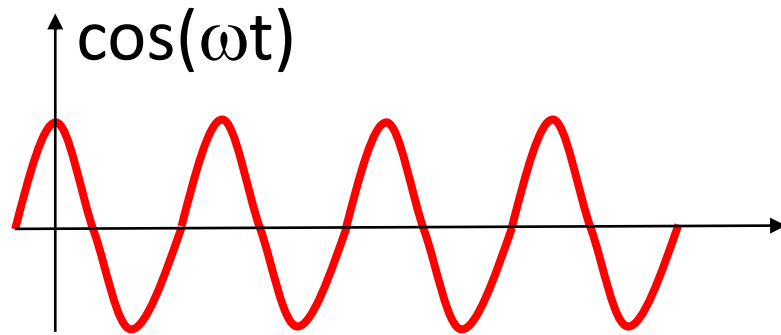
Reminder: Transmissions channels are modeled using **complex numbers** to model **amplitude atenuation** and **phase change** by a single multiplicative complex coefficient



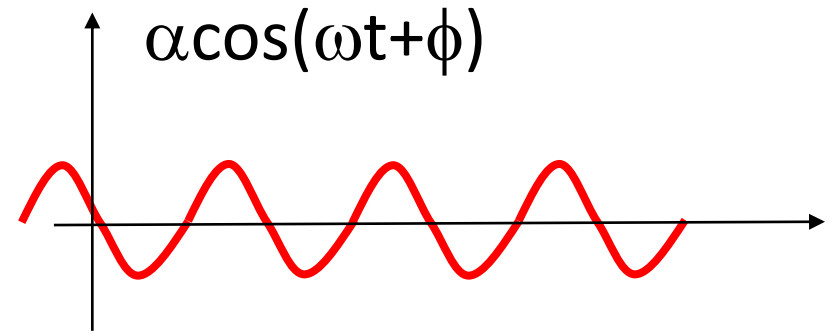
Reminder: Transmissions channels are modeled using **complex numbers** to model **amplitude attenuation** and **phase change** by a single multiplicative complex coefficient



Transmissions channels are modeled using **complex numbers** to model atenuation and phase changes **by a single multiplicative complex coefficient**

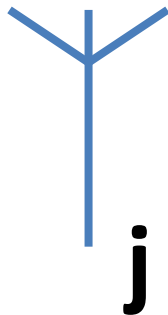


$$s(t) = \cos(\omega t) + i \sin(\omega t) \\ = e^{i\omega t}$$

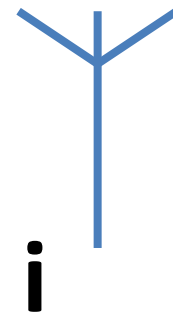


$$r(t) = \alpha \cos(\omega t + \phi) + i \alpha \sin(\omega t + \phi) \\ = \alpha e^{i(\omega t + \phi)} = \alpha e^{i\phi} e^{i\omega t}$$

Transmitting
antenna

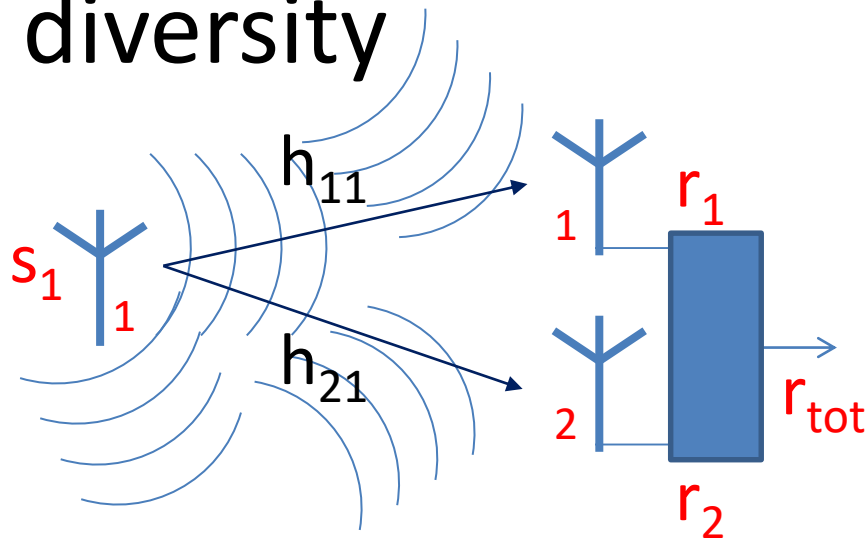


$$h_{ij} = \alpha e^{i\phi}$$



Receiving
antenna

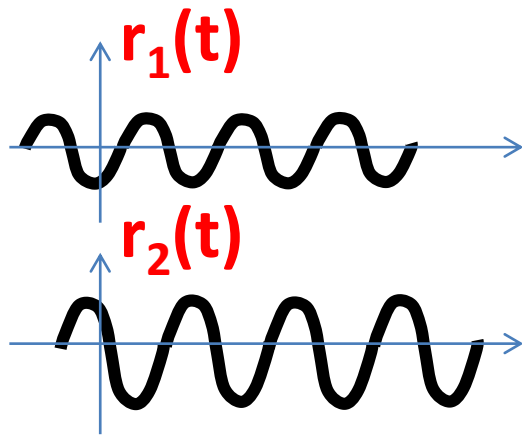
Receiver antenna diversity



$$r_1 = h_{11}s_1 + n_1$$

$$r_2 = h_{21}s_1 + n_2$$

Antenna separation must be larger than half wavelength

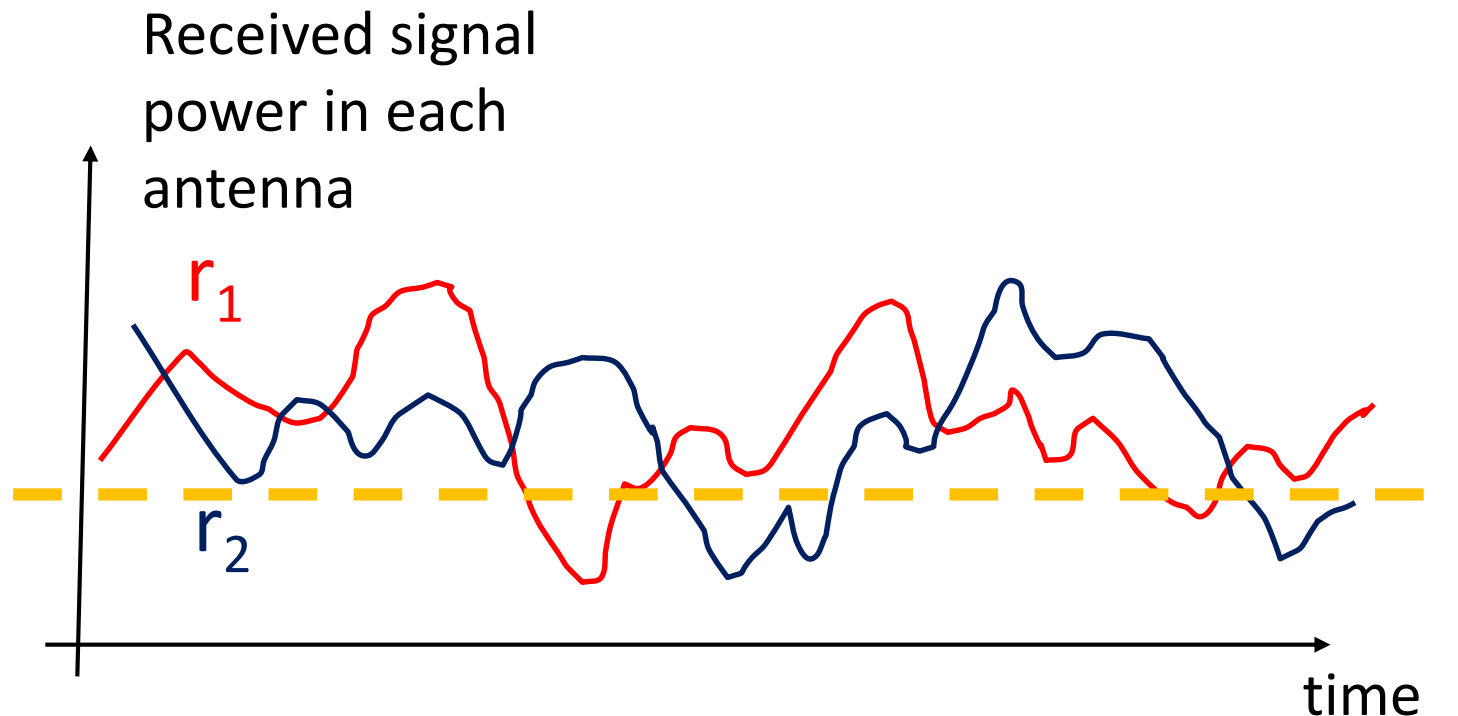


Noise not included in the drawing...

Due to the receiver antenna separation, there are different travel times for the interfering paths. This means that **the signals received in antennas 1 and 2 suffer in general different multipath fading.**

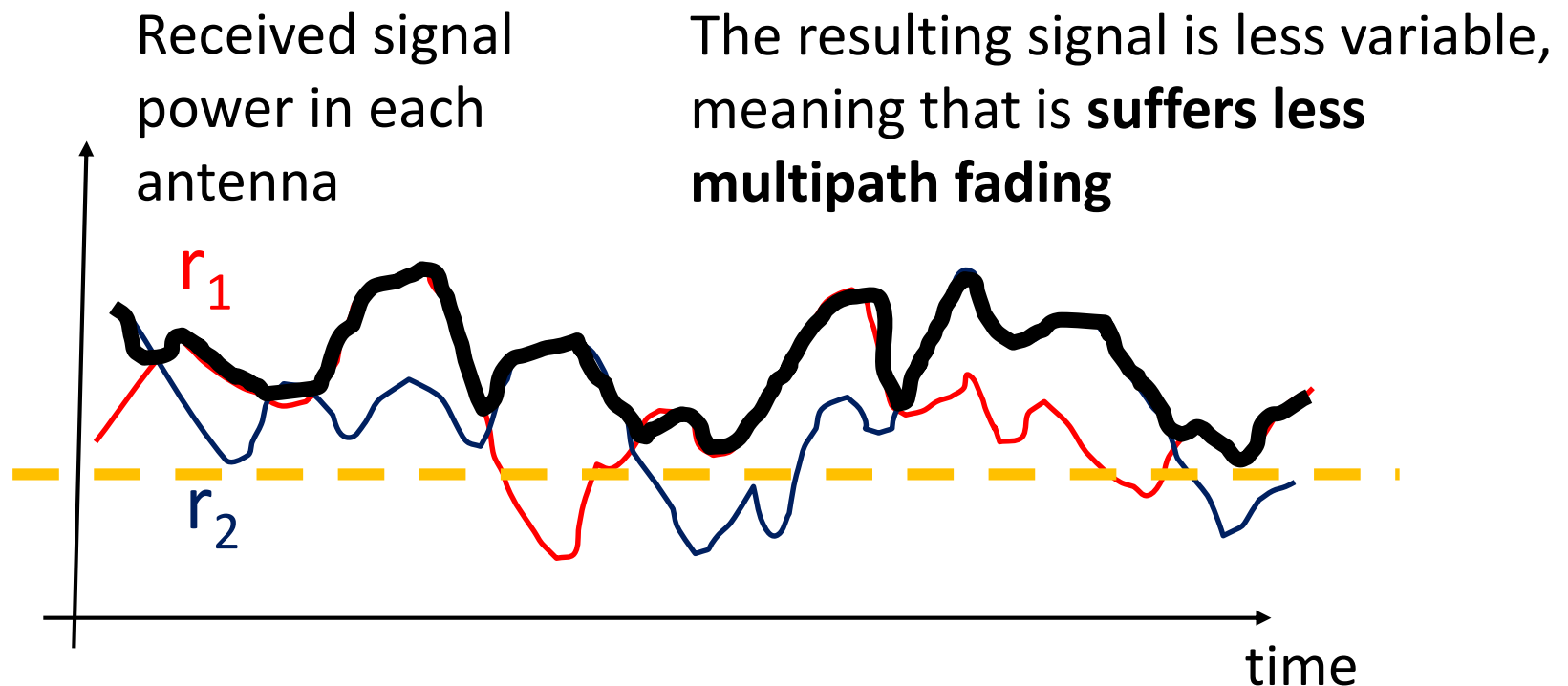
Receiver antenna diversity: Selection combining

Use the antenna with the **strongest signal** (i.e. largest SNR) to receive the packet and ignore the others.

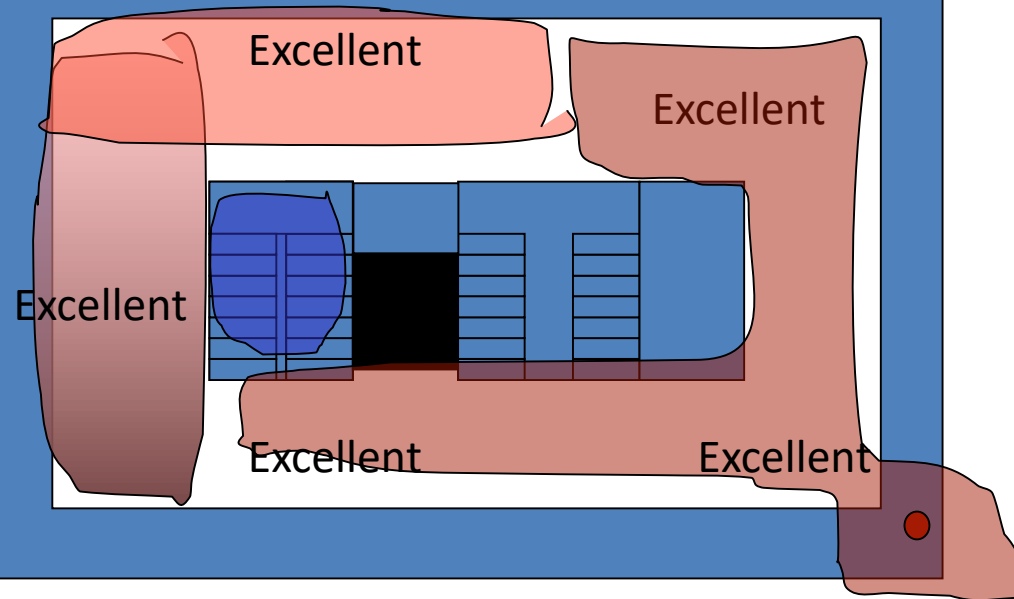


Selection combining: Diversity gain

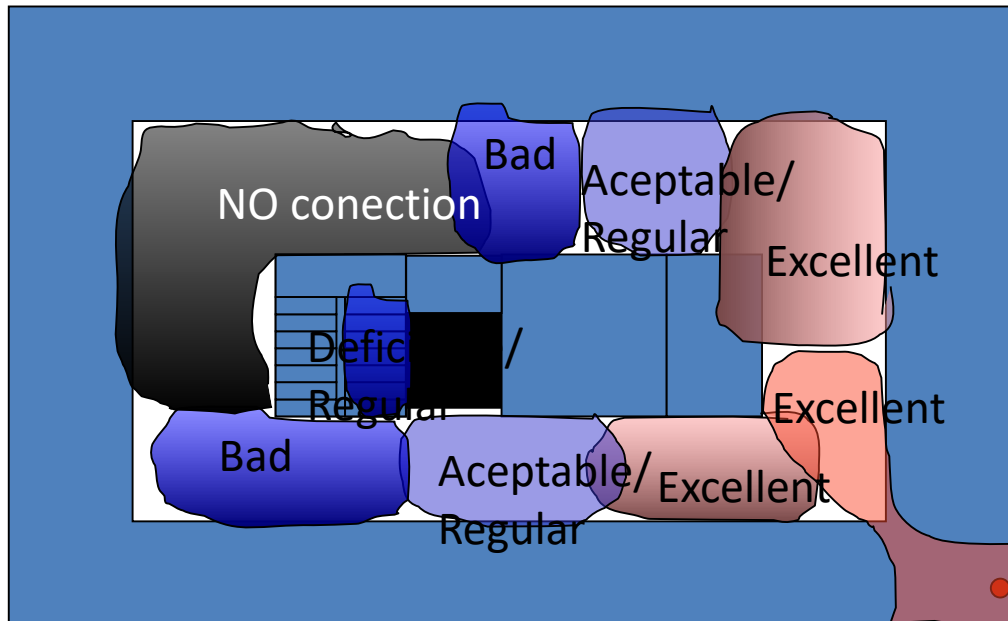
Use the antenna with the **strongest signal** (i.e. largest SNR) to receive the packet and ignore the others.



C6 2nd floor

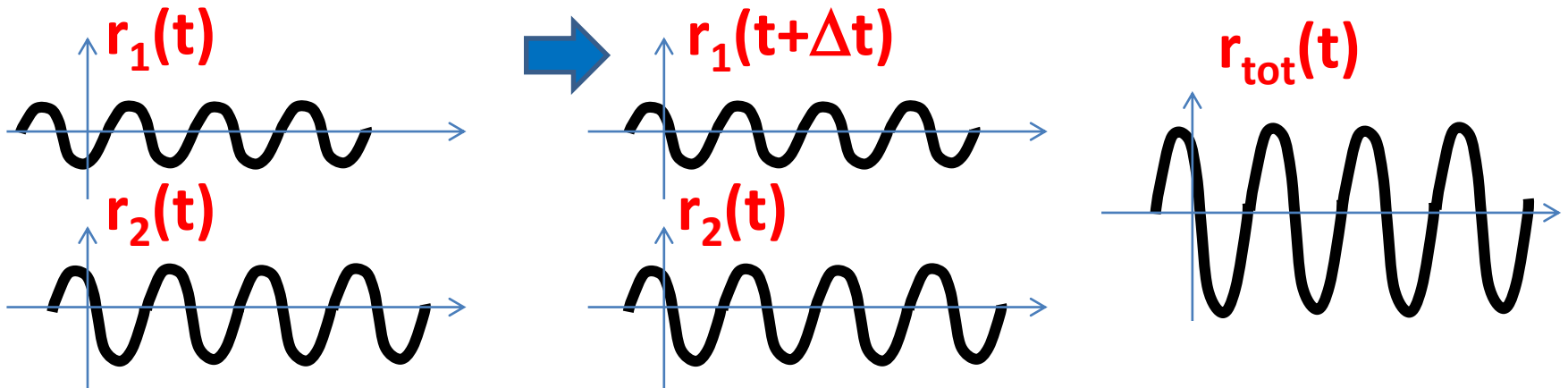


AP Cisco 1200,
2 antennas, diversity
(Selection Combining)
30mW



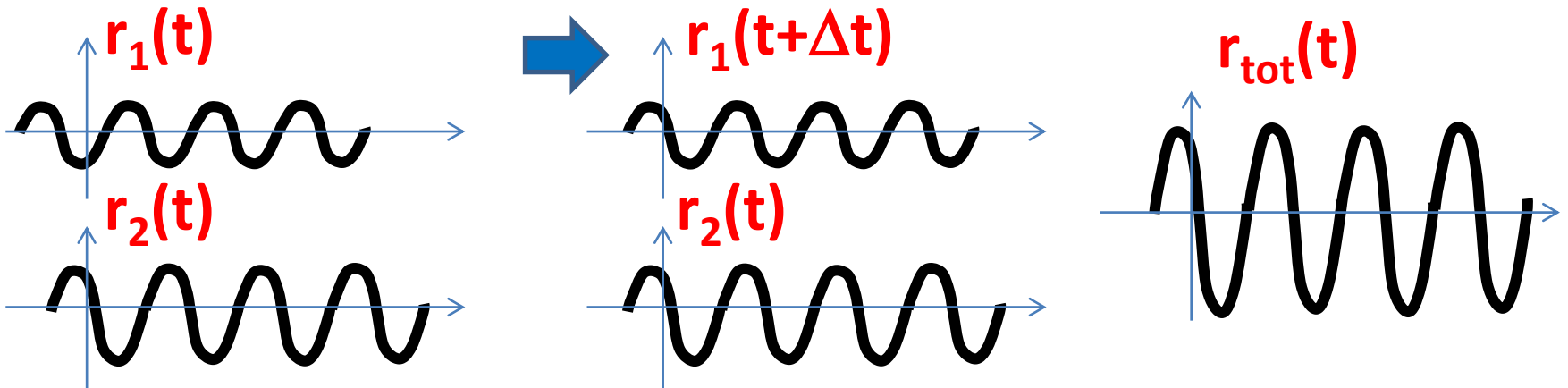
AP Cisco 1200,
1 antenna, (no diversity)
30mW

Receiver antenna diversity: Maximal Ratio Combining (MRC)

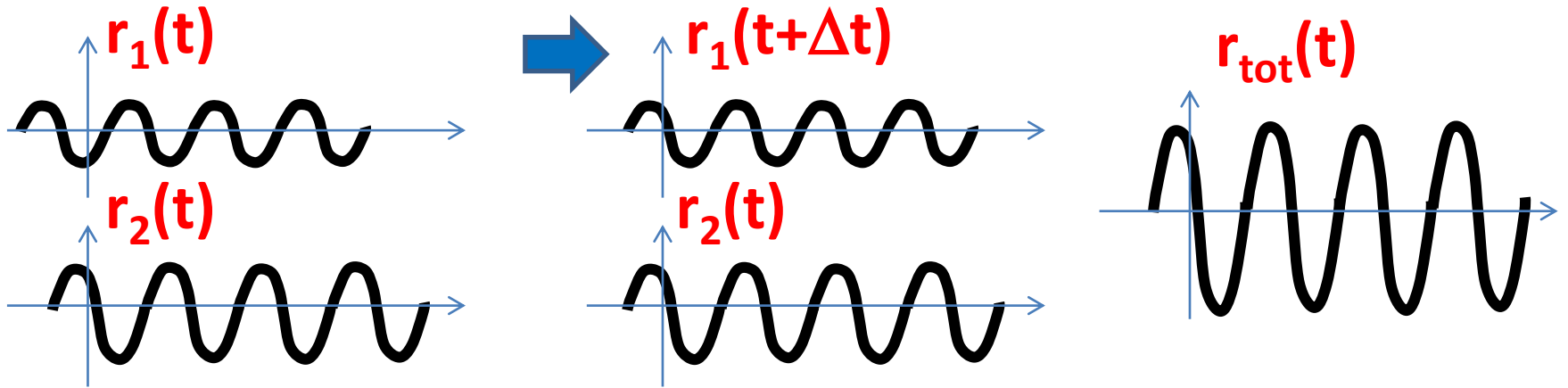


Better option: Add both signals **in phase** (coherently) :
delay one signal so that both signals are in phase before
they are added

MRC: Diversity gain



When we add both signals, we also obtain a less variable signal, meaning that we also have a **diversity gain**



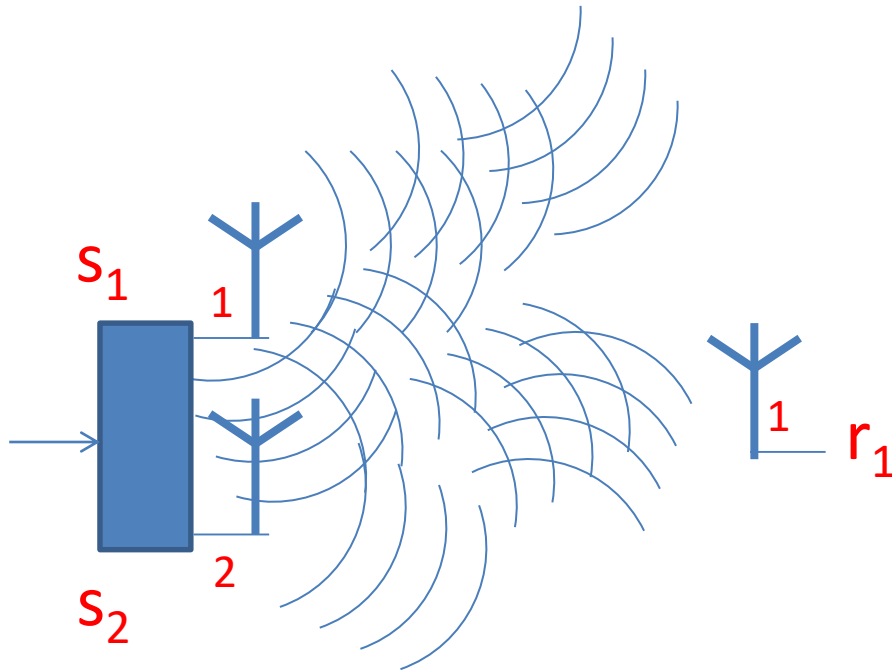
We add both signal **in phase** (coherently) $\Rightarrow$ We add amplitudes. e.g_ when $A_1 = A_2 \Rightarrow A_{\text{tot}} = 2 A_1 \Rightarrow \mathbf{P_{tot} = 4 P_1}$
 For the noise, we have **incoherent sum** (add powers) $\Rightarrow$
 $\mathbf{P_{N_{tot}} = 2P_{N1}}$

$$\mathbf{SNR_{tot} = SNR_1 + SNR_2}$$

Additionally, we can give more weight in the sum to the signal with better SNR

Transmitter antenna diversity

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$

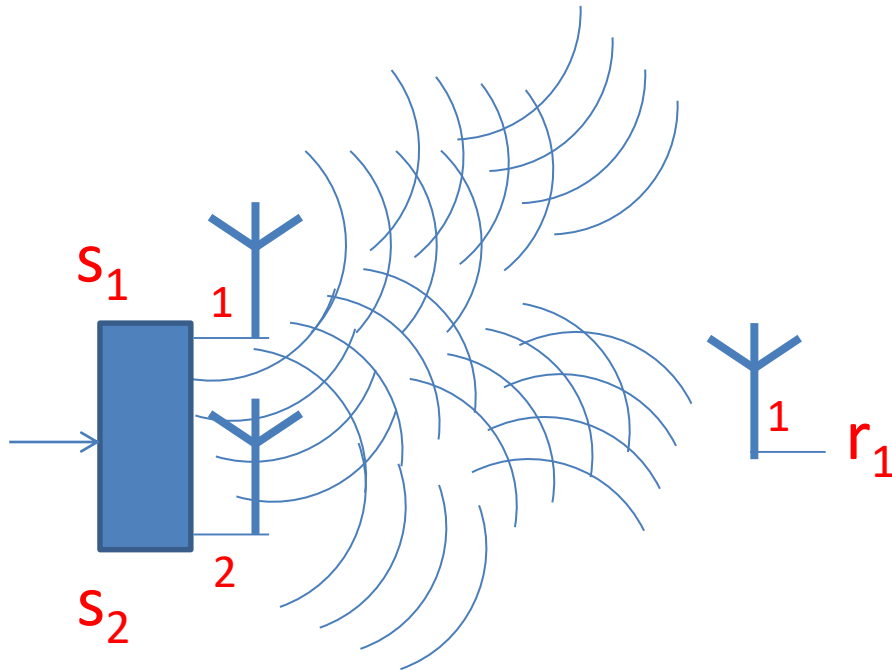


When we have several antennas in the transmitter, we can select the best antenna to transmit the packet (equivalent to Selection Combining)

Noise not included in the drawing...

Transmitter antenna diversity

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$



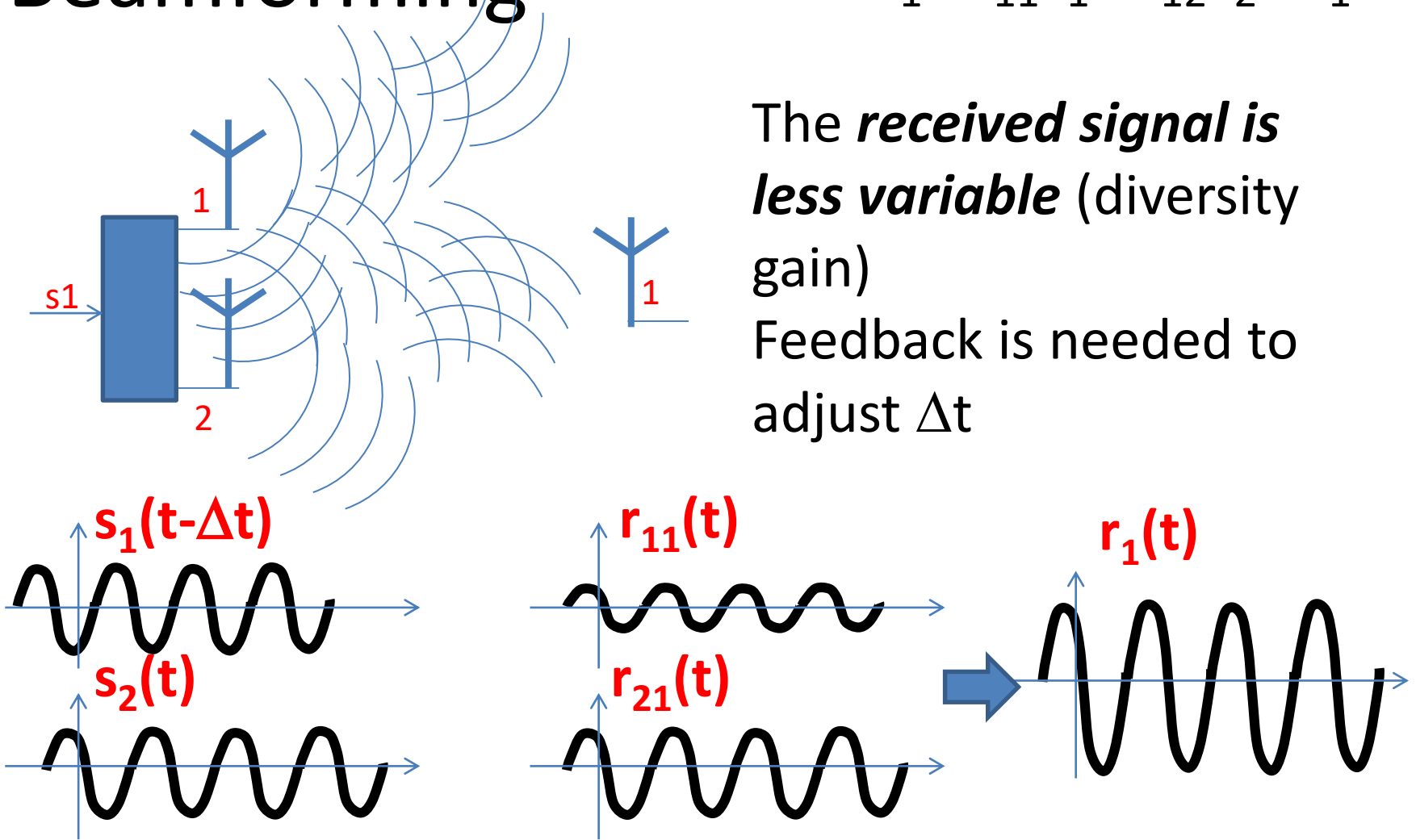
We can also use an equivalent to MRC, called **beamforming**. The transmitter delays one of the signal to change the phase such that they combine constructively at the receiver's antenna,

Noise not included in the drawing...

Beamforming

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$

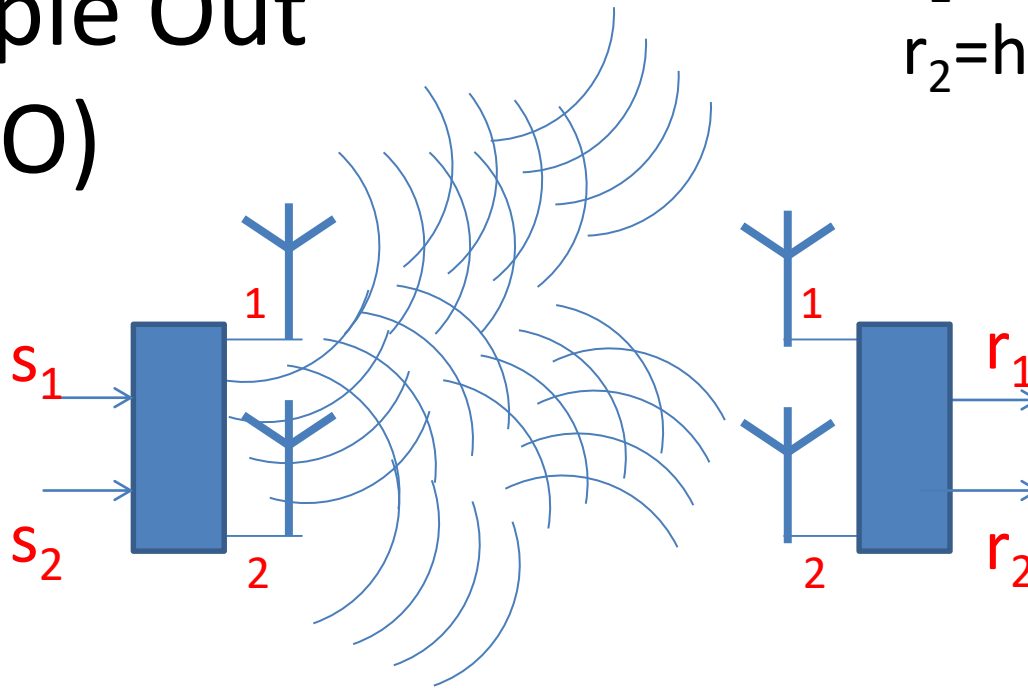
The *received signal is less variable* (diversity gain)
Feedback is needed to adjust Δt



Noise not included in the drawing...

Multiple In Multiple Out (MIMO)

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$
$$r_2 = h_{21}s_1 + h_{22}s_2 + n_2$$



Transmit same symbol in both Tx antennas, meaning that received signal is less variable (**diversity gain**)

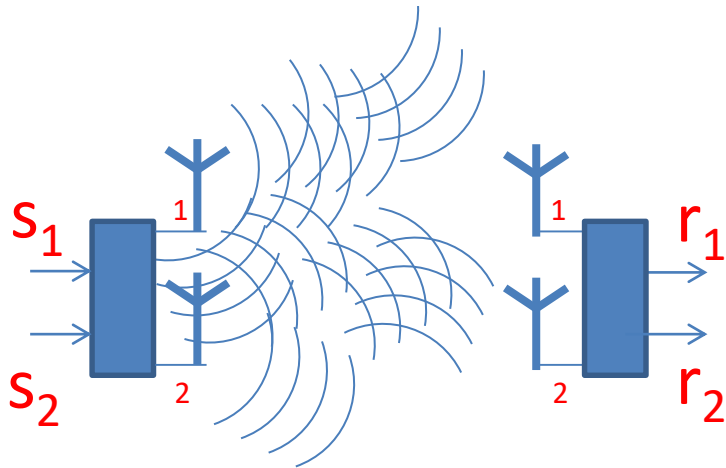
Transmit different symbol through each Tx antenna (**multiplexing gain**)

Direct mapping

MIMO

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$
$$r_2 = h_{21}s_1 + h_{22}s_2 + n_2$$

$$r = Hs + n$$



$$s = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} a_1 \cos(\omega t) \\ a_2 \cos(\omega t) \end{bmatrix}$$

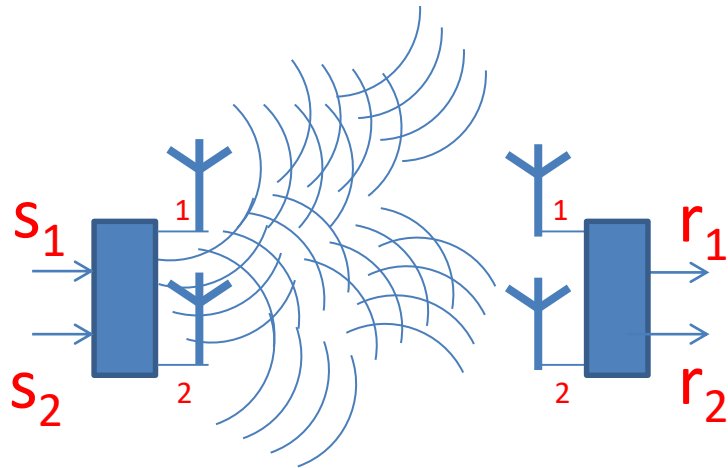
$$r = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$H = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

In general the elements of H
are complex numbers



Direct mapping MIMO



$$s = \begin{bmatrix} (+1)\cos(wt) \\ (-1)\cos(wt) \end{bmatrix}$$

$$H = \begin{bmatrix} 0.1 & 0.2 \\ -0.3 & 0.1 \end{bmatrix}$$

$$r = \begin{bmatrix} (-0.1)\cos(wt) \\ (-0.4)\cos(wt) \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 1.429 & -2.857 \\ 4.286 & 1.429 \end{bmatrix}$$

$$\text{Estimation of } s = \begin{bmatrix} 0.9999 \\ -1.0002 \end{bmatrix}$$

Here we are not taking noise into account

Direct mapping MIMO

$$s = \begin{bmatrix} (+1)\cos(wt) \\ (-1)\cos(wt) \end{bmatrix}$$

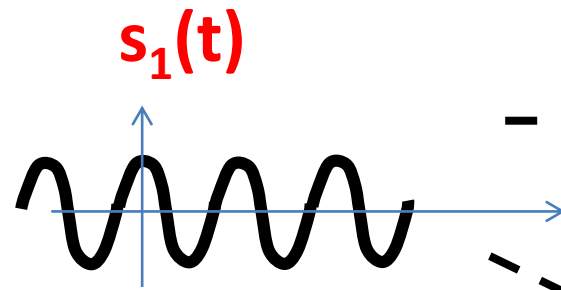
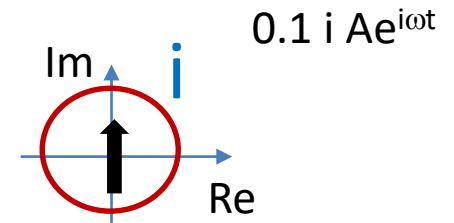
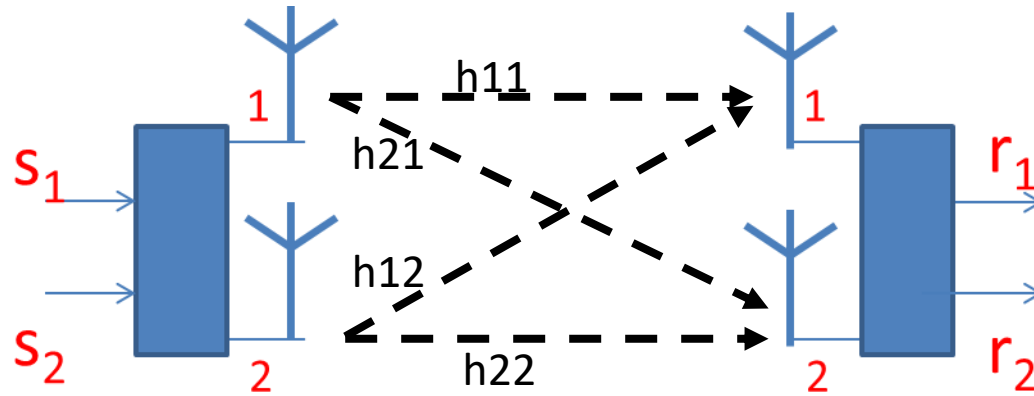
$$H = \begin{bmatrix} 0.1 & 0.2 \\ -0.3 & 0.1 \end{bmatrix}$$

$$r = \begin{bmatrix} (-0.1)\cos(wt) + (-0.01)\cos(wt) \\ (-0.4)\cos(wt) + (0.02)\cos(wt) \end{bmatrix}$$

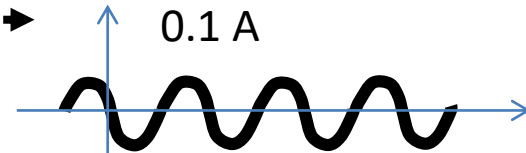
$$H^{-1} = \begin{bmatrix} 1.429 & -2.857 \\ 4.286 & 1.429 \end{bmatrix}$$

$$\text{Estimation of } s = \begin{bmatrix} 0.928 \\ -1.014 \end{bmatrix}$$

Taking noise into account

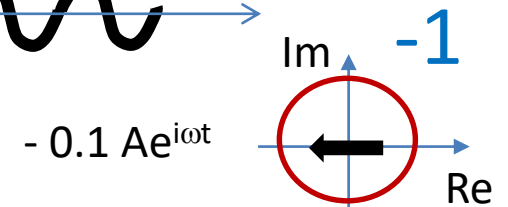
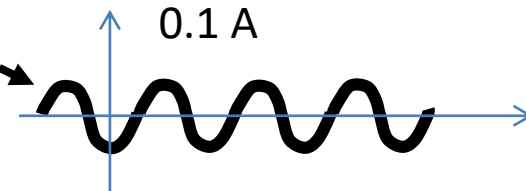


$$h_{11} = 0.1 i$$

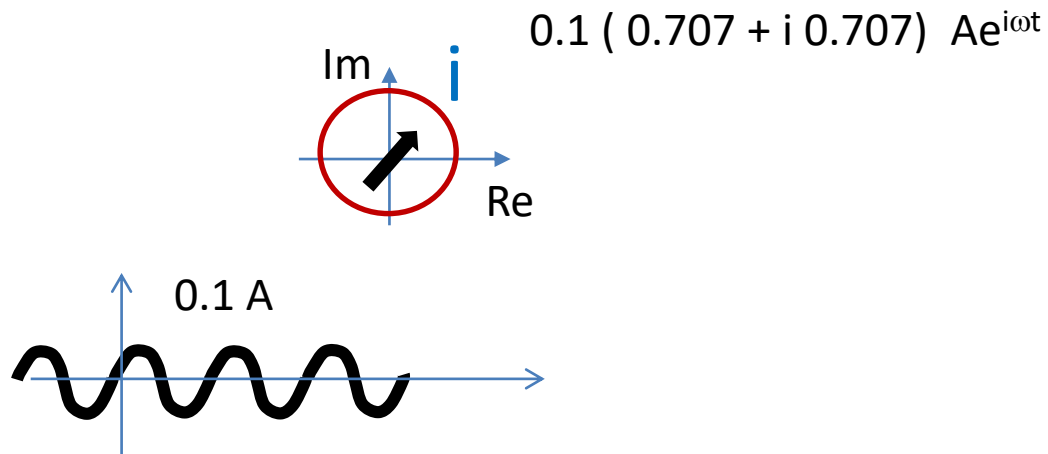


$$\text{Re}(0.1 A i e^{i\omega t}) = 0.1 A \text{Re}[(-\text{sen}(\omega t) + i \cos(\omega t))] = -0.1 A \text{sen}(\omega t)$$

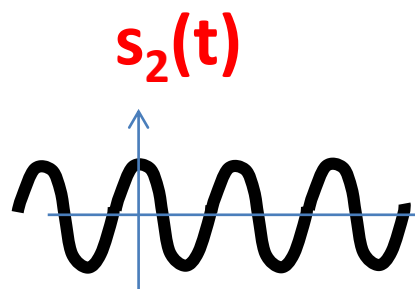
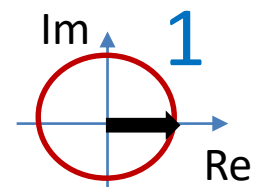
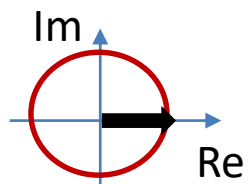
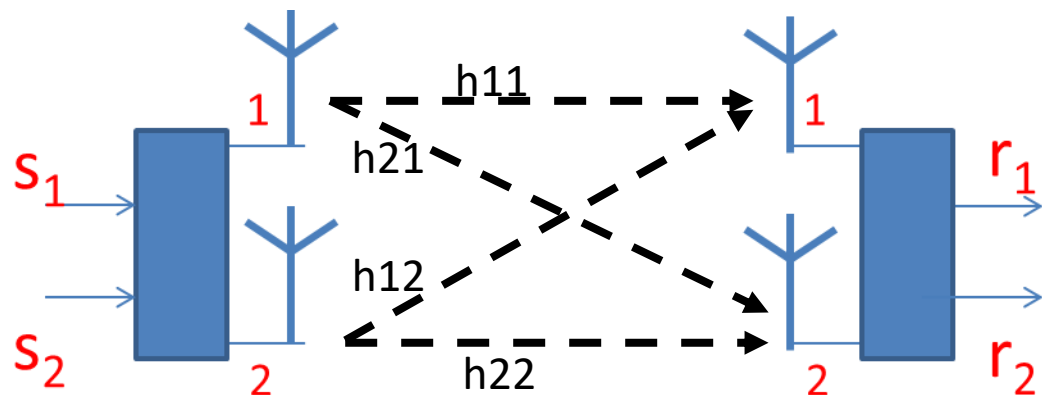
$$h_{21} = -0.1$$



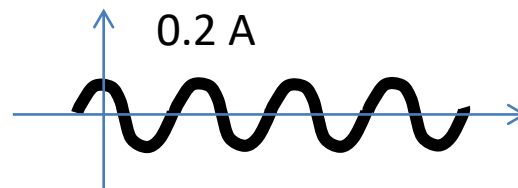
$$S_1 = \text{Re}(A e^{i\omega t}) = A \text{Re}[\cos(\omega t) + i \text{sen}(\omega t)] = A \cos(\omega t)$$



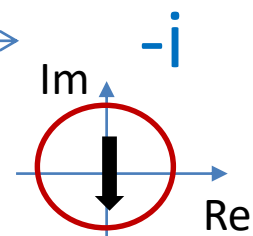
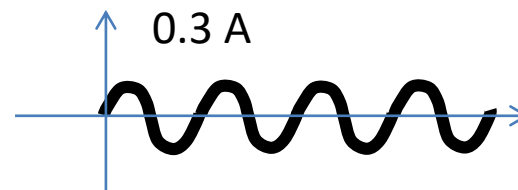
$$\begin{aligned}
 &\text{Re}(0.1 A (0.707 + i 0.707)(\cos + i \sin))= \\
 &0.1A \text{Re}(0.707 \cos + 0.707 i \sin + i 0.707 \cos - 0.707 \sin)= \\
 &0.1 A (0.707\cos-0.707\sin)
 \end{aligned}$$



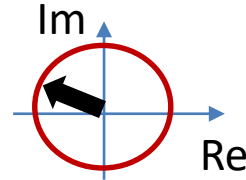
$$h_{12} = 0.2$$



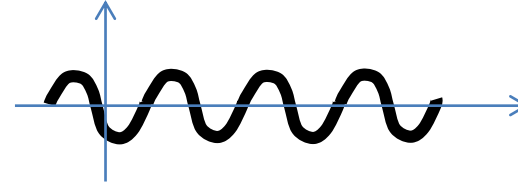
$$h_{22} = -0.3 i$$



$$(-0.2 + 0.1 i) A e^{i\omega t}$$



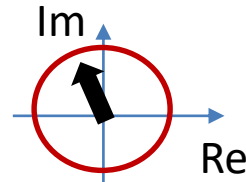
$$0.1 \sqrt{5} A$$



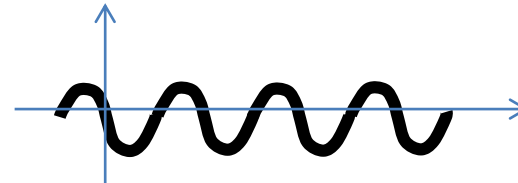
$$\mathbf{r1} = -0.2 + 0.1i$$

$$\mathbf{r2} = -0.1 + 0.3i$$

$$(-0.1 - 0.3 i) A e^{i\omega t}$$



$$0.1 \sqrt{10} A$$



$$\begin{cases} -0.2 + 0.1i = 0.1i s_1 + 0.2 s_2 \\ -0.1 + 0.3i = -0.1 s_1 - 0.3i s_2 \end{cases} \quad H = \begin{bmatrix} 0.1i & 0.2 \\ 0.1 & -0.3i \end{bmatrix}$$

$$\begin{cases} -0.2 + 0.1i = 0.1i s_1 + 0.2 s_2 \\ -0.1i - 0.3 = -0.1i s_1 + 0.3 s_2 \Rightarrow -0.5 = 0.5 s_2 \end{cases}$$

Due to noise we cannot solve exactly the linear system of equations => Use SVD

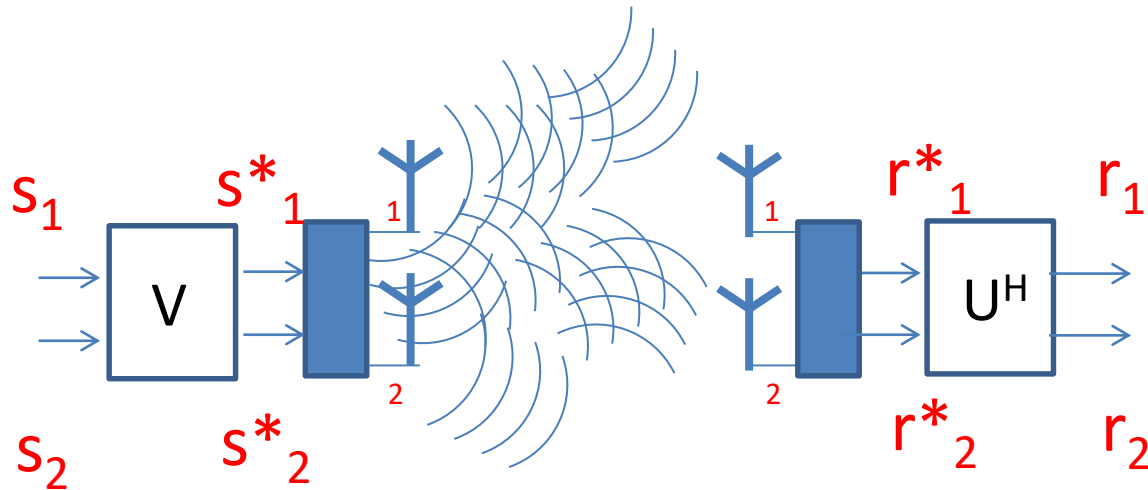
Assume that Tx and Rx know the channel (H) => Singular Value Decomposition

SVD: $H = U \Sigma V^H$

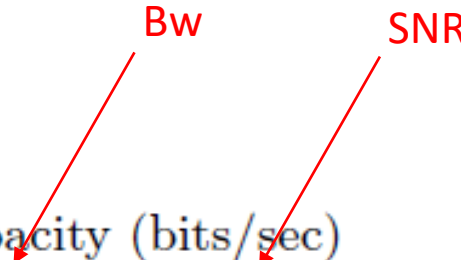
$$s^* = V s \quad \text{"Transmit precoding"}$$

$$r = U^H (r^* + n) \quad \text{"Receiver shaping"}$$

$$r = \Sigma s + n' \Rightarrow r_i = \sigma_i s_i + n'_i$$



MIMO Capacity increase, ideal case



Method	Capacity (bits/sec)
SISO	$B \log_2 (1 + \rho)$
Diversity (1xN or Nx1)	$B \log_2 (1 + \rho N)$
Diversity (NxN)	$B \log_2 (1 + \rho N^2)$
Multiplexing	$BN \log_2 (1 + \rho)$

} Improve SNR

Table 1: Capacity for wireless links in an ideal channel. Diversity and multiplexing respectively improve SNR and capacity. For an $N \times N$ link, diversity is better to use at low SNR, multiplexing at high SNR. These best-case gains will vary in real channels.

MIMO Capacity increase

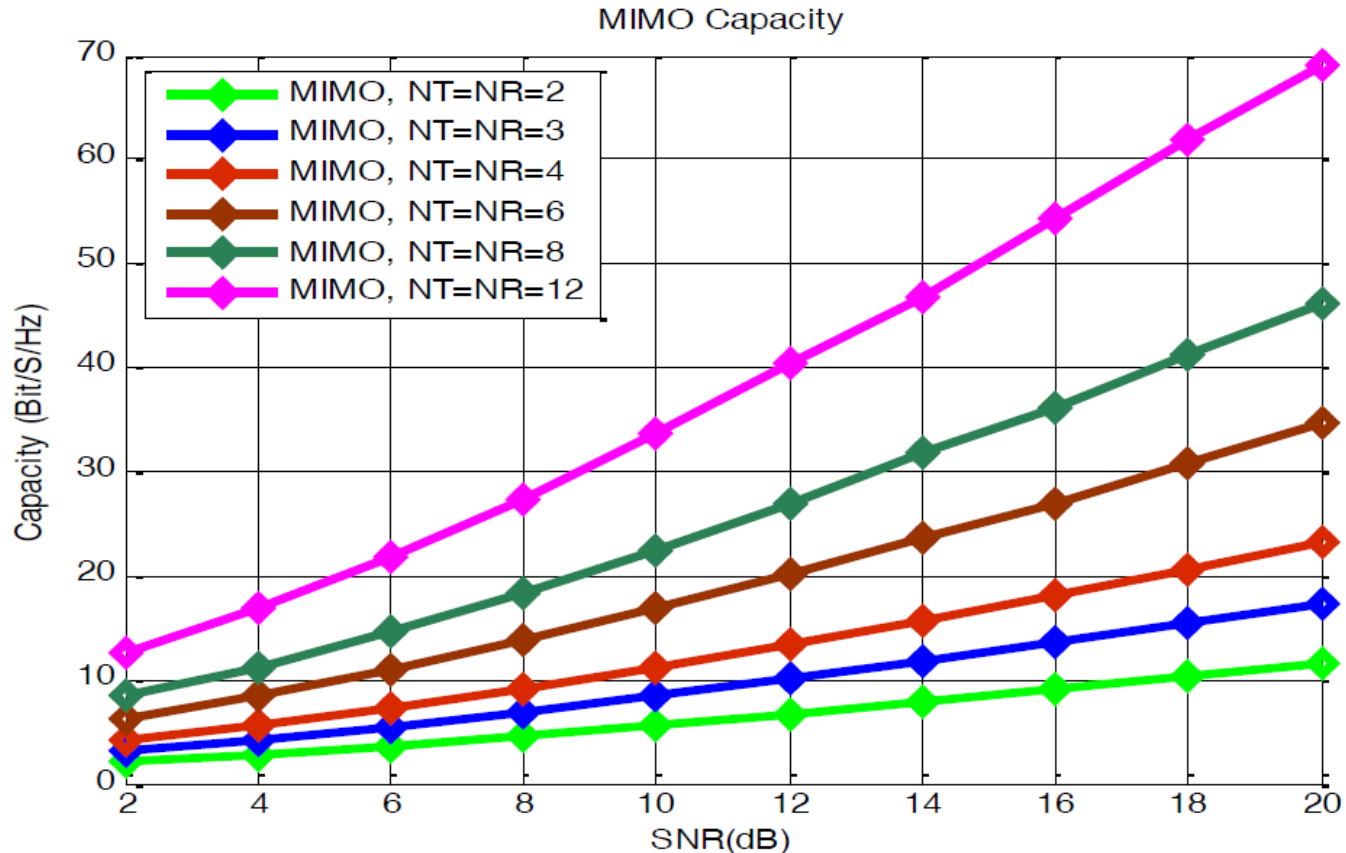
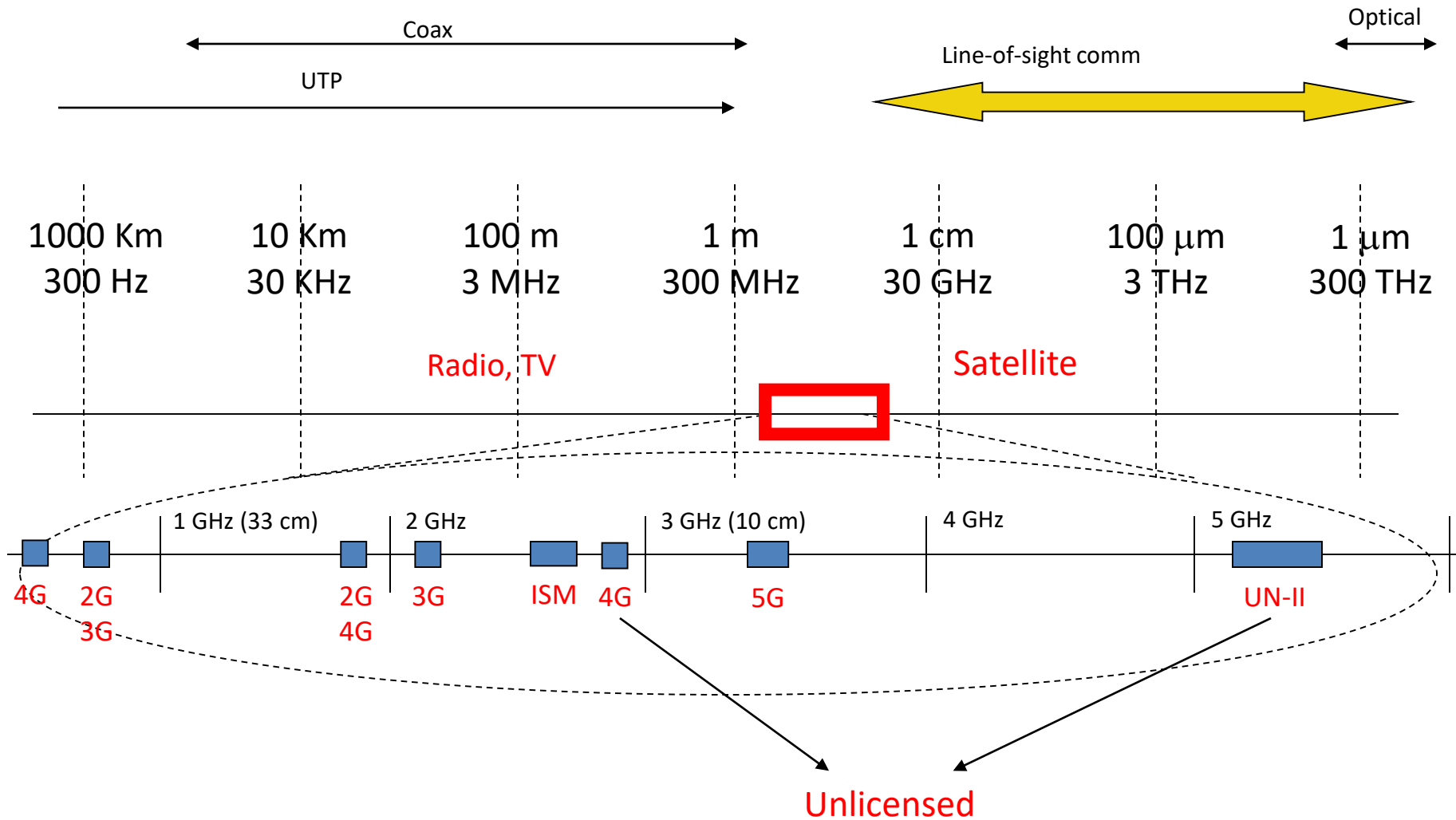


Fig.5. Capacity vs number OF MIMO antennas.

Frequency Bands



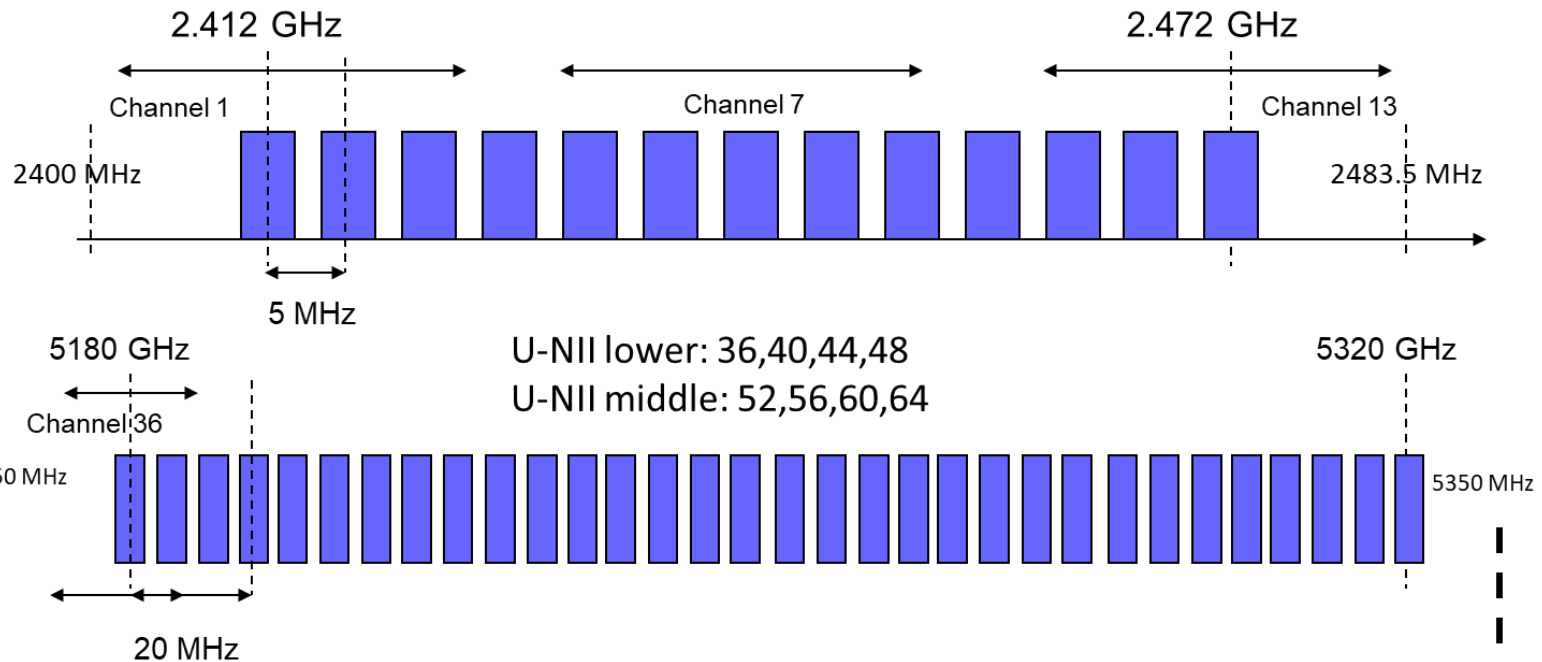
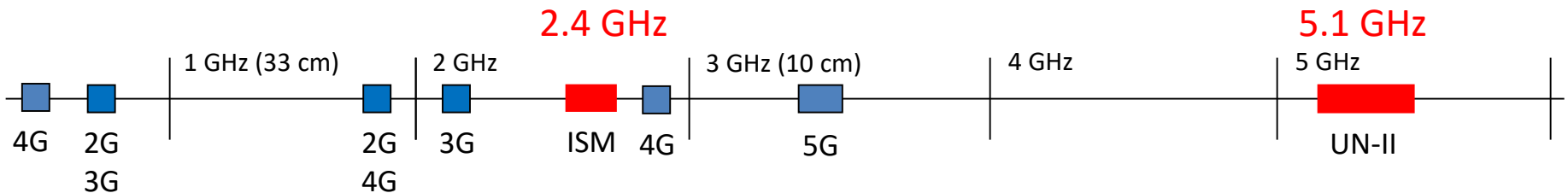
Legislación española (Ley General Telecom., '03)

- El espectro radioeléctrico es un *bien de dominio público*, cuya titularidad, gestión, planificación, administración y control corresponden al Estado.
- El Gobierno reglamentará:
 - Niveles de emisión radioeléctrica tolerables
 - Cuadro Nacional de Atribución de Frecuencias (CNAF)
 - Procedimientos de otorgamiento de derechos de uso del dominio público radioeléctrico
 - Habilitación para el ejercicio de los derechos de uso
 - Adecuada utilización del espectro radioeléctrico

Legislación española (Ley General Telecom., '03)

- Licencias y asignación de espectro:
 - ITU -> CEPT y ETSI -> Ministerio Ind Tur & Com- > SETSI - > AER,CMT
 - Cuadro Nacional de Atribución de Frecuencias (CNAF) español
 - NOTA: La CMT NO regula el espectro, sino los servicios y la calidad con la que deben ser provistos
- Bandas de frecuencia que no requieren licencias
 - dispositivos domésticos: hornos microondas, mandos a distancia, Bluetooth, Cordless phones, etc.

Unlicensed Bands



ETSI/U-NII: 5470-5725: 11 channels with Bw= 20 MHz
(NO included in 802.11a standard)

Sub-GHz bands

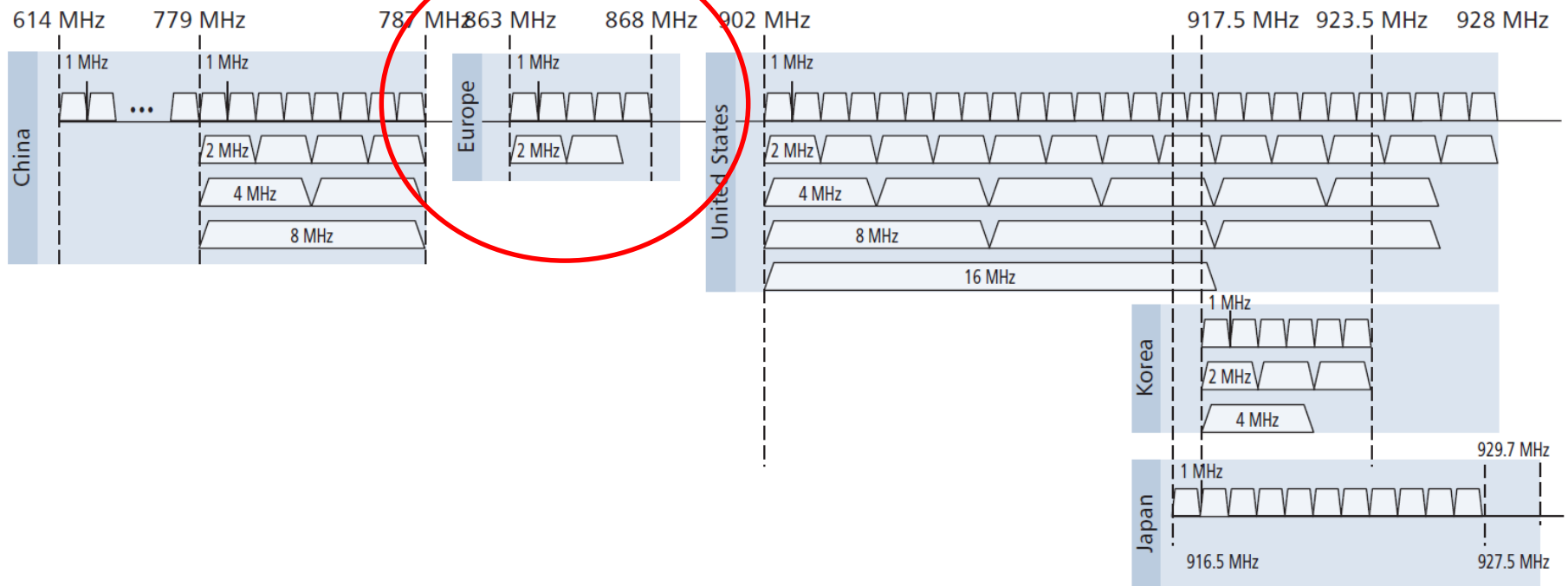


Figure 1. Frequency spectrum availabilities in sub-1-GHz and channelization in key geographies.

Licencias 3G/4G/5G en España

Digital
dividend



Refarming



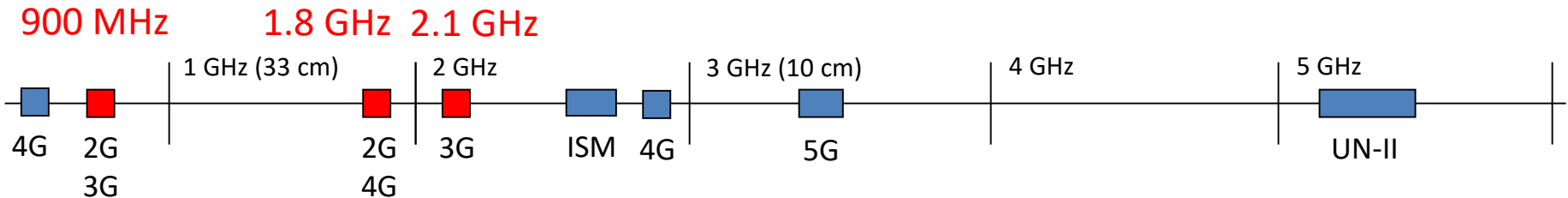
Capacity



	800 MHz 4G banda 20	900 MHz 2G/3G banda 8	1800 MHz 2G/4G banda 3	2100 MHz 3G banda 1	2600 MHz 4G banda 7	3500 MHz 5G banda 42
Movistar	10 MHz	14,8 MHz	20 MHz	15 MHz FDD 5 MHz TDD	20 MHz	90 MHz
Vodafone	10 MHz	10 MHz	20 MHz	15 MHz FDD 5 MHz TDD	20 MHz FDD 20 MHz TDD	90 MHz
Orange	10 MHz	10 MHz	20 MHz	15 MHz FDD 5 MHz TDD	20 MHz	100 MHz
MásMóvil			14,8 MHz	15 MHz FDD 5 MHz TDD	10 MHz TDD (autonómicos)	80 MHz

Source: <https://www.adslzone.net/bandas-5G-4G-3G-2G-frecuencias-telefonía-movil>

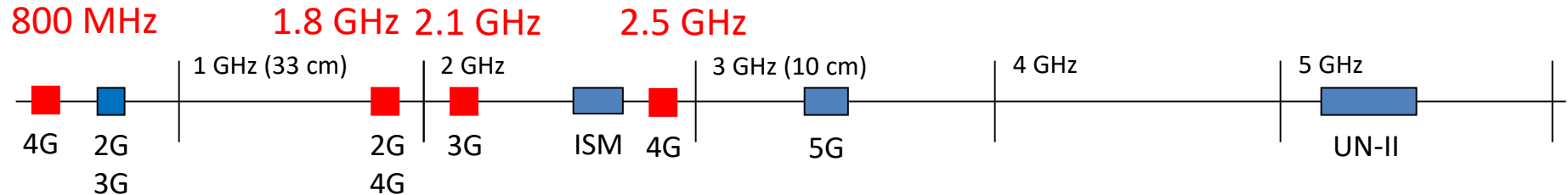
2G/3G Bands



	Año	Mecanismo asignación	Ingresos (aprox.)	Número de licencias licitadas
Alemania	2000	subasta	50000 M€	6
Austria	2000	subasta	671 M€	6
Bélgica	2001	subasta	456 M€	3
Dinamarca	2001	subasta	516 M€	4
España	2000	concurso	456 M€	4
Finlandia	1999	concurso		4
Francia	2001	concurso	1200 M€	3
Grecia	2001	subasta	441 M€	3
Holanda	2000	subasta	3000 M€	5
Irlanda	2002	concurso	175 M€	3
Italia	2000	subasta	12000 M€	5
Luxemburgo	2002	concurso	9 M€	3
Noruega	2000	concurso	36 M€	4
Portugal	2000	concurso	400 M€	4
Reino Unido	2000	subasta	37000 M€	5
Suecia	2000	concurso	12000 M€	4
Suiza	2000	subasta	128 M€	4

Figura 6: Asignaciones primarias del espectro para redes 3G en Europa (Fuente: Gretel 2007³¹)

4G Bands



3GPP New Spectrum lining up with LTE availability

2.5-2.6 GHz

Potential for up to 20MHz per operator

Capacity Band

Digital Dividend (800MHz)

Potential for up to 10MHz per operator

Coverage Band

Refarming 1800MHz & 2100MHz

5-10MHz in early days & up to 15MHz eventually

2.5-2.6 GHz Early Spectrum Auction

Country	Spectrum	Date
Austria	2500 MHz – 2690 MHz	Q2 2009
Germany	2500 MHz – 2690 MHz	Q3 2009
Norway	2500 MHz – 2690 MHz	Completed
Sweden	2500 MHz – 2690 MHz	Completed
United Kingdom	2500 MHz – 2690 MHz	Q3 2009
Hong Kong	2500 MHz – 2690 MHz	Completed

Digital Dividend (800MHz) Spectrum Auction

Country	Spectrum	Date
France	790 MHz – 862 MHz	2009
Germany	790 MHz – 862 MHz	2009 - 2010
Switzerland	790 MHz – 862 MHz	2013
Sweden	790 MHz – 862 MHz	2009 - 2010
United Kingdom	790 MHz – 862 MHz	2009 - 2012

ZDNET: 4G in Europe: How far, how fast and how much is LTE in Germany?

“...Germany's spectrum auction ended on 20 May, 2010 after 27 days and 224 bidding rounds. Up for grabs were 41 frequency blocks in the 800MHz, 1.8GHz, 2GHz and 2.6GHz bands; 2GHz and 2.6GHz were available as both paired and unpaired spectrum. Vodafone secured 12 blocks of spectrum for €1.42bn; Telefonica's O2 paid €1.38bn for 11 blocks; E-Plus bought eight blocks for €283.6m; and Deutsche Telekom's T-Mobile acquired 10 blocks for €1.3bn.

The auction raised almost €4.4bn in total, which was far shy of original estimates. Reports at the time suggested the amount raised was below target because the market was still spooked by the aftermath of the costly 3G auctions — and the onset of the global financial crisis doubtless did not help matters...”

5G in EU

- 700 MHz, 3,5 GHz y 26 GHz
- Banda 3.5GH en España:
 - Xfera Móviles, S.A 3420-3500 MHz;
 - Telefónica de España, S.A.U, (40 MHz) y Telefónica Móviles España, S.A.U., (60 MHz): 3500-3600 MHz
 - Orange Espagne, S.A.U., 3600-3710 MHz;
 - Vodafone España, S.A.: 3710-3800 MHz.
 - Al mismo tiempo introdujo en el mismo proceso la posibilidad de
- Posibilidad de utilizar las bandas de frecuencias de 900 y 1.800 MHz (refarming), hasta entonces de uso exclusivo por GSM.

El 5G obliga a resintonizar la TDT y sustituir las televisiones más antiguas

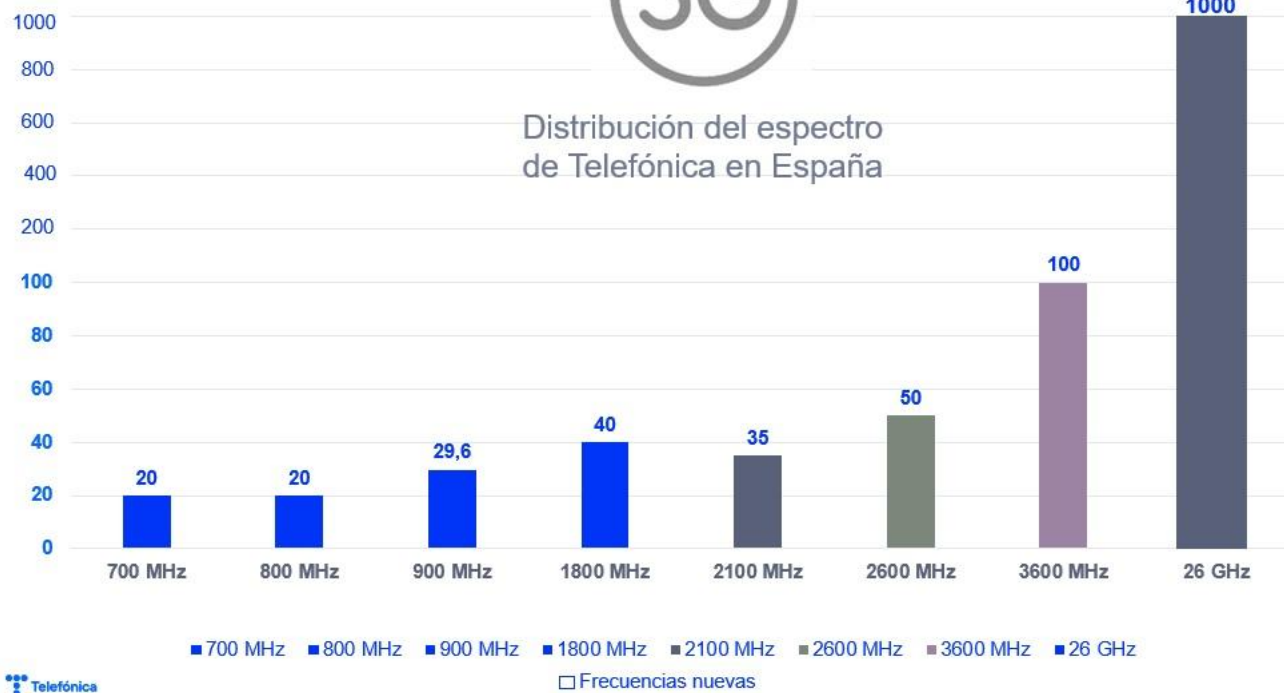
La liberación de la banda de 700 megahercios para la nueva telefonía móvil obliga a reprogramar amplificadores y televisiones. Los receptores que no soporten HD quedarán obsoletos en 2023

RAÚL LIMÓN

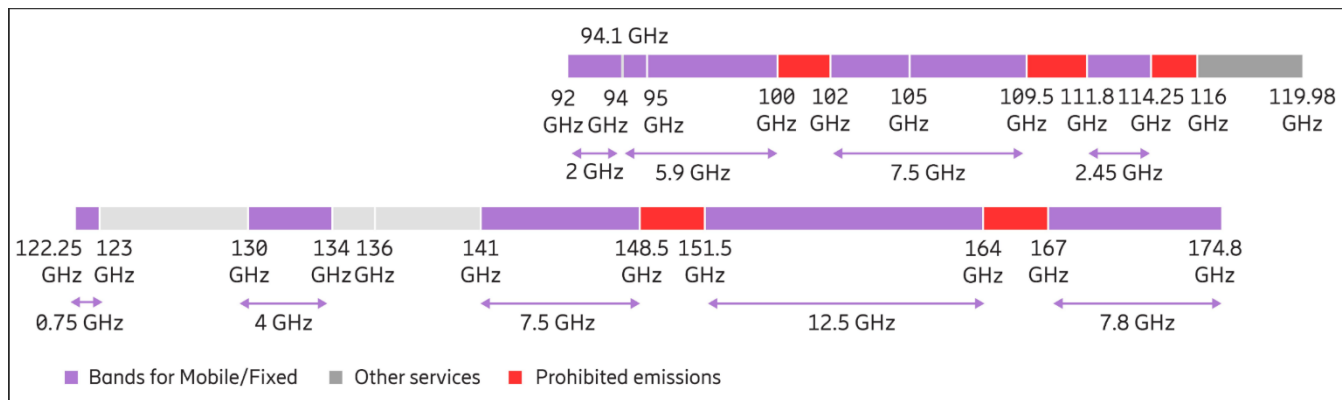
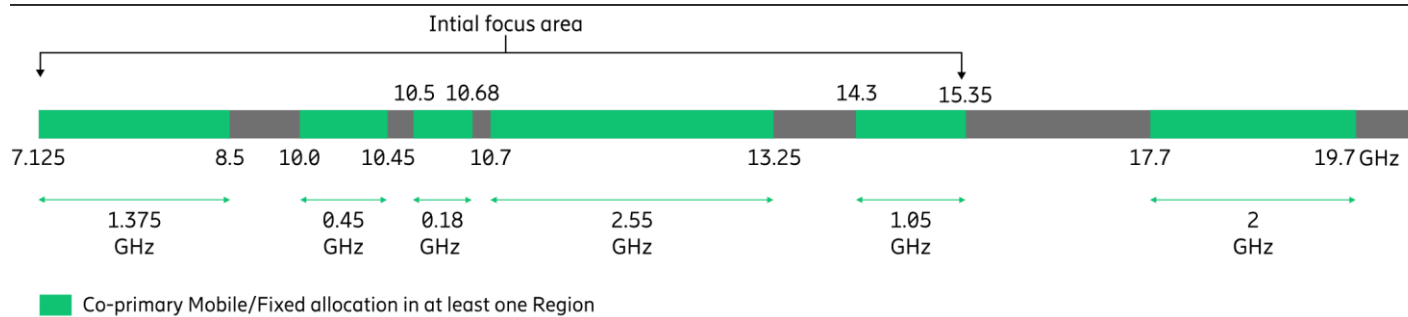
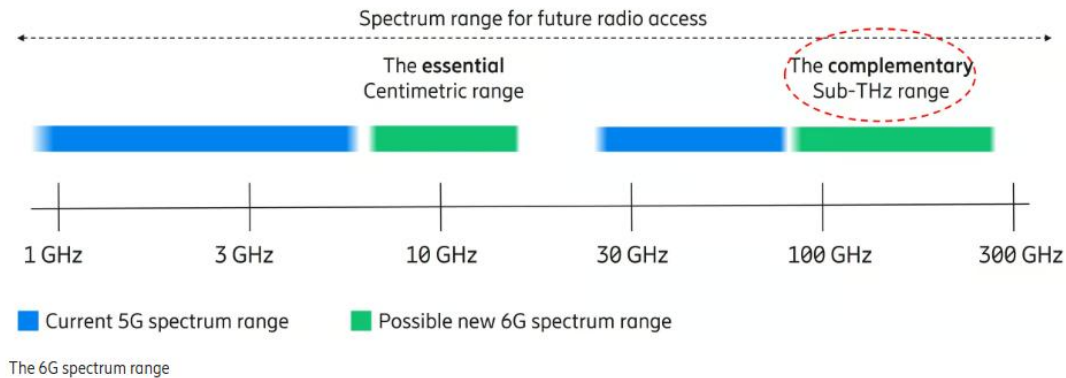
1 JUL 2019 - 13:36 CEST



Distribución del espectro de Telefonía en España

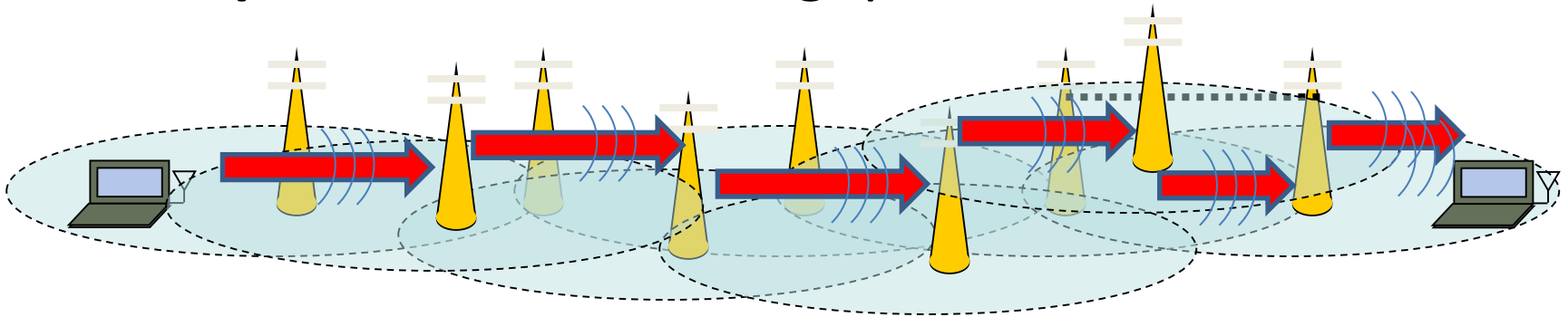


6G (under study)

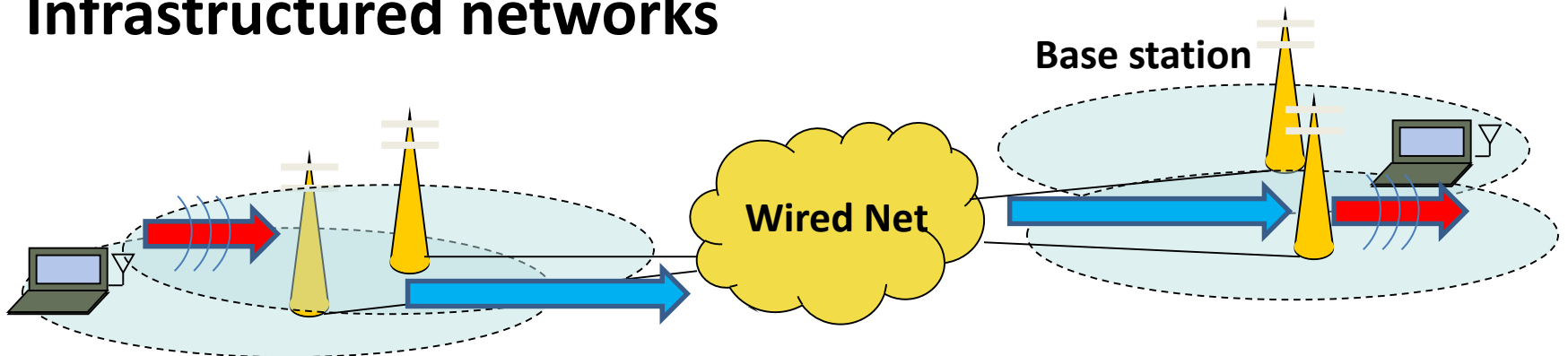


Wireless Nets

Multihop: Small & low throughput nets

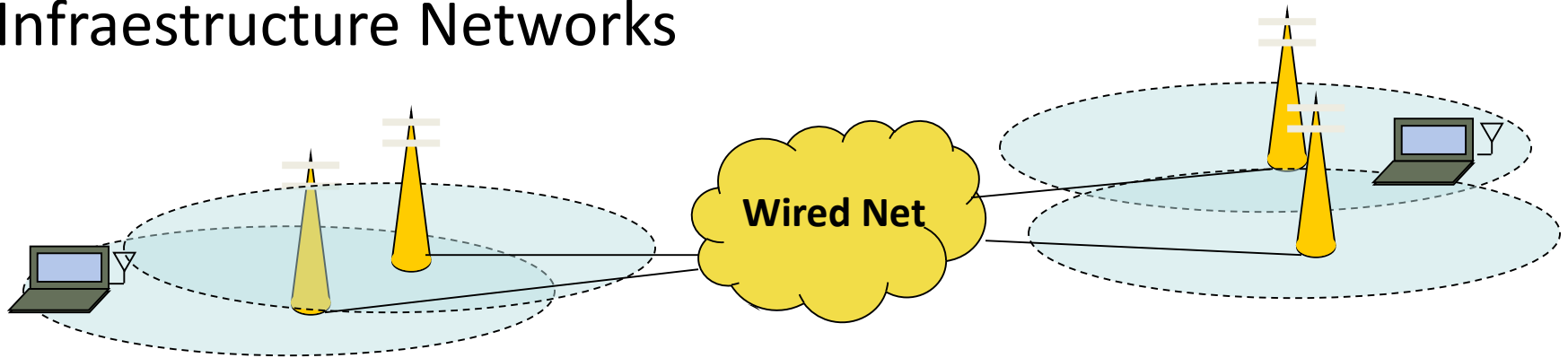


Infrastructured networks



Wireless Nets

Infrastructure Networks



GSM (2G), GPRS (2.5G), UMTS (3G), HSPA (3.5G), LTE (4G)

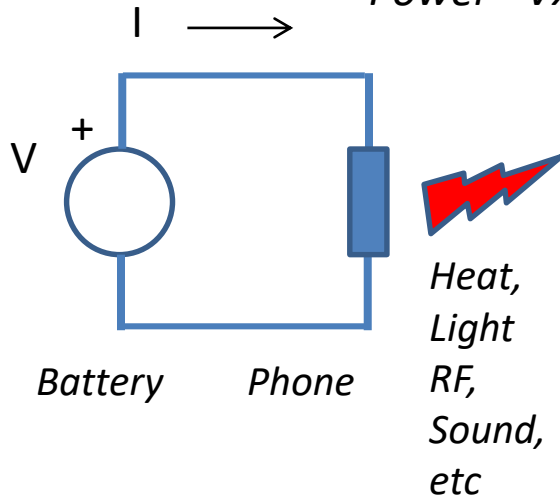
LoRa/LoRaWAN

WiFi Hotspots

Reminder: Energy versus power

- Energy: Joules (J)
- Power: J/s = Watts (W)
- Batteries: mA $\times$ h (Note: It does not give the energy $\Rightarrow \times 4.5$ V)

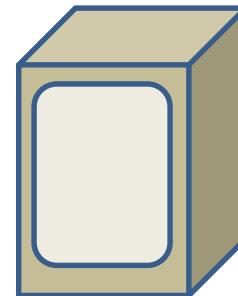
$Power = V \times I$ (Volts $\times$ Amperes = Joules)



$E = 4,5 \text{ V} \times 1200 \text{ mA} \times \text{h} = 4,5 \times 1,2 \times 3600 \text{ J} = 19,4 \text{ KJ}$

If power consumption is 1,5 W (say) $\Rightarrow$

$T = 19400 \text{ J} / 1,5 \text{ W} = 12960 \text{ s} = 3.6 \text{ h}$



Heat dissipation:
 ΔT , surface

Performance versus power

Modern electronic systems:

- low power and
- high performance.

Battery operated portable platforms.

- Extend the battery service life
- Reduce power dissipation

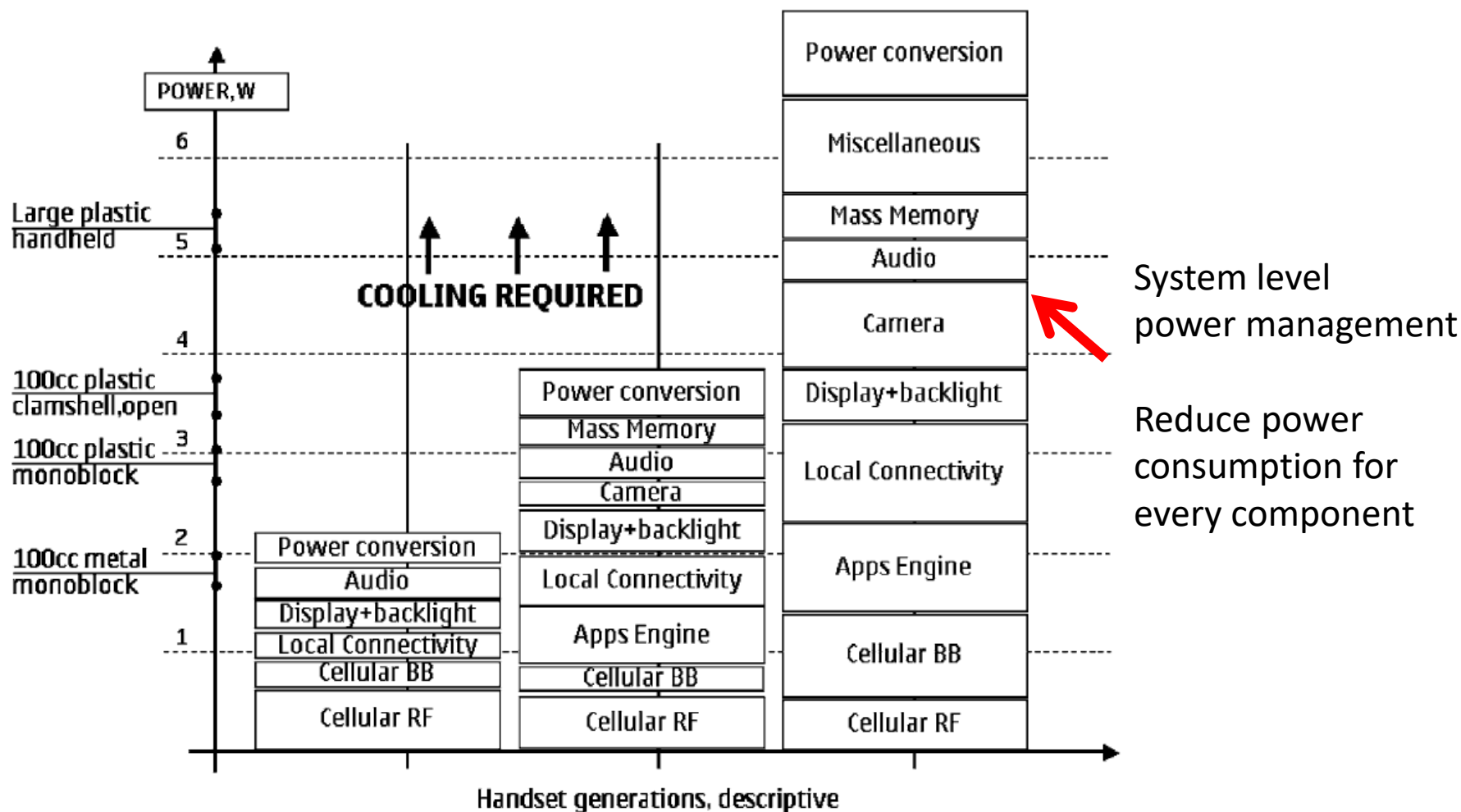


Figure 5. Power consumption and heat dissipation of three different device classes.

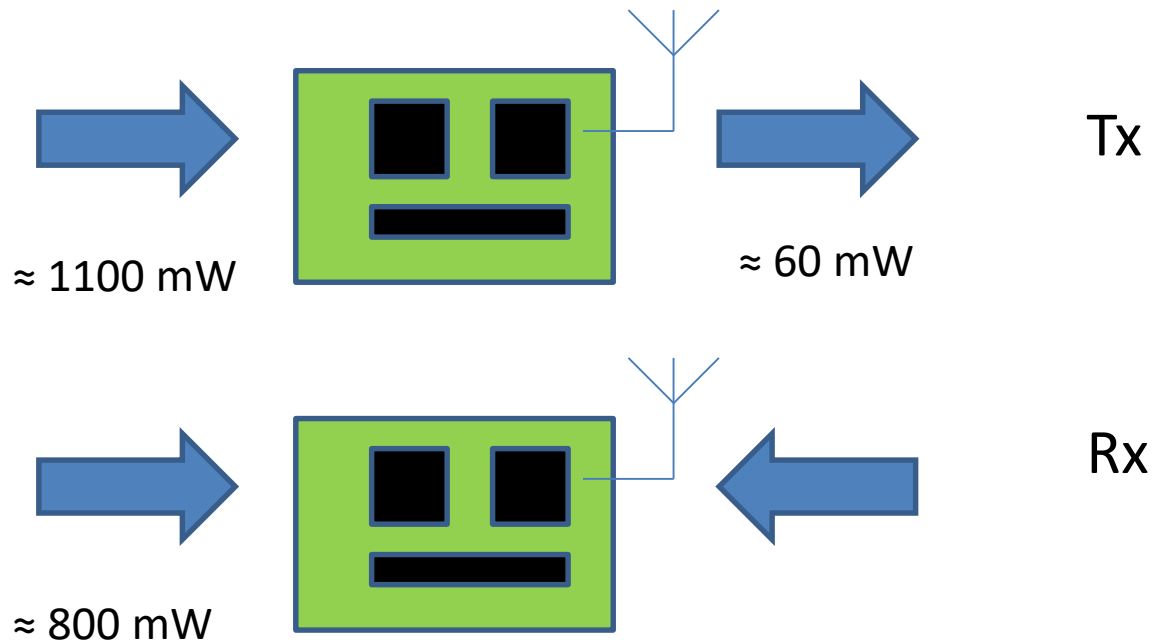
Power Consumption; Measured on Nokia n78



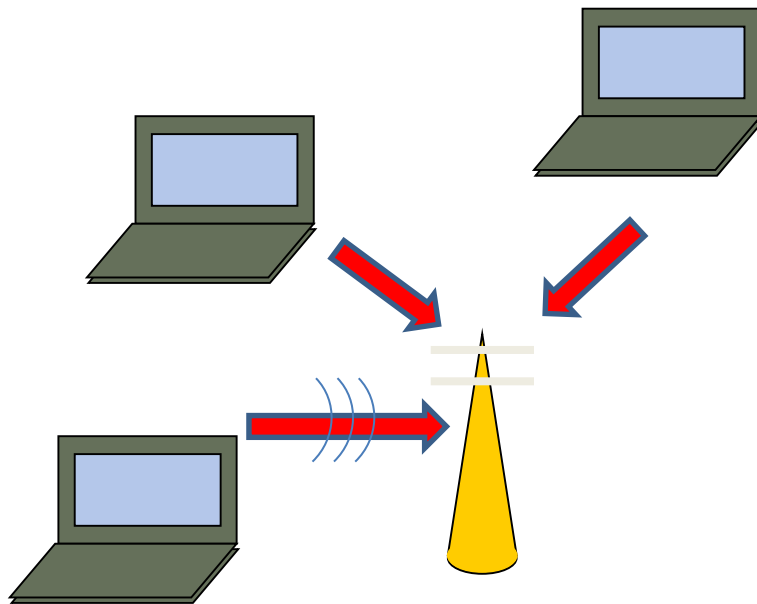
State	No screen	Only screen on	3G, Screen on	3G, ftp download Screen on	GPS Screen on	Bluetooth screen on
Power (W)	0,15	0,40	0,8	1,5	0,67	0,46

Component	system	screen	GPS	BT	Wifi (Rx)	WiFi (Tx)	3G (Rx/Tx)
Power (W)	0,15	0,25	0,27	0,06	0,75	1,1	1,1

Power consumption, e.g. WiFi

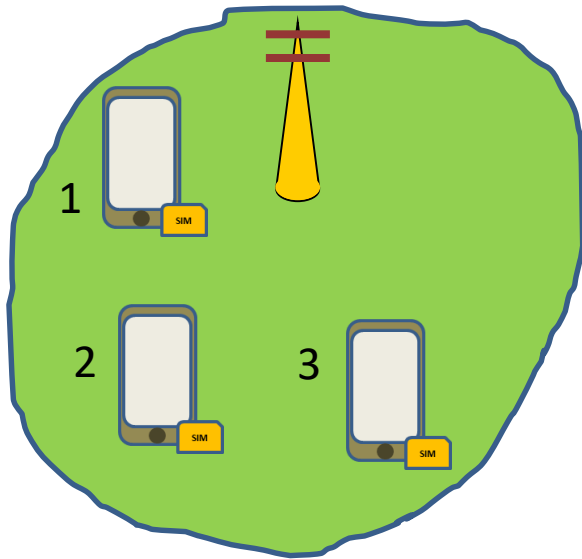


Multiple access protocols

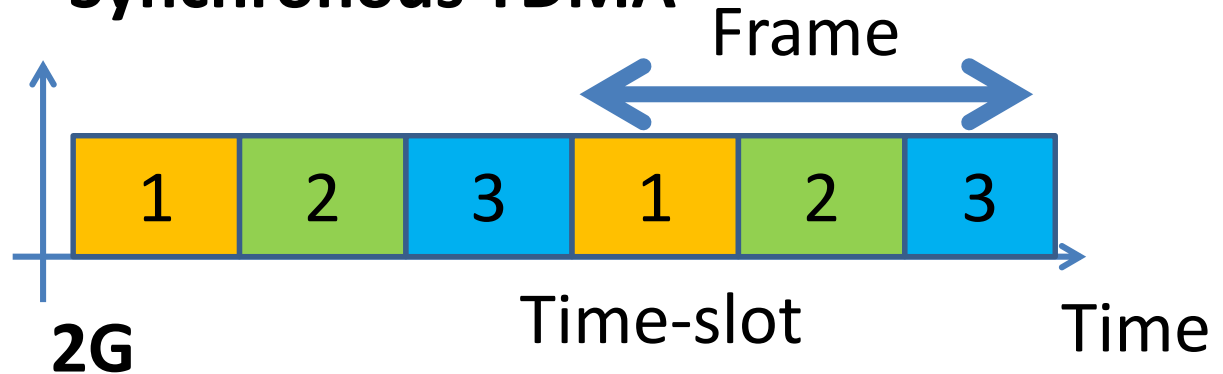


Simultaneous transmission at the same time, frequency and code, can lead to a collision

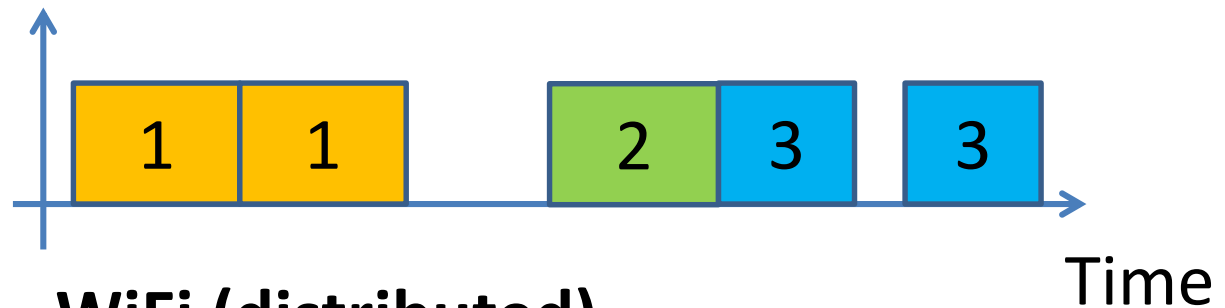
Time Division Multiple Access (TDMA)



Synchronous TDMA

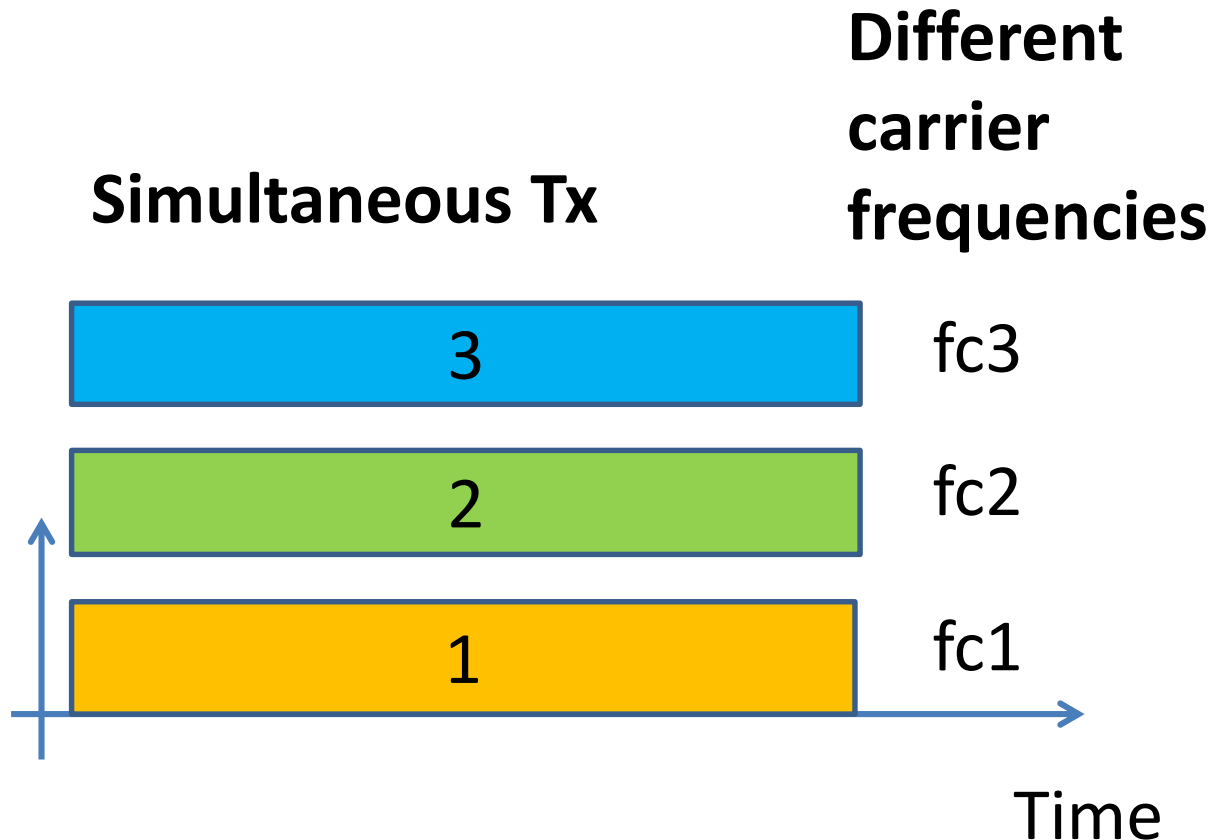
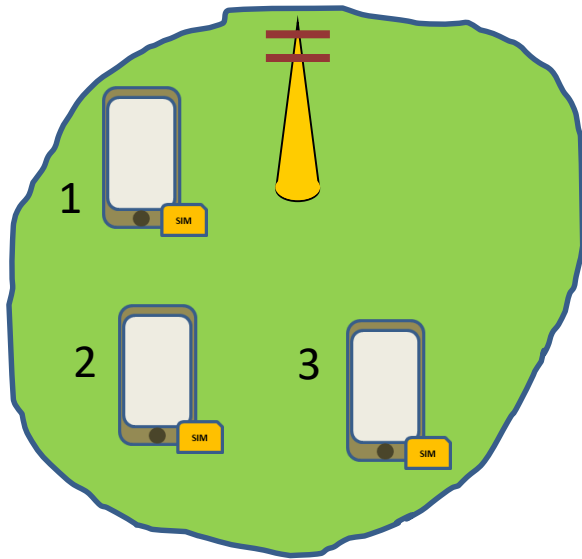


Statistical TDMA



WiFi (distributed),
2.5G (centralized)

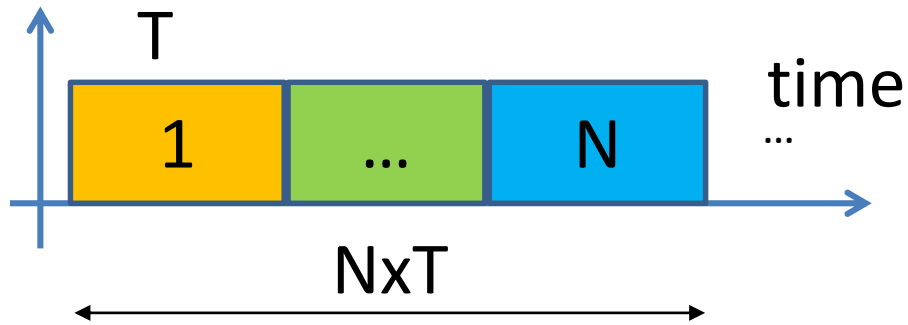
Frequency Division Multiple Access (FDMA)



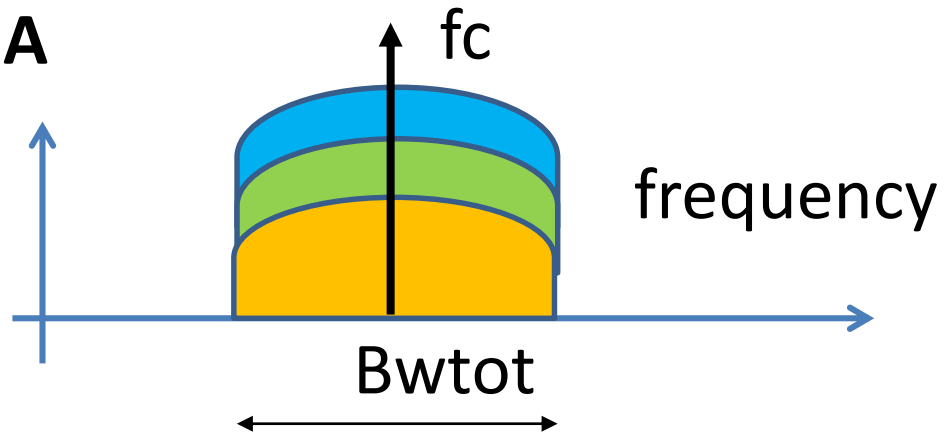
2G, 2.5G: FDMA+TDMA

4G: OFDMA, SC-FDMA

TDMA

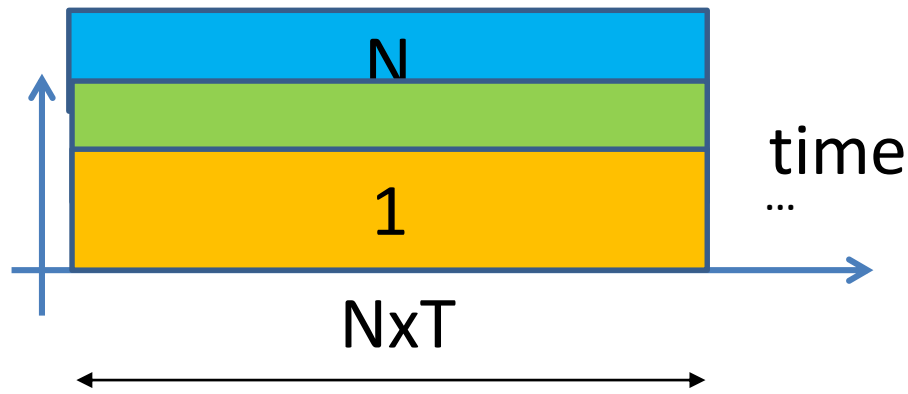


$$R_{\text{tot}} = N \times R = N(1/N \times T) = 1/T$$

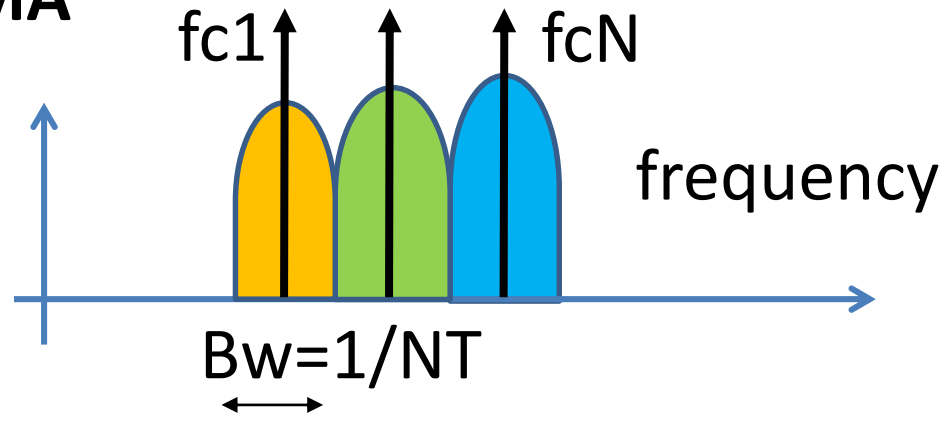


$$B_{\text{wtot}} = 1/T$$

FDMA

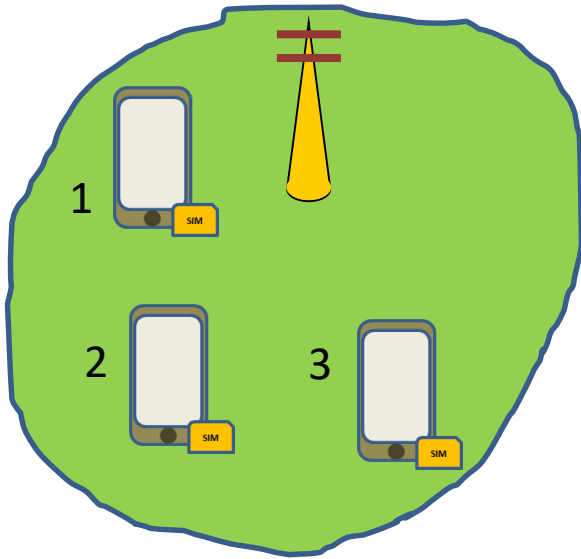


$$R_{\text{tot}} = N \times R = N(1/N \times T) = 1/T$$



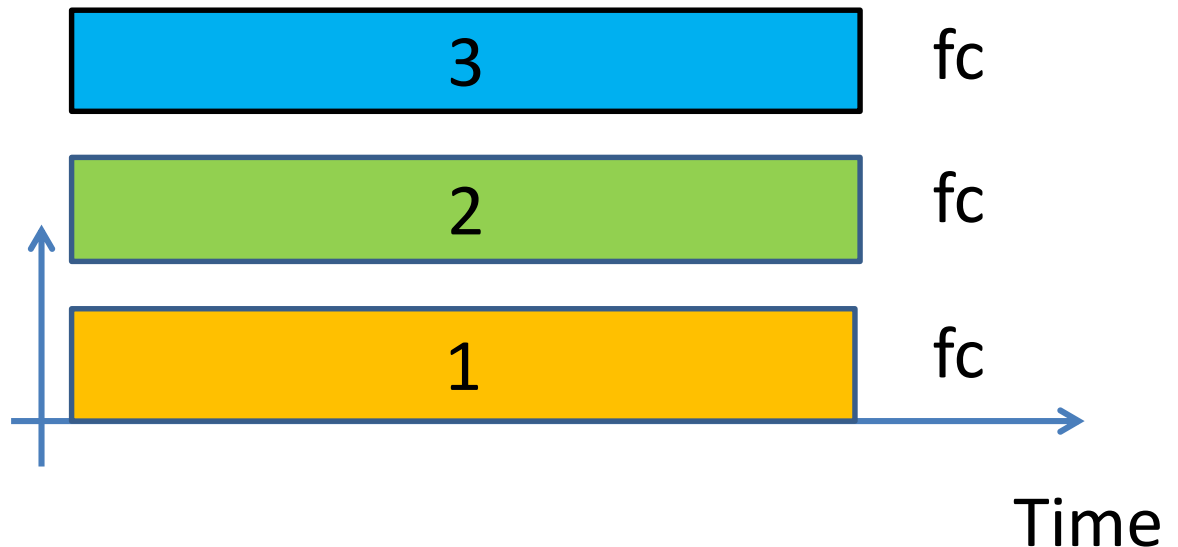
$$B_{\text{wtot}} = N \times (1/NT) = 1/T$$

Code Division Multiple Access (CDMA)

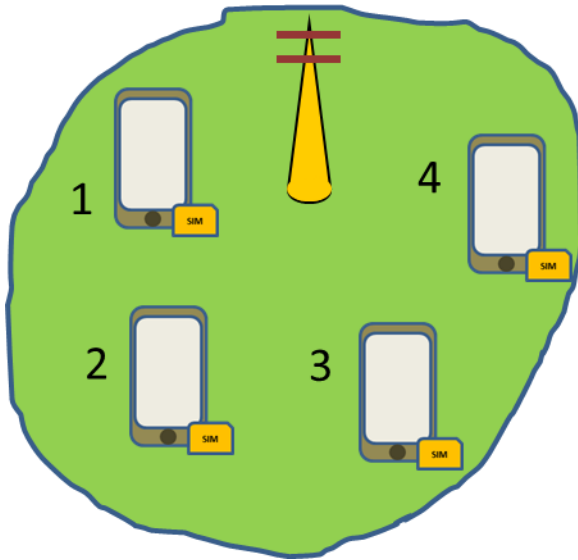


Orthogonal codes

**Same
carrier
frequencies**



3G: WCDMA



$$R = (2, 2, 2, -2)$$

$$R = s_1 c_1 + s_2 c_2 + s_3 c_3 + s_4 c_4$$

$$C_1 = (1, 1, 1, 1)$$

$$C_2 = (1, 1, -1, -1)$$

$$C_3 = (1, -1, 1, -1)$$

$$C_4 = (1, -1, -1, 1)$$

$$\langle R, c_3 \rangle = s_1 \langle c_1, c_3 \rangle + s_2 \langle c_2, c_3 \rangle + s_3 \langle c_3, c_3 \rangle + s_4 \langle c_4, c_3 \rangle$$

$$\langle R, c_3 \rangle = 2 - 2 + 2 + 2 = 4$$

$$\langle c_1, c_3 \rangle = 0 \quad \langle c_2, c_3 \rangle = 0 \quad \langle c_3, c_3 \rangle = 4 \quad \langle c_4, c_3 \rangle = 0$$

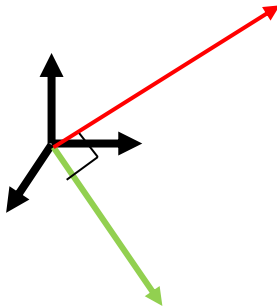
$$4 = 0 + 0 + s_3 4 + 0 \Rightarrow s_3 = 1$$

CDMA: scalar product

- Reminder

Assume $\mathbf{x}$ and $\mathbf{y}$ are two vectors of N components $\mathbf{x}=(x_1, x_2, \dots, x_N)$ and $\mathbf{y}=(y_1, y_2, \dots, y_N)$

Scalar product: $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_N y_N$



Fundamental properties:

- $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
- $\langle \mathbf{x}, a\mathbf{y} \rangle = \langle a\mathbf{x}, \mathbf{y} \rangle = a\langle \mathbf{x}, \mathbf{y} \rangle$
- If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, $\langle \mathbf{x}, \mathbf{y} \rangle = 0$

CDMA: scalar product

- E.g. Check that these 3 vectors are mutually orthogonal (we will call them “**codes**”)

c1=<1,1,1,1>, **c2**=<1,-1,1,-1>, **c3**=<1,1,-1,-1>

- Try to find **c4** which is orthogonal to **c1**, **c2**, and **c3**
(Note, as $N=4$, we cannot find more than N mutually orthogonal codes)

Scalar product, trick

- Check that $\mathbf{c4}=(1,-1,-1,1)$ is a solution of the question in previous slide
- Think on 4 numbers, a,b,c,d (restrict to values $+1$ or -1 , although the trick will work in general)
- Find now $\mathbf{x}=a\mathbf{c1}+b\mathbf{c2}+c\mathbf{c3}+d\mathbf{c4}$
- For instance, if we choose $a=1, b=-1, c=1, d=-1$, we have: $\mathbf{x} = 1\mathbf{c1}+(-1)\mathbf{c2}+1\mathbf{c3}+(-1)\mathbf{c4} = \mathbf{c1}-\mathbf{c2}+\mathbf{c3}-\mathbf{c4} = (0,4,0,0)$
- Question: how to find a,b,c and d if you only know $\mathbf{x}$?
- E.g. find a,b,c , and d if $\mathbf{x}=(0,0,0,-4)$

Answer

If you can find $a, b, c,$ and d solving a 4×4 linear system of equation, but we can use a trick that makes things much easier:
find the scalar product between one of the codes and the vector x :

$$x = ac_1 + bc_2 + cc_3 + dc_4$$

$$\begin{aligned} \langle c_1, x \rangle &= a\langle c_1, c_1 \rangle + b\langle c_1, c_2 \rangle + c\langle c_1, c_3 \rangle + d\langle c_1, c_4 \rangle = \\ &a\langle c_1, c_1 \rangle + 0 + 0 + 0 = a\langle c_1, c_1 \rangle \end{aligned}$$

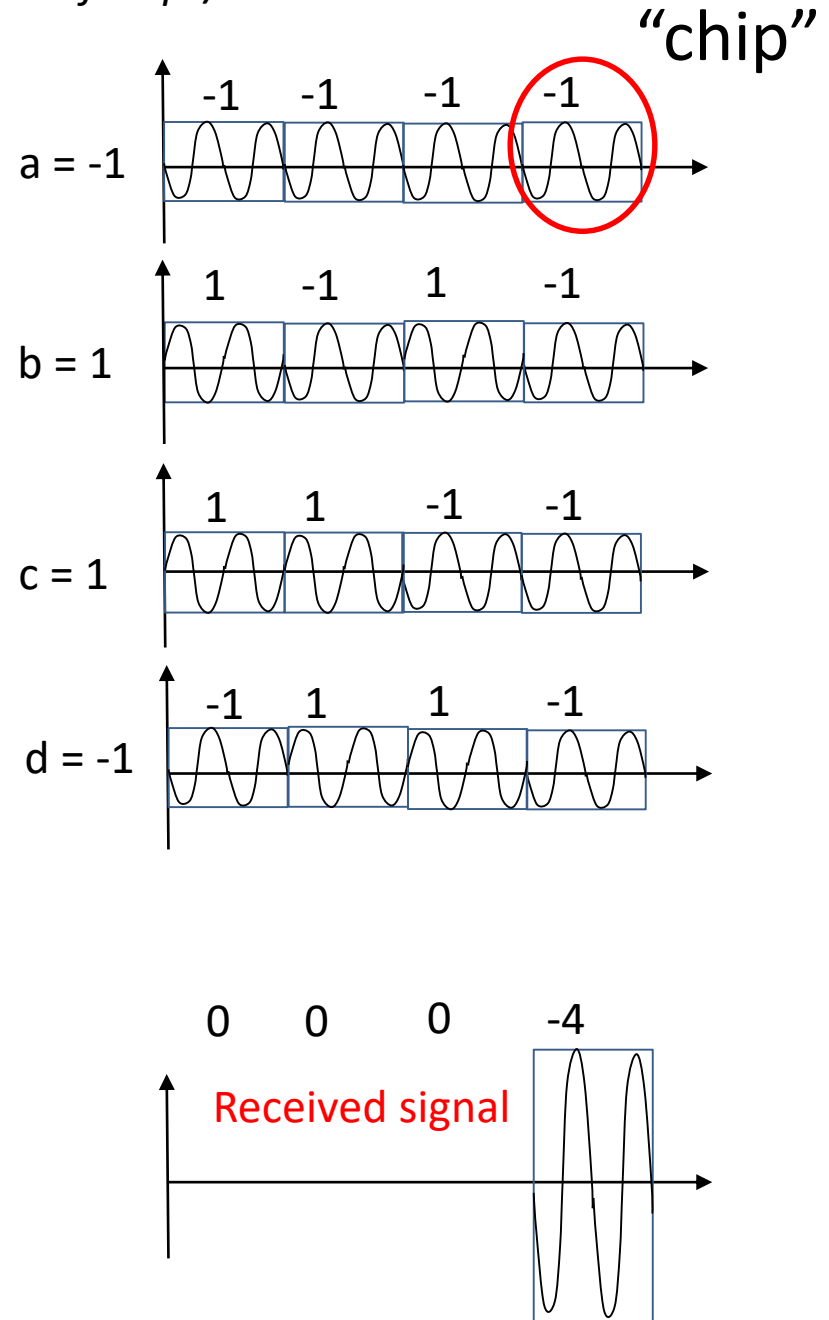
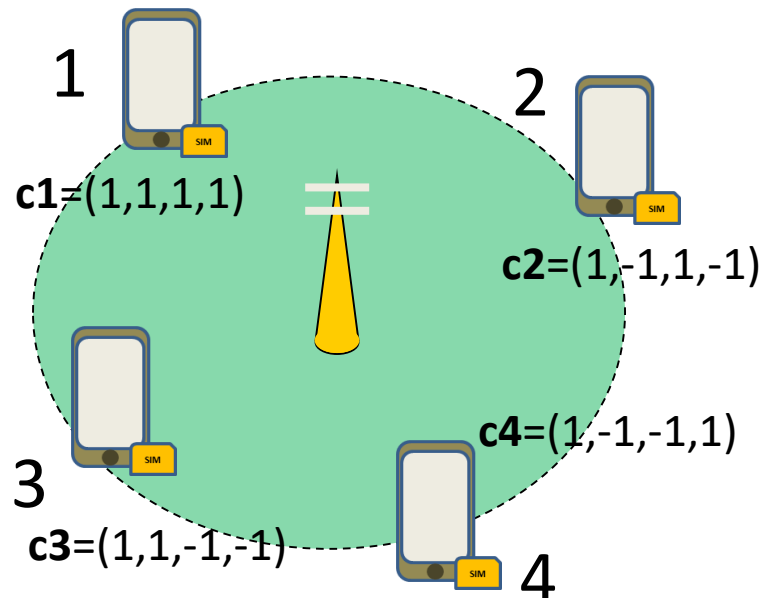
$$\text{e.g. } x = (0, 0, 0, -4) \Rightarrow \langle c_1, x \rangle = -4, \langle c_1, c_1 \rangle = 4 \Rightarrow a = -1$$

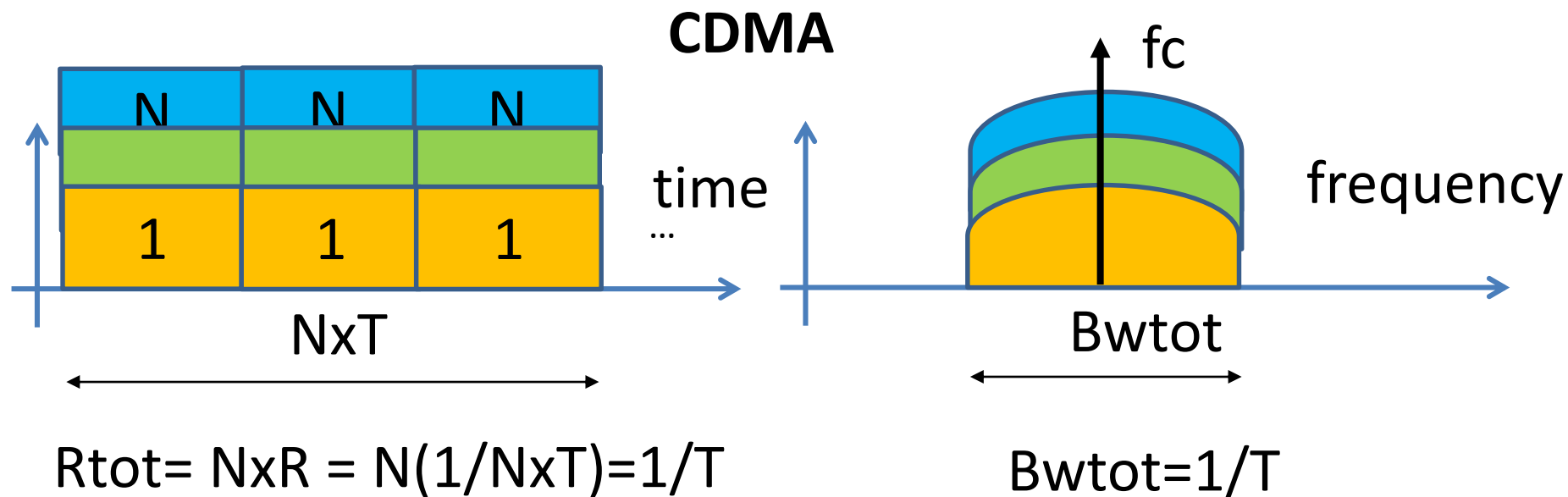
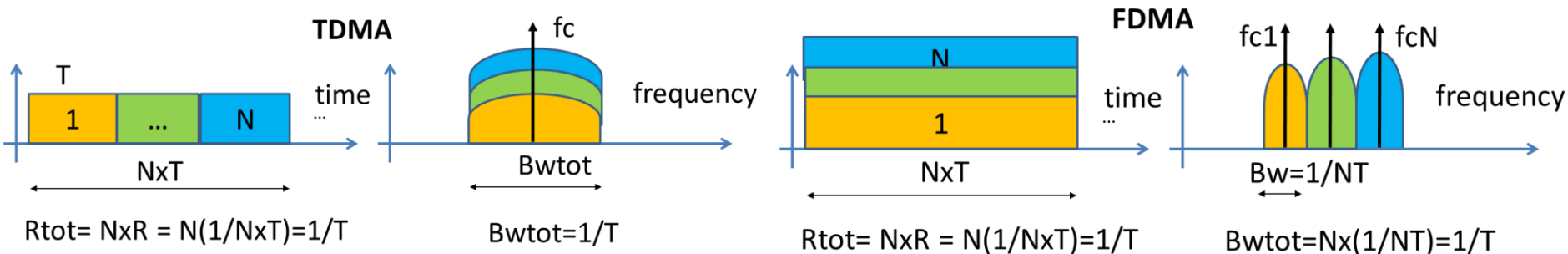
Correct! The same trick can be used to find b, c and d if we multiply x by c_2, c_3 and c_4 . Check this, you should obtain $b=1, c=1, d=-1$

$$R_c \text{ (Chip rate)} = \text{number of chips/second} = N \times R$$

CDMA

This trick can be used by a base station in a multiaccess channel, assuming it knows the codes used by each user:

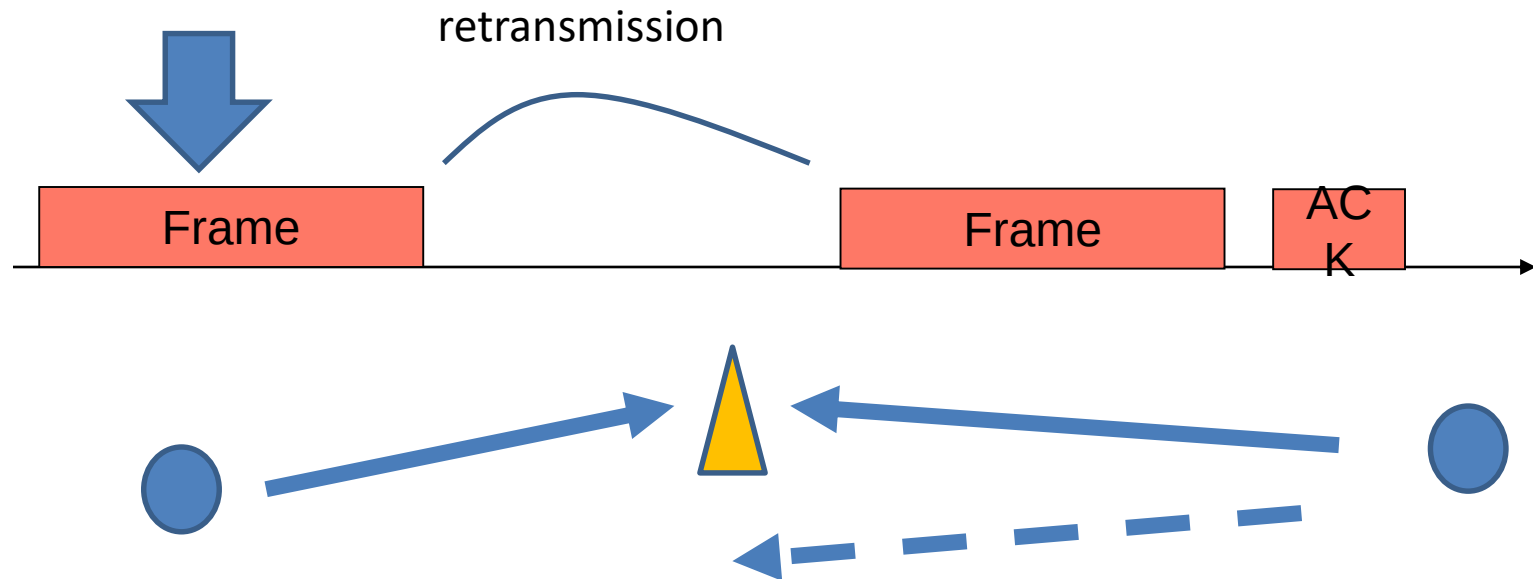




Access protocols Aloha (e.g. LoraWAN class A device, RFID, RACH)

The node sends a packet and wait for an ACK. IN case no ACK is received, it waits a random time and retransmits the frame

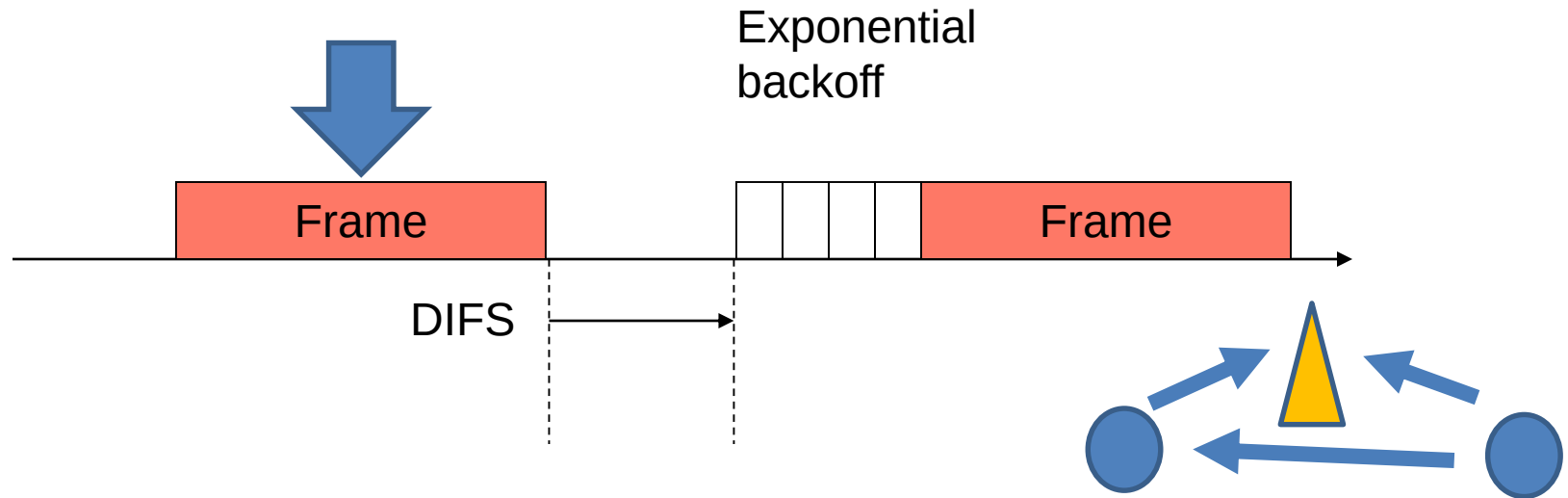
NO! Carrier sense (CSMA), simple!



Access protocols CSMA/CA (e.g. 802.11, 802.15.4)

If the medium is busy, the station must wait until it becomes idle for longer than a DIFS, and then it performs an exponential backoff...

Carrier sense (CSMA)

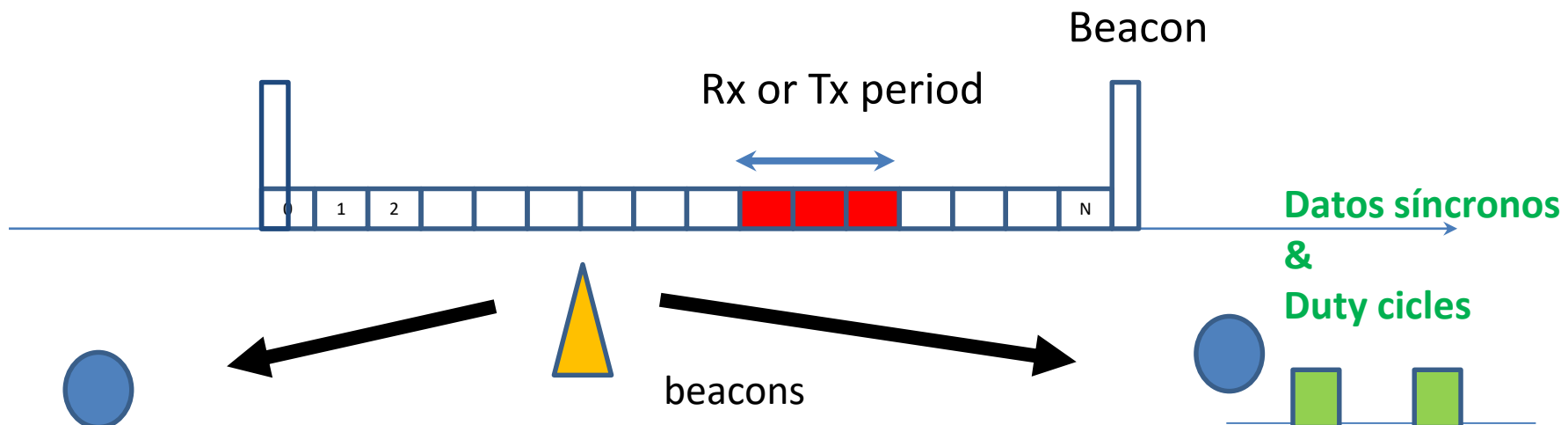


In the first attempt, the Contention Window (CW) measures CW_{min} (e.g. 7 slots).
In case of collision, in attempt n -th, the CW is set to $CW_{min} \times 2^{n-1} - 1$, until reaching a maximum value of CW_{max} ...

Scheduled transmission (Beacons)

2G, LoraWan Class B, 802.15.4)

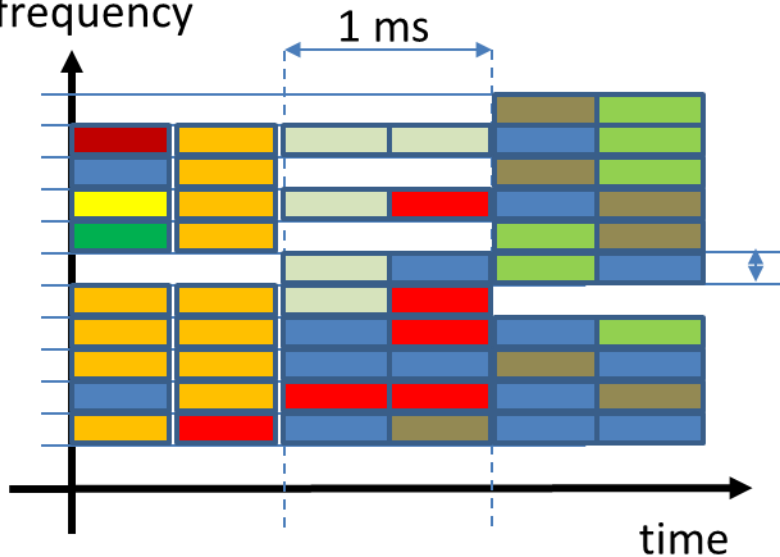
- A dedicated network coordinator transmits beacons in predetermined intervals
- All nodes switch awake/sleep periods (net nodes need to receive periodic beacons)



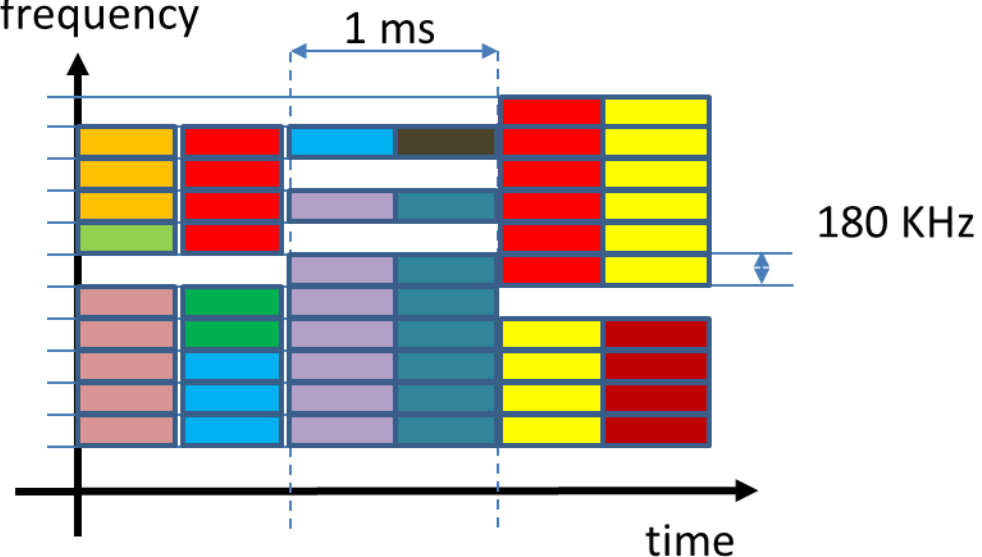
Frequency diversity frames (2G, 4G, 15.4 TSCH)

4G

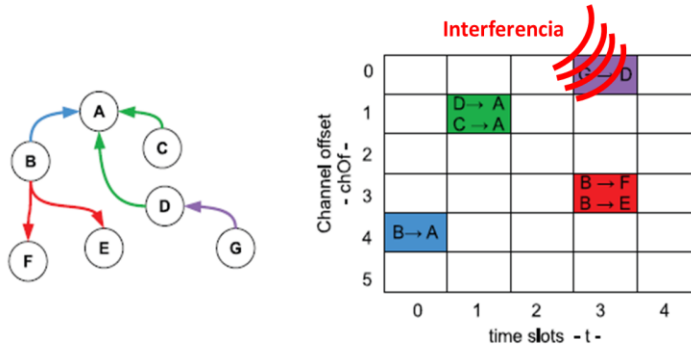
Downlink
frequency



Uplink
frequency

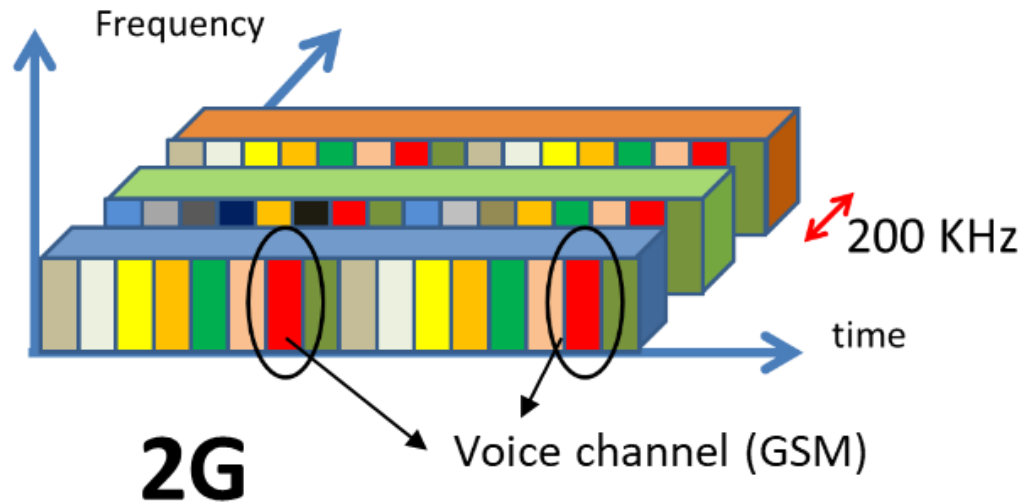
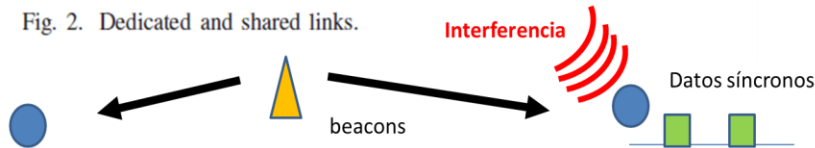


Frequency diversity frames (2G, 4G, 15.4 TSCH)



IEEE 802.15.4 TSCH

Fig. 2. Dedicated and shared links.



CDMA (3G)

